

In some cases, because of a recent evaluation for a previous project, the documentation may include verification that no major changes, detrimental to the maintenance program, have occurred since the previous evaluation, and that the municipality is fulfilling the obligations of its existing signal maintenance agreement(s).

4.9.5 Quality Assurance Review Form

I. Evaluation Process Review

A. District Procedures

1. Checklist to municipality by letter (Reference Publication 191)
2. Analysis of returned checklist
3. Field evaluation
4. Interview/meeting
5. Consider effect of new equipment and required changes to municipality's maintenance program
6. Recommendations discussed with municipality

B. District Records

1. Completed checklist
2. Narrative of field review
3. Minutes of interview/meeting
4. List of action items

II. Signal Maintenance Agreement Review

A. Completed properly

1. Prior to final design
2. Format correct
3. Item #4 completed
4. Exhibit D: Personnel requirements and training – completed

5. Contractor's name, address, and its maintenance contract with municipality
6. Signatures and other information completed on signature page
7. Signature resolution by municipality authorizing the signees to execute the agreement, original copy required

B. Modification to Agreement

III. Q/A Inspection (field) of Municipality's Maintenance Effort

- A. Signal lenses clean, colors correct, and light output acceptable
- B. Signal heads aligned properly
- C. Sensors maintained and operating (including pushbuttons)
- D. Controller operation satisfactory

IV. District Comments on Guidelines, Evaluation Process, Agreements, and Municipality's Reaction

4.10 Removal of Traffic Signals

4.10.1 General

At intersections, traffic control devices regulate the flow of conflicting traffic streams. Since traffic signals provide the strongest form of at-grade intersection control, the public has sometimes erroneously assumed that traffic signals are a panacea for intersection operational or safety concerns. Therefore, due to lack of transportation engineering expertise, or public and political pressure, or both, unwarranted traffic control signals may exist at intersections in some communities. Additionally, at other intersections, changing traffic patterns may have caused signals that were originally warranted to no longer meet warrants. Moreover, the removal of a traffic signal often involves political and institutional considerations as well as technical factors. In keeping with these practical realities, the approach to signal removal justification is a sequential screening process in which a series of criteria must all be satisfied, and the various impacts predicted, before removing a signal.

Knowing the probable impacts on intersection safety, traffic operations (e.g., delay, stops, and fuel consumption), and costs, the traffic engineer can make a sound technical decision on the question of signal removal. Because the removal of a traffic signal often involves institutional and/or political constraints, temper the technical element with professional judgment when making the final decision. The decision process allows for including these institutional and political considerations.

For a complete explanation of the removal process, refer to FHWA IP-80-2, Users Guide for Removal of Not Needed Traffic Signals, November 1980.

When the Department initiates the process to remove any traffic signal, no one shall instruct the municipality to begin the removal process until the Secretary of Transportation reviews the study data and approves the removal of the signal. Therefore, when the District Traffic Unit is recommending the removal of a signal, as a minimum, submit the following study data and analysis to the Bureau of Highway Safety and Traffic Engineering for review and submission to the Secretary of Transportation:

- Intersection geometrics
- A 24-hour vehicle volume summary
- Peak hour turning movement counts
- Pedestrian volumes
- Side street sight distance data
- Crash experience and impact analysis
- Completed Traffic Signal Report Form (from Publication 149)

- Special site conditions

Requests to remove traffic signals that originate from the local authorities do not need the Secretary's approval, but District Traffic Units should follow/consult the FHWA IP-80-2 report.

As noted in [Section 4.3.3.1\(h\)](#), State and federal funds may be used to remove unwarranted traffic signals.

4.11 Certificates of Approval

4.11.1 General

The Department maintains a program to approve power-operated traffic control devices. It requires testing the equipment under field conditions to evaluate the functional and operational features of the equipment at an intersection. Although the Department supervises these tests, the tests are a cooperative effort between the municipality, the manufacturer, and the Department.

All data concerning equipment approved by the Department is computerized and is available to District personnel. Publication 35 also lists approved equipment.

4.11.2 Application Procedure

Manufacturers that wish to have a device considered for possible approval must submit a completed application to BHSTE. All completed applications must be accompanied by an operation or instruction manual, a copy of specifications for the product, and 15 copies of descriptive literature for the device. In addition, the manufacturer must submit independent laboratory test reports to show that the product complies with applicable Department specifications, NEMA Standards, and other requirements.

Since the Department is not equipped to test power-operated traffic control devices, the equipment must be installed under actual field conditions to permit functional and operational testing under the Department's supervision. Prior to this testing, a Provisional Certificate of Approval will be issued to the manufacturer, provided the equipment appears to perform the functions and operation required by policies and regulations. The provisional approval will be in effect for a period of at least 6 months, but not to exceed 18 months, after initial installation of the equipment to permit the necessary evaluation and testing of the equipment. This approval will permit the sale of the equipment to any municipality in Pennsylvania provided the municipality has been advised in advance that the equipment has provisional approval and the manufacturer or its representative has first obtained concurrence of the ITS and Congestion Management Division, Department of Transportation, for each installation of such equipment.

By accepting the Provisional Certificate of Approval, the manufacturer agrees to replace all equipment installed under that provisional approval should the Department or the municipality determine that its operation is unacceptable. The replacement equipment shall be approved equipment that is acceptable to the Department and the municipality, and may be from another manufacturer. The manufacturer or its

representative shall keep an accurate record of all maintenance and repairs to the equipment during the period of provisional approval. If the equipment operates satisfactorily for the prescribed period, the Department will issue a Sale Certificate of Approval.

Pennsylvania Department of Transportation



pennsylvania
DEPARTMENT OF TRANSPORTATION

**Publication 46
Chapter 5**

Low-Volume Roads

Pennsylvania Department of Transportation
Chapter 5 of Publication 46
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5.1 General

5.1.1 Purpose

Part 5 of the *MUTCD* is entitled “*Traffic Control Devices for Low Volume Roads*,” and it provides for a reduction in the number and size of traffic-control devices on “*low-volume roads*.” In general, these roadways are farm-to-market roads, recreational roads, resource management and development roads, and local roads.

Moreover, Section 5A.01 of the *MUTCD*, which defines “*low-volume road*,” as follows, precludes State highways from being a low-volume road:

- (a) A facility lying outside of built-up areas of cities, towns, and communities, with a traffic volume less than 400 AADT.
- (b) Not a freeway, expressway, interchange ramp, freeway service road, or a roadway on a State highway system.
- (c) Classified as either a paved or an unpaved road.

Therefore, some municipal roads may satisfy the definition of “low-volume road” and thereby use fewer and smaller traffic-control devices, but this definition never applies to State highways.

5.1.2 Laws, Regulations and Other Publications

Manual on Uniform Traffic Control Devices (MUTCD), FHWA. Specifically, Part 5 addresses low-volume roads, and it is available at <http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/Ch5.pdf>.

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Temporary Traffic Control

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6.1 General

6.1.1 General

The *Manual on Uniform Traffic Control Devices (MUTCD)* defines the term “temporary traffic control” as:

“Temporary Traffic Control Zone—an area of a highway where road user conditions are changed because of a work zone or incident by the use of temporary traffic control devices, flaggers, uniformed law enforcement officers, or other authorized personnel.”

Most Department employees refer to this term as “work zone traffic control,” which is the most common type of temporary traffic control.

This chapter provides internal policy for the Department relative to temporary traffic control, supplementing the following other documents in general order of importance:

1. Part 6 of FHWA’s *Manual on Uniform Traffic Control Devices (MUTCD)*, entitled “Temporary Traffic Control.”
2. Subchapter E of 67 Pa. Code, Chapter 212, entitled “Temporary Traffic Control.”
3. Publication 213, entitled “Work Zone Traffic Control Guidelines.”

6.1.2 Laws, Regulations, and Publications

- Approved Construction Materials (Publication 35).
- Authorizing Appropriately Attired Persons to Direct, Control, or Regulate Traffic (67 Pa. Code, Chapter 101).
- Engineering and Traffic Studies (67 Pa. Code Chapter 212, and available as Publication 212 with an Appendix).
- Flagging Handbook (Publication 234).
- Handbook of Approved Signs (Publication 236M).
- *Manual on Uniform Traffic Control Devices (MUTCD)*.
- Specifications (Publication 408).
- Traffic Control Signing Standards, TC-8700 Series (Publication 111M).
- Traffic Control Standards – Pavement Markings, RPMs & Delineators, TC-8600 Series.
- The Vehicle Code (75 Pa. C.S.).
- Work Zone Traffic Control (Publication 213).

6.1.3 Definitions

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

ADT – Average daily traffic – The total volume of traffic during a number of whole days, more than 1 day and less than 1 year, divided by the number of days in that period.

Active work zone – The portion of a work zone where construction, maintenance or utility workers are on the roadway or on the shoulder of the highway, and workers are adjacent to an active travel lane. Workers are not considered adjacent to an open travel lane if they are protected by a traffic barrier and no ingress or egress to the work zone exists through an opening in the traffic barrier.

Advisory speed – The recommended speed for vehicles operating on a section of highway based on the highway design, operating characteristics and conditions. When posted, the speed is displayed as a warning sign; that is, either a black-on-yellow or a black-on-orange sign.

Delineator – A retroreflective device mounted on the road surface or at the side of the roadway in a series to indicate the alignment of the roadway, especially at night or in adverse weather.

Department – The Department of Transportation of the Commonwealth.

Divided highway – A highway divided into two or more roadways and so constructed as to impede vehicular traffic between the roadways by providing an intervening space, physical barrier or clearly indicated dividing section.

Expressway – A divided arterial highway for through traffic with partial control of access and generally with grade separations at major intersections.

Freeway – A limited access highway to which the only means of ingress and egress is by interchange ramps.

MUTCD – The current edition of the Manual on Uniform Traffic Control Devices, as adopted by the Federal Highway Administration (FHWA), and available on the FHWA website. Specifically, Part 6 addresses temporary traffic control.

Night, nighttime – The time from 1/2 hour after sunset to 1/2 hour before sunrise.

Normal speed limit – The regulatory speed limit or the 85th percentile speed that existed before temporary traffic control was established, for example, prior to the beginning of a work zone.

Roadway – That portion of a highway improved, designed or ordinarily used for vehicular travel, exclusive of the sidewalk, berm or shoulder. If a highway includes two or more separate roadways, the term refers to each roadway separately but not to all roadways collectively.

Shadow vehicle – A vehicle positioned in advance of a work area or work vehicle to provide advance information to approaching drivers or protection for the workers or work vehicle.

State highway – A highway or bridge on the system of highways and bridges over which the Department has assumed or has been legislatively given jurisdiction.

TTC, Temporary traffic control – An area of a highway where road user conditions are changed because of a work zone or incident by use of temporary traffic-control devices, flaggers, police officers or other authorized personnel.

TTC plan – A plan for maintaining traffic through or around a work zone.

Work zone – The area of a highway where construction, maintenance or utility work activities are being conducted, and in which traffic-control devices are required in accordance with this chapter.

6.2 Safety

6.2.1 Safety First

Safety in work zones is the Department's highest priority. Therefore, discuss protection of the traveling public and the worker at all meetings with construction, maintenance, and permit/utility personnel.

All traffic control devices used in a work zone shall be clean and in good condition, and all signs must be legible. On Department operations, do not permit employees to work without proper flagging as required in the *MUTCD*.

Moreover, in accordance with 67 Pa. Code §212.412 (relating to flagger signaling devices), the red flag may only be used as follows:

“A red flag shall only be used to control traffic in emergencies when a STOP/SLOW Paddle (R21-10) is not available or at intersections when a flagger is used within the intersection.”

Before assigning any new employees as a flagger, including summer help, it is mandatory that they first complete the flagger-training course available at [P:\penndot shared\Flagger Safety](#) Training latest update.

6.2.2 Pennsylvania State Police (PSP) Cooperation

The State Police Administration has indicated they will cooperate with the Department and provide police assistance for Department maintenance operations on high-volume, high-speed highways within the limits of their manpower.

Give as much advance notice as possible (at least 2 weeks) to the State Police when requests are made for their assistance. It is equally important to notify the State Police of any changes in maintenance schedules if police assistance has been requested for the operation. More information on police assistance is available in [Subchapter 6.15](#) of this publication.

6.2.3 Work on Limited Access Highways

The Department has advised AFSCME that in consideration of employee safety, it is the Department's intention to avoid assigning employees to limited access highways during periods when the Department anticipates unusually high traffic volumes.

Specifically, the Department will attempt to avoid working on all four-lane limited access highways on Fridays and the day before the holidays listed in Article 10 of the Master Agreement. In addition, this policy may be applied at other times and on

other highways when specific local circumstances create an increased risk to employee safety. However, this general policy shall not prohibit work during emergencies.

6.3 Work Zone Safety and Mobility

6.3.1 Work Zone Safety and Mobility Policy (WZSM)

The Department is responsible for designing and maintaining work zones that limit the impact to motorists, while maintaining safety for highway workers and the traveling public.

6.3.1.1 Policy Scope

This policy applies to all projects that receive Federal and/or State funds and the following conditions apply:

- Project is located on any of these fully-controlled, limited access highways: interstate, PA Turnpike, or freeway (interstate look-a-like),

and
- Project occupies a location for more than 3 days with either intermittent or continuous lane closures.

Also, any project meeting the above conditions, along with additional travel times greater than 20 minutes for two consecutive hours, is deemed as a “Significant Project” and must meet further criteria as defined later in this policy.

These conditions meet the “sustained work zone impacts” intent referenced in FHWA’s Work Zone Safety & Mobility Final Rule (23 CFR 630 Subpart J). While the Final Rule centers around interstate projects within the boundaries of a Transportation Management Area (TMA), this policy simplifies the process and encompasses all interstate/freeway projects regardless of being within a TMA or not.

6.3.1.2 Policy Objectives

The intent of this policy is to provide safe work zones that minimize delay for the traveling public and reduce community impacts while maintaining fiscal responsibility. This policy also is intended to bring the Department into compliance with FHWA’s Work Zone Safety and Mobility Final Rule (23 CFR 630 Subpart J).

The goals of this policy are to:

- Establish a systematic approach for improving work zone safety and mobility that will include the consideration of transportation management and congestion mitigation from the project planning stage throughout the life of the project.
- Expand work zone management beyond typical traffic safety and control within the project footprint to address corridor, network, and regional issues including day-to-day operations and the effective dissemination of public information.

6.3.1.3 Responsible Organizations

The Bureau of Highway Safety and Traffic Engineering (BHSTE) and the Bureau of Design (BOD) are responsible for administering this policy for PennDOT. This policy is cross-cutting with all organizations involved in the planning, design, and construction of highway projects. Organizations responsible for implementing this policy include:

- Bureau of Highway Safety and Traffic Engineering
- Bureau of Design
- District Offices
- Bureau of Construction and Materials
- Bureau of Maintenance & Operations
- Center for Program Development and Management

6.3.1.4 Program Management

BHSTE will monitor the effectiveness of this work zone policy through Independent Oversight Program (IOP) process reviews, an assessment of work zone performance measures, and coordination with District Traffic Engineers.

BHSTE will serve as the lead Bureau for the PennDOT Work Zone Safety and Mobility Committee which will include membership from all Central Office Engineering Bureaus, District representation, and the FHWA. The Committee shall be responsible for reviewing the effectiveness of this policy and making changes as necessary.

6.3.2 WZSM Implementation

This section provides direction for implementing the requirements of PennDOT's Work Zone Safety and Mobility Policy.

6.3.2.1 Objectives & Goals

PennDOT's Work Zone Safety and Mobility Policy includes six broad objectives:

- Develop and implement systematic procedures to assess work zone impacts during planning and project development, and effectively manage safety and mobility during project implementation.
- Mitigate traffic and mobility impacts for the project area and for the surrounding network.
- Use field observations, available work zone crash data, and operational information to mitigate work zone impacts for specific projects during implementation. Continually pursue improvement of work zone safety and mobility by analyzing work zone crash and operational data from multiple projects to improve processes and procedures.
- Train personnel involved in the development, design, implementation, operation, inspection, and enforcement of work zone related transportation management and traffic control in a manner appropriate to the job decisions each individual is required to make. Update the training periodically to reflect changing industry practices, processes, procedures, and technologies.
- Perform FHWA/PennDOT Independent Oversight Review of work zones to evaluate the effectiveness of this policy and work zone management on projects.
- Provide guidance in the development and implementation of Transportation Management Plans (TMP) on "Significant Projects".

PennDOT's Work Zone Safety and Mobility Policy includes three goals:

- Strive to keep work zone-related additional travel delays under 20 minutes when compared to the base condition.
- 10% annual reduction in work zone-related fatalities & crashes (current year vs. last 5-year running average).
- Strive to implement appropriate level TMP elements on all projects listed in [Section 6.3.1.1](#), regardless of whether a formal TMP submission is required.

6.3.2.2 Determination of Significant Projects

All highway construction projects defined in [6.3.1.1](#) shall be considered as Significant Projects unless a work zone analysis shows that added travel delays do not exceed 20 minutes for 2 consecutive hours.

Refer to the following **Exhibits 6.3-A, 6.3-B & 6.3-C** for more information.

Exhibit 6.3-A Determination of Significant Projects - Flow Chart

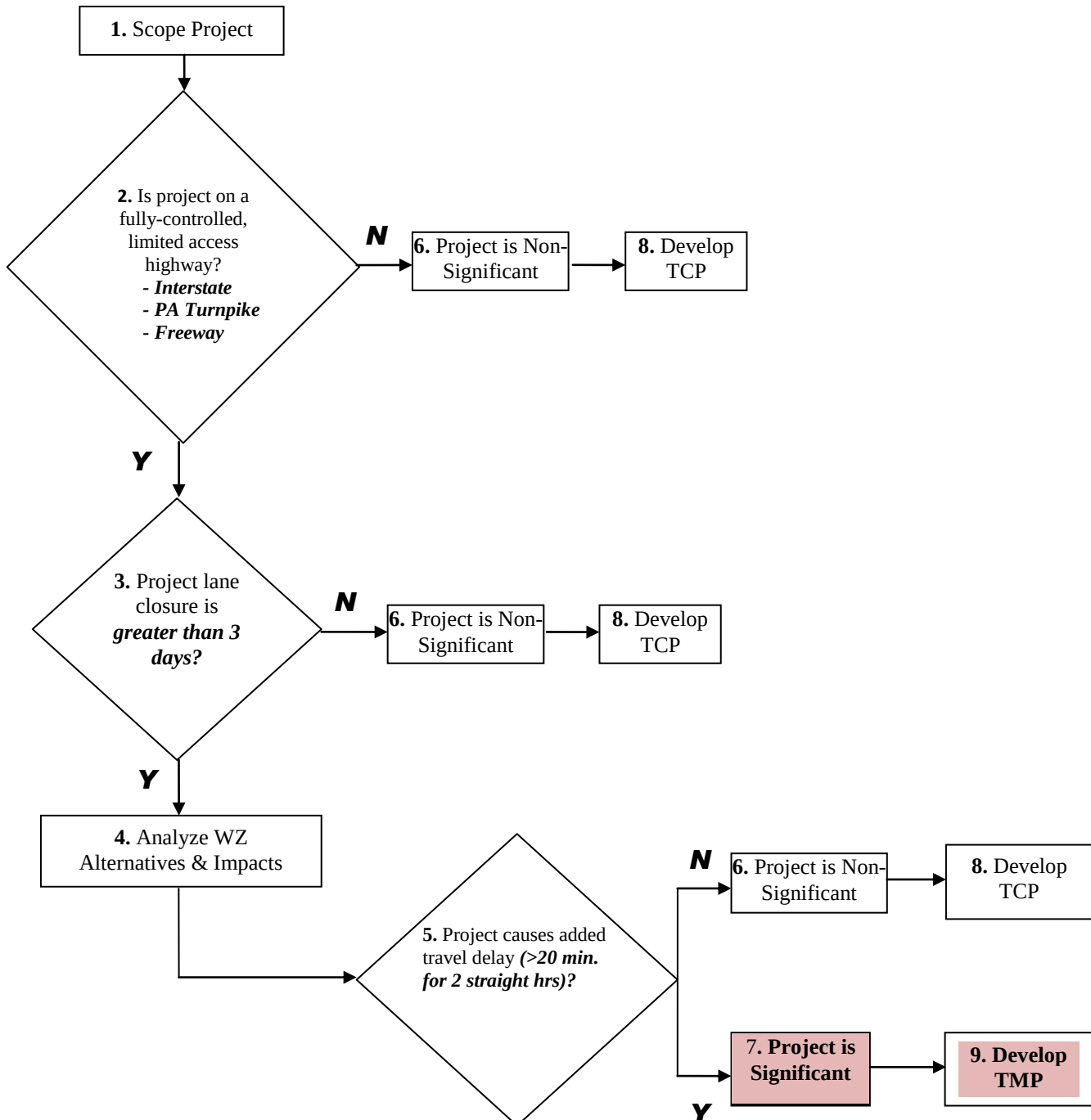


Exhibit 6.3-B Determination of Significant Projects - Flow Chart Step Descriptions

Step	Process Title	Comments
1	Scope Project	Refer to Design Manual (DM) 1A, Chapter 5
2	Project Location	Is the project located on an interstate (including the PA Turnpike) or a freeway with fully-controlled, limited access?
3	Project lane Closures	Will the project have either intermittent or continuous lane closures (including detours) for more than 3 days?
4	WZ Alternatives Analysis & Impacts	<u>WZ Alternative Analysis</u> - identify potential, practical (viable) WZ alternative setups which fit the nature of the project and will minimize traffic impacts to motorists (could be 1 or more). - Conduct WZ impacts analysis for these viable WZ alternatives as indicated below. <u>WZ Impacts</u> - use Delay Analysis Workbook (DAWB), QuickZone, Synchro, HCS or other computer modeling programs. - Note: prior analysis done for a recent similar project (≤ 3 yrs old) may be used to estimate impacts in lieu of new model development, if project characteristics are similar (WZ setups, location, traffic volumes, grades, access, etc.); if this prior analysis & impacts meets acceptable delays, there is no need for additional impact analysis.
5	Traffic Delay Thresholds	Acceptable Project Delay Impacts (Project is Non-Significant) - Additional (project-related) travel time through the project area (includes detours) is ≤ 20 minutes. Unacceptable Project Delay Impacts (Project is Significant) - Additional (project-related) travel time through the project area (includes detours) is > 20 minutes for time periods of 2 or more consecutive hours. <i>Note: Added travel time shall be determined as per Step 4 (under WZ Impacts).</i>
6	Non-Significant Project	Project does not meet policy parameters for being a Significant Project <u>For selected WZ approach</u>, request approval as a Non-Significant Project. (See Exhibit 6.3-C). - Traffic delay analysis and preliminary project mitigation strategies (for the selected WZ approach) must be submitted & approved as part of the Non-Significant determination process.
7	Significant Project	Project meets policy parameters for being a Significant Project <u>For selected WZ approach</u>, request approval as a Significant Project. (See Exhibit 6.3-C). - Work zone alternative analysis and preliminary project mitigation strategies (for the selected WZ approach) must be submitted & approved as part of the Significant Project determination process.
8	Develop TCP	Must develop and complete a Traffic Control Plan (TCP) for the approved WZ approach. - While a formal TMP submission isn't required, appropriate-level TMP elements need considered on these Non-Significant Projects as well. - Consider the development of an appropriate level TMP for the project, even though not required by policy.
9	Develop TMP	Must develop and complete a Transportation Management Plan (TMP) for the approved WZ approach. Includes a Traffic Control Plan (TCP), Transportation Operations Plan (TOP) and Public Information Plan (PIP). (See Section 6.3.3)

Exhibit 6.3-C Determination of Significant Project – Submission Approval Process

Approval Process	PennDOT Oversight (PennDOT has approval authority)	Federal Oversight (FHWA has approval authority)
Non-Significant Project Determination	Step 6 • District Executive must approve the traffic delay analysis & prelim. Project mitigation strategies. • BOD receives a copy of the DE approval along with the delay analysis.	Step 6 • BOD must review/concur with the traffic delay analysis & prelim. Project mitigation strategies. • FHWA must approve.
Significant Project Determination	Step 7 • District Executive must review/concur with the work zone alternatives analysis & prelim. project mitigation strategies. • BOD must approve.	Step 7 • BOD must review/concur with the work zone alternatives analysis & prelim. project mitigation strategies • FHWA must approve.

Note: Significant Projects approval submissions should include the following types of information:

- List of alternative TCPs considered.
- Reasons for selection of preferred TCP and why thresholds can't be met.
- Documentation for submitted TCP.
 - Results of computer modeling.
 - Impacts on transportation system.
 - Impacts on community.
 - Estimated costs.
 - Any municipal official input.
- Other mitigation strategies (may be qualitatively assessed such as public information elements because delay savings could be difficult to determine).

6.3.2.3 Applying WZSM in Project Development Process

Engineering District offices shall address transportation management and congestion mitigation issues in all phases of project delivery from the project planning stage throughout the life of the project to improve work zone safety and mobility, refer to **Exhibit 6.3-D**.

Exhibit 6.3-D WZSM in Project Development Process - Key Items

Project Development Stage	Key Items Due
Planning	- Potentially Significant Projects shall be identified as early in the project development process as possible so that work zone impacts may be fully evaluated, appropriate funding allocated and overall network & region-wide impacts considered. - Districts shall work with MPOs and RPOs in the programming of projects to consider minimization of road user impacts and appropriate sequencing of projects. The cumulative impacts of multiple projects in the region's TIP shall be considered.
Scoping Field View (SFV)	Preliminary considerations shall be given to potential work zone alternatives & impacts and to what degree those considerations may influence the evaluation and selection of a build alternative. Additional studies and information needs shall be identified that will assist in determining whether the project is significant. Determine the status of the project as: - Non-Significant - Significant - To be determined (TBD)
Preliminary Engineering	An analysis of work zone alternatives & impacts is required prior to Design Field View, this analysis shall be conducted using prior analysis or prior documented delays for another project, QuickZone, Highway Capacity Manual/Software (HCS), Synchro, Delay Analysis Workbook (May 2008 Final).xls or other pre-approved modeling methods.
Prior to Design Field View (DFV)	Make final determination of the project as: - Non-Significant - Significant Note: Approval of either the delay analysis or work zone alternatives analysis must occur prior to the DFV. Attach final approvals to ECMS project development checklist.
Design Field View (DFV) Submission	- Non-Significant Projects – provide a DFV level TCP. - Significant Projects – provide a draft TMP, including DFV level TCP
Final Design Office Meeting (FDOM) Submission	- Non-Significant Projects – provide a Final TCP. - Significant Projects – provide a Final TMP, including a Final TCP.
Plans, Specifications, and Estimates (PS&E)	- At the time the PS&E is submitted to Central Office, the TMP must be attached to the Project Development Checklist in ECMS. The PS&E shall include appropriate provisions for implementing the TMP. This may include: - Allowable working hours, transportation operation requirements, public information requirements, pay items, performance data collection items, special provisions, etc. - The contract special provisions in each Significant Projects shall specify that a Work Zone Traffic Control Supervisor (or similar designation) be provided by the Contractor. This person has the primary responsibility and sufficient authority for implementing the TMP and other safety and mobility aspects of the project.

Project Development Stage	Key Items Due
Construction	• PennDOT must designate a person trained in the fundamentals of this policy at the project level. This person will usually be the PennDOT or consultant Inspector-in-Charge. • Contractor must designate a Work Zone Traffic Control Supervisor (or similar designation); this person has the primary responsibility and sufficient authority for implementing the TMP and other safety and mobility aspects of the project.
Conduct Performance Assessment & Process Reviews	• Work Zone Independent Oversight Program. • Produce WZ Annual Report. • Assess performance of work zones by analyzing crash & operational data. • Use the performance assessment to improve WZ processes & procedures.

6.3.2.4 Applying WZSM in the Design-Build Process

For projects defined in [Section 6.3.1.1](#), the Department must prepare sufficient documentation regarding anticipated temporary traffic control and project phasing so that the design-build team can develop final plans and specifications for the project. In order to carry out the requirements of the work zone safety and mobility policy on design-build projects, the Department must complete the work zone alternatives analysis (including analysis of delay impacts) prior to design field view and/or advertisement (**complete steps 1 thru 7 in Exhibit 6.3-B**).

The Department must provide the design-build team with enough details to allow the team to fully develop the required elements of the TMP for the project. Therefore, information such as number of lanes required, when lane closures are permitted, detour requirements, expected project completion date, and other key parameters shall be provided to the design-build team. The Department shall also identify specific contract bid items relating to maintenance and protection of traffic that the design-build team should include in the contract.

If allowed by the contract, when developing the PS&E the contractor may propose changes to the Department's selected work zone approach; however, the contractor must analyze work zone impacts expected from their proposed alternative and receive approval from PennDOT (and FHWA as appropriate) before any changes are authorized.

PennDOT's procedures for implementing design-build projects are found in Publication 448, Innovative Bidding Toolkit.

6.3.2.5 Performance Measures

Central Office will select and evaluate a group of projects that may consist of “Significant Projects” and “Non-Significant Projects”. The evaluation will measure the effectiveness of the approved work zone alternatives for each selected project. The evaluation will focus on two main areas: 1) additional project-related travel time through the project area, and 2) work zone-related fatalities & crashes.

Additional Travel Time/Delays (Mobility):

The objective is to evaluate additional project-related travel time (delays) through the project area, including baseline data prior to the construction being implemented. The Department’s 511PA Travel Information System (511PA) and Road Condition Reporting System (RCRS) will be used to capture the needed data and information.

- The 511PA System operates 24/7 on selected major highways. The baseline data generated for this evaluation is currently being collected and archived through 511PA. This will allow the evaluation process to incorporate information from established time periods before, during, and after construction. Any work zone project within the 511PA coverage could potentially be selected and evaluated for effectiveness.
- As projects are selected for this performance measure evaluation, Central Office will coordinate with the District Office (Traffic & Construction Unit staffs) to ensure that all traffic restrictions are documented accurately in the RCRS.
- The RCRS data indicating the actual times when the lane restrictions are placed and removed are very essential for this evaluation. Therefore, it is imperative that this information is accurately documented in RCRS by District Traffic and Construction Unit representatives.
- The data and information used to evaluate a project’s work zone safety and mobility performance will be retained electronically for future use by the Department on similar projects (scope, location, etc.).
- The evaluation findings will be shared with both FHWA and the District Offices.

Work Zone-Related Fatalities & Crashes (Safety):

Fatality and crash data will be collected at the project level and District level. The construction representative shall ensure that:

- A diary is kept in accordance with the Project Office Manual.
- The inspections and findings related to work zone reviews are documented.
- All crash reports are obtained and forwarded to the District Traffic Engineer.
- A complete file of inspections, crashes and correspondence related to work zones is maintained by the construction project manager. Verification of fatal and crash statistics will be cross-checked with the Department's Crash Record System.
- This information will be available to FHWA and appropriate PennDOT personnel.

6.3.3 WZSM Transportation Management Plan (TMP)

A Transportation Management Plan (TMP) shall be developed for each Significant Project and submitted as per **Exhibit 6.3-D**. A TMP lays out a set of coordinated strategies and describes how these strategies will be used to manage the approved work zone alternative of the project. The level of detail included in the TMP will be appropriate for the size and complexity of the project. A sample TMP is discussed at the end of this section, and included in [Appendix 6A](#). A TMP shall contain the elements listed in **Exhibit 6.3-E**. While a formal TMP submission isn't required for Non-Significant Projects, appropriate-level TMP elements need considered on these projects as well.

Exhibit 6.3-E TMP Elements

Element	Requirement
1. Introduction	Cover page, table of contents, list of tables, list of abbreviations and symbols, and terminology.
2. Executive Summary	Overview of each of the TMP components.
3. Roles and Responsibilities	• TMP (Work Zone) manager. • Stakeholder/review committee. • Approval contact(s). • TMP implementation task leaders (e.g., ADE Construction, Press Officer, Portfolio Manager, etc.).
4. Project Description	Information such as: • Project type. • Project background. • Project area/corridor. • Project goals and constraints. • Proposed construction phasing. • General schedule and timeline. • Related project information.
5. Traffic Conditions	For the project area: • Including data collection and modeling approach. • Existing roadway characteristics (history, roadway classification, number of lanes, geometrics, urban/suburban/rural). • Existing and historical traffic data (volumes, speed, capacity, volume/capacity, percent trucks, queue length, peak traffic hour). • Existing traffic operations (signal timing, traffic controls). • Local community and business concerns/issues. • Traffic growth rates (for future construction dates). • Prior documentation of work zone delays on similar projects.
6. Work Zone Impact Assessment	• Description of only the approved WZ strategy, including the potential WZ impacts of this strategy. Note, do not reiterate the whole WZ alternatives analysis in the TMP!
7. Work Zone Impact Management Strategies	For project, detour routes, corridor, region by construction phase, including TCP strategies, PI strategies, and TO strategies. Findings and recommendations.
7a. Traffic Control Plan (TCP)	Always required for both Significant and Non-Significant Projects. See Design Manual Part 3, Chapter 4. A Traffic Control Plan addresses traffic flow and control through the work zone (required on all projects).
7b. Transportation Operations Plan (TOP)	See Section 6.3.4. Addresses sustained operations and management of the work zone impact area (including impacts to adjacent roadways in the network and impacts to other traffic movements).
7c. Public Information Plan (PIP)	See Section 6.3.5. Addresses communication with the public and concerned stakeholders.
8. TMP Performance	Strategies to monitor TMP performance.
9. Contingency Plans	Potential problems and corrective actions to be taken, standby equipment or personnel.
10. Break out of TMP costs	Itemized costs.

The typical responsibility for development of the Transportation Management Plan including the Traffic Control Plan (TCP), Traffic Operations Plan (TOP), and Public Information Plan (PIP) is as follows:

Project Design By	TMP Development Responsibility
Consultant	Consultant
Design-Build	Contractor *
In House	District

*** Note: unless traffic control excluded as a design/build item.**

Contractors who propose an alternate TCP must submit a new or revised TMP including an impact analysis, TOP, and PIP.

Sample Transportation Management Plan (TMP)

[Appendix 6A](#) is a sample Transportation Management Plan (TMP) created for a project on a rural interstate highway with low traffic impacts. The level of detail included in the TMP for each project will be appropriate for the size and complexity of the project. The sample TMP was drafted using information that is readily available from other sources such as ECMS.

6.3.4 WZSM Transportation Operations Plan (TOP)

The Transportation Operations Plan (TOP) is a required part of the TMP on Significant Projects and is a description of the strategies that are being used to address safety and mobility. It shall be coordinated with the stakeholders (i.e., other DOTs, police, fire, emergency medical services, traffic control centers, transit, schools, etc.). The level of detail included in the TOP will be appropriate for the size and complexity of the project. A checklist of items that may be included in the TOP follows:

6.3.4.1 Traffic Demand Mitigation Strategies

- ☐ Transit service improvements
- ☐ Transit incentives
- ☐ Shuttle services
- ☐ Ridesharing/carpooling incentives
- ☐ Park-and-ride promotion
- ☐ HOV lanes
- ☐ Toll/congestion pricing

- ☐ Ramp metering
- ☐ Parking supply management
- ☐ Variable working hours
- ☐ Telecommuting

6.3.4.2 Corridor/Network Management Strategies

- ☐ Signal timing/coordination
- ☐ Temporary traffic signals
- ☐ Street/intersection improvements
- ☐ Bus turnouts
- ☐ Turn restrictions
- ☐ Parking restrictions
- ☐ Truck restrictions
- ☐ Separate truck lanes
- ☐ Reversible lanes
- ☐ Pennsylvania Late Merge
- ☐ Dynamic Late Merge
- ☐ Ramp Metering
- ☐ Suspension of ramp metering
- ☐ Ramp closures
- ☐ Street closures
- ☐ Incorporate Permanent ITS
- ☐ Railroad crossing controls
- ☐ Coordination with other construction sites

6.3.4.3 Work Zone Safety Management Strategies

- ☐ Work Zone speed limits
- ☐ Temporary traffic signals
- ☐ Temporary concrete barrier
- ☐ Movable barrier
- ☐ Attenuators
- ☐ Rumble strips
- ☐ Warning signs
- ☐ Automated Flagging Assistance Device
- ☐ Multidisciplinary taskforce
- ☐ QA reviews
- ☐ Dedicated traffic control supervisor
- ☐ Training
- ☐ Safety meetings

6.3.4.4 Traffic/Incident Management and Enforcement Strategies

- ☐ Formal Incident Management Plan (include copy)
- ☐ ITS

- ☐ Transportation Management Center
- ☐ Aircraft
- ☐ Traffic Screens
- ☐ Call boxes
- ☐ Mile-post markers
- ☐ Service patrols
- ☐ Supplemental police assistance/enforcement
- ☐ Automated enforcement

6.3.5 WZSM Public Information Plan (PIP)

The Public Information Plan is a required part of the TMP for Significant Projects. Public information is a cooperative effort with the Central Office Press Office (when necessary), the District Press Office, Community Relations Coordinator, Traffic Unit, Design Unit, and Construction Unit. The PIP shall be developed well in advance of the start date of the project, since much of the communications should be undertaken prior to start of construction. Therefore, initiation of the Public Information Plan shall begin with the scoping field view. The level of detail included in the PIP will be appropriate for the size and complexity of the project.

6.3.5.1 Checklist for Public Information Plan

Each PIP should consist of the following elements:

1. Brief summary of project (situation analysis)
 - ☐ Project purpose and need
 - ☐ Project cost and funding source breakdown
 - ☐ Start Date
 - ☐ Completion date
 - ☐ Description of the Traffic Control Plan. Begin with conceptual plan and update as plan develops.
 - ☐ Closures and detours
 - Official detour
 - Time of day for lane drops and closures
2. List of affected stakeholders to be targeted
 - ☐ Municipalities, counties, townships, villages, MPOs/RPOs, recreational facilities, and special event sponsors
 - ☐ Local and state law enforcement
 - ☐ Emergency services
 - ☐ Affected business community
 - ☐ Schools

3. Communications plan and timeline for each part of the plan
 - ☐ Standard communications tools
 - ☐ Project-specific communications tools
 - ☐ Contact information for person responsible for each aspect
4. Milestones for updates
5. Means of evaluation of communications plan
 - ☐ In-progress
 - ☐ Post-construction

Some examples of communications tools are listed below:

- Press releases, media alerts, public service announcements
- Focus group/stakeholder meetings
- Brochures, maps
- Public involvement meetings
- Editorial board meetings
- Dedicated project web page (e.g. I-70 org)
- Changeable message board communications
- Community and group informational talks
- Local government and legislator meetings
- Dedicated phone number for information (District PIO line or other)
- Targeting tourist areas with maps and brochures (hotels, restaurants, chambers, visitor centers, etc.)
- Business letters/brochures/maps packets to major businesses affected (hospitals, colleges, large corporations)
- Business letters to any company along a ramp or route closing (Ex: SR 256 – businesses claimed they were not given enough warning)
- Dedicated am-band radio station with construction info on loop
- Billboards along affected routes
- Post “future detour” signs to allow motorists to become familiar with the route
- Regular updates to trucking companies and largest freight carriers in area

Please refer to Publication 295, Public Involvement Handbook, for further general guidance and tools that can be used in developing the PIP.

6.3.6 WZSM Definitions

1. Freeways – are fully-controlled, limited access arterial highways, as per AASHTO Geometric Design of Highways & Streets (Green Book), Chapter 8.
2. Modes of Transportation – motor vehicles (including buses and trucks), transit, bicycles or pedestrians.
3. Public Information Plan (PIP) – a plan that informs the public of the impacts on traffic and the general area during or prior to construction.
4. Significant Projects – is defined as one that, alone or in combination with other concurrent projects nearby, is anticipated to cause sustained work zone impacts that are greater than what is considered tolerable based on State policy and/or engineering judgment.
5. Traffic Control Plan (TCP) – a supplemental plan currently required for all projects that addresses traffic flow and control through the work zone which must conform to the MUTCD and Publication 213.
6. Transportation Management Area (TMA) - an area designated by the Secretary of Transportation, having an urbanized area population of over 200,000, or upon special request from the Governor and the MPO designated for the area.
7. Transportation Management Plan (TMP) – lays out a set of coordinated strategies and describes how these strategies will be used to manage the impacts of a project during construction.
8. Transportation Operations Plan (TOP) – a plan that includes Demand Management Strategies, Corridor/Network Management Strategies, Work Zone Safety Management Strategies and Traffic/Incident Management and Enforcement Strategies.

6.4 District Responsibilities

6.4.1 District Traffic Unit

6.4.1.1 Field Reviews

1. In general, the District Traffic Units should have a District-wide program of inspecting work zone traffic control on construction projects, maintenance operations, and permit/utility work within the District. Nighttime inspections of work zone traffic control should be included in the program.
2. The inspection of work zone traffic control and the reporting of findings to the appropriate District unit should be a major item for the employee performing the maintenance and protection of traffic function.

6.4.1.2 Crash Report Reviews

1. The District Traffic Unit should review crash reports received from the Construction Unit and maintain a file of individual crash reports for each construction project. The “Construction Zone Vehicular Crash Report Form” is in Publication 2, “Project Office Manual”.
2. When a recurring crash problem arises on a project, the District Traffic Unit should inspect the work zone traffic control to see if any changes are necessary. If changes should be made, the District Traffic Unit should work with the Construction Unit to see that the changes are implemented as soon as possible. Any approved changes should be documented on the TCP, dated, and signed.

6.4.1.3 Training

The District Traffic Unit should, in cooperation with the District Training Coordinator, participate in the training of:

- District Construction Inspectors
- Maintenance Managers, Assistant Managers, Foremen, and all other District Maintenance employees that work on the highway
- Permit Personnel
- Survey Crews

- Bridge Inspection Teams

6.4.1.4 Annual Report

Each year, the Traffic Engineering and Operations Division will ask the District Traffic Units to provide District data for the Annual Report on Work Zone Traffic Control.

The report should include the following:

1. For each category (construction, maintenance, and permit/utility), provide a summary of any District Q/A reviews. Include the scores and the number of failures; the identified deficiencies and the frequency of occurrence; and what actions the District took to correct them.
2. Any new work zone policies or procedures implemented within the District. Include any District policy letters written on this subject, and the reasons for implementing these policies and procedures.
3. Any recommendations to revise Department standards, practices, policies and procedures.
4. District “best practices” that you would recommend on a statewide basis.
5. The positive or negative results of novel or innovative traffic control.
6. A listing and description of all training efforts that were conducted by the District and counties for Department or outside personnel. Use the following general format:

Date	Duration (hrs.)	Number Trained	Agency Trained	Audience	Topic/ Subject
2/2/07	2.5	24	PennDOT	Foremen	Flagging
6/21/07	4	10	Nat'l Fuel Company	Supervisors	Work Zone

7. Problems with permit/utility operations and related activities, and any recommendations for improvement.
8. ITS deployment in construction work zones. Include the location (project, county, SR, segment) and type of ITS (queue detection, motorist information/travel time and delay, variable speed limits, etc.).

9. Suggestions for needed research on either the State or national level.
10. Any other information that may be of interest to the other Engineering Districts, Counties, Central Office or FHWA.

6.4.2 District Construction Unit

The Construction Unit is directly responsible for the administration of Department construction projects including the implementation of the TCP, routine inspection of work zone traffic control setups and devices.

Through routine inspection of work zone traffic control setups and devices, field inspectors will use a standardized form (Maintenance and Protection of Traffic Inspection Checklist) to record whether or not all temporary traffic control devices are either compliant or deficient as per the criteria in Publication 408, Section 901.3(v). The standardized form will also show the dates and approximate times when the contractor was notified of the deficiency and when the deficiencies were corrected. The form can be used for documentation to assess Work Zone Liquidated Damages (as described within Sections 901.3(t) and 105.01(b) of Publication 408). The standard form is available at the following location
<ftp://ftp.dot.state.pa.us/public/PubsForms/Forms/CS-901.pdf>.

The Construction Unit is also responsible to collect crash data on construction projects since this information is a valuable aid in reducing future work area crashes. Therefore, the following procedures are established

- (a) The Construction Unit should send a letter to the State Police station or to local police advising them of each upcoming project and asking them to call the inspector-in-charge whenever a traffic crash has occurred within the specified limits of the construction project.
- (b) When the State Police or local police call to notify the inspector-in-charge of the occurrence of a construction zone crash, the inspector should complete a copy of the “Construction Zone Vehicular Crash Report Form” which is available in Publication 2, “Project Office Manual”. The use of this form should provide all pertinent information. The inspector-in-charge should also request a copy of the police report.
- (c) The inspector-in-charge should provide a copy of the completed “Construction Zone Vehicular Accident Report Form” and any police reports to the Project Engineer and the District Traffic Unit.
- (d) After receiving each accident report, the Project Engineer should inspect the work zone to determine if changes or revisions are necessary. However, the Project Engineer should not change the TCP until they receive approvals from

the District Traffic Unit. The project Engineer should document and date all changes or revisions made because of a crash.

- (e) Maintain all individual crash reports and project crash analysis in the project file.

6.5 Quality Assurance

6.5.1 Independent Oversight Program (IOP) with FHWA

- (a) In June, 2007 PennDOT and FHWA updated their “Stewardship and Oversight Agreement” that administers the Federal-aid Highway Program in Pennsylvania, resulting in PennDOT assuming certain FHWA oversight roles and approval responsibilities. To ensure Quality Management in producing superior products and services, the agreement set up a joint Independent Oversight Program (IOP) to perform selective reviews of PennDOT processes. As a result, a team including a representative from PennDOT’s Central Office Work Zone Section, Construction Quality Review Section, and FHWA will jointly conduct work zone IOP reviews on construction projects, one-third of the Engineering Districts each year. (Maintenance operations will not be a part of these reviews.)
- (b) These reviews will include District policies, procedures, and field reviews of work zones. The team will use PennDOT’s existing QA forms or a similar document for work zone traffic control field reviews. The team will give the District Work Zone Manager one week notice prior to the start of the IOP review. If possible, the Work Zone Manager or another representative from the District Traffic Unit should accompany the team during the review (a representative of the District’s Construction Unit may also accompany the team).
- (c) On the first day, the review team will meet with the Work Zone Manager to discuss the scope of the review and go over District policies. After completing the field reviews, the team will conduct a closeout conference with the Assistant District Executive for Construction or their representative to discuss the findings. A formal report will be written and submitted at a later date.

6.5.2 Other Quality Assurance Reviews

In addition to the quality assurance reviews conducted by the Central Office Review Team, additional QA reviews are as follows:

- Construction projects — Bureau of Construction and Materials, Quality Assurance Division.
- Maintenance projects — Bureau of Maintenance and Operations, Quality Assurance Review field personnel (part of the accreditation process).
- Permit/utility projects — District Maintenance Offices, Permit Section.

The Engineering and Maintenance Districts should also conduct work zone QA reviews on construction, maintenance, and permit/utility operations.

6.5.3 Quality Assurance Review Form

The Quality Assurance Review form is a general comprehensive checklist used for construction projects, but may be used on any type of operation. The Quality Assurance Review form and database use by the Central Office Review Team are available at the following locations:

- Form – P:\bhste_shared\workzones\QA Database\QA Blank Short Form.doc
- Database – P:\bhste_shared\workzones\QA Database\QA State Wide Database.mdb

6.6 Traffic Signs

6.6.1 General

6.6.1.1 Fluorescent Orange Sheeting Materials

There currently are many types of fluorescent orange sheeting materials approved in Pennsylvania, including Types III, VI, VII and VIII. Since all of these products are acceptable, there is no reason to specify a specific type.

6.6.1.2 Breakaway and Yielding Sign Supports

Department guidelines for breakaway and yielding sign installations are contained in Publication 111. Approved Typed B sign supports are listed in Publication 35.

Everyone should refer to Publication 111. In particular, TC-8702B applies to the small sign supports normally used for post-mounted work zone signs.

6.6.1.3 Covering Inappropriate or Conflicting Signs

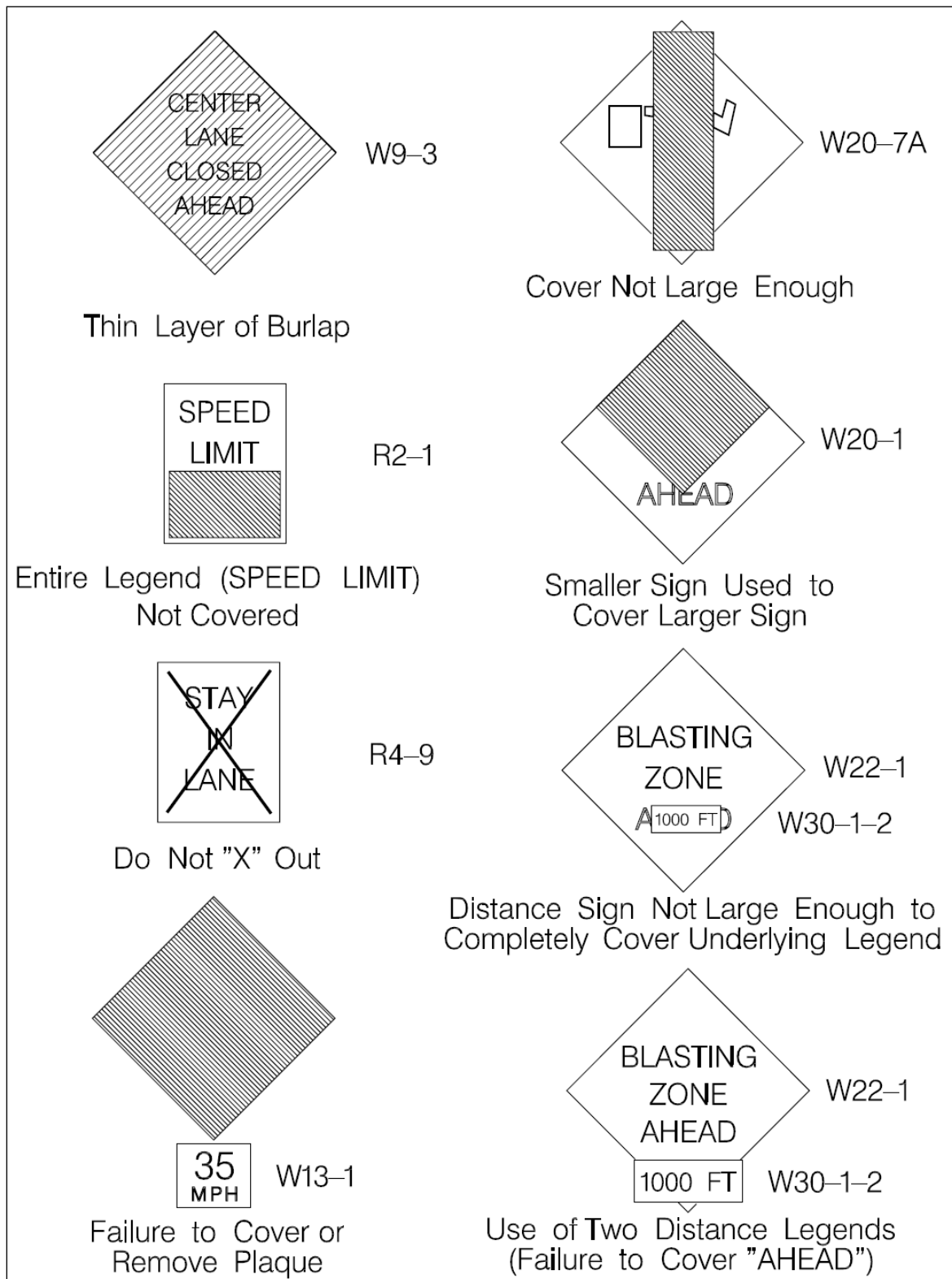
Remove, cover, or fold work zone signs that are conflicting, inappropriate, or are not applicable to existing conditions so that they are not readable by oncoming traffic.

When covering signs, completely cover them with a material that will prevent the sign from being read during all conditions of light and weather. Normally, a single layer of burlap and non-opaque materials should not be used to cover a sign since those materials may let the underlying sign message be seen at night because headlights reflect the message through the material. [Exhibit 6.6-A](#) shows examples of improperly covered signs.

Remove or deactivate any warning lights during those periods when the signs are covered or folded.

Note: Conflicting or inappropriate signs may not be turned away from approaching traffic unless studies indicate that this does not compromise the crash worthiness of the sign.

Exhibit 6.6-A Improperly Covered Signs



6.6.2 Work Zone Speed Limits

6.6.2.1 General

Posting a reduced work zone speed limit within a work zone enables police to issue citations under §3362 of the Vehicle Code (relating to maximum speed limits). In order to be enforceable, the speed limit must be established and posted in compliance with Publication 212.

However, some past practices of posting reduced speed limits in work zones were unreasonable, e.g., long sections of reduced speed limits when no activity is present and to all appearances, a reduced speed limit is not justified.

Moreover, studies indicate that unless police enforcement is actually present, a reduced speed limit does not necessarily cause traffic to slow to that speed. Many studies have shown that drivers will not usually slow down just because of a posted speed limit unless they perceive the need to do so. For example, motorists adjust their speed based on geometrics, lateral clearance, type of activity, and traffic control devices more than on speed limit signing.

This emphasizes the need to design the travel way through the work zone for the speeds vehicles will travel, not for the speed one hopes they will travel. Therefore, in the planning and design of traffic control schemes, the goal should be to route traffic through the work zone with geometrics and traffic control devices as close as possible to those for normal situations. Moreover, only establish a reduced speed limit when conditions dictate a need for a reduction (such as restrictive geometrics, construction quality, worker protection, or the presence of potential hazards).

6.6.2.2 Construction Projects

The Department is continually searching for ways to reduce the speed of vehicles in work areas. To date, the only effective method of reducing speeds through a work area is to establish a reduced work zone speed limit and have police enforcement. While this method is effective, only use it where a reduced speed is critical. Long sections of reduced work zone speed limits in work areas where no worker activity is taking place and the drivers can see no reason for the speed reduction is not only ineffective but also counter-productive and must be avoided. Examples are:

- A crossover project where traffic is separated by a concrete median barrier and all work is taking place on the other side of a wide median.

- A pavement patching operation where one lane is closed with work taking place along a relative short section of highway in the closed lane but a reduced work zone speed limit has been posted for the entire 5-mile construction area.

Policy

Districts are encouraged to establish reduced work zone speed limits with police enforcement on construction projects where high-speed traffic is immediately adjacent to workers that are not protected by a positive barrier, the use of reduced work zone speed limits on long-term construction projects must be used sparingly. Therefore, the following is relative to reduced work zone speed limits and advisory speeds on long-term construction projects:

- (a) Reduced work zone speed limits should be established only where high speed (45 mph or higher) traffic is immediately adjacent to workers and the workers are not protected by a temporary concrete median barrier. This reduced work zone speed limit should only be established in the area(s) where work is under way and workers are present. During non-working hours when workers are not present, the reduced work zone speed limit signs should be removed, folded, or covered, so that the signs are not readable by drivers.
- (b) Reduced work zone speed limits may also be established if a study is conducted (and permit received) by District Traffic personnel which indicates that potential hazards exist in the open travel lane(s), (e.g., raised manholes or inlets). A 24-hour reduced work zone speed limit may be established throughout the work area containing the hazard(s) in accordance with the study.
- (c) For all other long-term construction areas where the District wishes to advise drivers of a recommended speed due to potential hazards or a reduced design speed for temporary geometric features, use advisory speeds throughout the area containing the potential hazards. However, when workers are present for short periods and the workers are immediately adjacent to high speed traffic without a positive barrier, a reduced work zone speed limit may be installed during that time.

Placement of Signs

- (a) When reduced work zone speed limit signs are used, the first sign(s) should be installed approximately 500 to 1,000 feet in advance and throughout the reduced speed area at intervals not to

exceed one-half mile. Where more than one “activity area” exists within the construction project, a separate reduced work zone speed limit must be established and posted for each “activity” area if the activity areas are more than one-half mile apart. Reduced work zone speed limit signs should be installed as shown in Publications 212 and 213. All conflicting speed limit signs must be removed, covered, or folded so that the signs are not readable by drivers when the reduced work zone speeds are in effect.

- (b) When Advisory Speed Signs are used, they shall be in accordance with Publications 212 and 213.

The concept of establishing a reduced work zone speed limit is intended to improve the safety of the highway worker when performing duties on high speed highways. Each project should be examined to determine why and when reduced speed zones should or should not be considered. We need to take this action in order to promote a greater degree of credibility to our reduced work zone speed limits in work areas, to avoid possible entrapment of drivers by installing unrealistic speed limits (double fines), and to promote greater safety in long-term construction projects by reducing the speed differential between vehicles.

- (c) The District Executive must approve any deviation from this policy in writing on a project-by-project basis.

6.6.2.3 Department Maintenance Operations

- (a) All maintenance operations involving moving short-term operations on divided highways, one-way highways, or two or more lane approaches of undivided highways, where the operation moves intermittently or continuously at an average speed of more than 1 mph (88 feet/minute), shall consider establishing a reduced “Work Zone Speed Limit” in accordance with Publications 212 and 213.
- (b) All maintenance operations on multi-lane, high-speed highways using work zone traffic control, in accordance with Publication 213, may install work zone speed limit signs reducing the work zone speed limit 10 mph. This reduced speed limit should not be installed for the entire length of the work area if work is confined to only a portion of the total area at any one time. The reduced speed limit should only cover the area approaching and the area through which work is actually in progress.

- (c) When a maintenance operation implements a work zone speed limit, the first work zone speed limit sign must be installed approximately 500 to 1,000 feet in advance of the work area in order to give drivers time to slow down before entering the work area. The first work zone speed limit sign must be installed on the right side of the roadway, and if the reduced speed limit extends more than one-half mile, additional work zone speed limit signs must be installed. (The Vehicle Code mandates that speed limit signs for speed limits less than 55 mph be installed at intervals not greater than one-half mile.) For the reduced speed limit to be legal all conflicting speed limit signs must be covered and an “End Work Area” sign must be installed beyond the work area. In order to provide a greater target value to the work zone speed limit signs, orange flags may be installed with the signs in the same manner as with warning signs.

6.6.2.4 Drone Radar

Drone radar should be considered at the beginning of the confined lanes and at approximate 2-mile intervals. Drone radar may also be added to moving construction vehicles since this seems to be very effective.

6.6.3 Act 229 Signs

On December 23, 2002, Act 229 modified the Vehicle Code to improve safety in work zones (including but not limited to adding §§1378, 4309 and 6123.1, and modifying §§102, 1373, 1535, 3326, 3365, 3368, 3372 and 6123). Initially, the Department developed very broad guidelines as intended by the legislation, but with the approval of the legislative architects, eventually reduced the scope to apply only for long-term, high-speed projects.

Specific elements of Act 229 include:

9. Installation of signs to have drivers turn headlights on in “active work areas” (a legal opinion indicates that daytime running lights are acceptable).
10. Need for speed display signs on larger Interstate construction projects.
11. Department to withhold \$1,000 from contractors for each day that they are in non-compliance with contract safety provisions.
12. A 15-day suspension of operating privileges for drivers committing violations in an active work zone.
13. Stronger suspension of motor carrier vehicle registrations for safety problems.

14. Emphasize good management principles, i.e., keep work zones as short as practical, minimize lane restrictions, and remove traffic control devices as soon as possible.

Department regulations relative to Act 229 are in 67 Pa. Code §212.419 (relating to special controls in work zones), and a copy is included herein as [Appendix 6B](#).

6.6.4 Temporary Business Signs in Work Zones

When a highway or street is under construction, or when a detour is established forcing motorists onto different routes, temporary business signs should be installed to guide motorists to the affected businesses. Do not automatically assume that the location of the business, office complex, shopping center, or their entrances are intuitively obvious. The general rule is to sign for elderly drivers who are not familiar with the area.

(a) Criteria.

1. Road closed with detour established.

Install signs for major shopping centers, offices, industrial sites and general street locations (e.g., Locust Street businesses) with directional arrows in advance of locations where a left or right turn is required to direct drivers along the proper route to the location. Sign individual businesses if there is a possibility of a problem in locating them. A “ROAD OPEN TO _____” sign may also be necessary.

2. Road closed but traversable.

Install sign at the road closure saying “Road Open to _____,” using the name of the furthest office complex, shopping district, plaza or mall, street, or geographic location.

3. Road under construction.

If the access to a business is changed, moved, or not easily identified, then place a sign with the business name and a directional arrow at the business entrance.

4. Use other appropriate sign schemes to fit unique conditions.

- (b) Due to the numerous variables involved (type of business or office, number of businesses, length of words in the business name, number of locations, etc.), it is not possible to specify a single sign standard that would be correct for every signing situation, but the following general guidelines are offered.

1. Use black legend on an orange background. Do not install business logos.
2. Whenever possible, use 6-inch legend on 72-inch wide sign panels on multilane roadways and on all roadways where the assumed travel speed is 40 mph or higher.
3. Show the business sign layouts and locations on the Traffic Control Plan.

(c) Cost/Installation

Sign costs and installation are to be included as part of the construction project, businesses are not to be charged. A separate pay item should be established on a square foot basis.

6.6.5 Winter Shutdown

During the fall, construction projects that will be continuing into the next year begin shutting down for the winter. To maintain our work zone credibility, all signs (including work zone speed limit signs) and channelizing devices are to be removed or covered, except if the unfinished project poses a hazard to motorists.

6.6.6 Portable Changeable Message Signs (PCMS)

Changeable message signs (CMSs) can be effective on high-speed highways by furnishing drivers with real time information of problems and unexpected conditions, and telling them the best course of action. CMSs can be particularly useful at highway construction and maintenance work sites that frequently present drivers with unexpected traffic or detour situations.

Their primary purpose in work zones should be to advise drivers of unexpected traffic and routing conditions; however, repeat drivers become accustomed to the situation and may not read the message. Therefore, consider changing the message slightly to help keep the drivers' attention.

When using CMSs, take care to be sure that the messages are correct and displayed at the proper time. Drivers will normally have a negative attitude about the sign if it:

- Displays information that is contrary to existing conditions
- Displays information that is not easily understood or read in ample time to make the appropriate maneuvers.
- Tells drivers something they already know.

When using multiple signs to convey information, maintain 1,000 feet between signs. The message must be short enough so that an approaching driver can read the entire message. The message must be displayed long enough so that the message is readable at the travel speed of the highway. Normally, a driver needs at least one second per short word (up to eight characters) and two seconds per unit. A 12-inch message is normally readable from approximately 450 feet and an 18-inch message from approximately 650 feet.

6.6.7 Advance Notice of Pending Work

Districts should consider using portable changeable message signs (PCMS) or the ROAD WORK TO BEGIN NEXT WEEK (G20-1-2) signs to advise motorists of future lane closures on freeways, expressways, and other high-volume highways. Use these signs during the week before a maintenance activity begins to alert motorists of the upcoming project. The G20-1-2 sign may be on a Type III barricade or a portable sign support.

If a PCMS is used, Districts are encouraged to use them throughout the duration of the activity as a supplemental warning device.

6.7 Pavement Markings and Delineation

6.7.1 General

In accordance with 67 Pa. Code §212.407 (relating to markings for temporary traffic control), when lane line and centerline pavement markings on more than 250 linear feet of highway are covered or destroyed by construction, maintenance, utility, permit, or other work, the lines must be replaced before terminating work each day. The replacement markings may be standard pavement markings or temporary markings as included in Section 901.3(k) of Publication 408 (Work Area Pavement Markings) or in Section 6F.78 of the *MUTCD* (regarding temporary pavement markings).

6.7.2 Compliance by Maintenance Districts

- (a) Replace centerlines and lane lines that were covered or destroyed as a result of the work operation with standard pavement marking patterns or pavement marking patterns meeting the requirements shown in the TC-8600 Standard prior to terminating work each day, except that:
 - Markings are not required if the surface texture of the roadway prohibits the installation of markings. In this case, install pavement markings as soon as practical.
- (b) Replace edge lines when the roadway is scheduled for repainting. Schedule school zone markings, railroad markings, stop lines, and any other markings required along the roadway as soon as possible after installing the edge lines.
- (c) In order to comply with the new FHWA requirements for “short-term” pavement markings in work zones, each Engineering and Maintenance District should require the contractor to install all required pavement markings, including “short-term” and/or standard pavement marking patterns on all construction projects including Purchase Order Contracts.
- (d) For the installation of required pavement markings on Department Force projects, the Engineering and/or Maintenance District may select one or more of the following methods or may select another method not listed below:
 1. Schedule the work in each county so that the required “short-term” and permanent pavement markings required in the TC-8600 Standard may be installed with the District’s large truck-mounted paint machines and/or small paint machines.

2. Use approved temporary pavement marking tape for all required “short-term” pavement markings, or install standard pavement markings using approved cold plastic or other approved durable materials. These options are especially viable if a county only does short sections of motor paving or skin patching.
3. Purchase a small truck-mounted paint machine (pick-up truck size) with a two-color system that can paint a double line, paint on the left and right side at the same time, use heated paint for rapid drying, and dispense beads onto the painted line. Rotate the truck throughout the District by having the counties schedule all of their resurfacing, motor paving, skin patching, and seal coating projects during the time they will have the truck-mounted paint machine. The Engineering District could assign personnel to this truck at their discretion, and could possibly use the equipment to help complete their small paint machine work.
4. The District or County could investigate the possibility of using contract painting.

6.7.3 Pavement Markings in Work Zones

This should clarify several issues associated with the Department’s policy on pavement markings used in work zones.

- (a) Refer to Section 901.3(k) of Publication 408:
- (b) It is permissible to use any approved, non-removable, temporary pavement marking tape in lieu of paint.

6.7.4 Delineation of Temporary Concrete Median Barriers

To improve visibility of the barrier, Districts shall provide delineation of the barrier and a temporary edge line.

Delineation should generally conform to Sheet 2 of 4 of the TC-8604 Standard, except you are encouraged to install barrier-mounted delineation at 20-foot spacing within transitions and curves and at 4-foot spacing on tangent sections.

Edge lines should be any of the following:

1. Install temporary pavement marking tape or paint on the roadway at a distance of 6 inches or more from the base of the barrier.
2. Apply an edge line (paint and glass beads, or temporary pavement marking tape) on the lower sloping surface of the barrier instead of on the road surface. Before applying the line, the barrier must be thoroughly cleaned by high-pressure water blasting. Moreover, because traffic does not wash the line by running on it, add a special provision to require the contractor to clean the edge line with high-pressure water at least once every 2 months.
3. Attach temporary RPMs to the road surface at a minimum distance of 6 inches from the base of the barrier, and at 40-foot spacings within transitions and curves, and at 80-foot spacings on tangent sections.

The advantage of the last two types of “edge lines” is that these methods eliminate the need to eradicate the edge line, thereby reducing costs and eliminating eradication scars.

6.7.5 Temporary “Shadow Lines” and “Standard Lines”

Shadow Lines

Some Districts have installed temporary pavement markings using half the normal application rate for paint and glass beads.

These 7- or 8-mil thick temporary “shadow lines” generally allow the final durable marking to mechanically bond with concrete or asphalt road surfaces, thereby eliminating the need for costly surface preparation or pavement marking removal. Conversely, when using a full thickness (12 or 15 mils) temporary line, the final durable markings will not bond properly with the roadway surface and removal of the temporary lines is required.

Districts whose past practice has been to allow the application of “shadow lines” will have to work with their contractors to ensure a minimum retroreflectivity of 125 mcd/m²/lux for “shadow lines” installed prior to the installation of final durable markings. The procedure is as follows:

1. Allow the contractor to apply a thinner temporary line that still provides the required minimum retroreflectivity for the duration of the construction phrase and may not require eradication before the final markings are applied.

2. The contractor will conduct and document initial and weekly retroreflectivity measurements of the temporary lines to guarantee that the minimum retroreflectivity requirements are being met to ensure that FHWA will participate in the funding of the pavement marking items.
3. The contractor is responsible for corrective action should the final markings fail to adhere to the temporary markings and/or not meet the required retroreflectivity for a period of 6 months following placement.

“Shadow lines” are the only temporary marking that requires a minimum retroreflectivity.

Standard Lines

Apply the standard (15-mil thick) temporary line and eradicate the temporary line before installing the final durable markings.

6.8 Traffic Signals

6.8.1 Traffic Control Signals

Refer to Publication 213 for traffic control figures and appendix for temporary signal requirements, application, and permit.

6.9 Barriers, Drums and Fences

6.9.1 Temporary Concrete Barrier Types

This section provides policy and guidance for using appropriate temporary concrete barriers as part of the work zone setup. If temporary concrete barriers are used, it is important to follow the following guidelines.

The temporary concrete barriers available for use are as follows:

- (a) NCHRP 350 or MASH approved barriers:
 - 1. Use the PennDOT standard F-Shape concrete median barrier in accordance with the current RC-Standards.
 - 2. Other approved NCHRP 350 or MASH barriers may be used in work zones as appropriate provided they are crash tested and approved by the FHWA and PennDOT.
- (b) All new temporary concrete barriers purchased by the Department or contractors must be NCHRP 350 or MASH approved.

See [Section 6.14.2.3](#) for other guidance on usage.

6.9.2 Drums

Under normal conditions, all channelizing devices listed in Section 901 of Publication 35 are considered equally effective in guiding motorists through the work area and around obstacles. Therefore, if Districts direct the use of drums (or the use of some other specific type of channelizing device), it should be specified in the 901 special provision. In addition, a clear and sufficient reason for specifying the specific type of channelizing device should document the decision in writing and kept in the project file.

6.9.3 Reflectorized Safety Fence

Reflective plastic fence currently does not require a Certificate of Approval for use in work areas. This fence comes in several colors; however, only orange may be used in construction areas within the Commonwealth.

This reflective fence is not a traffic control device but may supplement traffic control devices required by Chapter 212 or the Traffic Control Plan by delineating areas within the work area. When used, mount the fence on non-metallic drums or

approved breakaway or base bending posts. Install enough supports to keep the fence in a vertical position.

6.9.4 Non-Conforming Traffic Control Devices

The use of non-conforming devices requires permission to experiment by the FHWA. This may be obtained by submitting a Request For Permission to Experiment in accordance with the MUTCD, Section 1A.10. The Bureau of Highway Safety and Traffic Engineering (BHSTE) will coordinate requests.

6.9.5 Certification of Compliance – Materials and Products

Contractors may use either Form CS-201, Source of Supply-Traffic Items (available in ECMS), or [Exhibit 6.9-A](#) to summarize the specific types of traffic control devices used on a project.

Exhibit 6.9-A Document the Types of Traffic Control Devices

Subject: Certification of Traffic Control Devices.

This is to advise that the traffic control devices that will be used for maintenance and protection of traffic on _____, have been obtained from the following approved sources:

Signs

Company _____

Approved No. _____

Arrow Board

Company _____

Approved No. _____

Type "A" Low Intensity Flashers

Company _____

Approved No. _____

Traffic Cones

Company _____

Approved No. _____

Type "B" High-Intensity Flashers

Company _____

Approved No. _____

Tubular Markers

Company _____

Approved No. _____

Type "C" Steady-Burn Lights

Company _____

Approved No. _____

Temporary Pavement Mkg. Tape (Non-Removable)

Company _____

Approved No. _____

White High-Intensity Flasher (Act 229)

Company _____

Approved No. _____

Temporary Pavement Mkg. Tape (Removable)

Company _____

Approved No. _____

Non-Metallic Drum

Company _____

Approved No. _____

Speed Display Trailer

Company _____

Approved No. _____

Vertical Panels

Company _____

Approved No. _____

Temporary Raised Pavement Markers

Company _____

Approved No. _____

Modular Glare Screen

Company _____

Approved No. _____

Other Device: _____

Company _____

Approved No. _____

Portable Changeable Message Sign

Company _____

Approved No. _____

Other Device: _____

Company _____

Approved No. _____

Signed: _____

Date: _____

6.10 Arrow Panels and Warning Lights

6.10.1 Arrow Panels

6.10.1.1 Display

- (a) Approved arrow panels, listed in Publication 35, Section 901, are required, as a minimum, to display each of the following messages:
 - 1. Left flashing arrow.
 - 2. Right flashing arrow.
 - 3. Simultaneous left and right flashing arrows (flashing double arrow).
 - 4. A caution mode consisting of four or more lamps arranged in a rectangular pattern, or horizontal pattern of four or more lamps in a straight line or “bar” that will not indicate a direction.
- (b) Some arrow panels will also display a left or right sequential arrow or sequential stem mode. The use of the sequential arrow and sequential stem message modes should not be used for the following reasons:
 - 1. An FHWA sponsored research study found that drivers clearly prefer the flashing arrow and sequential chevron modes, and these modes were generally superior to the sequential arrow and sequential stem modes in promoting lane changing and driver understanding.
 - 2. The sequential arrow and stem must go through four pulses as opposed to two pulses for the flashing arrow.
 - 3. The arrowhead in the sequential stem mode only appears during the third pulse.
- (c) Only use the flashing arrow mode for a lane closure on multilane-lane roadways. The caution mode may be used for shoulder work or for temporarily closing one lane of a two-lane roadway.

6.10.1.2 Dimming

The Department's specification for arrow panels requires that all arrow panels have an automatic dimming circuit actuated by a photocell at a light level of approximately 5 foot-candles to provide a minimum of 50 percent dimming from the rated lamp voltage. If this dimming does not take place, arrow panels are much too bright during hours of darkness. Periodically inspect all Department-owned arrow panels to verify that the photocell and dimming circuit are operating properly. Also, make periodic checks of all arrow panels used on construction projects to determine if the proper dimming takes place during hours of darkness. If problems occur, notify the contractor and require the necessary repairs made.

6.10.1.3 Placement

Arrow panels should generally be located as shown in the typical applications in Publication 213; however, this location should be field adjusted to optimize visibility of the device. Each District should require all Department users of arrow panels to become familiar with Section 6F.56 of the *MUTCD* and all the figures relating to arrow panels in Publication 213. Project Manager on construction projects should review all arrow panels to determine if arrow panels are properly located.

6.10.2 Warning Lights

6.10.2.1 Application of Flashing Warning Lights, Types A and B

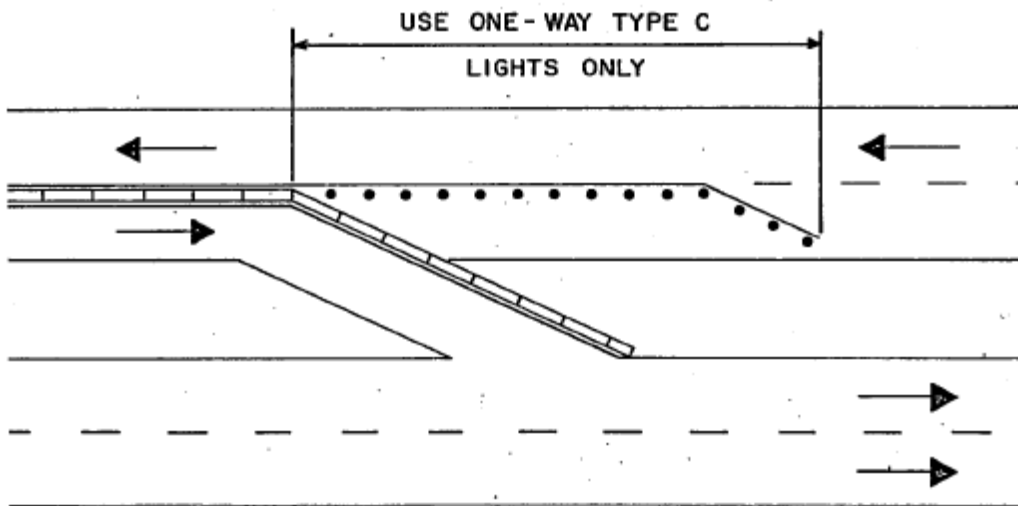
- (a) Flashing warning lights may be used to call drivers' attention to signs, obstructions, or potential hazards.
- (b) Flashing warning lights may be used to mark obstructions or potential hazards. They are not permitted to be used for delineation since a series of flashers would tend to obscure the desired vehicle path. Flashing warning lights are not permitted to be used in a series unless the spacing between successive flashers is at least 250 feet.
- (c) Two-way Type A lights are not permitted to be used where they would be visible and confusing to drivers in other directions of travel.
- (d) Type A and B lights shall be mounted on signs in accordance with the *MUTCD*. When used, the lights shall have a minimum mounting height of 30 inches to the bottom of the lens, and they shall be mounted at or above the midpoint of the sign face without

obscuring any part of the sign face. When a Type A or B light is mounted on a STOP (R1-1) sign, the color of the light shall be red.

6.10.2.2 Application of Steady Burn Lights, Type C

- (a) Type C lights may be used to delineate the proper vehicle travel path through a work zone. They may be mounted on barricades, drums, vertical panels, barriers, or other supports. See Exhibit 6.10-A.
- (b) Type C lights may be used when a series of lights is required.
- (c) Two-way Type C lights are not to be used where they would be visible and confusing to drivers traveling in the opposite direction to that of which the lights are meant to serve.

Exhibit 6.10-A Use of Type C Lights Through Work Zone



6.10.2.3 Flashing Warning Lights on Type III Barricades

- (a) When using a flashing warning light in conjunction with a barricade, TC 8716 requires the Type B light assembly to be attached to the sign, but the battery case must rest on the pavement surface and be fastened to a vertical leg. The distance between the top of the battery case and the pavement surface shall not exceed 20 inches.

6.11 Shadow Vehicles and Impact Attenuators

6.11.1 Shadow Vehicle(s) Required

General Pennsylvania Statewide Policy (from Publication 213). This applies to all entities (contractors, utilities, municipalities, etc.) performing roadwork in Pennsylvania. As shown in the PATA figures of Publication 213, shadow vehicles are required with a Truck Mounted Attenuator (TMA) on all freeways and expressways, but are optional on other facilities as noted in the specific PATA drawings.

Specific Policy for PennDOT Maintenance. Use shadow vehicles in the following short-term maintenance work areas:

- (a) On freeways and expressways. Use a shadow vehicle for all full-lane or partial-lane closures and for all work on the shoulder. If the shoulder is not wide enough to safely park the shadow vehicle, then the adjacent lane should be closed.
- (b) On other roadways.
The use of a shadow vehicle is optional as stated in Pub 213 and in Pub 23 (Chapter 11 under Shadow Vehicle).

6.11.2 Occupancy of Shadow Vehicles

- (a) When protecting a stationary or stop-and-go operation where the operation will be stopped for an estimated 5 minutes or more, turn the ignition off, set the emergency brake, put the transmission in reverse, and turn the front wheels to the right. However, when the shadow vehicle is in the extreme left lane or left shoulder where a minimum 30-foot wide median exists, then turn the front wheels to the left.

Moreover, the shadow vehicle shall be unoccupied. The operator of the shadow vehicle shall not remain in the vicinity of the stationary shadow vehicle but shall assist the work crew or function as a watcher as assigned by the foreman.

- (b) When protecting a continuous moving or stop-and-go operation where the operations will be stopped for less than 5 minutes, the driver of the shadow vehicle shall be the only occupant in the vehicle.

6.11.3 Posting of Shadow Vehicle(s)

- (a) In general, all shadow vehicles that are not behind other work area protection devices should be visible from behind for a distance, in feet, of ten times the legal speed limit.

- (b) When only one shadow vehicle is used, it shall be located behind the work area at an approximate distance of 100 to 250 feet.
- (c) When two shadow vehicles are used, except for stationary short-term operations in a “two-way, left-turn lane, (i.e., except for PATA 15 in Publication 213), the first shadow vehicle shall be located as in (b) above and the second shadow vehicle shall be located completely on the shoulder and a minimum distance upstream from the first shadow vehicle as follows:
 - 1. All freeways and expressways: 1,000 feet.
 - 2. All other highways:

25 mph	130 feet
30 mph	200 feet
35 mph	250 feet
40 mph	320 feet
45 mph	540 feet
50 mph	600 feet
55 mph	660 feet
- (d) When two shadow vehicles are required on short-term stationary operations in a “center-two-way-left-turn-lane” (PATA 15 in Publication 213), one shadow vehicle shall be located on each side of the work area at an approximate distance of 100 to 250 feet. (NOTE: If wheels are turned on the shadow vehicles in this scenario, the wheels should be turned to the right.)
- (e) All shadow vehicles shall be positioned so that the back of the vehicle is facing oncoming traffic.

6.11.4 Size, Weight, and Equipment for Shadow Vehicles

When used with a truck-mounted attenuator (TMA), the size and weight of the vehicle should be in conformance with the TMA manufacturer’s specifications.

6.11.5 Attenuator System Design

- (a) Publication 35 lists approved attenuator systems. For impact attenuator specifications, refer to Publication 408, Sections 696 and 697.
- (b) The individual design of attenuators shall be in accordance with the information in the Design Manual, Part 2, and with the manufacturer’s catalogs and recommendations.

-
- (c) The nose of all impact attenuators used in construction projects shall have delineation in accordance with RC 57M, Sheet 3 of 8, Detail A, and Note 2.
 - (d) If using sand barrels for attenuation, use the layouts as displayed in the [Appendix 6C](#).

6.12 Workers' Attire

6.12.1 Required Attire

FHWA Visibility Rule, November 24, 2006

- All workers within the right-of-way of a Federal-aid highway who are exposed either to traffic (vehicles using the highway for purposes of travel) or to construction equipment within the work area shall wear high-visibility safety apparel.
- *Worker* - a person on foot whose duties place him or her within the right-of-way of a street, highway, or pathway, such as street, highway, or pathway construction and maintenance forces, survey crews, utility crews, responders to incidents within the street, highway, or pathway right-of-way, and law enforcement personnel when directing traffic, investigating crashes, and handling lane closures, obstructed roadways, and disasters within the right-of-way of a street, highway, or pathway.
- Responders to incidents and law enforcement personnel may wear a "Public Safety Vest within the highway right-of-way
- Firefighters or other emergency responders working within the right-of-way and engaged in emergency operations that directly expose them to flame, fire, heat, and/or hazardous materials may wear retroreflective turnout gear that is specified and regulated by other organizations, such as the National Fire Protection Association.
- High-visibility safety apparel means personal protective safety apparel that is intended to provide conspicuity during both daytime and nighttime usage and that meets the Performance Class II or III requirements of the ANSI/ISEA 107-2004 or newer publication entitled "American National Standard for High-Visibility Safety Apparel and Headwear."

Department Employees

- (a) Employees who are exposed to moving vehicles (live traffic) or construction equipment shall wear a department-issued hard hat (helmet) along with high-visibility attire.
- (b) Employees exposed to moving vehicles and equipment must always wear a high-visibility Class II fluorescent yellow-green (or fluorescent yellow-green-orange) vest.
- (c) Employees exposed to moving vehicles and equipment at night must always wear fluorescent yellow-green (or fluorescent yellow-green-orange) vests in combination with fluorescent high-visibility pants or leggings that together equals Class III.

- (d) Department employees operating as flaggers must wear fluorescent yellow-green-orange (or fluorescent yellow-green) Class II high-visibility reflective vests and fluorescent high-visibility pants or leggings that together equals Class III may be worn but not required.
- (e) Fluorescent yellow-green Class II high-visibility/reflective rainwear can be substituted in any situation.
- (f) All high-visibility attire must be Department approved materials.
- (g) Effective March 9, 2010, Class II high-visibility reflective T-shirts, sweatshirts or jackets will be prohibited (HR Employee Safety Division Memo 3/9/10).

6.12.2 Ordering High Visibility Garments

- (a) High visibility pants and leggings are available through the PIBH. These garments are made with the same lightweight (2.3 oz) material as used in the vests. There are two horizontal retroreflective stripes on each pant leg or legging. Use of these garments will increase visibility of the wearer both day and night due to the continual movement of an employee's legs while walking/working.
- (b) High visibility leggings were developed as an option due to comfort concerns. The leggings slip on the lower leg between the knee and ankle. High visibility pants are loose fitting and should be slipped over the employee's regular pants. In no case should these garments be substituted as regular pants the employees wear to work on a daily basis.
- (c) Questions regarding high visibility clothing should be forwarded to the Employee Safety Division.

6.12.3 Reflective Tape on Hard Hats

Department employees may place one-inch wide strips of silver-colored reflective tape on their hard hats since this may improve the visibility of workers during hours of darkness.

6.13 Specialized Operations

6.13.1 Counter Installation

6.13.1.1 General

This is the Safety and Work Zone Traffic Control Policy for the installation, maintenance, repair, or removal of traffic counting equipment for the purpose of recording traffic volumes, classification, and weight. While this policy encompasses most situations encountered during the installation, maintenance, repair, or removal of traffic counting equipment and is intended to highlight and clarify key safety and work zone issues, it is ultimately the responsibility of those engaged in this activity to comply with the policy set forth in the appropriate regulations and publications.

This policy applies to all Department personnel, Metropolitan Planning Organizations (MPOs), Regional Planning Organizations (RPO), contracted vendors and others engaged in the installation, maintenance, repair, or removal of traffic counting equipment on highways within the Commonwealth.

6.13.1.2 Vehicle

The vehicle(s) used during the installation, maintenance, repair, or removal of traffic counting equipment shall be equipped with either a flashing or revolving yellow strobe light or a bar of lights. The light or bar of lights shall be at a location on the vehicle where it is visible by approaching traffic from all directions (see 67 Pa. Code, Chapter 173, “Flashing or Revolving Lights on Emergency and Authorized Vehicles”).

6.13.1.3 Personal Protection Equipment and Attire

The following are regulations regarding equipment and attire for personal protection as specified in the Department’s “Safety Policy Manual,” Publication 445.

- (a) High Visibility Safety Apparel. All personnel involved in the installation, maintenance, repair, or removal of traffic counting equipment shall wear a high-visibility vest, shirt, or jacket. For nighttime conditions, similar outerwear shall be reflectorized and for PennDOT personnel Leggings are required.
- (b) Head Protection. During the installation, maintenance, repair, or removal of traffic counting equipment, all personnel shall wear a hard hat. The hard hat may only be removed at the point of and

during any task that makes it difficult to keep the hard hat on the head, at which time the hard hat may be removed and placed next to the worker to complete the task. The hard hat must be replaced immediately after completing the task. These tasks may include but are not limited to: kneeling to install portable counters and/or road tubes, in-pavement sensor installations, and routing of sensor lead wires.

- (c) Eye Protection. Wear safety goggles/glasses with side-impact protection during any installation, maintenance, repair, or removal of traffic counting equipment that may cause an object or material to become airborne. Eyewear must meet ANSI Z87.1 standards.
- (d) Hand Protection. Wear gloves during any installation, maintenance, repair, or removal of traffic counting equipment that may cause abrasions, lacerations, blisters, or punctures to the hand(s).

6.13.1.4 Work Zone Safety and Signing

Most activities performed during the installation and removal of portable traffic counters, and the installation, maintenance, or repair of permanent traffic counting facilities are short-term operations. Therefore, employees shall comply with figures from Publication 213.

- (a) Portable Counter Installation/Removal. Since the installation and removal of portable traffic counters normally takes less than 60 minutes. Therefore, if applicable, refer to notes on the figures in Publication 213 to eliminate the signs and channelizing devices. The vehicle and traffic counting personnel shall be seen by approaching traffic for a distance, in feet, equal to ten times the posted speed limit. In addition, if the counting equipment cannot be safely installed due to a narrow shoulder, insufficient sight distance, heavy traffic volumes, or any other unsafe condition, use a flagger or a uniformed police officer to assist the traffic-counter personnel. If this assistance is not available, do not attempt to set the counter. Notify the immediate supervisor.
- (b) Permanent Traffic Counting Facility Installation/Maintenance/Repair. Permanent traffic counter facilities may include, but are not limited to, Automatic Traffic Recorder (ATR), Continuous Automatic Vehicle Classifier (CAVC), or Weigh-in-Motion (WIM) sites. Use figures from Publication 213 to establish work zone traffic control during the installation, maintenance, or repair of permanent traffic counting equipment and site components located adjacent to the roadway. Any operation that blocks a

portion of the travel lane must comply with Department of Transportation Regulation, 67 Pa. Code, Chapter 212, “Official Traffic Control Devices.”

- (c) Signing. When warning signs are required, consider using a COUNTER INSTALLATION AHEAD (W21-15) sign instead of the ROAD WORK AHEAD (W20-1) sign.

6.13.1.5 Additional Safety Precautions

The work performed by personnel during the installation, maintenance, repair, or removal of traffic counting equipment demands a heightened sense of alertness due to exposure to passing motorists. It is for these reasons that everyone is required to adhere to this policy and observe all possible safety precautions to prevent injury to one’s self and to prevent hazardous conditions for the motoring public.

- (a) Carefully plan the location of traffic counters — use routes that restrict numerous directional changes. A tangent section of highway is best for setting traffic counters. This allows for additional sight distance and helps to ensure the longevity of road tube by avoiding hard steering or braking traffic.
- (b) Pull the vehicle(s) onto the shoulder and turn on the four-way flashers, flashing, or revolving yellow strobe light or light bar, and headlights to give additional warning to approaching motorists.
- (c) Wear personal protective equipment, including a hard hat, safety goggles/glasses, gloves, and a high visibility safety vest as required.
- (d) Allow enough time to travel between counter locations. Look for a stable (but not too hard) surface to strike nails or spikes into and be careful to strike the center of the nail head or spike to avoid ricochet. Carefully secure the “dead end” of the road tube far enough away from the path of travel to reduce the possibility of injury by a passing vehicle.
- (e) Wait until all cars in a row have passed and there is no sound of approaching traffic. Allow enough time to set counters safely and be sure to have enough “slack” in hand before starting across the highway. Tie off and secure the road tube to avoid having the road tube and nails pulled up by traffic.

- (f) Avoid setting traffic counters in areas of tall grass where ticks and other flying insects may be harboring. Wear a good pair of hiking shoes, long sleeve shirt, and durable jeans that protect the legs.
- (g) Drive defensively. Other drivers are often impatient as you turn or slow down to set up the counters. Use turn signals, mirrors, and avoid backing up whenever possible. Be aware that the typical van creates blind spots to the rear and to the side.
- (h) If an accident or injury should occur, immediately report it to the supervisor.
- (i) Do not attempt to set a counter in an obviously dangerous area — look for a safer location. If this is not possible, do not attempt to set the counter, but inform the supervisor of the problem.

6.13.1.6 PennDOT Traffic Counter Training

All persons that will be engaged in installing and removing portable traffic counters shall view PennDOT's Traffic Counter Training video, which includes safety and installation/removal best practices. The PennDOT Traffic Counter Training video is available on the Bureau of Planning and Research's internet "Traffic Partners Page" at:
<http://www.dot.state.pa.us/Internet/bureaus/pdPlanRes.nsf/infoBPRvideochoice>.

Copies of PennDOT's Traffic Counter Training video are also available on compact disc from the Bureau of Planning and Research by calling 717-787-5796.

6.13.2 Backing Up or Driving in a Counter-Flow Direction

6.13.2.1 Legal Opinion

§15.3(b)(1)(ii) of 67 Pa. Code, Chapter 15, "Authorized Vehicles and Special Operating Privileges," authorizes maintenance vehicles to back up on limited access highways and to drive in a counter-flow direction. Moreover, the Office of Chief Counsel supports this position in the following legal opinion:

"It is our opinion, and you are hereby advised, that your interpretation is correct, and the vehicles you describe (Department maintenance vehicles) would, in performing the necessary maintenance activities, be exempted from Section 3702. Section 6107 of the Vehicle Code (75 PA C.S. Section 6107) provides that the Department may designate any vehicle as

authorized upon a finding that the vehicle is used in the performance of public service or government functions, and that duly authorized vehicles are exempted from certain provisions of the Vehicle Code as should be specified in regulations promulgated by the Department. These regulations were duly promulgated and are applicable to the activity discussed.”

6.13.3 Mobile and Stationary Maintenance Operations

Based on recommendations from the Work Zone Traffic Control Safety Task Force, the following figures for Department force apply to short-term, mobile, or stationary operations, and comply with the *MUTCD*.

- (a) Figures PATA 11A and 11C of Publication 213 generally apply to the following operations, and other similar operations for roadways with an ADT of 1,500 or less:
 - Shoulder grading
 - Shoulder stabilization
 - Shoulder cutting
 - Mechanized patching
 - Surface treatment
 - Sub-base repair
 - Skin patching
 - Scratch coat
- (b) Figure PATA 11D of Publication 213 applies to pipe replacement.

6.13.4 Mowing Operations

- (a) The “MOWING NEXT ___ MI” (W21-14) sign is optional for either contract or Department mowing operations. However, Districts are encouraged to:
 - (1) use the “whip type antenna” with the fluorescent orange flag on all mowing equipment; and
 - (2) paint the roll bar framework on Department mowing equipment a bright orange. See the figures in Publication 213 for additional information.
- (b) The requirements of PennDOT Publication 213 will still apply and must be followed for mowing operations. These guidelines allow for the use of a shadow vehicle. Consider the use of a shadow vehicle when the mowing operation is on the travel portion of the highway and the sight distance (in feet) to the work vehicle (mowing equipment) from behind is less than ten times the speed limit (in miles per hour).

6.13.5 Department Surveying Operations

Highway surveying operations that block any portions of a travel lane shall comply with the following guidelines. Traffic control schemes shall comply with the appropriate typical figures. The SURVEY CREW (W21-6) sign may be used in lieu of the ROAD WORK AHEAD (W20-1) sign for surveying operations. Department Survey Crews are subject to the requirements of Chapter 212, Publication 213, and the *MUTCD*.

6.14 Traffic Control Plan (TCP) Design Considerations

6.14.1 General

- (a) Refer to [Section 6.3](#) for the Work Zone Safety and Mobility Policy for specific requirements and procedures on the determination of “Significant Projects” and the development of Transportation Management Plans (TMP) for projects classified as “Significant.” The Traffic Control Plan (TCP) will be included by reference within the TMP when a TMP is developed for a specific project. Additional guidelines intended to supplement Section 6.3 is included in this Section.
- (b) Consider the maintenance and protection of traffic on construction projects during the pre-design and design phases of the project. It is the District’s responsibility to consider appropriate traffic control with the aim to avoid as much disruption and inconvenience to the motorists as is practical while providing adequate safety to motorists and construction workers. Therefore, designers should coordinate this effort with the District Construction and District Traffic Units.
- (c) The designer is primarily to consider achieving maximum traffic control through the project and then to develop a design accordingly. Consider road user costs in the maintenance and protection of traffic schemes. In many instances, the type of design may not change, but the method or scheduling of construction may prove significant in the traffic control plan and have an effect on the community.
- (d) Where the impacts on the community would be significant, proper coordination with the community and civic leaders are necessary. Their concerns and input must be a part of the evaluation of the maintenance and protection of traffic schemes. Additional guidance on this is provided in Section 6.3 (Work Zone Safety and Mobility).
- (e) An evaluation of maintenance and protection of traffic should be included in all phases of design development.

6.14.2 Work Zone Safety Design Considerations

6.14.2.1 All Highways

The following guidelines apply to all highways (including freeways; expressways; two-lane, two-way highways; arterials; and streets).

- (a) Portable changeable message signs (PCMS) combined with active radar can be effective in reducing speeds through work zones.

When the radar detects a speeding vehicle, the sign will display a message to inform the driver that he/she is speeding. Districts are encouraged to use one of the following word messages instead of a numeric speed display:

“YOU ARE SPEEDING // SLOW DOWN”

“TOO FAST // SLOW DOWN”

Place the sign(s) at the beginning of the reduced speed area or in advance of other active work activities.

- (b) Determine the maximum length of the queue (see [Section 6.14.3.4](#)). Approximately one mile upstream of this point (depending upon the speed limit); use a PCMS to provide the estimated distance to the queue. Repeat “BE PREPARED TO STOP” (W3-4) signs at 1-mile intervals to the lane closure or reduction.
- (c) To be of any value, signs must be in good condition, convey a clear, concise message, and be placed appropriately for current conditions. Anything less creates disrespect and causes the driver to question the validity of the signs. As a result, the driver may ignore messages that are critical to safety. Regularly review work zone signs for relevance. Signs that are irrelevant or misleading will cause driver confusion and damage our credibility. Immediately remove or cover irrelevant or misleading signs.
- (d) Whenever possible, construction vehicles should be restricted from entering or exiting the work area from travel lanes.
- (e) Use the 50-inch temporary concrete barrier as the standard for separating traffic through crossovers and in two-way operations. If the higher barrier would create a sight distance problem, use the 32-inch high barrier.
- (f) Whenever traffic is restricted to narrow travel lane without good shoulders, wheels of large vehicles tend to track beyond the paved area creating pavement edge drop-offs. Therefore, plans should provide for shoulder widening and reconstruction before confining traffic to narrow travel lanes.

6.14.2.2 Freeways and Expressways

- (a) On major freeway construction projects, develop a formal incident management plan as part of the design phase of the project. If one

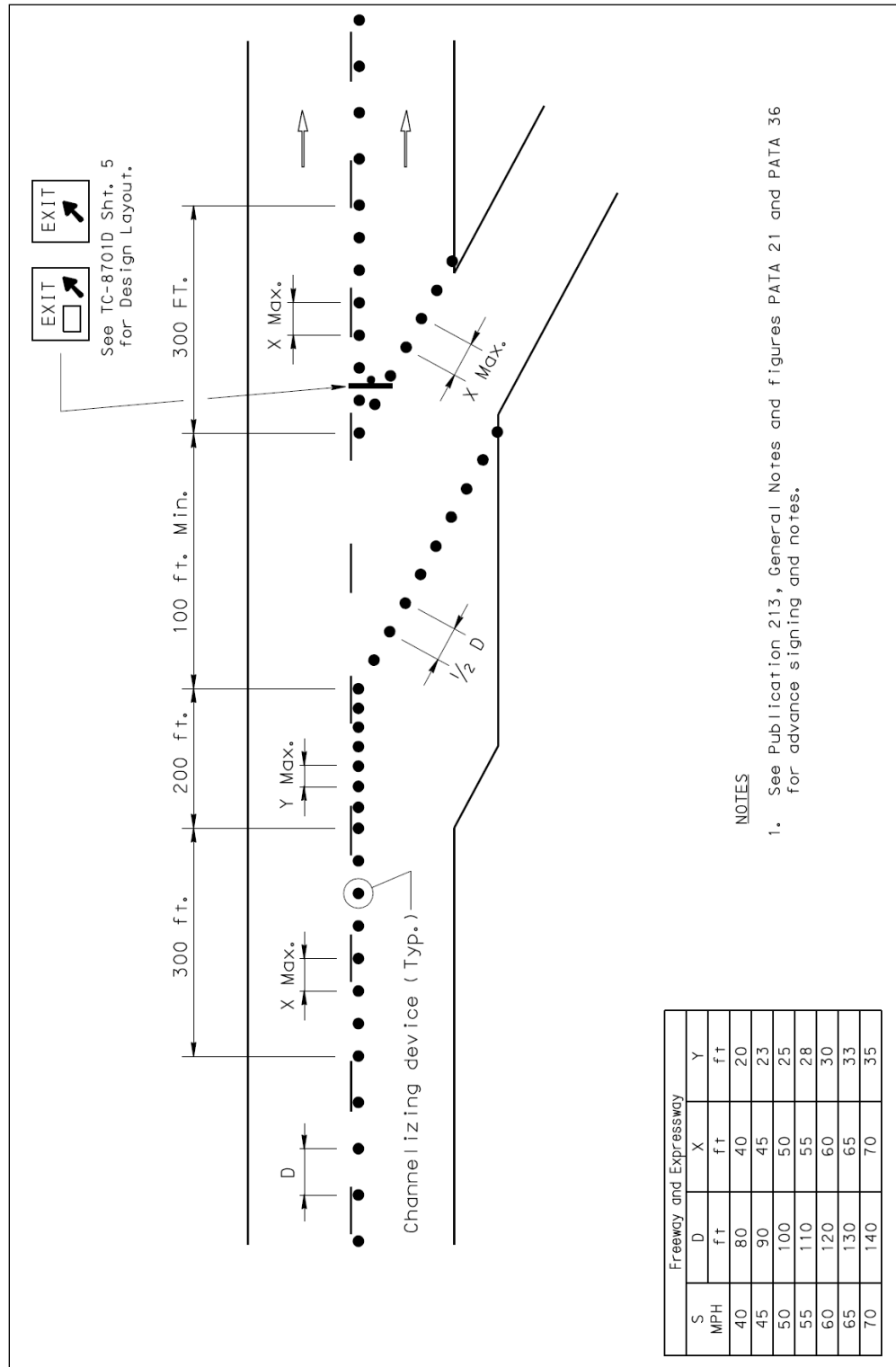
already exists, update it if necessary. It is important to involve local community representatives, emergency response units, and local police agencies along with the Department in the preparation of the incident management plan, so that everyone understands their responsibilities. See [Section 6.3](#) for further requirements and guidance on the development of incident management plans as part of the Transportation Operations Plan (TOP) and TMP for “Significant Projects.”

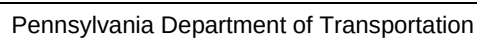
- (b) When closing the right lane through an interchange area, reduce the spacing between the channelizing devices at the exit ramp to highlight the exit path. In addition, to make the exit sign more visible on long-term operations, use a green-and-white EXIT sign to provide contrast with the orange channelizing devices as illustrated in [Exhibit 6.14-A](#). (A 48"x48" black-on-orange EXIT sign, i.e., W25-4, may continue to be used for short-term operations.)
- (c) Consider installing pole-mounted “cobra head” type luminaries on freeway crossovers. Avoid floodlights since they only illuminate a small area with a bright light that may cause drivers to lose their night sight adaptation. Cobra head streetlights illuminate a larger area and are not as distracting to the driver.
- (d) Provide emergency pull-off areas at nominal one-half-mile intervals, adjusted to meet field conditions. Specify standard dimensions for the pull-off areas per [Exhibit 6.14-B](#). Also require the area to be paved since gravel pull-offs are more difficult to maintain and provide poor traction for acceleration and deceleration. To highlight the area, install flexible delineator posts or other channelizing devices at 25 to 40 foot intervals.

It is recommended that Emergency Pull-Off” signs be used instead of “Emergency Parking Area” signs which have been used in some Districts, since “parking” gives the connotation that the area can be used for causal stoppages.

- (e) Highway Advisory Radio (HAR) can be an effective method of giving motorists up-to-date information. The project office should have a cellular phone hook-up to change the message as necessary. Remotely controlled flashing lights, and the “URGENT

Exhibit 6.14-A Alternate Lane Closure near an Exit Ramp





MESSAGE WHEN FLASHING” (G60-1A) sign are also effective. Special information regarding FCC licenses is as follows:

- The Radio System Section in the Bureau of Maintenance and Operations (BOMO) manages all HAR call signs and radio station licenses for PennDOT. They will also register these signs with FCC, but the Engineering District Office must retain the original FCC license.
 - During construction, the contractor would normally complete the tasks associated with securing the FCC license for the HAR. Once the HAR is operational, the contractor turns the license over to the District who must maintain it. At that point, the District should forward a copy of the FCC license to the Radio System Section so that they can register the new HAR call signs with the FCC for a Taxpayer Identification Number (TIN).
 - If a District has not already done so, they should forward a copy of all FCC licenses to the Radio System Section. Additionally, please forward copies of new FCC licenses within 30 days after receipt.
 - The FCC only recognizes a few contacts within any agency for centralized coordination, so this is an activity that the contractor or District cannot complete. The FCC requires that each FCC license have a TIN assigned to it. Failure to do so will cause the FCC to suspend the FCC license until a TIN is obtained.
- (f) Use the Work Zone Alert and Information Radio, commonly called the “*Wizard*,” especially when there are a high percentage of commercial vehicles. When used, prerecord up to three messages on the “*Wizard*” in the office before taking it to the field.
- (g) Consider handing out brochures at truck stops. The Bureau of Highway Safety and Traffic Engineering will provide your Community Relations Coordinator with materials.
- (h) Continue the practice of providing enhanced work zone police enforcement, including the practice of stationing an officer in a marked police car with its lights flashing, upstream of any queue buildup.

- (i) Whenever possible, construction vehicles should be restricted from entering or exiting the work area from the travel lanes.
- (j) Use the 50-inch temporary concrete barrier as the standard for separating traffic through crossovers and in two-way operations.
- (k) Consider installing the modified rumble strip pattern upstream of the lane closure or reduction (see [Exhibit 6.14-C](#)).
- (l) On major projects, the contractor (or subcontractor) should assign an employee the full-time job of patrolling the work zone and looking for “knock downs,” damaged, missing, confusing, or irrelevant traffic control devices.

6.14.2.3 Enhanced Worker Protection

Consider the use of temporary concrete barriers to protect workers in all freeway and multi-lane work zones when all of the following apply:

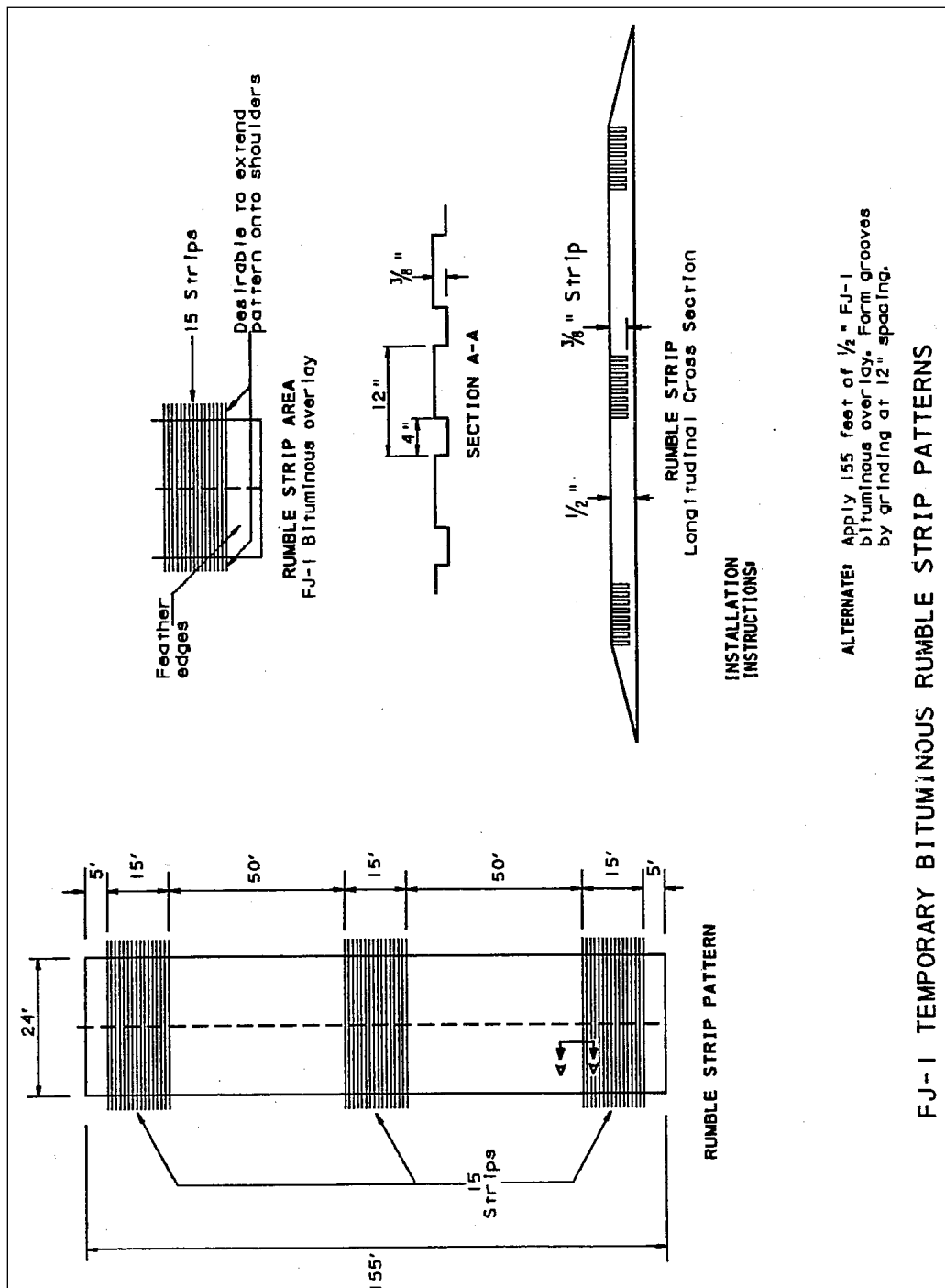
- The speed limit is 45 mph or greater.
- When workers would be present within one lane width from a live travel lane.
- The closure applies 24 hours per day for more than 2 weeks.

For operations that meet the above requirements where it is not feasible to provide portable concrete barriers, shadow vehicles as defined in [Section 6.11](#) of this document should be specified at individual work sites to protect workers.

6.14.2.4 Bicyclist and Pedestrian Accommodation

- (a) In the areas where bicycle or pedestrian use exists, traffic control plans (TCPs) should include provisions to accommodate bicyclists and pedestrians. The TCP may consist of a modified Publication 213 figure that reflects these special provisions.
- (b) Protect pedestrians from work vehicles, equipment, and operations. Some circumstances may require establishing a detour.
- (c) Do not expose bicycle and pedestrian pathways to unsafe conditions, such as unprotected excavations, open utility access, overhanging equipment, or other similar conditions.

Exhibit 6.14-C Temporary Bituminous Rumble Strips



6.14.2.5 Reporting Management and Oversight Costs

Act 229 of 2002 requires that “*an amount totaling at least one half of 1% of the total dollar amount of projects let annually by the Department of Transportation shall be allocated for the purposes of management and oversight of work zones in order to enhance work zone safety.*” Therefore, the Department must track the following costs and annually report them to the transportation committees:

1. State Police Overtime Safety Assistance. BHSTE will track this cost; therefore, no action is required on the part of the Engineering Districts.
2. Design of Work Zone Traffic Control. Use the Cost Function “82427” (i.e., “WORK ZONE TRAFFIC CONTROL”) for in-house design of all WZTC items for all Programs. (Note, based on reviews of a cross-section of consultant design expenditures, on a statewide basis BHSTE will use a very conservative estimate of 4 percent of all contract design expenditures.)
3. Review of the Design of Work Zone Traffic Control. Use the Cost Function “82457” (i.e., “WORK ZONE TRAFFIC CONTROL”) for this effort for all Programs.
4. Construction Inspection of Work Zone Traffic Control. Use the Cost Function “89967” (i.e., “WORK ZONE TRAFFIC CONTROL”) for this effort for all Programs
5. Local Police Safety Assistance. District should tabulate this cost on a project-by-project basis, as applicable.
6. Cost of Advanced Work Zone Technologies. Districts should tabulate construction line expenditures for queue detection systems, variable speed limits, ramp-metering, technology-assisted enforcement, camera surveillance through the work area, etc., as applicable.
7. Work Zone Training. Districts should tabulate personnel and training costs for design, construction, and traffic personnel.

Of the seven different costs, Districts should provide the annual costs for Items 5, 6, and 7 by July 31 of each year for the fiscal year that ended in the previous month. To help coordinate these expenditures, all Districts are to use the database located at the “Mgmt./Oversight Cost” tab located

at [P:\bhste_shared\workzones\WZ Annual Reports\WZTC Database \(Annual Report\).mdb](P:\bhste_shared\workzones\WZ Annual Reports\WZTC Database (Annual Report).mdb) (see Exhibit 6.14-D).

Exhibit 6.14-D Costs to Enhance Work Zone Safety

Microsoft Access - [Data Form]

Record Number: 55, Calendar Year: 2006, District: 10-0

Click on Tabs below

QA INFO | WZ Crash Info | Mgmt./Over-sight Cost | Best Practices | WZ Training | Measures/Technology | Policy/Procedures | ITS in Work Zones | BHSTE

District QA Reviews

Total	Avg. Score	Failures
0	0.00	0 (<65%)

Most Common Deficiencies

Deficiencies	Comments
1 N/A	N/A
2	
3	
4	
5	
6	
7	
8	
9	

BHSTE Use ONLY

Record: 1 of 10
Form View

6.14.3 Work Zone Congestion Reduction

6.14.3.1 General

- Refer to [Section 6.3.2](#) to determine if a project is a “Significant Projects,” and to [Section 6.3.3](#) for the development of Transportation Management Plans (TMP) for “Significant Projects.”
- It is incumbent on all of us to incorporate congestion mitigation measures in all construction projects (I4R, 3R, Bridge, Preventive Maintenance, etc.). The goal is to minimize delays to motorists as much as practical.
- Consider the following three principles in the design of projects to minimize construction delays:
 - Avoid recurrent congestion (Level of Service E) if possible by maintaining all travel lanes or by specifying the work hours for times when congestion will be minimal. When

this is not possible, recurrent congestion should be minimized by using innovative contracting procedure incentive/disincentive clauses such as the “A+Bx concept” and/or the lane rental methods.

2. Delay times (stoppage) due to construction operations (normally encountered on resurfacing of two-lane highway projects) should be minimized to no more than 5 minutes, if practical.
 3. Minimize delay problems associated with incidents in the work zone by proper planning and, where practical, develop an incident management plan.
- (d) Refer to [Section 6.3.3](#) for the Work Zone Safety and Mobility Policy for specific requirements and procedures on the development of Transportation Management Plans (TMP), for “Significant Projects,” and proper development and application of TMP provisions to mitigate work zone congestion issues. It is desirable to identify and address congestion issues as early as possible in the project development process. Also, scrutinize and adjust maintenance operations on freeways and other high-volume arterials to avoid congestion.
- (e) To address congestion, Districts should put together teams consisting of one or more representatives of Traffic, Design, Maintenance, and Construction to address the above situations.
- (f) Examples of congestion and delay mitigation provisions to consider are as follows:
1. On freeways and expressways, first consideration should be to maintain all travel lanes, if practical, particularly if a capacity analysis indicates significant congestion associated with lane reductions will occur during several consecutive hours. (Generally assume 1,200 to 1,600 passenger car equivalents per hour, per lane, but the actual value should be refined based on local conditions.)
 2. Restrict work hours to nighttime, weekend, or other off-peak hours. The use of accelerated cure concrete can help ensure the quick reopening of lanes. Also consider special provisions restricting operations that take out a lane temporarily (e.g., setting barriers in off-peak periods).

3. Letting of projects so that as much work as possible can be accomplished before or after the “Memorial Day to Labor Day” period.
4. Tighter CPM schedules to force overtime and get work off the road quicker.
5. Using incentive/disincentive clauses such as, the “A+Bx concept,” “lane rental,” and other innovative contract procedures.
6. When appropriate, use provisions in 3R and preventative maintenance projects, particularly on two-lane, rural highways to minimize the length of stopped delay time due to construction operations.
7. Coordination of projects so that corridors with a high volume of through traffic do not have several construction projects underway at the same time. This is very important if there will be substantial delays at each project.
8. Prior to construction, establish incident management plans for all freeway and expressway projects involving the Department, police, contractors, and emergency responders to minimize the length and impact of incidents. Districts should also review alternate routes for the preplanned detour routes in the event that the alternate route is not available when needed due to an incident or routine maintenance.

6.14.3.2 Improved Traffic Flow

- (a) Consider maintaining all lanes of traffic, by shoulder widening or some other means, to remove traffic completely from the work area. Although material costs may be higher, the removal of traffic may simplify the construction of the project and reduce unit prices and the maintenance and protection of traffic.
- (b) For concrete patching operations on freeways and multi-lane highways which have a speed limit of 45 mph or greater, adhere to the following:
 1. Use high, early strength concrete for pavement repairs whenever traffic flow will be impaired by a long-term lane closure. Include special provisions that minimize the closure time.

2. On high volume facilities (25,000 ADT or greater), establish construction sequencing to close the lane after the evening peak period and reopen it prior to the morning peak period. Schedule work so that excavations can be replaced prior to reopening the roadway, or use temporary material in the excavated patch area to reopen the lanes during daylight hours. Allow exceptions on a case-by-case basis, based on the justification.
- (c) Use innovative contracting to reduce the duration of the work zone (e.g., night work, A+Bx, lane rental, and incentive/disincentive) on a minimum of 80 percent of all projects where State Police protection is requested due to anticipated traffic queues.
 - (d) The length of lane closures should be as short as possible. Whenever possible, use a short “rolling work zone.” On limited access highways, the maximum length of lane closures should be 3 miles unless documentation shows that a longer length minimizes road user costs, or substantially shortens the overall duration of a lane closure. Permit exceptions only on a case-by-case basis.
 - (e) Design freeway projects for a work zone speed limit of not more than 10 mph below the normal speed limit. To accomplish this, long transitions, good sight distance, and excellent delineation are essential. To assist motorists on wet nights, use temporary RPMs or “wet reflective” temporary pavement marking tape throughout the project unless drums or vertical panels are used along the travel lanes to provide a high level of nighttime delineation. (The approved temporary removable pavement marking tape, i.e., Type 750 white and Type 751 yellow, is capable of providing high retroreflectivity even when under water.)
 - (f) Give attention to speed zone restrictions to ensure that the restriction only applies to the area of need. For example, if the speed limit is to improve safety for workers, confine the restriction to the active work area and not throughout the limit of work or during those times when workers are not present. Therefore, it will be necessary to both cover and uncover speed limit signs, or to include special provisions in the contract to require remotely-activated speed limit signs.

6.14.3.3 Late Merge Concept

- (a) The “late merge” concept, uses a unique traffic control setup on freeway construction projects when it is necessary to close a travel

lane. Instead of encouraging motorists a mile or more before the lane closure to form a single lane, this setup encourages motorists to use both travel lanes up to the merge point. Other states are also now using this setup.

(b) Potential benefits of the “late merge” concept include the following:

1. Queued vehicles can stack in two lanes thus making the length of any queue shorter. This may eliminate backups through nearby interchanges, thereby reducing delays for entering and exiting traffic.
2. Studies performed by a consultant and the University of Nebraska suggest that the capacity of a lane reduction may be up to 15 percent greater with the late merge concept, as opposed to using the conventional traffic control. If this is universally true, the “late merge” should reduce any queues to less than 50 percent of the queue length in a normal lane closure.
3. The potential to reduce road rage caused by some motorists passing the queue and butting in line. This allows motorists to drive at a more relaxed and uniform speed.

(c) [Exhibit 6.14-E](#) is a suggested traffic control plan for the “late merge” concept.

NOTE: Field experience in Pennsylvania has showed mixed results with the late merge concept across the state. In some cases it has worked well and in others it has not. There does not seem to be any apparent pattern as to the success of implementing the late merge concept. It has been tried in both urban & rural settings as well as different geographical regions of the state. Districts are still encouraged to consider this approach wherever the throughput capacity may be an issue, while at the same time taking into account past experience as to where the late merge concept has worked and where it has not.

Freeway and Expressway		
MPH	D	E
	f t	f t
50	100	425
55	110	495
60	120	570
65	130	645

1. See Publication 213, General Notes and figures for additional guidelines.

6.14.3.4 Projecting Hourly Traffic Volumes

- (a) Refer to [Section 6.3](#) for the Work Zone Safety and Mobility Policy for specific definitions, requirements, and procedures to determine “Significant Projects” status.
- (b) To plan for major construction projects, it is essential that the designer knows the projected hourly traffic volumes during construction. When they know the projected volumes and capacities, the designer can more accurately determine if incentives, disincentives, lane rental, or other innovative contracting methods are required.
- (c) Therefore, the following guidelines apply to all future construction projects involving lane closures during peak traffic conditions on expressways and freeways, as well as other major highways.
 - 1. Whenever possible, obtain hourly traffic classification counts during the prior year. Take traffic counts in each direction during the most critical month of the proposed construction project (see the tables in the website at <http://www.dot.state.pa.us/Internet/Bureaus/pdPlanRes.nsf/PlanningAndResearchHomePage?OpenFrameset>; click on “Transportation Planning Division” and then click on “Traffic Information”). If possible, obtain continuous classification counts for a Saturday and Sunday. If manual counts are necessary, include the period from 6 a.m. to 10 p.m. on a Friday, and other critical periods during the week based on the existence of nearby shopping or recreational facilities, amusement parks, etc.
 - 2. Use the county growth factors unless more site-specific information is available.
 - 3. Estimate the maximum percentage of traffic that will divert to other roadways during the peak hour. Take into consideration the availability of alternate routes and planned route-diversion techniques. Normally, the maximum route diversion is about 5 to 10 percent.
- (d) By using the results of the traffic counts and analysis, Districts will have a good idea if problems will exist during construction. If the District anticipates recurring severe backlogs, the District should seek methods to reduce the problems, (e.g., incentives, disincentives, route diversion techniques, etc.).

(e) Method #1

1. Convert the projected hourly traffic volume into passenger-car equivalents, using the techniques in the Highway Capacity Manual to show clearly where the projected demand exceeds capacity (see the tables on the website at <http://www.dot.state.pa.us/Internet/Bureaus/pdPlanRes.nsf/PlanningAndResearchHomePage?OpenFrameset>; click on “Transportation Planning Division” and then click on “Traffic Information”).
2. For most work zones, assume capacity to be 1,200 pcphpl. The District may use a projected capacity of up to 1,600 pcphpl if employing the “use both lanes to merge point” technique and based on local conditions. If the length of the work zone is more than 5 miles long, or the travel lane(s) is narrow, the capacity may be less.
3. Based on the projected passenger-car equivalents, determine when the projected demand (including previously queued vehicles) will exceed capacity.

(f) Method #2

Use FHWA’s QuickZone work zone delay estimation tool, which is a Microsoft Excel-based program that generates estimated anticipated delays and queue lengths at approaches to work zones. The program is located at P:\bhste_shared\QUICKZONE. However, when using QuickZone, it is important to comply with the following suggestions:

1. Limit the use of QuickZone to freeways and expressways.
2. Traffic volume data should come from traffic counts taken prior to the development of the Traffic Control Plan during the same time of year that the project will be underway. Weekend and holidays should be included.
3. Avoid using ATR counts and QuickZone program default values.
4. Using the knowledge of highways and habits of drivers in the area, select an appropriate work zone capacity for your District (within the range of 1,200 to 1,600 passenger-car equivalents per hour).

6.14.4 Detours

6.14.4.1 Establishing Detours

As currently written, Pennsylvania Statutes Annotated Title 36, “Highways and Bridges,” Section 670-423, always mandates the establishment of a detour whenever a State highway is closed. It also mandates signs at every intersection, which by definition would include intersections with alleys. Therefore, all road closures, regardless of ADT, shall have an established detour signed in accordance with Publication 213.

6.14.4.2 Construction Detours

- (a) [Exhibit 6.14-F](#) is a legal opinion dated May 19, 1993 concerning detours on local roads. Briefly, it states that Pennsylvania highway law gives the Department the duty to designate, layout, and maintain detours. Therefore, it is not necessary to have an agreement with the local municipality. The Department is obligated however, to repair the detour to a condition at least equal to its pre-detour condition. Further, Districts may make minor repairs to a local road prior to the establishment of the detour to make it safe and passable. It is appropriate for the Department to expend money for this purpose. The repair work is normally included in the contract and performed by the contractor.
- (b) Maintaining good relations between the Department and local officials is important. As stated in [Exhibit 6.14-F](#) first paragraph the Department does not need agreement to use a local road for a detour. It is highly recommended that the District obtain the concurrence of the municipality when establishing a detour on their facilities See [Appendix 6F](#) example agreement. If they are strongly opposed to the detour, use good judgment to determine whether to seek an alternate route. In any event, the District must keep local authorities fully informed.
- (c) When establishing a detour for motor vehicles, provisions are also required for bicyclists and pedestrians. For example, signs and markings to indicate that the bikeway/walkway is closed and detour directional signing. If the detour is longer than 0.5 mile for pedestrians or 2 miles for bicyclists, consider placing distance measurements at both ends of the detour route (when appropriate, use distances in blocks for pedestrians).

Exhibit 6.14-F Detours on Local Roads

From my review of the proposed detour use agreement, the Department has selected a township road as detour and will agree to not only bring the road into its pre-detour condition, but to make certain additional improvements to the township road. Please be advised that we do not need an agreement with the township in order to use the township road as a detour. Section 423 of the State Highway Law (, Act of June 1, 1945, P.L. 1242, as amended, 36 P.S. §670-423, imposes upon the Department the duty to designate and layout detours and to maintain such in a safe and passable condition. Section 423 of the act also statutorily obligates the department to repair the detour to a condition at least equal to its pre-detour condition. Thus, an agreement with the township to establish these responsibilities is not necessary or recommended.

An agreement is only necessary if the detour would involve a private road. Section 423 of the Act specified that “whenever necessary in the creation of the detour, the department may enter into agreement with the owners of private lands, privileges over private property for the period when the main highway shall be closed to traffic.” (emphasis added). It goes without saying that the township road being considered for detour use is not a private road.

With respect to the additional improvements requested by the township, it has been the Department’s policy in the past to require reimbursement of all costs associated with the additional improvements. However, if these improvements are necessary to bring the detour in safe and passable condition, then it is appropriate for the Department to incur these costs.

Please advise the township that we will not be executing the proposed detour use agreement.

Sincerely,

Audrey F. Miner
Assistant Chief Counsel
May 19, 1993

6.14.4.3 Detours vs. Maintaining Through Traffic

General

FHWA has provided information relative to detouring traffic or maintaining traffic through the construction zone. FHWA guidelines consider a policy of first weighing the advantages and disadvantages of “closing to traffic” versus “maintaining through traffic.”

The guidelines require an analysis of overall contract and public costs, as well as safety aspects, before making a decision. On some projects, you may easily dismiss the “closing to traffic” alternative, while on other projects; it may be the best alternative.

Some factors to consider in analyzing and deciding whether to close a particular project to traffic are in [Exhibit 6.14-G](#). The pros and cons of closing a roadway for the duration of construction is examined in detail in

Exhibit 6.14-G Detouring Traffic vs. Closing the Road

	Positive Factors	Negative Factors
Cost	Lower contract costs possible due to more efficient working conditions and reduced traffic control costs possible in many cases.	Possible increase in cost to the public including vehicles and transit systems and business loss along the closed section. Some businesses may be forced to close down. Advertising for closures and detours may be a substantial cost. Cost (strengthen or repair) for use of detours associated with heavy truck traffic.
Time	Contract production rates should increase due to lack of traffic interference. This should be reflected in the contract time allowed and the contract should contain provisions such as incentive/disincentive to assure this advantage.	Detour time to the public may be intolerable in some cases. Length and/or congestion of the detour may make the closing infeasible, inconvenient, or politically unacceptable. Reduction of emergency services may be a heavy elimination factor.
Energy	Contractor energy costs should be reduced as traffic conflicts are eliminated.	Traveling public may expend more energy in traveling the indirect route of a detour. This may more than offset savings in energy by the contractor. Closing a road to "through traffic" is not generally effective if local traffic must be maintained. If "through traffic" can use the project with less inconvenience than the detour, it normally will do so eliminating or reducing contractor savings.
Safety and Traffic Volumes (see Subchapter 6.3 for more information)	Contractor personnel crashes with vehicular traffic will be reduced. Vehicle crashes with the construction environment will be practically eliminated. Little or no impact to the highway user on low volume roads	On high volume routes, detouring may increase congestion and crashes on alternate routes. May cause mass transit system disruption or failure. May delay or eliminate emergency services.
General		Should unforeseen difficulties arise, it will not be easy to reverse the plans of closing a road after award of a contract. Local governments may not permit traffic to be detoured onto their system from another system for various reasons. Some reasons are noise and air pollution, congestion, school student safety, and reduced emergency services.

the Work Zone Safety and Mobility Policy. Specifically, refer to [Section 6.3](#) for specific requirements and procedures on the development of Transportation Management Plans (TMP), for “Significant Projects,” and proper development and application of TMP provisions, such as full road closures, targeted at mitigating work zone safety and congestion issues. However, in evaluating the type of maintenance and protection of traffic, the designer should use the *PUB 213*.

Bridge Construction

Because of negative comments from the public, business community, and the media when closing bridges during major rehabilitation or replacement, the Department reviewed the maintenance and protection of traffic on bridge construction projects. The review indicated that even though project cost was the major reason for specifying detours, other items such as ADT/truck traffic, impact on local economy and emergency services, environmental impact, and ease of getting the needed right-of-way for a temporary crossing should be considered prior to specifying detours. Therefore, Districts need to evaluate options for maintenance of traffic for bridge projects during the preliminary studies and/or environmental studies stage, and perform adequate public coordination to minimize adverse impacts.

Additional guidelines for the maintenance and protection of traffic were developed. The guidelines indicate that generally the selection of maintenance and protection of traffic should be based on the following hierarchy of options:

1. Detour
2. Half-Width Construction
3. New Bridge Adjoining the Existing Bridge
4. Temporary Crossing/Bridge and Approaches

Consider the following for each of the four options:

1. Detour: Consider a detour if any of the following apply:
 - Emergency conditions require closing of the bridge.
 - Moderate and tolerable impact on the local economy, emergency services, school bus routes, etc.

- The route is other than a high volume and detour length is less than approximately ten miles (local conditions may dictate a different length).
 - The detour does not generate any major controversy.
 - A temporary crossing would cause significant environmental impacts (need for extensive archaeological studies, wetland, 4(f) items, etc.) and/or right-of-way clearance problems.
 - At major river crossings where the new bridge has to be at the existing location and half-width construction is not possible.
 - The cost of improving and/or maintaining the designated detour is less than the cost of the half-width construction or temporary crossing options.
2. Half-Width Construction: Consider this option if:
- It is compatible with the scope of work and method of construction.
 - Half-width construction is a viable option and traffic signals can control one-lane traffic on the bridge.
3. New Bridge Adjoining the Existing Bridge: Consider this option when the new alignment at least maintains or improves the existing highway geometry and the extension of work limits does not significantly increase the cost of the project.
4. Temporary Crossing/Bridge and Approaches: Generally, this is the most costly alternative; and can be used if:
- The new bridge must be located on the same alignment as the existing bridge and half-width construction is not possible, and
 - Detouring is not a viable option.

If selecting a detour on a project that can be completed within one construction season, adjust the project letting date so that the detour will not be required over the winter months. If necessary, specify double shifts or a tight construction schedule with a heavy penalty for delays if the

impact on the traveling public and/or affected economic community is severe.

If a temporary crossing/bridge is selected:

1. FHWA must be consulted and approve its use if Federal funds are involved. (“Temporary Bridge Structure” will be added to the list of “Design Items Requiring FHWA Consultation and Approval” in the attachment to the CA document.)
2. A reevaluation of environmental effects will be necessary only if a detour was addressed in the original environmental evaluation process and wetlands, archaeological, or historically sensitive areas are involved. (This is based on the premise that using a temporary bridge in lieu of a detour will have a positive environmental effect.) In addition, DEP and Fish Commission concurrence/approval, if required, will be obtained prior to constructing such temporary bridges.
3. To reduce the cost of a temporary crossing, Districts should consider storing and supplying to contractors used beams and open steel decks removed from existing bridges for such temporary bridge construction.
4. The District should consider specifying, in the proposal, that the superstructure (beams, deck, etc.) elements, when provided by the contractor, will become the property of the Department after construction, in order to build up a stock for future temporary bridges.
5. The form D-4232 (authorization) for construction must include a separate line entry for “Temporary Bridge.” Use regular type code and add “Temporary Bridge” in parenthesis.
6. When the D-4232 for Final Design is prepared, if it is known that a temporary bridge will be used, the “Remarks” section will include a reference to the temporary bridge.
7. If, after the D-4232 for construction has been approved by the FHWA, it is determined that a temporary bridge will be used, an amended D-4232 (including the temporary bridge line entry) will be submitted with appropriate back up data. Obtain federal authorization prior to initiating construction of the temporary bridge.

6.14.5 Overwidth Vehicles

If the construction project is on a designated network highway or an approved access route that permits the operation of 102-inch wide vehicles, the maintenance and protection of traffic plan should include the continued operation of those vehicles during construction.

If the roadway or lane widths are restricted, consideration should be given to the installation of a sign indicating “VEHICLES OVER (___) FT. WIDE PROHIBITED” (R14-16-1). If lane widths are restricted because of construction, you should go through the permit office.

6.15 State Police Assistance

6.15.1 Memorandum of Understanding (MOU)

The safety of highway workers and the traveling public is a top priority with both the Department and the Pennsylvania State Police (PSP). Because of a high number of freeway fatalities, in 1994 the Department initiated a cooperative program where PSP troopers were used to alert motorists of traffic queues on freeway projects. The original name for this cooperative effort was the 'Pennsylvania State Police Work Zone Supplemental Safety Enhancement Program,' but this publication uses the term 'PSP Assistance.'

The concept is to position a trooper in a marked patrol car with activated lights on the shoulder of the roadway, and to have the trooper maintain a location about 1/4 to 1/2 mile in advance of the queue to 'wake up' any inattentive drivers. When the troopers use overtime, the Department reimburses the PSP in accordance with a Memorandum of Understanding (MOU) between the two agencies (the current MOU is at P:\bhste_shared\workzones\PSP).

The program was originally limited to expressways and freeways, but the Memorandum of Understanding (MOU) now allows more flexibility, including application on two-lane, two-way roadways. However, in the Post-9/11 Era, there are additional burdens placed on the PSP and other enforcement agencies. Therefore, there has been some concern over the number of troopers requested for construction projects and maintenance activities, especially on two-lane, two-way highways.

Therefore, Engineering Districts should review their requests for PSP Assistance on two-lane, two-way highways to determine if the number of programmed hours is excessive, or even if any PSP presence is warranted since a well-designed and maintained traffic control setup should provide excellent safety without the presence of a PSP trooper. Districts should also consider ITS technologies to improve safety in an effort to reduce the burden on the PSP. The MOU requires advance approval by both the Bureau of Patrol and the Bureau of Highway Safety and Traffic Engineering (BHSTE).

6.15.2 Budget

Initially, some of the PSP Assistance was at no cost to the Department because troopers used regular time. However, because of other labor needs, Districts should assume that overtime is required for all PSP assistance. In addition to labor charges, the Department reimburses PSP for the cost of the patrol vehicle.

The per hour cost for PSP assistance (includes the vehicle) can be found in the PSP database at P:\bhste_shared\workzones\PSP\Add PSP Projects Here\PSP Database\PSP Assistance Database.

Therefore, when developing a project, each District should budget for anticipated PSP Assistance using the hourly cost for each trooper from the database. Include this cost in the amount that goes for approval by the Program Management Committee (PMC). Districts should indicate the estimated cost for PSP Assistance on the Highway/Bridge phase costs/SPN information screen in the Multi-Modal Project Management System (MPMS). When preparing the request for the Federal Authorization (Form D-4232), include the amount estimated for PSP Assistance as a separate line item and include a comment in the State Remarks field that indicates “Line No. ___ includes \$ ___ for PSP assistance.”

Also, add the estimated amount for PSP Assistance to the Project Development Checklist screen in ECMS (this value must be consistent with the requested amount on the D-4232).

Note: PSP Assistance is not a bid item in the proposal and is not included in the Maintenance and Protection of Traffic items. However, to clarify both the Department and the contractor roles, include the “Pennsylvania State Police Construction Assistance” special provision.

6.15.3 Administrative Procedures

Establish the List of Eligible Projects

1. During project development, the Engineering District is to determine if the project warrants PSP Assistance in accordance with the current MOU. Generally, if the normal hourly traffic volume is more than 1,200 passenger cars or passenger-car equivalents, queuing will occur and the District should consider PSP assistance as necessary.)
2. For each warranted project, the Engineering District should estimate the required funding level using the PSP Database to determine the hourly rate for the PSP officer.
3. Each District is to designate a single work breakdown structure (WBS) code for the life of the project for PSP Assistance charges, even if the project crosses county lines.
4. Districts should add eligible projects to the statewide listing at P:\bhste_shared\workzones\PSP\Add PSP Projects Here\PSP Database\Shortcut to PSP Assistance Database as early as possible, but always

at least 3 weeks before PSP assistance is required. When adding new projects, Districts are to include the WBS and Fund codes.

Schedule and Track PSP Assistance

1. Project managers should advise the local PSP stations as far ahead as possible of the anticipated days and hours that assistance will be required. Project managers are also to maintain communication with PSP throughout the life of the project. Project managers should make it clear that contractors may not request PSP assistance.
2. On larger projects, project offices are encouraged to provide a cell phone to the PSP station for use by the troopers to improve communication with the project office.
3. A PSP trooper is to advise the project office of the number of officers that are on the project, their time of arrival and the time of their departure. If the project provides a cell phone for the PSP, the trooper should call at the times of arrival and departure. If the project does not provide a cell phone, the trooper may stop at the project office, have the dispatcher leave a telephone message, or provide the information by a mutually agreeable method. If the information is not provided, then a daily estimate of the number of hours is to be made. All project offices are to have an answering machine or an answering service that will allow PSP officers to leave messages during those times when the project office is not staffed. (In accordance with the MOU, PSP does not bill travel time to and from the project except for certain occasions.)

If the Department cancels scheduled PSP Assistance within a day of the proposed service, PSP will generally charge the project 3 hours for the call-out even though no service is provided.

4. Project offices should maintain a log in order to record the daily number of hours of assistance at the project office. Project offices are encouraged to use a spreadsheet such as the one at P:\bhste shared\workzones\PSP Spreadsheet Template.xls (please copy and save to another file name), which will automatically calculate costs and maintain a budget balance.

PSP Requests Secondary Cost Element (SCE) from Integrated Enterprise System (IES)

1. Coding for expenditure commitment item, cost center, and funds will be supplied by the PennDOT Bureau of Fiscal Management (BFM).

PSP Submits Preliminary List of Charges for PennDOT Review

1. PSP will prepare a detailed listing of project-by-project charges every month, and will send an electronic copy of the spreadsheet to the Bureau of Highway Safety and Traffic Engineering (BHSTE).
2. BHSTE will review the list and forward it to the Districts for verification of ECMS and WBS numbers. The District will respond back to BHSTE within 10 working days.
3. When all items are verified, BHSTE will notify PSP of any necessary changes, and/or give them authorization to submit the Pre-Note (invoice).

PSP Comptroller's Office Sends Pre-Note to Transportation Comptroller's Office

1. PSP Comptroller's Office sends the Pre-Note to the Transportation Comptroller's Office.
2. Transportation Comptroller's Office seeks BHSTE concurrence.
3. BHSTE requests Bureau of Fiscal Management (BFM) to make funds available in payment account.
4. If PennDOT does not object within 14 calendar days, SAP automatically pays the invoice. If an objection is made, the "14-day clock" stops until all issues are resolved.
5. BHSTE will perform the cost distribution. This will transfer expenditures from the initial payment account to the individual projects by the WBS codes and their corresponding cost centers and funds.

6.15.4 Coordination

The PSP Assistance program requires the cooperation of both the Department and the PSP. Any concern expressed by the PSP need to be addressed and the Department must take the necessary steps to ensure that the PSP Assistance program functions as it was intended. As a result, Districts need to comply with the following requirements:

1. Each Engineering District should establish a Program Coordinator to maintain the Approved Project List. This person should also be available as the contact for the PSP or Central Office in case any problems arise.
2. The Program Coordinator should make requests to add a project to the list at least 3 weeks prior to the initial proposed need (1 week is necessary to get the project on the Approved Project List, and PSP needs 2 weeks advance notice so that they can schedule their personnel).

The need for a 2-week advance notice is a major recurring issue that frustrates the Pennsylvania State Police. Therefore, all Districts must be cognizant of this requirement.

3. Department employees are the only persons authorized to call out the PSP. Authorized persons would generally be the Project Engineer or his/her designee for construction projects, and the County Maintenance Manager or Assistant for maintenance activities. However, these authorized persons cannot make any requests for PSP assistance unless the project is first on the Approved Project List.
4. Because of weather and construction scheduling, the authorized person should reconfirm their needs within the last 48 hours.

6.15.5 PSP Weekly Report of Queue Lengths

The District Work Zone Manager should request PSP to monitor queue lengths every 30 minutes while performing their PSP Assistance role using the form in [Appendix 6D](#). As noted on the form, PSP is to provide the information to the Project Manager at the beginning of every week with a copy sent to the work zone manager. Districts should use this data to help predict queue length on future projects.

6.15.6 SAP Procedures

To streamline the SAP process and reduce errors, the following procedures were developed:

- Use work breakdown structure (WBS) and Fund codes.
- For each new project that requires PSP Assistance, designate a single WBS code for PSP assistance.
- Request PSP officers to notify the project office when they arrive and when they leave so that the project office can track projected expenditures for PSP Assistance.
- When no one is in the project office, use an answering machine or answering service to record PSP calls.
- The project manager is to daily record the total hours of PSP Assistance.
- Provide a timely review and approval of the preliminary list of charges prior to the PSP submission of the Pre-Note.

6.15.7 Police Assistance in Adjacent States

When performing highway work at or near our state borders, congestion and backups sometimes spill over into the neighboring state. In these cases, the Department wants to continue the practice of stationing an officer in a marked patrol car upstream of any traffic queues. Since PSP troopers cannot perform this service, Districts must use a police agency from the neighboring state.

To obtain police services in other state, follow these guidelines:

1. Make the initial contact with the police agency as early as possible, i.e., as soon as it is determined that backups will occur in the adjoining state. Initially, make a telephone call and follow up with a letter. Normally, the services requested should be limited to providing an officer in a marked patrol car with its emergency lights activated stationed upstream of the queue. The other state may provide enforcement, but a police presence to warn approaching motorists is the Department's top priority.
2. To obtain service from the Delaware State Police, Ohio Highway Patrol, New Jersey State Police, or West Virginia State Police, use a separate pay item to establish a sum of money for the police assistance. Pay the work as a "Service by Others" in accordance with existing specifications governing the performance of extra work on a force account basis. Since each project is unique and requires this supplemental safety effort under different circumstances, it is not possible to provide a standard special provision; therefore, Districts need to develop a project-specific special provision.
3. To obtain services from the Maryland State Police, contact the Bureau of Highway Safety and Traffic Engineering (BHSTE) since a MOU is currently in place.
4. Districts are encouraged to have a face-to-face meeting with representatives of other police agencies to discuss the project, iron out any difficulties in the call-out procedure, and to meet the officers who will be providing the service. If the other state is using local police, invite a member of the PSP to attend the meeting by contacting the Patrol Lieutenant from the station in which the construction project is located.

6.15.8 Additional Guidelines for Maintenance Projects

PSP Assistance

The following additional guidelines and information apply to PSP Assistance for maintenance projects.

1. If the need for PSP Assistance is anticipated for maintenance, the District Work Zone Manager in conjunction with the County Maintenance Manager is to provide a blanket state project number to the Bureau of Highway Safety and Traffic Engineering (BHSTE). Complete this task at least 3 weeks before the first anticipated need.
2. BHSTE will include the county maintenance activity as a “project” in the statewide listing of projects for State Police assistance. In this listing, BHSTE will use a “dummy CMS number” consisting of the county code followed by three zeros (e.g., Allegheny County will always be “111000”) to distinguish these charge numbers from the CMS numbers used on construction projects.
3. At least 2 weeks in advance of the PSP Assistance, schedule it (date and hours) with the local commander. The County Maintenance District will be subsequently charged for all PSP overtime expenses in accordance with MOU (including an hourly rental rate for the patrol vehicles).
4. All requests for PSP assistance should be made to the appropriate PSP Troop Headquarters and come from either the County Maintenance Manager or one of the assistants. If making adjustments in the schedule, it is the county’s responsibility to keep the Troop Headquarters advised. The person requesting PSP assistance should also keep a record of each request, including the date and time, the route number, and the location. Therefore, it is very important that the County keep the Troop Headquarters informed in a timely manner.

If the County cancels scheduled PSP Assistance within a day of the proposed service, PSP will generally charge the County 3 hours for the call-out even though no service is provided.
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5. The trooper should be in the patrol car with lights flashing. If queues are anticipated, the patrol vehicle should be on the shoulder about 1/4 to 1/2 mile in advance of the back of the queue similar to the application on construction projects.
6. IF queues do not exist the troopers will perform enforcement operations in the 2-mile section of highway immediately in advance of the work area or a traffic queue.
7. Because of other commitments, PSP will not always be able to assist PennDOT. In fact, PSP may call the trooper away from the project to take care of an emergency.
8. The use of troopers is to provide added safety and is not to eliminate any normal safety devices.

Guidelines for Completing the “Information for Police Arrest Report”

Use the “Information for Police Arrest Report,” as included in [Appendix 6E](#), to assist police and report near misses in work zones. Please follow these guidelines when filling out this form.

1. Note as much information as possible – details are imperative.
2. List witnesses.
3. Call the police immediately after the incident.
4. Immediately after the incident, send a copy to the appropriate police jurisdiction.
5. Violations of §3102 (relating to obedience to authorized persons directing traffic) and §3326 (relating to duty of driver in construction and maintenance areas) of the Vehicle Code should also be reported to the police.

Chapter 6

Appendix

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SAMPLE

Transportation Management Plan

PennDOT District _- _

Project # _____

SR_____Section__

From Seg_____Offset_____ to Seg_____Offset_____
_____County

SAMPLE

Transportation Management Plan

OUTLINE OF CONTENTS

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6. WORK ZONE IMPACT ASSESSMENT
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9. CONTINGENCY PLANS
10. BREAKOUT OF TMP COSTS

1. INTRODUCTION, TABLE OF CONTENTS, LIST OF FIGURES

2. EXECUTIVE SUMMARY

This Transportation Management Plan (TMP) is being prepared in accordance with PennDOT Publication 46. The TMP lays out a set of coordinated transportation management strategies and describes how they will be used to manage the work zone impacts of a road project. Transportation management strategies for a work zone include temporary traffic control measures and devices, public information/outreach, and operational strategies such as travel demand management, signal retiming, and traffic incident management.

3. ROLES AND RESPONSIBILITIES

The Pennsylvania Department of Transportation (PennDOT) is responsible for the creation, implementation, and enforcement of this plan. The following individuals are responsible for specific activities to carry out the plan:

- Project Manager (design)—creates the plan and incorporates elements of the TMP into the plans, specifications, and estimate (PS&E)
- Inspector-in-Charge—monitors the day-to-day activities being carried out under the TMP and provides feedback to the District Traffic Unit
- District Press Officer—coordinates activities included in the Public Information Plan
- District Traffic Unit—reviews the activities of the TMP and compiles work zone performance data in conjunction with construction personnel/contractor
- Contractor—performs specific activities provided by the provisions included in the contract in accordance with the TMP. These activities will typically be carried out by the Traffic Control Coordinator.

Any changes to the TMP must be approved, in writing, by the District Traffic Engineer.

4. PROJECT DESCRIPTION

The description and location of the project is as follows: _____ COUNTY: From Segment 0410 Offset 0400 NB to Segment 0444 Offset 0925 NB and Segment 0411 Offset 0600 SB to Segment 0445 Offset 1357 SB. The scope of the project is concrete patching at selected areas. There are approximately 15 patches at this location. Additionally, the existing guide rail end treatments will be upgraded to current standards. From Segment 0444 Offset 0925 NB to Segment 0504 Offset 2100 NB and Segment 0445 Offset 1357 SB to Segment 0511 Offset 1500 SB. The scope of the project is concrete patching and a bituminous overlay. Additionally, the existing guide rail will be upgraded to current standards; the shoulders will be upgraded; the existing median barrier will be replaced; drainage will be improved, and bridge approach slabs will be replaced. Also, the NB on-ramp at the Pennsylvania interchange will be reconstructed all as indicated on the approved drawings included in the bid package for STATE ROUTE _____, SECTION ____ in _____ COUNTY, _____ TOWNSHIP.

This section of roadway is a four-lane, rural, limited-access Interstate highway that is not within the boundary of Interstate Management Area. Work will be performed under lane closures.

The project is to be constructed over two construction seasons with a winter shutdown. The anticipated Notice to Proceed date is 12/25/2005, and the completion date is anticipated by 07/25/2007. There are no milestone dates for this project.

The proposed construction phasing can be found on the following page.

CONSTRUCTION ACTIVITIES

PRE-PHASE 1- BEDFORD COUNTY SHORT TERM

RIGHT LANE CLOSURE

- UPGRADE SHOULDERS AS INDICATED

PHASE 1 - BEDFORD COUNTY LEFT LANE CLOSURE

- DRAINAGE UPGRADES
- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS.

PHASE 2 - BEDFORD COUNTY RIGHT LANE CLOSURE

- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS.
- DRAINAGE UPGRADES

PHASE 3 - BLAIR COUNTY (BELLWOOD TO GRAZIERVILLE)

LEFT LANE CLOSURE

- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS
- UPGRADE GUIDERAIL
- STRUCTURE REHABILITATION

PHASE 4 - BLAIR COUNTY (BELLWOOD TO GRAZIERVILLE)

RIGHT LANE CLOSURE

- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS
- UPGRADE GUIDERAIL
- STRUCTURE REHABILITATION

PHASE 5 - BLAIR COUNTY (GRAZIERVILLE TO TYRONE)

LEFT LANE CLOSURE

- INSTALL PAVEMENT BASE DRAIN.
- UPGRADE SHOULDERS AS INDICATED

PHASE 6 - BLAIR COUNTY (GRAZIERVILLE TO TYRONE)

RIGHT LANE CLOSURE

- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS.
- REPLACE BRIDGE APPROACH SLABS AND PAVEMENT RELIEF JOINTS AS INDICATED.
- INSTALL PAVEMENT BASE DRAIN.
- BITUMINOUS OVERLAY (EXCEPT WEARING COURSE) OF TRAVEL LANE AND SHOULDER.

PHASE 7 - BLAIR COUNTY (GRAZIERVILLE TO TYRONE)

LEFT LANE CLOSURE

- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS.
- REPLACE BRIDGE APPROACH SLABS AND PAVEMENT RELIEF JOINTS AS INDICATED.
- STRUCTURE REHABILITATION
- INSTALL PAVEMENT BASE DRAIN.
- DRAINAGE UPGRADES
- UPGRADE MEDIAN CROSS-OVER
- BIT. OVERLAY (ALL COURSES) TRAVEL LANE AND SHOULDER
- REMOVE & REPLACE EXISTING CONCRETE MEDIAN BARRIER
- PLACE SHOULDER BACK-UP & UPGRADE GUIDERAIL

PHASE 7A - BLAIR COUNTY (GRAZIERVILLE TO TYRONE)

RIGHT LANE CLOSURE

- PLACE WEARING COURSE
- PLACE SHOULDER BACK-UP
- UPGRADE GUIDERAIL

PHASE 8 - BLAIR COUNTY (TYRONE TO BALD EAGLE)

LEFT LANE CLOSURE

- INSTALL PAVEMENT BASE DRAIN.
- UPGRADE SHOULDERS AS INDICATED

PHASE 9 - BLAIR COUNTY (TYRONE TO BALD EAGLE)

RIGHT LANE CLOSURE

- CONCRETE PAVEMENT PATCHING AND RELATED ITEMS.
- REPLACE BRIDGE APPROACH SLABS AND PAVEMENT RELIEF JOINTS AS INDICATED.
- INSTALL PAVEMENT BASE DRAIN.
- BITUMINOUS OVERLAY (EXCEPT WEARING COURSE) OF TRAVEL LANE AND SHOULDER.
- DIAMOND GRIND CONC. AS INDICATED AND PAVE SHLDRS.

5. TRAFFIC CONDITIONS

Existing roadway traffic data follows:

TRAFFIC DATA			
HIGHWAY CLASSIFICATION	- RURAL INTERSTATE	CURRENT DATA	- 15,882 (2005)
DESIGN SPEED	- 70 M.P.H.	DESIGN YEAR A.D.T.	- 21,816 (2025)
PAVEMENT WIDTH	- 2 - 24 FT. LANES	D.H.V.	- 2618
SHOULDER WIDTH	- 4 FT. MEDIAN	D	- 65
	- 10 FT. OUTSIDE	T	- 19%

The local school district has requested that access be maintained at the _____ exit at all times from September 1 to June 1. Emergency service providers are located at the _____ exit. Other major traffic generators such as events, shopping, recreation, and tourism locations should be identified in this section.

6. WORK ZONE IMPACT ASSESSMENT

Multiple options for controlling traffic in the work zone were considered and analyzed:

- Long term lane closures
- Night work with short term lane closures
- Full Interstate closure

A. Long-Term Lane Closures – Long-term lane closures were selected because the added travel time is less than the threshold of 15 minutes as defined in Publication 46. A QuickZone analysis was conducted based on a capacity of 1500 vphpl. The table below provides a summary of the QuickZone results. The maximum queue length of 0.73 miles equates to an overall maximum delay of 9.16 minutes during the Sunday PM peak hour from 5:00 pm to 6:00 pm. This is consistent with delay experienced on similar, nearby projects on this State Route.

SR _____ QuickZone 2.0 Project Summary

Period with highest delay in After Case

Phase	1 - Stage 1
Direction	Inbound
Day/Time	Sunday 17:00

			Total Project User Cost (\$)				
	Max Queue (Miles)	Max Delay (min)	Passenger Cars	Truck	Detour	Econ/Misc	Total
Baseline	0	0	\$0	\$0	\$0	\$0	\$0
After	0.73	9.16	\$1,881,639	\$611,616	\$0	\$0	\$2,493,255
Total	0.73	9.16	\$1,881,639	\$611,616	\$0	\$0	\$2,493,255

With the relatively low volumes and lack of major traffic generators on this SR, the anticipated traffic impacts are minor. The contractor will be required to maintain access through the work zone for emergency vehicles and school buses. With 19% trucks, the contractor must maintain adequate lane widths at all times and ensure all temporary barrier, signing, and channelizing devices are maintained to meet their needs. Truck percentages are typically higher during the nighttime hours.

- B. Night Work with Short-Term Lane Closures** – Costs associated with loss of productivity and higher risk due to safety implications did not appear to outweigh the benefit to the traveling public. Estimated project cost when night work was estimated to be ___% higher than when long-term lane closures were used.
- C. Full Interstate Closure with Detour** – This alternative was not selected or analyzed because there was not a suitable detour route.

7. WORK ZONE IMPACTS MANAGEMENT STRATEGIES

- A. Traffic Control Plan (TCP)** – This project will be managed in accordance with the TCP which may be found at [insert ECMS link here]. Construction phasing is described in the Project Description section of this TMP.

B. Transportation Operations Plan (TOP)

- i. Traffic Control Supervisor** - The contractor's project traffic control supervisor is required to be adequately trained in accordance with PennDOT requirements. The traffic control supervisor is responsible for the following:
- Coordinate all traffic control installations, pattern changes, and removals.
 - Inspect long-term traffic control devices and patterns each working day.
 - Inspect each short-term traffic control pattern that is set up.

- Document the details of these inspections (items inspected, deficiencies discovered, and action taken to correct the deficiencies).
 - Program changeable message signs.
 - Notify immediately the District Traffic Management Center (TMC) of any highway incidents within the limits of the work zone.
 - Perform duties related to data collection, sharing, and performance monitoring.
- ii. **Changeable Message Signs (CMS)** - CMS will be used in the following manner:
- Notify motorists of the start dates of any detours, any major changes to the traffic control patterns, or any short-term road closures.
 - Provide messages specifically related to slowing traffic down or being prepared to stop during any short-term road closures or anticipated traffic queues
 - Provide messages for use during incident management.
- iii. **Work Zone Speed Limits** – This project will incorporate a reduced regulatory speed limit of 55 mph. Install and maintain reduced regulatory speed limit signs in work areas, as indicated on the Traffic Control Plan or accepted alternate plan. Temporarily remove or cover reduced regulatory speed limit signs, except when workers are present adjacent to a travel lane without a positive barrier separating vehicles and workers or when the Engineer determines one of the following conditions exist:
- Potential hazard exists on, or adjacent to, the roadway.
 - Physical restrictions exist; such as narrowed travel lanes, median crossovers, and lane shifts.
 - Reduced speeds are desirable to improve construction quality.

The Engineer will notify the Contractor, in writing, when one or more of these conditions exist. The notification will include locations, speed limits and a schedule of time periods when the reduced regulatory speed limit is to be in effect.

- iv. **Lane Closure Requirements** - Lane closures will not be permitted on the following dates and times:
- 12 Noon on April 13, 2006 to 9 AM on April 17, 2006
 - 12 Noon on May 26, 2006 to 9 AM on May 30, 2006
 - 12 Noon on June 30, 2006 to 9 AM on July 5, 2006
 - 12 Noon on September 1, 2006 to 9 AM on September 5, 2006
 - 12 Noon on November 22, 2006 to 9 AM on November 27, 2006
 - 12 Noon on April 5, 2007 to 9 AM on April 9, 2007
 - 12 Noon on May 25, 2007 to 9 AM on May 29, 2007
 - 12 Noon on July 3, 2007 to 9 AM on July 5, 2007

- No long-term lane closures on SR 0099 are permitted prior to March 1, 2006, or between November 30, 2006, and March 1, 2007.

Any traffic stoppages (emergencies excluded) or "rolling" road closures, regardless of their duration, must be approved by the District Traffic Unit.

Notification will be provided to the emergency services and school districts in the area and will be included in the contract by special provisions.

- v. **Incident Management Plan** – An incident management plan will be developed in accordance with the special provision in the contract. In this plan, a list of 24-hour emergency phone numbers (emergency services and organizations, contractor's representatives, Department representatives, County Maintenance Manager, etc.) will be provided. The incident management plan also requires that the contractor immediately notify the District Traffic Management Center for all lane closures and major incidents so that information may be provided to motorists and other stakeholders via district variable message signs, highway advisory radios, and the statewide road closure reporting and traveler information systems.

The contractor is required to submit an "Emergency Response Plan" at the pre-construction conference. The Emergency Response Plan should include, contact individuals for incidents that occur during non-working periods, alternate contact individuals, and maximum response time

Following an incident, the contractor will conduct an After Action Review with the Department and emergency services and organizations to evaluate the plan. The contractor will modify the plan as directed.

- C. **Public Information Plan (PIP)** – The contractor shall maintain ongoing communication with the Inspector-in-charge regarding phasing of operations that will impact traffic control and transportation operations in the project area. The contractor shall maintain lists of phone and fax numbers for school district, media outlets, emergency services, major businesses in the area, transit companies, nearby trucking firms, and other prominent traffic generators in the project vicinity. The contractor is responsible for notifying these groups of changes in the traffic control phasing at least 48 hours in advance of the phase change as directed by the Inspector-in-charge. The Inspector-in-charge shall coordinate with the District Press Officer to develop and issue press releases relating to traffic control and transportation operations on this project.

8. **TMP PERFORMANCE** – The performance of the transportation management plan will be regularly assessed from two standpoints:
- Monitoring the overall work zone performance
 - Evaluating individual TMP strategies

Crash, travel time data, and queue length will be the primary sources information used to determine the overall effectiveness of the work zone traffic management strategies. Since this project was not characterized as “significant” with respect to PennDOT’s Work Zone Safety and Mobility Policy, travel delay monitoring is not an automatic requirement. However, BHSTE determined that this project would be one in the sample of the “non-significant” project that would be monitored.

The contractor’s traffic control supervisor shall maintain a daily record of crashes, work zone incidents, and maximum daily queue lengths. This information shall be coordinated daily with PennDOT’s Inspector-in-charge. Feedback from the public through phone calls, in person, or in writing will also be captured.

The IIC will regularly inspect the manner in which traffic control installations, pattern changes, and removals are conducted and check to ensure that the portable changeable message signs are operational and being used appropriately. All this monitoring information will be documented in the project diary.

PennDOT’s IIC will share this information monthly with the District Traffic Engineer. The DTE will also arrange for periodic floating car runs through the project work limits to validate / corroborate work zone performance and to compare the actual queues and travel delays with those estimated as part of the project impact assessment.

The IIC, DTE, and TMC manager will participate in partnering meetings or conduct bi-monthly conference calls to discuss how well the TMP is performing and to verify that the PennDOT project manager, TMC manger, and public safety stakeholders have been receiving timely notifications of incidents as required by the incident management plan.

When the project is completed, the DTE (with cooperation from the contractor, IIC and ACE) will prepare a final evaluation report which will be forwarded to BHSTE. *[Note that daily records of delay and queue lengths and the annual reports can be completed by the contractor or a consultant.]* The evaluation will summarize the overall project travel delay impacts. Crash data will also be summarized and inclusion of crashes into Crash Records System will be verified. The TMP implementation costs and how they compare to budgeted costs will be noted. The decision to control traffic primarily through long-term lane closures will be revisited. Local emergency service providers and the schools will be contacted for their feedback on any impacts the project had on their operations. Finally, lessons learned will be documented.

9. **CONTINGENCY PLANS** *[Only include for high-impact Significant Projects]* – This section would specify activities that should be undertaken to

minimize traffic impacts when unexpected events such as crashes, inclement weather, unforeseen demand, etc., occur in the work zone.

The following should be included in the contingency plans (when in this section is included in the TMP):

- Trigger points that initiate contingency plan (fatal crashes, queues exceeding x miles, etc.)
- Contact names and 24-hour contact phone numbers of those in charge of the project or in charge of emergency response personnel that serves the project area (TMP manager, ACE, IIC, contractor, local police, and emergency response agency commanders, etc.)
- Decision tree with clearly defined lines of communication and authority. This section would also include coordination strategies between those involved.
- Specific duties of personnel during emergency lane closure operations
- Standby equipment (tow trucks, etc.) and list of personnel available on callout (for emergency response operations)

10. BREAKOUT OF TMP COSTS

Item	Description	Quantity	Unit	Unit Price	Total
0901-0001	MPT	1.00	LS	\$350,000	\$350,000
0901-0203	Arrow Panel	2.00	Each	\$4,000	\$8,000
0901-0205	CMS	2.00	Each	\$10,000	\$20,000
0901-0240	Additional Traffic Control Signs	300.00	SF	\$10.00	\$3,000.00
0901-0321	6" STANDARD PAVEMENT MARKINGS, PAINT & BEADS, YELLOW	75,298.000	LF	\$0.20	\$15,059.60
0901-0331	6" STANDARD PAVEMENT MARKINGS, PAINT & BEADS, WHITE	73,768.000	LF	\$0.20	\$14,753.60
0901-0334	24" STANDARD PAVEMENT MARKINGS, PAINT & BEADS, WHITE	320.000	LF	\$5.00	\$1,600.00
0901-0360	TEMPORARY NONPLOWABLE RAISED PAVEMENT MARKER, (ONE WAY W/B)	35.000	Each	\$12.00	\$420.00

Act 229 Requirements in Chapter 212

§212.419. Special controls in work zones.

(a) *General.* Special signing required in 75 Pa.C.S. §§[3326](#), [3365](#), [4309](#), [6123](#) and [6123.1](#) will be in addition to the traffic-control devices required by the *MUTCD* and shall be installed in accordance with this section.

(b) *Application.* Signing under this section is discretionary in the following work zones:

- (1) Short duration work, where the operation will be completed in less than 1 hour.
- (2) Mobile operations, where the work moves intermittently or continuously.
- (3) Stationary work where the daily duration of the construction, maintenance or utility operation is less than 12 hours and all traffic-control devices are removed from the highway at the completion of the daily operation, including all advance warning signs.
- (4) Work along highways other than expressways or freeways where the normal speed limit is 45 miles per hour or less.
- (5) Work in response to emergency work or conditions such as a major storm.

(c) *Work Zone – Turn on Headlights Sign (R22-1).* The Work Zone--Turn on Headlights Sign (R22-1) shall be erected as the first sign on each primary approach to the work zone, generally at a distance of 250 to 1,000 feet prior to the first warning sign. On high-speed roadways including all expressways and freeways, the larger advance distances should be used. If work begins at or near a border to this Commonwealth, the R22-1 signs should be installed within this Commonwealth.

(d) *Active Work Zone When Flashing Sign (W21-19).* The Active Work Zone When Flashing Sign (W21-19) shall be erected as close as practical to the beginning of the active work zone.

- (1) The sign should not be erected within a transition or at a location where workers are put at risk when they may need to turn the light on and off.
- (2) When a construction, maintenance or utility project has more than one active work zone and the active work zones are separated by a distance of more than 1 mile, signs for each active work zone shall be erected.
- (3) The W21-19 signs shall be installed on temporary sign posts or on Type III barricades, and a white Type B high-intensity flashing light must be attached to the upper portion of each W21-19 sign. The light shall be activated only when workers are present, and deactivated when workers are not anticipated during the next 60 minutes.

(e) *End Active Work Zone Sign (W21-20).* The End Active Work Zone Sign (W21-20) shall be erected immediately at the end of each active work zone, except this sign is not necessary if

either the End Road Work Sign (G20-2a) or the End Work Area Sign (G20-3) is installed at the end of the active work zone.

(f) *Work zones on expressways or freeways.* When the work zone is on an expressway or freeway, appropriate signs and lights identified in subsections (c), (d) and (e) at on-ramp approaches to the work zone shall be installed.

(g) *Portable changeable message sign.* A portable changeable message sign (PCMS) may be used in lieu of the R22-1, W21-19 or W21-20 signs.

(h) *Speed display sign.* In Interstate highway work zones with a project cost exceeding \$300,000, a speed display sign shall be installed on each mainline approach to the work zone to inform motorists of their speed.

(1) The speed display sign must display the motorist's speed in miles per hour in numerals at least 18 inches in height.

(2) As an alternative, a portable changeable message sign (PCMS) may be equipped with radar and programmed to display vehicles speeds.

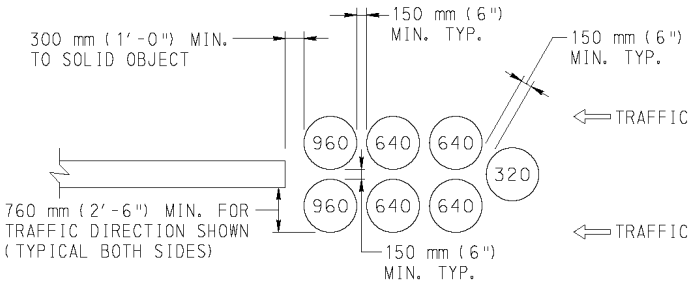
(3) PCMSs may also flash appropriate messages such as "YOU ARE SPEEDING" or "SLOW DOWN." The signs shall be placed 1/2 to 1 mile in advance of the physical work zone.

Sand Barrel Layouts

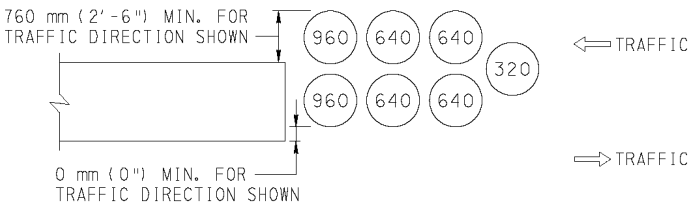
ENERGY_ABSORP. DGN

ARRAYS

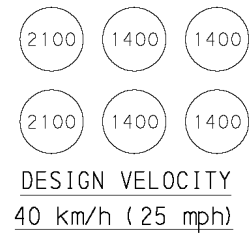
FOLLOWING ARE SEVERAL STANDARD ARRAYS FOR THE "ENERGITE III/FITCH UNIVERSAL MODULE AND TRAFFIX SYSTEMS". THEY WILL PERFORM AS INDICATED IN THE CALCULATIONS ONLY IF HIT HEAD-ON. ARRAYS ARE PROVIDED USING BOTH METRIC AND ENGLISH UNITS. BELOW ARE TYPICAL ARRAYS SHOWING UNIDIRECTIONAL AND BIDIRECTIONAL CONFIGURATIONS. THESE ARE MINIMUM ARRAYS AND MORE BARRELS ARE RECOMMENDED AND SHOULD BE USED WHEN CONDITIONS PERMIT.



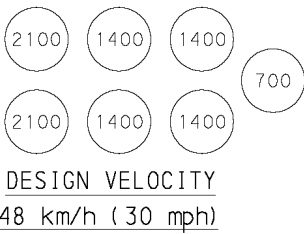
UNIDIRECTIONAL



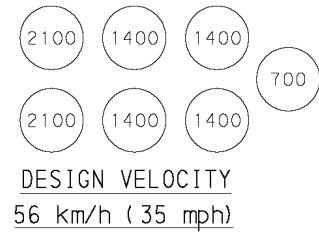
BIDIRECTIONAL



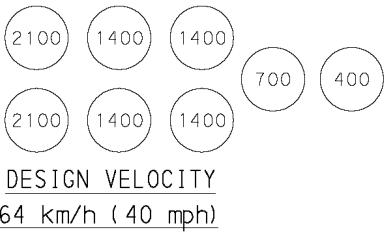
DESIGN VELOCITY
40 km/h (25 mph)



DESIGN VELOCITY
48 km/h (30 mph)

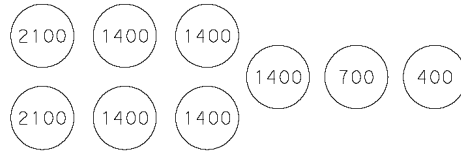


DESIGN VELOCITY
56 km/h (35 mph)

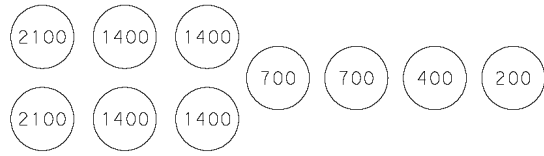


DESIGN VELOCITY
64 km/h (40 mph)

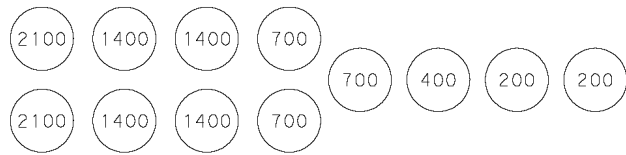
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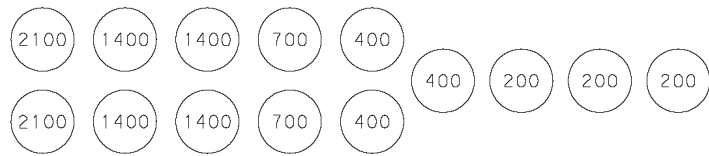
DESIGN VELOCITY
72 km/h (45 mph)



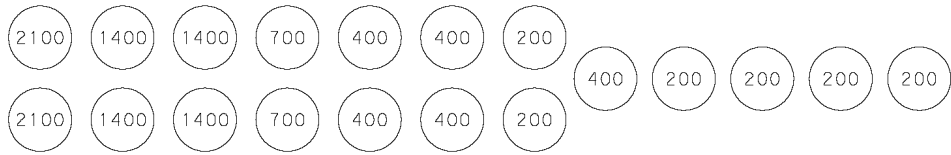
DESIGN VELOCITY
80 km/h (50 mph)



DESIGN VELOCITY
88 km/h (55 mph)



DESIGN VELOCITY
96 km/h (60 mph)



DESIGN VELOCITY
113 km/h (70 mph)

PSP Queue Length Reporting Form

Date	Officer	County	SR	Location (direction & mile marker)

Queue Info	Time		Length	Was Queue Caused By							
	Start			Crash			Disabled Vehicle		Heavy Traffic		
	Record every 1/2 hour				Yes	No	Incident #	Yes	No	Yes	No
End											

Was Crash/Disabled Vehicle in Construction Zone?
If Yes, Provide Breif Description

Yes	No

Were additional Troopers needed for Queuing?

Yes	No

If Yes, Number of additional Trooper

Remarks:

Please submit all completed forms to Project Manager at the beginning of each week

HS-2 (1-07)



Guidelines For Required Information For Police Arrest

(Check All Boxes That Apply)

Location of Incident:

County: _____ Township/Boro.: _____ Local Name: _____

S.R.: _____ Seg/Off: _____ Milepost: _____

Description of Vehicle:
☐ Car ☐ Truck ☐ Tractor Trailer ☐ Motor Home ☐ Motorcycle ☐ Other: _____

Truck Co. Name (if applicable) _____

Color: _____ Make: _____ Model: _____

Plate # (Vehicle/Trailer): _____ / _____ State: _____ Other Markings: _____

Driver: ☐ Male ☐ Female Age: _____ Hair Color: _____ Clothing: _____

Number/Description of Occupants: _____

Travel Direction: ☐ North ☐ South ☐ East ☐ West
Descriptive Statement of Incident: (Include: Who, What, When, Where, Why, and How)

Date: _____ Time: _____ AM _____ PM Weather: _____

Can any witnesses identify the driver? ☐ Yes ☐ No
Description of Work Zone:

Warning signs in place? ☐ Yes ☐ No **Flagger?** ☐ Yes ☐ No **Operation Type?** ☐ Moving ☐ Stationary

Regulatory Posted Speed? _____ MPH

WITNESSES	NAME	ADDRESS	TELEPHONE NUMBER

Reported By: _____ **Date:** _____ **Reported to Police:** ☐ Yes ☐ No

If Yes: Police Barracks: _____ **Officer's Name:** _____

Engineering District 8-0
2140 Herr Street
Harrisburg, PA 17103-1699
May 25, 20____

EXAMPLE

Dover Township
SR 4014-003 Project
Detour for SR 4014 (Harmony Grove Road)

Betty A. Shoemaker, Manager
Dover Township
2480 West Canal Road
Dover, PA 17315

Dear Ms. Shoemaker:

The Department is preparing a construction project on Harmony Grove Road (SR 4014). A portion of roadway will be totally reconstructed, therefore a detour is required while the road is closed.

The Department is proposing to use Blackberry Road (T-819) as a detour route to move motorists around the road closure. We have attached a Memorandum of Understanding along with the detour route that we are planning to use during construction.

Please review and sign the Memorandum of Understanding and return it to our office as soon as possible.

If you have any questions or require additional information, please feel free to contact _____ in our Traffic Unit at _____.

Sincerely,

District Executive

EXAMPLE

Pennsylvania Department of Transportation Engineering District 8-0

Municipal Roadway Detour

On this _____ day of _____ 20__ the Commonwealth of Pennsylvania, Department of Transportation and Dover Township of York County have entered into an understanding whereby the Department of Transportation may utilize a municipally owned roadway(s) for the purpose of establishing a detour. The roadway(s) are described below.

- ◆ *Blackberry Road (T-819) from PA 74 to Harmony Grove Road (SR 4014).*

The duration and time for the enactment of this detour is anticipated as: (12 weeks)

The Department of Transportation will provide the following services for establishment of the detour as shown on the attached plan.

- ◆ *Place and maintain all detour signing.*
- ◆ *Restripe of double yellow pavement markings on Blackberry Road (T-819) before detouring traffic.*

Upon removal of the detour, the roadway condition will be equal to the condition at the time of the placement of the detour.

Municipal Official	Title	Date
County Manager	Date	
Approved by: _____		Date
District Executive		

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Pennsylvania Department of Transportation



pennsylvania
DEPARTMENT OF TRANSPORTATION

**Publication 46
Chapter 7**

School Areas

Pennsylvania Department of Transportation

Chapter 7 of Publication 46

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7.1 General

7.1.1 Need for Standards

The best way to achieve safe and effective traffic control to protect school students is through the uniform application of realistic policies, and standards developed through engineering judgment. Therefore, whenever possible, the Department should always follow the *MUTCD*.

Signs and pavement markings should also conform to those policies established in Chapters 2 and 3 of this manual. All school warning signs shall use retroreflective sheeting, either yellow or fluorescent yellow-green, but local authorities should be encouraged to avoid mixing the two colors whenever possible.

7.1.2 Laws, Regulations and Other Publications

Hazardous Walking Routes (67 Pa. Code Chapter 447). Regulations issued under the authority of the Public School Code of 1949 (24 P.S. §§13-1362 and 25-2541) to help determine where student-walking routes are hazardous, which in turn affects the amount of reimbursement that school districts receive for busing school students. This regulation is available at <http://www.pacode.com/secure/data/067/chapter447/chap447toc.html>.

Manual on Uniform Traffic Control Devices (MUTCD). Part 7 is titled, “Traffic Control for School Areas”, and is available at <http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/ch7.pdf>.

Official Traffic-Control Devices (PennDOT Pub. 212). Regulations published as Chapter 212 of Title 67 of the PA Code. This regulation adopts the Federal Highway Administration’s *Manual on Uniform Traffic Control Devices (MUTCD)* and establishes additional study requirements, warrants, principles and guidelines not included in the *MUTCD*. Specifically, [§212.501](#) establishes the warrants and procedures for establishing 15-mph school zone speed limits in accordance with 75 Pa.C.S. §3365(b) (relating to special speed limitations). Publication 212 is available at <http://www.pacode.com/secure/data/067/chapter212/chap212toc.html>, and [§212.501](#) is included herein as [Appendix 7A](#).

Pennsylvania Drivers Manual. This manual provides guidance for drivers, and is available at https://www.dot3.state.pa.us/pdotforms/pa_forms_manuals/padriversman.pdf.

School Trip Safety Program Guidelines, 1984 Edition (ITE).

Traffic Control – Pavement Markings and Signing Standards (PennDOT Pub. 111M). Standard drawings specifying the types, dimensions, locations and lighting of signs on expressways and freeways, and the legend spacing and sign supports for signs on all highways. Available at

<ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20111M.pdf>

Vehicle Code (75 Pa.C.S.). The Pennsylvania Vehicle Code is law that typically defines actions required by drivers and the Department. Specifically, [§3365\(b\)](#) discusses the establishment of the 15 mph school zone speed limit. In addition, [§3345\(a\)](#) discusses the driver's responsibility when approaching a school bus.

7.1.3 Definitions

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

Divided highway – A highway divided into two or more roadways and so constructed as to impede vehicular traffic between the roadways by providing an intervening space, physical barrier or clearly indicated dividing section. Each roadway of a divided highway is a “separate roadway” as used in [75 Pa.C.S. §3345\(g\)](#).

Elementary students – School students in kindergarten or grades one through six.

Hazardous – An unsafe condition caused by potential incompatibility between vehicles and school students, while the students are walking between their home and their school or school bus stop.

School zone – A portion of a highway that at least partially abuts a school property or extends beyond the school property line that is used by students to walk to or from school or to or from a school bus pick-up or drop-off location at a school.

Secondary students – School students in grades 7 through 12.

Separate roadway – One of the roadways of a “divided highway.”

Shoulder – The portion of the highway contiguous to the roadway used for accommodation of stopped or parked vehicles, for emergency use or for lateral support of base and surface courses.

Sidewalk – That portion of a street or highway or other public right-of-way that is reserved exclusively for pedestrian travel and is normally protected by a minimum average 4-inch high, non-mountable curb, or is not immediately adjacent to the roadway. A sidewalk should have a minimum width of 2 feet; a gravel, brick, stone or paved surface; and be available for use during normal weather conditions.

Student-walking route – The system of streets, shoulders, sidewalks and crosswalks used by school students when walking between their homes and their school or school bus stop, officially designated by the school district or, where no official route has been designated, used by school students because of the unavailability of a reasonable alternate route.

7.2 School Zone Speed Limits

7.2.1 Criteria

The criteria to determine if a 15-mph school zone speed limit is applicable is in 67 Pa. Code §212.501 (see [Appendix 7A](#)).

PLEASE NOTE:

1. In accordance with §212.501(a), a 15-mph school zone speed limit is only applicable when at least one student walks to school.
2. The Department must approve all school zones, including the location and hours of operation of the speed limits on both State highways and on local roads, except as noted in §212.501(a)(2).

7.2.2 Signing Requirements

Posting requirements are included in §212.501(b).

7.3 Student-Walking Routes

7.3.1 Legislative Requirement

The Public School Code of 1949 (24 P.S.) requires the Department to take into account all relevant safety factors for student-walking routes when certifying whether or not walking constitutes a hazard to the students. Section 25-2541(c) of the Code is included as [Exhibit 7.3-A](#).

7.3.2 Department Regulations and Interpretations.

The regulations for Hazardous Walking Routes (Chapter 447 of Title 67, Pennsylvania Code) establish criteria for determining if student-walking routes are or are not hazardous for the students (see [Appendix 7B](#)). In addition to the regulations, the following interpretations provide guidance in evaluating the requests:

- (a) When requested by the school district, the Department will evaluate a student-walking route regardless if a student is walking from home to school or to a school bus stop, or if the student is being transported by either a private or a school district conveyance.
- (b) Crossing at a signalized intersection may be declared hazardous for an elementary school student if all of the following apply:
 - 1. The signal installation does not include an exclusive pedestrian walk phase.
 - 2. An adult crossing guard is not permanently assigned to the signalized intersection during the school year.
 - 3. Sight distance, traffic volumes, or roadway widths make it difficult for an elementary student to cross safely.
- (c) Crossing at a signalized intersection may be declared hazardous for all school students if an adult crossing guard is not permanently assigned to the signalized intersection during the school year and one or more of the following is satisfied:
 - 1. The complexity of the operation or design of the signal system is such that:

- signal indications do not readily provide a visible indication for the school student desiring to cross the intersection; or
 - a multi-phase signal operation exists and it may not be apparent what traffic is being given a green indication.
2. A 4.5-foot tall student using a crosswalk within the intersection may not be visible at a point that allows an approaching driver turning across the crosswalk to come to a safe stop.
 3. The number of approach lanes and/or the complexity of the geometries of the intersection makes it difficult for a secondary school student to traverse the intersection or to reach a safe refuge.

7.3.3 Field Study and Evaluation

The Department will determine if a student-walking route is hazardous when a written request is received from the school district. If a request is received from anyone other than the school district, the Engineering District will return the request with a letter explaining the Department's policy for evaluating student-walking routes. The District should also copy the school district, and include a copy of the original request. This will bring the potentially hazardous situation to the attention of the school district and serve as an official notification from the Department.

It is the responsibility of the school district to complete one or more Study and Data Sheets (see [Appendix 7C](#)) for each street or highway within the student-walking route. If the Department receives a request from a school district without the Study and Data Sheets, the request should be acknowledged by letter, asking the school district to provide the completed sheets. (The Department will provide a copy of Chapter 447 and one or more copies of the Study and Data Sheets.)

After receipt of the completed forms, the District Traffic Unit will evaluate the information on the Study and Data Sheets to determine if the student-walking route is or is not hazardous for the students. At the District Traffic Unit's discretion, they may field verify any of the information. The District Traffic Unit shall conduct a study to determine if the student-walking route is or is not hazardous.

If the Engineering District cannot issue a certification within 2 weeks, the District Traffic Unit should acknowledge the School District's request and advise them when they should anticipate the certification.

7.3.4 Certification

Justification for Hazardous Certification

The District Traffic Unit will prepare the certification (see [Appendix 7D](#) for a basic format). If the student-walking route (as defined on the Study and Data Sheets) is hazardous, the certification form shall cite the section(s) of Chapter 447 or the appropriate sections of this policy which was used to declare the route hazardous. For example:

The results of the investigation indicate that sidewalks do not exist, the shoulders are less than 4 feet wide, the roadway width is less than 20 feet wide and one or more trucks with three or more axles were observed using the roadway during the time the elementary students are enroute to or from school. Therefore, in accordance with the provisions of §447.4(b)(1)(i), this route is declared hazardous for elementary students.

Partial Hazardous Certification

If one or more portions of a designated walking route is determined to be hazardous and the balance is determined to be non-hazardous, certify the student-walking route accordingly. For example, a certification form could indicate:

The results of the investigation indicate that the section of Street “X” between “_____” and “_____” does not have sidewalks, the shoulders are less than 4 feet wide, the roadway width is less than 20 feet wide and one or more trucks with three or more axles were observed using the roadway during the time the elementary students are enroute to or from school. Therefore, in accordance with the provisions of §447.4(b)(1)(i), this section of Street “X” is declared hazardous for elementary students. The remaining sections of Street “X” between “_____” and “_____” are non-hazardous.

Approval of the Certification

The District Executive should sign the certification and forward copies to the school district and to the following address:

Pennsylvania Department of Education
Bureau of Budget and Fiscal Management
Division of Subsidy Data and Administration
333 Market Street, 4th Floor
Harrisburg, PA 17126-0333

**Exhibit 7.3-A Payments on Account of Pupil Transportation
(24 P.S. §25-2541(c))**

(c) Payments for pupil transportation on account of the school year 1979-1980 and every school year thereafter shall be made only in the following cases:

*(1) To all school districts for the transportation to and from school of elementary school pupils, including kindergarten pupils, residing one and one-half (1½) miles or more by the nearest public highway from the school in which the pupils are enrolled and to which transportation is authorized under section 1361 of this act or residing in areas where the road or traffic conditions are such that walking constitutes a hazard to the safety of the child **when so certified by the Department of Transportation**. The Department of Transportation shall take into account the presence of sidewalks along the highway, but such presence or lack thereof shall not be controlling and the department shall consider all relevant safety factors in making its determination as to whether or not walking constitutes a hazard to pupils. Such elementary school pupils shall include nonresident children who are placed in the home of a resident, or who are residents of an orphanage, or home or children's home or other institution for the care and training of orphans or other children.*

*(2) To all school districts for the transportation to and from school of secondary school pupils residing two (2) miles or more by the nearest public highway from the school in which the pupils are enrolled and to which transportation is authorized under section 1361 of this act or residing in areas where the road or traffic conditions are such that walking constitutes a hazard to the safety of the child **when so certified by the Department of Transportation**. The Department of Transportation shall take into account the presence of sidewalks along the highway, but such presence or lack thereof shall not be controlling and the department shall consider all relevant safety factors in making its determination as to whether or not walking constitutes a hazard to pupils. Such secondary school pupils shall include nonresident children who are placed in the home of a resident, or who are inmates of an orphan asylum or home or children's home or other institution for the care and training of orphans or other children.*

*(3) To all school districts for pupils transported to and from approved consolidated schools or approved joint consolidated schools living one and one-half (1½) miles or more from the school of attendance or residing in areas where the road or traffic conditions are such that walking constitutes a hazard to the safety of the child **when so certified by the Department of Transportation**. The Department of Transportation shall take into account the presence of sidewalks along the highway, but such presence or lack thereof shall not be controlling and the department shall consider all relevant safety factors in making its determination as to whether or not walking constitutes a hazard to pupils.*

Consolidated schools or joint consolidated schools shall so long as they are approved as to organization, control, location, equipment, courses of study, qualifications of teachers, methods of instruction, condition of admission, expenditures of money, methods and means of transportation and the contracts providing therefore, constitute approved consolidated schools or approved joint consolidated schools.

(4) To all school districts for the transportation of exceptional children regularly enrolled in special classes approved by the Department of Education or enrolled in a regular class in which approved educational provisions are made for them.

(5) To all school districts for pupils transported to and from area technical schools.

Chapter 7

Appendix

<u>Appendix</u>	<u>Page</u>
7A 67 Pa. Code School §212.501, Zone Speed Limits	<u>7A-1</u>
7B 67 Pa. Code Chapter 447, Hazardous Walking Routes	<u>7B-1</u>
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67 Pa. Code §212.501 School Zone Speed Limits

Sec.

212.501. [School zone speed limits.](#)

§212.501. School zone speed limits.

(a) *Establishment.* A 15 miles per hour school zone speed limit may be established in a school zone during the normal hours that walking students are arriving at or leaving school, under 75 Pa.C.S. §3365(b) (relating to special speed limitations).

(1) To establish a school zone, local authorities shall be responsible to prepare and submit a drawing showing the locations where students walk along or across roadways that are adjacent to school property, the hours that students are going to or from school and the proposed limits for the school zone to the Department for approval.

(2) The Department is responsible for approving the establishment of all school zones, including the locations and hours of operation, except local authorities shall be responsible for approving school zones at the following locations:

(i) On local highways when the municipality has received municipal traffic engineering certification under Chapter 205 (relating to municipal traffic engineering certification).

(ii) On State-designated highways when the municipality has entered into an agreement with the Department thereby transferring to the local authorities the authority to install traffic-control devices without specific Department approval.

(iii) On highways in cities of the first and second class, except not on expressways.

(3) The duration of a 15 miles per hour school zone speed limit should be only long enough to include the time that walking students routinely arrive at or leave school.

(b) *Posting.* A school zone speed limit shall be posted on official traffic-control devices as follows:

(1) At the beginning of the school zone speed limit, one of the following signs or groups of signs shall be posted either on the right side of the roadway or over the roadway:

(i) A Speed Limit Sign (R2-1) with the appropriate school zone speed limit, with a School Panel (S4-3) mounted above the Speed Limit Sign (R2-1) and a When Flashing Sign (S4-4) mounted below the Speed Limit Sign (R2-1), with two flashing speed limit sign beacons.

(ii) A Speed Limit Sign (R2-1) with the appropriate school zone speed limit, with a School Panel (S4-3) mounted above the Speed Limit Sign (R2-1) and a Restricted Hours Panel (R10-20A) mounted below the Speed Limit Sign (R2-1).

(iii) A School Speed Limit When Flashing Sign with a blank-out “15” and flashers as illustrated in the *Traffic Signal Design Handbook* (Department Publication 149M).

(2) An End School Zone Sign (S5-2) shall be posted on the right side of the roadway to define the end of the school zone speed limit.

(3) The limits of a school zone may extend beyond the school property lines to improve the sight distance or to encompass a school crosswalk, except that the length of the zone may not be greater than 1,600 feet.

67 Pa. Code Chapter 447 Hazardous Walking Routes

Sec.

- 447.1. [Purpose.](#)
- 447.2. [Definitions.](#)
- 447.3. [General policy.](#)
- 447.4. [Criteria.](#)

Authority

The provisions of this Chapter 447 issued under sections 506 and 2001 of The Administrative Code of 1929 (71 P. S. §§186 and 511); and sections 1362 and 2541 of the Public School Code of 1949 (24 P. S. §§13-1362 and 25-2541), unless otherwise noted.

Source

The provisions of this Chapter 447 adopted August 1, 1980, effective August 2, 1980, 10 Pa.B. 3191, unless otherwise noted.

§447.1. Purpose.

This chapter establishes guidelines for determining if a designated school student walking route along a public highway is hazardous, as the defined term is used in sections 1362 and 2541 of the Public School Code of 1949 (24 P. S. §13-1362 and §25-2541).

Source

The provisions of this §447.1 adopted August 1, 1980, effective August 2, 1980, 10 Pa.B. 3191; amended August 7, 1981, effective August 8, 1981, 11 Pa.B. 2777.

§447.2. Definitions.

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

Elementary students—School students in kindergarten or grades one through six.

Hazardous—An unsafe condition caused by potential incompatibility between vehicles and school students, while the students are walking between their home and their school or school bus stop.

Safe-running speed—The official speed limit as posted by signs or, in the absence of a posted speed limit, the average speed as determined by making a minimum of five test runs in each direction and periodically recording the operating speed at different locations while

driving at a speed which is reasonable and prudent considering the spacing of intersections, roadside development and sight distance.

Secondary students—School students in grades 7 through 12.

Shoulder—The portion of the highway contiguous to the roadway used for accommodation of stopped or parked vehicles, for emergency use or for lateral support of base and surface courses.

Sidewalk—That portion of a street or highway or other public right-of-way which is reserved exclusively for pedestrian travel and is normally protected by a minimum average 4-inch high, nonmountable curb, or is not immediately adjacent to the roadway. A sidewalk should have a minimum width of 2 feet; a gravel, brick, stone or paved surface; and be available for use during normal weather conditions.

Student walking route—The system of streets, shoulders, sidewalks and crosswalks used by school students when walking between their home and their school or school bus stop, officially designated by the school district or, where no official route has been designated, used by school students because of the unavailability of a reasonable alternate route.

Source

The provisions of this §447.2 adopted August 1, 1980, effective August 2, 1980, 10 Pa.B. 3191; amended August 7, 1981, effective August 8, 1981, 11 Pa.B. 2777.

§447.3. General policy.

(a) A request for review of student walking routes should be referred to the appropriate engineering district as listed in Appendix A [*NOTE: Appendix A is not included in Publication 46*]. Personnel of the engineering district will make the necessary study upon receipt of a written request from a school district and the district engineer will certify whether the route is or is not hazardous. The certification will be forwarded to the school district and to the Department of Education.

(b) The Vehicle Code sets forth certain rights and duties for pedestrians and vehicular traffic. These rights and duties have been considered in the development of these guidelines. Accordingly, if a hazard exists solely because of failure of drivers or school students to obey the provisions of the Vehicle Code, the student walking route may be declared hazardous; however, the basis for the hazardous walking route determination shall be so noted on the certification and the problem brought to the attention of the municipality.

(c) Road and traffic conditions shall be evaluated before any highway or section of highway is declared hazardous. The presence or absence of side walks shall be a factor in the evaluation but may not be the controlling factor. The criteria for road and traffic conditions may apply only to student walking routes, as defined in this chapter.

(d) This chapter may not be construed to require school buses to stop at every dwelling in the event that a student walking route or a portion thereof is declared hazardous, since such a policy would increase the probability of bus-related accidents. A student may be required to walk up to 500 feet on a roadway designated as a hazardous walking route when the route is designated as hazardous in accordance with §447.4(b) (relating to criteria).

(e) If changes occur in the condition of a walking route that was previously inspected, a reevaluation of the route may be requested.

Source

The provisions of this §447.3 adopted August 1, 1980, effective August 2, 1980, 10 Pa.B. 3191; amended August 7, 1981, effective August 8, 1981, 11 Pa.B. 2777.

§447.4. Criteria.

(a) A student walking route shall be considered hazardous if any one of the following three conditions exist:

(1) Two or more pedestrian-related accidents have occurred during the last 3 years while the pedestrians were walking along the student walking route during hours students are normally going to or from school.

(2) It is necessary for a student to cross a roadway; either daily or intermittently, at a location where vehicular traffic is not controlled by either traffic control signals or a stop sign, or where students are not protected by an adult crossing guard; provided vehicular traffic on roadway is in excess of the values given in the table below for any 15-minute period during which students are enroute to or from school:

TABLE I

<i>Roadway Width (ft)*</i>	<i>For Elementary Students Number of Vehicles</i>	<i>For Secondary Students Number of Vehicles</i>
20 or less	155	175
24	130	150
30	100	120
36	80	100
48	40	60

* If the roadway is divided by a raised median which is at least 8 feet wide and has nonmountable curbs, the roadway should be considered as two separate roadways.

(3) It is necessary for students to cross a railroad-highway grade crossing which has two or more tracks and the following three qualifications are met:

(i) Trains normally — not necessarily with regularity — use the crossing at the time the students cross the tracks going to or from school.

(ii) The crossing is not protected by a flashing light signal or a crossing guard.

(iii) The speed of the trains and the available sight distance are such that students walking at a speed of 3.5 feet per second cannot safely cross the tracks.

(b) A student walking route shall be considered hazardous if a sidewalk does not exist and either paragraph (1) or (2) applies:

(1) The shoulders are less than 4 feet wide and for either:

(i) Elementary students, the roadway surface is less than 20 feet wide and one or more trucks with three or more axles, not including garbage trucks or other types of trucks making house-to-house stops, normally use the roadway during the time the elementary students are enroute to or from school.

(ii) Streets and highways with an average traffic volume of at least ten vehicles per hour during the time students are walking, a 3.5-foot tall elementary school student or a 4.5-foot tall secondary student is not visible by approaching drivers from at least the following minimum distances:

TABLE II

<i>Safe-running Speed</i>	<i>Minimum Distance (ft.)</i>
30 or less	200
35	240
40	275
45	315
50	350
55	410

(2) The normal vehicular traffic volume during any 15-minute period that students are enroute to or from school exceeds the following values for the appropriate safe-running speed range:

(i) Safe-running speed is 35 mph or less:

<i>Shoulder Width</i>	<i>For Elementary Students</i>	<i>Number of Vehicles For Secondary Students Only</i>
less than 4 ft.	30	45
4 ft. – 6 ft.	60	100

(ii) Safe-running speed is over 35 mph:

<i>Shoulder Width</i>	<i>For Elementary Students</i>	<i>Number of Vehicles For Secondary Students Only</i>
less than 4 ft.	20	30
4 ft. – 6 ft.	40	65

Source

The provisions of this §447.4 adopted August 1, 1980, effective August 2, 1980, 10 Pa.B. 3191; amended August 7, 1981, effective August 8, 1981, 11 Pa.B. 2777.

Cross References

This section cited in 67 Pa. Code §447.3 (relating to general policy).

Student-Walking Route Study and Data Sheet

County _____ Municipality _____

School District Name _____ IU _____

Address: _____

_____ Zip Code _____

FOR PURPOSES OF THIS REVIEW, THE FOLLOWING DEFINITIONS APPLY:

Elementary students – School students in kindergarten or grades one through six.

Hazardous – An unsafe condition caused by potential incompatibility between vehicles and school students, while the students are walking between their home and their school or school bus stop.

Safe-running speed – The official speed limit as posted by signs or, in the absence of a posted speed limit, the average speed as determined by making a minimum of five test runs in each direction and periodically recording the operating speed at different locations while driving at a speed which is reasonable and prudent considering the spacing of intersections, roadside development and sight distance.

Secondary students – School students in grades 7 through 12.

Shoulder – The portion of the highway contiguous to the roadway used for accommodation of stopped or parked vehicles, for emergency use or for lateral support of base and surface courses.

Sidewalk – That portion of a street or highway or other public right-of-way that is reserved exclusively for pedestrian travel and is normally protected by a minimum average 4-inch high, non-mountable curb, or is not immediately adjacent to the roadway. A sidewalk should have a minimum width of 2 feet; a gravel, brick, stone or paved surface; and be available for use during normal weather conditions.

Student-walking route – The system of streets, shoulders, sidewalks and crosswalks used by school students when walking between their homes and their school or school bus stop, officially designated by the school district or, where no official route has been designated, used by school students because of the unavailability of a reasonable alternate route.

PLEASE NOTE: A map or detailed sketch of the area must accompany this study and data sheet, highlighting the school student-walking route. This map or detailed sketch should be large enough to include nearby streets and roadways, and should show the location of all adult crossing guards. Also, provide any additional supporting data.

1. Location of school student-walking route: _____

2. Local street name, Township Road No., or State Route No. _____

Beginning location _____

Ending location _____

Approximate length _____

Any general comments regarding the location: _____

3. Typical roadway width is _____ feet. Shoulder width is _____ feet.

4. Are sidewalks present? _____ Are shoulders present? _____

5. Is this a request for a re-evaluation of a previously inspected route? _____

If yes, when was it last reviewed and what was the finding? _____

6. During what time periods are students using the subject route?

Elementary Students

Secondary Students

(a) Morning _____ to _____

(a) Morning _____ to _____

(b) Mid-day _____ to _____

(b) Mid-day _____ to _____

(c) Afternoon _____ to _____

(c) Afternoon _____ to _____

7. Which 15-minute time period has the greatest vehicular traffic volume while:
- (a) Elementary students are enroute?
 _____ to _____ 15-minute volume: _____
- (b) Secondary students are enroute?
 _____ to _____ 15-minute volume: _____
8. How many pedestrian-related accidents occurred in the study area in the last 36 months during the hours students are normally going to or from school? _____

(If any pedestrian accidents occurred, please attach a copy of each police accident report and indicate the location of the accident on the accompanying map. The walking route between two or more accident locations is hazardous.)

9. Does the student-walking route cross the roadway at any location where vehicular traffic is not controlled by either a STOP sign or traffic-control signal, or an adult crossing guard? _____ If yes, what is the roadway width? _____ and, is the crossing by:
- (a) Elementary students? _____ Secondary students? _____
- (b) Number of vehicles using the road during a 15-minute period while students would ordinarily be attempting to cross the road? _____

(If the number of vehicles exceeds the appropriate values in Table 1 in §447.4(a)(2) of the regulations, the crossing is hazardous.)

10. Does the student-walking route cross a highway-rail grade crossing that has two or more tracks? _____ If yes,
- (a) Do trains normally use the crossing during the time students are going to or from school? _____
- (b) Is the crossing unprotected? _____ Question (b) is answered "no" when:
- A flashing light signal (i.e., two alternately flashing red light units) is installed at the crossing, or

- A "flagger is employed by the railroad company to stop highway vehicles and pedestrians, is present whenever a train moves over the crossing.

(c) Is the speed of the trains and the available sight distance such that students walking at a speed a normal pace of approximately 3.5 feet per second cannot safely cross the tracks? _____

(If the answers to all four questions are "yes," crossing the rail-highway grade crossing is hazardous.)

11. Is the roadway less than 20 feet wide and without either sidewalks or minimum 4-foot wide shoulders at any location? _____ If yes, how many trucks with three or more axles (excluding garbage trucks or other types of trucks making house-to-house stops) normally use the roadway during the time elementary students are enroute?

(If the first answer is "yes," and one or more trucks normally uses the roadway during this time, the section of highway or street on which the trucks travel is hazardous.)

12. What is the safe running speed (see the definition on Page 7C-1)? _____ mph.
13. Do at least 10 vehicles use the roadway during the hours students are going to or from school, and is the roadway without either sidewalks or minimum 4-foot wide shoulders at any location? _____ If yes, are there any sections of the roadway where the visibility of the student(s) is a problem for approaching drivers? _____ If yes, how far away can drivers see the shortest student? _____ feet.

(If the distance is less than the appropriate value in Table II in §447.4(b)(ii) of the regulation, the section of street or highway on which the sight distance deficiency exists is hazardous.)

14. If the roadway has no sidewalks, how wide are the shoulders? _____ feet
- During any 15-minute period that students are enroute to or from school, how many vehicles normally travel on the roadway? _____

(If the number of vehicles exceeds the values in §447.4(b)(2) for the appropriate speed, the route is hazardous for elementary and secondary students.)

15. Do elementary students have to cross at a signalized intersection that does not have an exclusive pedestrian walk phase or an adult crossing guard? _____ If yes, is sight distance, traffic volumes, or roadway widths such that it may be difficult for an elementary student to cross the intersection safely? _____

(If both answers are "yes" the route is hazardous for elementary students.)

16. Do secondary students who use the student-walking route have to cross a signalized intersection which is not routinely protected by an adult crossing guard? _____

If yes, is one or more of the following is satisfied? _____

- Students cannot readily see visible signal indications when desiring to cross the intersection.
- The signal is a multi-phase operation where it may not be apparent what traffic has a green indication.
- A 4.5-foot tall student using a crosswalk within the intersection may not be visible at a point that will allow an approaching driver turning through the crosswalk time to come to a safe stop.
- The complexity of the geometrics of the intersection makes it difficult for a secondary school student to traverse the intersection or reach a safe refuge.

(If both answers are "yes" the route is hazardous for secondary students.)

17. Can the school bus stop or the student-walking route be relocated to avoid a hazardous certification? _____

18. Are there any other extenuating circumstances that you believe would qualify this route as being hazardous? _____

PERSON RESPONSIBLE FOR COMPLETING THIS FORM:

I hereby certify that I personally examined this student-walking route and, to the best of my knowledge, the information I have supplied on this Study and Data Sheet is true and correct.

Name _____

Signature _____ Date _____

Title _____ Telephone No. _____

SCHOOL SUPERINTENDENT:

Signature _____ Date _____

Student Walking Route Certification

On _____, the Pennsylvania Department of Transportation
(Date)
investigated _____, in Intermediate Unit No. _____,
(SR, or Road or Street)
_____ School District, between _____
_____ and _____
_____, in _____ County.

The results of the investigation indicate the following:

Certified by:

District Executive
Engineering District _____

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Pennsylvania Department of Transportation



**Publication 46
Chapter 8**

Highway-Rail Grade Crossings

**Pennsylvania Department of Transportation
Chapter 8 of Publication 46
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8.1 General

8.1.1 Purpose

Highway-rail grade crossings differ from a highway-highway intersection in that the train always has the right-of-way. Therefore, considering the disastrous nature of crashes that occur when a highway vehicle gets hit by a train, it is extremely important that everyone comply with requirements of Publication 371 and the *MUTCD*, FHWA's *Railroad-Highway Grade Crossing Manual*, and other national guidelines.

The Public Utility Commission (PUC) is vested with exclusive jurisdiction to determine and prescribe the manner in which, highway-railroad crossings may be constructed, altered, relocated, suspended, or abolished, and the manner and conditions in or under which such crossings shall be maintained, operated, and protected to effectuate the prevention of crashes and the promotion of the safety of the public. The PUC is also authorized to appropriate property for construction or alteration of crossings, to allocate costs, and to assign maintenance to any party under its jurisdiction. See 66 Pa. C.S., Chapter 27, §§2702 and 2704.

8.1.2 Laws, Regulations and Other Publications

[*Grade Crossing Manual*](#), PennDOT Publication 371, latest edition.

[*Guidance on Traffic Control Devices at Highway-Rail Grade Crossings*](#). This is a collaboration of research and accepted practices, published by the U.S. DOT, Highway-Rail Grade Crossing Technical Working Group (TWG).

[*Manual on Uniform Traffic Control Devices \(MUTCD\)*](#). FHWA's manual that establishes national guidelines for traffic-control devices. Specifically, [Part 8](#) addresses highway-rail grade crossings.

[*Official Traffic-Control Devices \(Publication 212\)*](#). Regulations published as 67 Pa. Code Chapter 212. This regulation adopts the Federal Highway Administration's *Manual on Uniform Traffic Control Devices (MUTCD)* and establishes additional study requirements, warrants, principles and guidelines not included in the *MUTCD*..

[*Traffic Control -- Pavement Markings and Signing Standards*](#). This is PennDOT Publication 111M. Specifically, the TC-8600 standard prescribes the pavement markings at highway-rail grade crossings.

[Railroad-Highway Grade Crossing Handbook](http://safety.fhwa.dot.gov/xings/07010/index.htm), Revised Second Edition, August 2007, Publication No. FHWA-SA-07-010, which is available at <http://safety.fhwa.dot.gov/xings/07010/index.htm> FHWA.

8.1.3 Definitions

The following words and terms, when used in this policy shall have the following meanings, unless the context clearly indicates otherwise:

Abandonment of a rail line – This occurs when the Railroad files with the Surface Transportation Board (STB) and through a hearing process ask to give up their rights to serve customers.

Abolishment of an at-grade crossing – This typically occurs when abandonment of a rail has occurred for which at-grade crossings exist. Once the crossing has been ordered by the PUC to be abolished, all passive and active warning devices are to be removed and the crossing surface can either be paved over or totally removed from the roadway. Once a crossing has been abolished the PUC has no further jurisdiction over said crossing.

Exempt signs – The Exempt Railroad Crossing (R15-3) sign and the Exempt Railroad Warning (W10-1A) sign shall be authorized for use at those railroad grade crossings having “exempt” status as determined by the PUC. The PUC can order Exempt signs to be installed should the railroad wish not to suspend a crossing, but agrees to stop and flag the crossing due to very minimal rail traffic (e.g., seasonal rail traffic, less than one train per week during daylight hours, rail traffic is suspended for an extended length of time, etc.). Placement of “Exempt” signs below the crossbucks and the advance warning signs allows school buses, trucks carrying hazardous material, and vehicles carrying passengers for hire (commuter buses) to cross without stopping.

Suspension of an at-grade crossing – This typically occurs when the rail line has not been abandoned and the railroad concurs with suspending the crossing because there is not current or immediate future rail service for the line. A suspended crossing involves the removal of the traffic control devices (crossbucks and advance warning signs), and crossing gates if they exist, and the flashing lights can either be removed, turned away from view, or hooded. Until the rail lines are either removed or paved over the TRACKS OUT OF SERVICE (R8-9) sign shall be installed. Once the crossing is paved over, removed, or rail line is placed back in service this sign shall be removed. The Public Utility Commission (PUC) has jurisdiction over suspended crossings, meaning that no alterations can occur without proper PUC approval.

8.2 Miscellaneous Provisions

8.2.1 Abandoned Railroad Grade Crossings

[§3342\(c\)\(4\)](#) of the Vehicle Code (relating to abandoned railroad crossings) requires that abandoned railroad grade crossings be marked by the former rail operator with a sign prescribed by the Department indicating that the rail line is abandoned. The appropriate sign for this application is the TRACKS OUT OF SERVICE (R8-9) sign.

8.2.2 Placement of the Advance Warning Sign

The Highway-Rail Grade Crossing Advance Warning (W10-1) sign (or any of the W10-series signs) shall be used on each highway approach in advance of every highway-rail grade crossing except as noted in Section 8B.04 of the *MUTCD*.

Note 22 on Sheet 3 of 8 of the Pavement Marking Standards (TC-8600), provides guidance on the location of the advance railroad warning sign (i.e., dimension “C”). The values in the standard are greater than the “0-mph” values for Condition B in Table 2C-4 of the *MUTCD*, and should be used to help compensate for the following:

1. The advance distances in the *MUTCD* uses the assumption that drivers have a 2.5-second reaction time, and then will use their brakes to decelerate at a rate of 11.2 feet/second². For example, on a 55-mph road, the advance distance for the warning sign in the *MUTCD* would be 325 feet, whereas the TC-7800 standard now uses 450 feet. Since queuing may occur at the crossing, the Department believes it is prudent to use larger values.
2. Research suggests that the current rule-of-thumb for legibility distance should be reduced from 40 feet to 30 feet per inch of legend height to more accurately conform to minimum eyesight requirements for some state driver licensing. Therefore, it is likely that the values in Table 2C-4 of the *MUTCD* may be increased in the near future.

It is important to note that these are minimum distances and they are measured from the stop bar and not the railroad track.

8.2.3 Pavement Markings

In accordance with Section 8B.20 and Figure 8B-6 of the *MUTCD*, railroad crossing pavement markings are to be placed in advance of the highway-rail grade crossing unless the posted/statutory highway speed is less than 40 mph, or in an urban area, if

an engineering study indicates that the other installed devices provide suitable warning and control. The stop line, which shall be placed on each approach to a highway-rail grade crossing, is to be placed no closer than 15 feet from the nearest rail. On multilane highways, the stop line shall traverse across all approach lanes and individual RXR symbols should be placed in each approach lane.

8.2.4 Maintenance Responsibilities

The ordering of the installation and future maintenance responsibilities for passive warning devices, advance warning signs, and railroad crossing pavement markings at public at-grade crossings are under the jurisdiction and approval of the PUC. With regards to maintenance:

- The railroad is responsible for the passive warning devices at the crossing, i.e., Railroad Crossbuck (R15-1), Tracks (R15-2), Exempt Railroad Crossing (R15-3), Emergency Notification, and TRACKS OUT OF SERVICE (R8-9) signs, and any signs attached to the flashing light supports masts/cantilevers.
- The road owner is responsible for the advance warning (W10-series) and Exempt Railroad Warning (W10-1A) signs, and railroad crossing pavement markings (including the stop bar).

Typically the installation of such warning devices would be the same party responsible for the future maintenance. The PUC will include in their PUC Orders the parties responsible for maintaining the active and passive warning devices, advance warnings signs, and pavement markings; and that the installation of such be in accordance with Part 8 of the *MUTCD*.

8.2.5 Retroreflective Strips on Crossbuck Signs and Supports

The *MUTCD* has established a compliance date of January 17, 2011 (see Page I-6 in the 2003 Edition) for the installation of retroreflective sheeting material in compliance with Section 8B.03. Specifically, a minimum 2-inch wide strip of retroreflective sheeting material is required as follows:

1. On the back of each blade of each Crossbuck (R15-1) sign for the length of each blade except when the R15-1 sign is installed back-to-back.
2. On each support at passive highway-rail grade crossings for the full length of the front and back of the support from the Crossbuck (R15-1) sign or the Number of Tracks (R15-2) sign to within 2 feet above the elevation of the edge of roadway, except on the side where a STOP sign or YIELD sign have been installed or on the backside of the support on a one-way street.

The railroad company is responsible to provide and maintain the Crossbuck (R15-1) sign and the retroreflective strips on the R15-1 signs and their supports. In light of the January 17, 2011 deadline, if Districts are aware that the retroreflective strips are not in place, they should remind the appropriate railroad company of their responsibilities.

In 2004, PennDOT initiated a limited-time program using railroad safety funds, to reimburse railroad companies a statewide fixed unit price to add retroreflective strips to their crossbucks and supports.

The program also used fixed unit prices for the installation of Emergency Notification Signs as discussed in Section 8B.12 of the *MUTCD*.

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**Publication 46
Chapter 9**

Bicycle Facilities

**Pennsylvania Department of Transportation
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9.1 General

9.1.1 Purpose

The purpose of this chapter is to supplement Part 9 of the *MUTCD* by providing additional policies designed to enhance bicycle travel within the Commonwealth.

9.1.2 Integration of Bicycle Travel into the Planning and Design Process

Department policy now requires the evaluation of the access and mobility needs of pedestrians and bicycle users in all highway and bridge transportation corridors. This revised policy mandates that highway and bridge projects must evaluate the existing, latent, and projected needs of pedestrians and bicycle users. It requires the integration of the identified needs into project planning and design processes. This revised policy also clarifies that bicycle users are vehicles and that bicyclists and pedestrians are traffic in accordance with the Vehicle Code, and bolsters the importance of pedestrians and bicycle travel as viable and connective modes of transportation.

This Department policy is in SOL No. 432-07-02 (subject is “Integration of Pedestrian and Bicycle Modes of Transportation into Planning and Design Processes”) and will be in Design Manuals 1 and 1A. The primary evaluation tool of access and mobility needs of pedestrians and bicycle users is the “Bicycle and Pedestrian Checklist,” which will be in Design Manual 1A, Appendix J, with supporting documentation in Appendix A.

9.1.3 Laws, Regulations and Other Publications

AASHTO Guide for the Development of Bicycle Facilities, American Association of State Highway and Transportation Officials (AASHTO), ISBN 1-56051-102-8, 1999. Available at <http://www.communitymobility.org/pdf/aashto.pdf>.

Americans with Disabilities Act and Architectural Barriers Act Accessibility Guidelines, United States Access Board, July 23, 2004. Available at <http://www.access-board.gov/ada-aba/final.pdf>.

Design Manual, Parts 1, 1A and 2, PennDOT Publications 10, 10A and 13M. See <ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%2012.pdf>.

Manual on Uniform Traffic Control Devices (MUTCD), FHWA. Specifically, Part 9 addresses traffic control devices for bicycle facilities and is available at <http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/Ch9.pdf>.

Official Traffic-Control Devices, Regulations at 67 Pa. Code Chapter 212 – available at <http://www.pacode.com/secure/data/067/chapter212/chap212toc.html>, which adopts the *MUTCD* and establishes additional study requirements, warrants, principles and guidelines not included in the *MUTCD*. The regulation is also available as PennDOT Publication 212, which includes an appendix with some additional guidance on engineering and traffic studies, is also available online at <ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20212.pdf>.

Standards for Roadway Construction, PennDOT Pub. 72M, April 2004. These standard drawings are for common highway features, and are available at <ftp://ftp.dot.state.pa.us/public/Bureaus/design/PUB72M/PUB72COV.pdf>. (NOTE: the RC-25M standard provides details for standard-type shoulder rumble strips, but not the bicycle tolerable shoulder rumble strips referenced in [Section 11.11.3](#) and detailed in [Appendix 11B](#))

Vehicle Code (75 Pa.C.S.). Title 75 of the Pennsylvania Consolidated Statutes, which sets forth the laws relative to travel on streets and highways, and specific actions required by the Department. Of specific importance, §§3501 through 3513 address the operation of pedalcycles – and is available at http://www.dot10.state.pa.us/pdotforms/vehicle_code/chapter35.pdf.

9.1.4 Definitions

The following word, when used in this chapter, shall have the following meaning, unless the context clearly indicates otherwise:

Freeway – A limited access highway to which the only means of ingress and egress is by interchange ramps.

9.2 Bicycle Routes

9.2.1 BicyclePA Routes

The PA Pedalcycle and Pedestrian Advisory Committee and PennDOT jointly developed various cross-state designated bicycle routes that are referenced as “BicyclePA Routes.”

When erecting signs for these routes, every turn is to have a BicyclePA Route Marker Turn Assembly (M1-8A) and a confirmation BicyclePA Route Marker (M1-8) in each direction of travel. Also, install the M1-8 markers at distances of approximately 3 to 5 miles along the route, including beyond major intersections, so that bicyclists approaching the route will recognize that they have intersected it.

9.3 Miscellaneous Provisions

9.3.1 Bicycles on Freeways

In 1992, §3511 was added to the Vehicle Code (relating to pedalcycles prohibited on freeways) to prohibit pedalcycles on freeways except through the establishment of a permit process.

Considerations for possible approval:

1. Length of an alternate route versus distance on freeway.
2. Absence of potentially hazardous conditions on an alternate route, including but not limited to the following:
 - ADT of 500 or greater
 - Speed limit of 35 mph or higher
 - Poor pavement condition
 - Driveways
 - Intersections
 - Narrow or unpaved shoulders
 - Narrow vehicular travel lanes
 - Pedestrian traffic
 - Presence of railroad tracks, especially skewed crossings
 - Narrow bridges
 - Poor sight distance
 - Parked vehicles
 - Turning trucks
 - High percentage of trucks, buses, and recreational vehicles
3. Freeway has continuous paved shoulders of sufficient width and acceptable quality to accommodate bicycles (absence of rumble strips).

Application approval process:

1. The applicant completes an application (see [Appendix 9A](#)) and submits it to the Central Office Bicycle/Pedestrian Coordinator, who redistributes it to the District Bicycle/Pedestrian Coordinator for review.
2. The Engineering District verifies that the road in question is a freeway as defined herein. If it is a freeway, the District Bicycle/Pedestrian Coordinator and representatives from the District Traffic and Maintenance Units (Plans

Unit is optional) should then field view the roads in question. As a minimum, the field view should determine:

- a. Alternatives – length, condition, geometrics, etc.
- b. Freeway section – condition, shoulder type, width, should condition (pavement integrity).

The Engineering District should make every reasonable attempt to work with the applicant to solicit all pertinent information so that the Department can make a well-informed decision. Judge each application on a case-by-case basis.

3. Within 45 days after receipt of the application, the District should make one of the following recommendations to the Central Office Bicycle/Pedestrian Coordinator:
 - a. Approve. There are no reasonable route available based on length or safety, and the shoulder can be used as a designated bikeway.
 - b. Approve in part. Section of requested route meets criteria, and an alternate route may satisfy the remainder.
 - c. Disapprove.
 - i. Requested route cannot safely accommodate bicycles (e.g., the shoulder may be too narrow or its condition may be of such poor quality that it cannot accommodate bicycles).
 - ii. An alternate route is available that is relatively safe and presents only minimal added distance and terrain requirements.
 - iii. Insufficient information, e.g., additional information must be provided before a decision can be rendered.
4. The Bureau of Highway Safety and Traffic Engineering (BHSTE) will then make a final recommendation. If the recommendation is to approve the request or part of the request, BHSTE will prepare a notice of any proposed approval for publication in the *Pennsylvania Bulletin* and submit it to the Deputy Secretary for Highway Administration for approval.
5. If the Secretary or Deputy Secretary approves the request, the notice is published in the *Pennsylvania Bulletin* and the freeway section is open to any pedalcyclist who is 18 years of age or older, or is accompanied by a pedalcyclist who is 18 years of age or older.

6. The applicant is notified in writing after a decision is rendered. If any part of the request is denied, the applicant may submit a request in writing within 20 days to the Administrative Docket Clerk, Office of Chief Counsel, 9th Floor, Commonwealth Keystone Building, 400 North Street, Harrisburg, PA 17120. The applicant will pay a minimum \$100 fee for reconsideration.
7. If any part of the request to use a freeway is approved, the District will have the following traffic signs installed at the locations indicated on the sign standard:
 - MOTOR VEHICLES AND BICYCLES ONLY (R5-3-1A)
 - BICYCLES MUST USE SHOULDER (R5-3-1B)
 - ALL BICYCLES MUST EXIT (R5-3-1C)
 - Bicycle Restrictions (R5-3-1D)

9.3.2 Longitudinal Rumble Strips

The purpose of installing longitudinal rumble strips is to alert drowsy and inattentive motor vehicle drivers, but the standard type as included in the RC-25M Standard, Sheets 4 and 5 may pose a control threat to bicyclists who may inadvertently steer into them.

Therefore, Districts are encouraged to either use edgeline rumble strips (ELRS) or bicycle tolerable shoulder rumble strips (BTSRS) as discussed in [Section 11.11.3](#) and depicted in [Appendix 11B](#), instead of the standard rumble strips addressed in the RC-25M Standard.

Chapter 9

Appendix

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Request to Operate a Pedalcycle on a Freeway

Name _____

Organization _____

Address _____

Daytime Phone No. (____) _____

Section of Freeway Requested for Use (Reference by interchange name or number on freeway or intersecting traffic route or interchange name:

From _____

To _____

Reasons Why an Alternate Route is Not Suitable (Consider length and safety considerations of alternate route. Use continuation sheets, if necessary. Attach maps, photographs, and other documentation, if appropriate):

NOTE: Return the completed form to: PennDOT Bicycle/Pedestrian Coordinator, Bureau of Highway Safety and Traffic Engineering, P.O. Box 2047, Harrisburg, PA 17105-2047. Telephone No. (717) 783-8444

Pennsylvania Department of Transportation



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**Publication 46
Chapter 11**

Traffic Studies

Pennsylvania Department of Transportation

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11.1 General

11.1.1 Engineering Study and Engineering Judgment

Always base the decision to use a particular device at a specific location on engineering study and the application of engineering judgment. Thus, while this manual provides guidance on performing some traffic studies, it is not a substitute for engineering judgment.

Exercise engineering judgment in the selection and application of traffic control devices, as well as in the location and design of the roads and streets that the devices complement. Therefore, a qualified traffic engineer should make the final decision concerning the application of traffic restrictions.

11.1.2 Definitions

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

ADT, average daily traffic – The total volume of traffic during a number of whole days, more than 1 day and less than 1 year, divided by the number of days in that period.

Advancing volume – The total hourly volume of traffic approaching the intersection from the same direction as the turn movement under consideration (includes the turn volume).

Advisory speed – The recommended speed for vehicles operating on a section of highway based on the highway design, operating characteristics and conditions. When posted, the speed is displayed as a warning sign; that is, either a black-on-yellow or a black-on-orange sign.

Angle parking – Parking, other than parallel parking, which is designed and designated so that the longitudinal axis of the vehicle is not parallel with the edge of the roadway.

Corner sight distance –

- (i) Available corner sight distance—The maximum measured distance along a crossing highway which a driver stopped at a side road or driveway along that highway can continuously see another vehicle approaching. For the purpose of measuring the available sight distance, the height of both

the driver's eye and the approaching vehicle should be assumed to be 3.5 feet above the road surface. In addition, the driver's eye should be assumed to be 10 feet back from the near edge of the highway or the near edge of the closest travel lane if parking is permitted along the highway.

- (ii) Minimum corner sight distance—The minimum required corner sight distance based on engineering and traffic studies, to ensure the safe operation of an intersection. The minimum value is a function of the speed of the approaching vehicles and the prevailing geometrics.

Crash –

- (i) A collision involving one or more vehicles.
- (ii) Unless the context clearly indicates otherwise, the term only includes those collisions that require a police report; that is, the collision involves one of the following:
 - Injury to or death of any person.
 - Damage to any vehicle involved to the extent that it cannot be driven under its own power in its customary manner without further damage or hazard to the vehicle, to other traffic elements, or to the roadway, and therefore requires towing.

Delineator – A retroreflective device mounted on the road surface or at the side of the roadway in a series to indicate the alignment of the roadway, especially at night or in adverse weather.

Department – The Department of Transportation of the Commonwealth.

Divided highway – A highway divided into two or more roadways and so constructed as to impede vehicular traffic between the roadways by providing an intervening space, physical barrier or clearly indicated dividing section.

85th percentile speed – The speed on a roadway at or below which 85 percent of the motor vehicles travel.

Engineering and traffic study – An orderly examination or analysis of physical features and traffic conditions on or along a highway, conducted in accordance with this chapter for the purpose of ascertaining the need or lack of need of specific traffic restrictions, and the application of traffic-control devices.

Expressway – A divided arterial highway for through traffic with partial control of access and generally with grade separations at major intersections.

Freeway – A limited access highway to which the only means of ingress and egress is by interchange ramps.

Grade – The up or down slope in the longitudinal direction of the highway, expressed in percent, which is the number of units of change in elevation per 100 units of horizontal distance. An upward slope is a positive grade; a downward slope is a negative grade.

Highway –

- (i) The entire width between the boundary lines of every way publicly maintained when any part thereof is open to the use of the public for purposes of vehicular travel.
- (ii) The term includes a roadway open to the use of the public for vehicular travel on grounds of a college or university, or public or private school, or public or historical park.

Local authorities –

- (i) County, municipal and other local boards or bodies having authority to enact regulations relating to traffic.
- (ii) The term includes airport authorities except when those authorities are within counties of the first class or counties of the second class.
- (iii) The term also includes State agencies, boards and commissions other than the Department, and governing bodies of colleges, universities, public and private schools, public and historical parks.

MUTCD – The current edition of the *Manual on Uniform Traffic Control Devices*, as adopted by the Federal Highway Administration (FHWA), and available on the FHWA website.

Narrow bridge or underpass – A bridge, culvert or underpass with a two-way roadway clearance width of 16 to 18 feet, or a roadway clearance less than the width of the approach travel lanes.

Numbered traffic route – A highway that has been assigned an Interstate, United States or Pennsylvania route number, consisting of one, two, or three digits, sometimes with an additional designation such as business route, truck route or other similar designation.

One-lane bridge or underpass – Any bridge or underpass with a clear width between the curbs, parapets, bridge railings, trusses, etc., of less than 18 feet and which does not satisfy the definition of “narrow bridge or underpass.” In addition, if the

geometrics of the approach roadway, the skew of the bridge, etc., create a situation where the effective roadway width on the bridge or through the underpass may be less than 18 feet, consider the bridge or underpass to be a one-lane facility.

Operational analysis – The analysis of transportation operations utilizing an approved traffic engineering software package as prescribed in Chapter 12 of PennDOT Publication 46, *Traffic Engineering Manual*.

Opposing volume – The total hourly volume of traffic opposing the advancing volume.

Percent turns – The percentage of the advancing volume that turns; equal to the turn volume divided by the advancing volume.

Procession –

- (i) An organized group of individuals, or individuals with vehicles, animals or objects, moving along a highway on the roadway, berm or shoulder in a manner that interferes with the normal movement of traffic.
- (ii) The term includes walks, runs, parades and marches.

Roadway – That portion of a highway improved, designed or ordinarily used for vehicular travel, exclusive of the sidewalk, berm or shoulder. If a highway includes two or more separate roadways, the term refers to each roadway separately but not to all roadways collectively.

School – A public, private or parochial facility for the education of students in grades kindergarten through 12.

School zone – A portion of a highway that at least partially abuts a school property or extends beyond the school property line that is used by students to walk to or from school or to or from a school bus pick-up or drop-off location at a school.

Special event – An assemblage, procession, special activity or other planned event held within the right-of-way of a public roadway, which requires its closure or partial closure, for which express written approval is needed from the Commonwealth and/or the affected municipality.

Speed – The legal speed limit of the roadway. If an engineering study has determined that the 85th percentile speed of the roadway is higher than the legal speed limit, then the 85th percentile speed may be used at the discretion of the Engineering District.

State highway – A highway or bridge on the system of highways and bridges over which the Department has assumed or has been legislatively given jurisdiction.

Traffic calming – The combination of primarily physical measures taken to reduce the negative effects of motor vehicle use, alter driver behavior and improve conditions for non-motorized street users. The primary objectives of traffic calming measures are to reduce speeding and to reduce the volume of cut-through traffic on neighborhood streets.

Traffic-control devices – Signs, signals, markings and devices consistent with this chapter placed or erected by authority of a public body or official having jurisdiction, for the purpose of regulating, warning or guiding traffic.

Traffic restriction – A restriction designated by a traffic-control device to regulate the speed, direction, movement, placement or kind of traffic using any highway.

Traffic signal –

- (i) A power-operated traffic-control device other than a sign, warning light, flashing arrow panel, or steady-burn electric lamp.
- (ii) The term includes traffic-control signals, pedestrian signals, beacons, in-roadway warning lights, lane-use-control signals, movable bridge signals, emergency traffic signals, firehouse warning devices, ramp and highway metering signals and weigh station signals.

Travel lane –

- (i) A lane of a highway which is used for travel by vehicles.
- (ii) A lane in which parking is permitted during off-peak hours but is restricted for use as a travel lane during peak hours to obtain greater traffic movement.

11.1.3 Customer Service

One of the most frequent controversies that the Department gets involved in is the approval or disapproval of a traffic control device to solve a given safety or congestion problem. Sometimes when the Department's response is different than what the municipality wants, they become frustrated and turn to their legislators. The legislators in turn contact the Secretary or one of the assistants for intervention.

The Department's goal is to minimize these controversies and reach good solutions to complex and emotional concerns. This is difficult because, in these situations, the Department has two customers to serve — the municipality and the driving public. The key is to find solutions that the municipality can embrace and will benefit the driving public.

Therefore, each Engineering District should have an established process for those situations where a municipality and the District Traffic Unit disagree on how to address these issues. As a minimum, the District Executive needs to be directly involved in reaching a satisfactory solution, and if it cannot be resolved at the District level, issue needs to be elevated to either the Director of the Bureau of Highway Safety and Traffic Engineering.

Situations to Avoid

The following are situations that the Department wants to avoid in the future:

1. A signalized intersection with left-turn lanes of adequate length where a municipality wants either a separate left-turn phase or a protected/prohibited left-turn phase because of a documented history of near hits or a substantial number of non-reportable crashes that would be susceptible to correction by this treatment. In some cases, the Department turns down the request solely because it does not meet the conflict factor criteria, even though the introduction of the requested left-turn phase will not plunge the intersection into Level of Service E and the municipality is willing to accept the additional delays to improve safety.
2. The denial of a municipal request that does not quite meet the warrant threshold for reportable crashes, but where the municipality can justify their request by providing verifiable documentation that a significant number of non-reportable crashes have occurred that would be susceptible to correction by the approval of the request. (Note: §212.106(c) and §212.302(b), respectively allow the use of non-reportable crashes for multiway stop control and traffic control signals.)
3. An intersection with an obvious sight distance problem that cannot easily be remedied in a timely fashion, which is not taken into consideration when determining whether a traffic control device should be installed.
4. Prohibiting traffic from using a wide, paved shoulder (that presently is in good condition) to turn right at a congested intersection because the shoulder actually is structurally inadequate or will be damaged, and there are no immediate future plans to upgrade the shoulder so that traffic can use it.
5. A catastrophic or high profile crash at a site where the municipality, legislators, or the media expect action, but the Department's only response is the installation of an additional warning sign that has minimal safety benefits, thereby leaving the customers frustrated. However, if the characteristics of the crash indicate that driver error was the predominant cause and that changing the roadway would have little impact or safety benefits, the Engineering District needs to do a more effective job communicating this with the customers. In addition, consider partnerships with the enforcement

community to initiate targeted, high profile, well-publicized enforcement operations.

Using Engineering Judgment

[§212.4\(b\)\(2\)](#) of Chapter 212, Official Traffic Control Devices, allows flexibility to exercise sound engineering judgment based upon a combination of all data sources in making a determination of how best to solve a problem. Therefore, a determination need not be solely based upon the specific criteria in the warrants, if it can be substantiated by other valid factors.

Warrants in Chapter 212 provide the threshold for consideration of the installation of a traffic control device, but are not a substitute for engineering judgment. Moreover, the fact that a warrant for a particular traffic control device is met is not conclusive justification for the installation of the device.

There are two key principles to follow when applying sound engineering judgment as it relates to the application of a traffic control device that does not meet the specified warrants:

1. The site has at least one significant attribute or characteristic that warrants do not adequately consider, and the application of a specific traffic control device will most likely improve safety or otherwise be beneficial. In this case, there can be no degradation in safety, and if there would be small, undesirable impacts on traffic flow, delay, and other parameters, the municipality must understand and concur with the trade-off.
2. The application will not result in a significant precedent-setting situation that can be broadly applied in the area with an overall negative impact on safety. An example would be the approval of a multiway stop control in a community to control speed, since studies document that motorists tend to increase their mid-block speeds between multiway stops and the overall compliance with STOP signs degrades, thereby increasing the potential for crashes at intersections with legitimate stop control.

When using [§212.4\(b\)\(2\)](#), the basis for using sound engineering judgment needs to be documented in writing and placed in the project file.

Unconventional Solutions

In addition to conventional traffic control devices, Engineering Districts should consider other solutions including an increase in enforcement or the judicious application of appropriate traffic calming devices. Traffic calming measures that may be applied include raised crosswalks, speed tables (not bumps), roundabouts, bulb-outs, diverters, chicanes, and other such measures. Most of these measures would

only be appropriate for local roads and collectors that are not numbered traffic routes and are in a residential area or business district.

11.1.4 Traffic Impact Studies – Seal by a Professional Engineer

Developers of major site developments generally provide traffic access and impact studies with their applications for driveway permits. This includes all high-volume driveways and those medium volume driveways where the site-generated traffic could adversely affect roadway capacity or safety, as defined in Publication 441, *Access to and Occupancy of Highways by Driveways and Local Roads*.

These studies serve as an effective tool for evaluating the effects of site-generated traffic on roadway network and for assessing the traffic needs for both developments and the transportation system. In most cases, consulting engineers who have training and experience in traffic and transportation engineering prepare these traffic impact studies. However, there have been cases where studies were prepared by individuals with backgrounds in other fields, such as, architecture, surveying, etc.

This matter has been reviewed by the State Registration Board for Professional Engineers who indicated that a traffic impact report must be sealed by a registered professional engineer when independent engineering judgment is required while conducting the study or analyzing the data. Therefore, the Department requires a registered professional engineer to seal all traffic impact studies submitted to the Department.

11.1.5 Performing Engineering and Traffic Studies on Local Roads

The Office of Chief Counsel has advised that it is inappropriate for the Department to perform engineering and traffic studies on local roads with Motor License Funds. In addition, performing these studies may expose the Department to additional tort liability.

The only exceptions are those cases in which the locals agree to enter into an agreement reimbursing the Department and the District can commit personnel to the effort without compromising work requirements on State highways. Therefore, it is acceptable to perform studies pursuant to agility agreements.

However, the Department should continue to provide guidance to municipalities on engineering and traffic study methodology and requirements. Many municipalities are not familiar with the requirements in the regulations and it is appropriate for the Department to provide technical advice.

11.1.6 Release of Studies to the General Public

It is imperative that the Department keep all engineering and traffic studies as privileged information. Therefore, the Department should deny all requests from the general public for engineering and traffic study or a compilation of the results of the study.

The Torts Litigation Section of the Office of Attorney General, has determined that engineering and traffic studies should not be made available to the general public because of the provisions of [§3754](#) of the Vehicle Code (75 Pa. C.S. §3754), as well as the provisions of federal law found at 23 U.S.C. §409.

Specifically, [§3754\(b\)](#) of the Vehicle Code (relating to confidentiality of reports) provides that in-depth crash investigations and other safety study records and information is prohibited in any legal action or proceeding and it protects such investigations and studies from disclosure in the discovery process as well, so that the Department can object to their release even if subpoenaed.

The language in the Federal Highway Law, found at [23 U.S.C. §409](#) even more broadly provides:

“Notwithstanding any other provision of law, reports, surveys, schedules, lists, or data compiled or collected for the purpose of identifying, evaluating, or planning the safety enhancement of potential accident sites, hazardous roadway conditions, or railway-highway crossings, pursuant to sections 130, 144, and 152 of this title or for the purpose of developing any highway safety construction improvement project which may be implemented utilizing Federal-aid highway funds shall not be subject to discovery or admitted into evidence in a Federal or State court proceeding or considered for other purposes in any action for damages arising from any occurrence at a location mentioned or addressed in such reports, surveys, schedules, lists, or data.”

The clear purpose of both the state and federal statutory language is to encourage the investigation and study of potentially hazardous roadways so that improvements can be made. To that end, both the General Assembly of the Commonwealth and the United States Congress have provided that such investigations and studies can be withheld, not only from use as evidence in legal proceedings, but even from mere disclosure to other parties in litigation.

A citizen may be seeking access to an engineering and traffic study for purposes other than bringing a lawsuit against the Department. However, the Department could not logically object to the disclosure of studies in litigation if it otherwise releases such information to the public at large. Moreover, it is doubtful that any court would agree that information could be withheld in a lawsuit when that information is otherwise made available to the general public.

Yet both the state and federal legislatures have acted so that access to such information and studies is not available to parties involved in a lawsuit. Therefore, it must have been the intent of both the state and federal legislatures that such information and analysis would also not be available to the general public.

While the Department is reluctant to deny citizen access to any study funded with public funds, the general release of these kinds of studies could so greatly disadvantage the Department in litigation as to discourage the effort to undertake them. The resultant could be that studies are not undertaken for fear of their use in litigation, and that hazards go unidentified and the traveling public is put at additional risk. Studies, however, can be released to PennDOT business partners conducting transportation planning as long as the business partner agrees to maintain confidentiality.

To emphasize the Department's position, the Engineering District shall place the following notice in the front of any traffic engineering and safety study completed by or for the Department.

Confidential – Traffic Engineering and Safety Study

This document is the property of the Commonwealth of Pennsylvania, Department of Transportation. The data and information contained herein are part of a traffic engineering and safety study. This safety study is only provided to those official agencies or persons who have responsibility in the highway transportation system and may only be used by such agencies or persons for traffic safety-related planning or research. The document and information are confidential pursuant to 75 Pa. C.S. §3754 and 23 U.S.C. §409 and may not be published, reproduced, released or discussed without the written permission of the Pennsylvania Department of Transportation.

In addition, the Engineering District shall affix the following stamp on each page of any traffic engineering and safety document that is released.

This traffic engineering and safety study is confidential pursuant to 75 Pa. C.S. §3754 and 23 U.S.C. §409 and may not be disclosed or used in litigation without written permission from PennDOT.

Each Engineering District should have a minimum of two stamps for each of the above.

11.1.7 Laws, Regulations, Forms and Other Publications

Other useful sources of information include the following:

1. *Manual on Uniform Traffic Control Devices (MUTCD)*, 2003 Edition, FHWA – available at http://mutcd.fhwa.dot.gov/pdfs/2003r1r2/pdf_index.htm.
2. *Mechanical Electrical, and Electronic Speed Timing Devices*. (67 Pa. Code Chapter 105). See <http://www.pacode.com/secure/data/067/chapter105/chap105toc.html>
3. *Municipal Traffic Engineering Certification*. (67 Pa. Code Chapter 205). See <http://www.pacode.com/secure/data/067/chapter205/chap205toc.html>.
4. *Official Traffic-Control Devices*, (67 Pa. Code Chapter 212), a regulation that adopted the *MUTCD* and establishes additional study requirements, warrants, principles and guidelines for the Commonwealth, and is available at <http://www.pacode.com/secure/data/067/chapter212/chap212toc.html>. The regulation with an appendix, which contains some additional guidance on engineering and traffic studies, is also available as PennDOT Publication 212, at <ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20212.pdf>.

5. Engineering and Traffic Study Forms are available at (p:\penndot_shared\Engineering and Traffic Studies\TE Forms)
6. Pavement Marking Standards, that specify the types, dimensions, and locations of pavement markings (TC-8600), snowplowable raised pavement markers (TC-8602), and delineators (TC-8604) – available at <ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%2011M.pdf>.
7. Roadside Design Guide, 2002 Edition (AASHTO).
8. Traffic Signal Design Handbook. (PennDOT Publication 149M).
9. Vehicle Code (75 Pa.C.S.), Pennsylvania's Motor Vehicle Code – available at http://www.dot4.state.pa.us/vehicle_code/index.shtml.

The Bureau of Highway Safety and Traffic Engineering maintains a more comprehensive list of hyperlinks for traffic engineers at P:\bhste_shared\hyperlinks\common-hyperlinks.doc. Submit any suggestions for additions or changes to the list to the Chief, Traffic Engineering and Operations Division.

11.2 Approvals

11.2.1 References

(a) The Pennsylvania Vehicle Code. (75 Pa. C.S.).

- [§3307](#). No passing zones.
- [§3308](#), One-way roadways and rotary traffic islands.
- [§3313](#) Restrictions on use of limited access highways.
- [§3331](#), Required position and method of turning.
- [§3353](#), Prohibitions in specified places (parking, standing, stopping).
- [§3754](#), Additional parking regulations.
- [§3363](#), Alteration in maximum limits (speed limits).
- [§3364](#), Minimum speed regulations.
- [§3365](#), Special speed limitations.
- [§3367](#), Racing on highways.
- [§3541](#), Obedience of pedestrians to traffic-control devices and regulations.
- [§4902](#), Restrictions on use of highways and bridges.
- [§6109](#), Specific powers of department and local authorities.
- [§6122](#), Authority to erect traffic-control devices.
- [§6123](#), Erection of traffic-control devices while working.
- [§6124](#), Erection of traffic-control devices at intersections.

(b) Official Traffic Control Devices. (67 Pa. Code Chapter 212, Publication 212).

- [§212.5](#), Installation and maintenance responsibilities.

- [§212.108](#), Speed limits.
- [§212.114](#), Stopping, standing and parking.
- [§212.701](#), Processions, assemblages and special activities.

(c) Municipal Traffic Engineering Certification, (67 Pa. Code Chapter 205).

(d) Traffic Signal Design Handbook, Section 21.0, Permits (Publication 149M).

(e) Pavement Marking Handbook.

11.2.2 Department Approval on State Highways

The District Executive, an Assistant District Executive, or the District Traffic Engineer may issue or cancel permits or approvals for traffic restrictions and the installation and removal of traffic control devices on State highways and at the intersection of a State highway and local highways, providing:

- The individual has a Pennsylvania Professional Engineer's License, and
- Department regulations do not specify otherwise.

Moreover, [§212.5](#) of Publication 212 (relating to installation and maintenance responsibilities) defines what agencies are responsible for installing and maintaining traffic control devices.

11.3 Speed Restrictions

11.3.1 References

- (a) The Vehicle Code. (75 Pa. C.S.).
 - [§3368](#), Speed Timing Devices.
 - [§6109](#), Specific powers of Department and local authorities.
- (b) Mechanical, Electrical and Electronic Speed-Timing Devices ([67 Pa. Code Chapter 105](#)).

11.3.2 Crash Rates

In accordance with [§212.108\(b\)](#) of Chapter 212 (relating to engineering and traffic studies), speed limits should be established based on the 85th-percentile speed unless:

- there are sight distance restrictions (stopping and corner sight distance), or
- the majority of crashes are related to excessive speed and the crash rate is greater than the applicable rate in the most recent high-crash rate or high-crash severity rate table.

Information on calculating crash rates and the homogeneous crash rate tables is included in Item 2(1) in the Appendix of Publication 212.

To determine if a crash was related to excessive speed, consider the causation factors in the following two lists, recognizing that this is only a tool and is not intended to be all-inclusive:

Probably Related to Excessive Speed

- 21 Tailgating
- 25 Failure to heed stopped school bus
- 26 Failure to heed pedestrian on roadway
- 27 Failure to heed stopped vehicle
- 29 Passing in a no-passing zone
- 30 Careless passing
- 38 Speeding in excess of posted limit
- 39 Too fast for conditions
- 47 Driver lost control

May or May Not Be Related to Excessive Speed

- | | |
|----|--|
| 02 | Driver drinking (charged or indicated) |
| 19 | Failure to respond to traffic control device |
| 22 | Proceeding without clearance after stopping |
| 23 | Reacting to an obstacle on the roadway |
| 34 | Sudden slowing/stopping |
| 36 | Making improper entrance to highway |
| 37 | Making improper exit from highway |
| 42 | Driving on the wrong side (part. / fully) |
| 43 | Careless lane change |
| 46 | Hit and run |
| 48 | Other operator performance failure |
| 49 | Forced movement |
| 75 | Sudden entrance by non-drinking pedestrian |
| 76 | Sudden entrance by bicyclist |

11.3.3 Electronic Ball Bank Instruments

An electronic digital ball bank instrument is available that provides consistent results with the standard curved-tube ball bank indicators historically used by the Department. The Department evaluated one from the Rieker Instrument Company, Inc., but other manufacturers may have similar units.

The electronic model reviewed by the Department has a number of advantages over the curved tube models, including the following:

- It has a programmable trigger that can be set at any degree, including the 12-, 14-, or 16-degree readings (see [Section 2.5.3.2](#)), that will cause a red light or an audible warning to activate when the reading exceeds the preset value.
- After a test run, the operator can press the function button and display the high reading on the curve, and also link the unit to a printer for the file.
- It should not be necessary to have a passenger to observe the reading, thereby making personnel more productive.

Engineering Districts are encouraged to consider purchasing an electronic model. However, when using an electronic model, it is very important to maintain a smooth turn to obtain accurate results.

11.3.4 25-mph Speed Limit in “Residence Districts”

On December 21, 1998, Act 1998-151 added §3362(a)(1.2) to the Vehicle Code (relating a 25-mph speed limit in residence districts). This new section provides for a statutory 25-mph speed limit in a “residence district” if the highway is functionally classified by the Department as a local highway, and the highway is not a numbered traffic route as defined in §102 (relating to definitions). The purpose of the change is to eliminate the need for an engineering and traffic study on many of the highways under control of local authorities.

This change has major impacts since, in 1999, approximately 84,000 of the 119,000 miles of highway in the Commonwealth were functionally classified as “local highways.” This classification included 70,883 mile (96 percent) of the highways under the jurisdiction of local authorities, and 8,817 miles (22 percent) of the State highways.

Two additional factors complicate matters. First, the definitions of residence district and urban district sometimes overlap with resulting conflicts in statutory speed limits. Second, the definitions in the Vehicle Code for residence district and the subsequently referenced business district previously had very few applications and are subject to interpretation.

The Vehicle Code defines the following terms:

“Residence district” – The territory contiguous to and including a highway not comprising a business district when the property on the highway for a distance of 300 feet or more is in the main improved with residences or residences and buildings in use for business.

“Business district” – The territory contiguous to and including a highway when within any 600 feet along the highway there are buildings in use for businesses or industrial purposes, including but not limited to hotels, banks, or office buildings, railroad stations and public buildings which occupy at least 300 feet of frontage on one side or 300 feet collectively on both sides of the highway.

“Urban district” – The territory contiguous to and including any street which is built up with structures devoted to business, industry or dwelling houses situated at intervals of less than 100 feet for a distance of a quarter of a mile or more.

Interpretations and Guidelines

1. “Residence district” is the territory contiguous to and including a highway that has structures:

- designed for use as residences and having direct access to the highway, and
 - which cumulatively accounts for at least 150 feet of frontage on one or both sides of the highway within every 300-foot section of the highway.
2. If a section of highway qualifies as both a residence district and an urban district, the local authorities shall determine the appropriate speed limit by the erection of signs at maximum one-half mile intervals as required by §3362(b) (relating to posting of speed limits).
 3. Short sections of reduced speed limits less than 500 feet in length should not be posted because they cannot be enforced per §3368(e) (relating to distance required for mechanical, electrical and electronic devices).
 4. §6109(e) (relating to engineering and traffic investigations required) was amended to delete the requirement for an engineering and traffic study when establishing a 25 mph speed limit in a residence district. However, nothing prohibits local authorities on highways under their jurisdiction, or the Department on State highways, from completing an engineering and traffic study and thereby justifying a different speed limit.
 5. As allowed under §6109(d) (relating to prior approval of the department), your Engineering District should approve all speed limits on State highways before a local authority is allowed to erect speed limit signs.
 6. Except in cities of the first and second class, on State highways the Engineering Districts should confirm that a location is a qualifying residence district prior to approving 25 mph speed limits and authorizing local authorities to erect speed limit signs. The District may, at their discretion, perform an engineering and traffic study and reach consensus with the municipality on any other appropriate speed limit determined from the study.

11.3.5 65-mph Speed Limits

§3362 and §6110 of the Vehicle Code respectively address the maximum speed limit on State highways and on highways under the jurisdiction of the Pennsylvania Turnpike Commission.

In light of recent legislative changes and a commitment that the Department would restudy some of the existing 55-mph sections of rural freeways, Engineering Districts (or the Pennsylvania Turnpike Commission, as applicable) should establish 65-mph speed limits on all freeways unless there is strong justification for leaving it at a lower speed limit. In particular, Engineering District were requested to review the following sections of highway for possible 65-mph speed limit:

- I-70 in Fulton and Bedford Counties
- I-76 between Exits 48 and 57 (PA Turnpike)
- I-78 between Exits 29 and 60
- I-95 in Bucks County
- I-276 between Exits 328 and 358 (PA Turnpike)
- I-476 between Exits 20 and 31 (PA Turnpike)
- US 30 from PA 24 in York County to PA 340 in Lancaster County
- US 222, the Kutztown Bypass
- US 322 between Lewistown and Seven Mountains
- US 322 between the Clarks Ferry Bridge and Harrisburg
- PA 33 from I-78 to I-80

In the review, Districts should consider the following:

1. One goal of establishing speed limits is to minimize the speed variance. An article written by Nicholas J. Garber and Ravi Gadiraju entitled, “*Factors Affecting Speed Variance and Its Influence on Accidents*” (Transportation Research Record 1213), is available at P:\bhste_shared\traffic_restrictions\speed_limits\Factors Affecting Speed Variance and Its Influence on Accidents.doc. In the article, the authors note that crash rates increase with an increase in the speed variance, and speed variance will be minimized if the posted speed limit is 5 to 10 mph lower than the design speed. The article also notes that drivers will drive at increasing speeds as the roadway geometric characteristics improve, regardless of the speed limit.
2. The average fatality rate on freeways is about one-third of the fatality rate on conventional highways. Therefore, by raising some of the speed limits, some motorists may change their travel patterns from conventional highways to the safer freeways, thereby improving the overall safety.
3. A 60-mph speed limit could legally be established on a freeway under §3363 of the Vehicle Code (relating to alteration of maximum limits) providing speed limit signs were posted at maximum intervals of one-half mile. However, Districts should not use 60-mph speed limits because the difference between a 65-mph and a 60-mph speed limit is minor, and most speed limits are currently posted at 10-mph increments, e.g., 25, 35, 45, 55 and 65 mph.
4. Maintenance concerns should normally not be a controlling factor to reduce speed limits, but rather the District should address the maintenance problems, and do so as soon as possible. In addition, either advisory speed limits or a short section of reduced regulatory speed is preferred for isolated problem locations in lieu of establishing an extensive length of reduced regulatory speed limit. The goal is to avoid unrealistic speed limits, which are likely to breed disrespect on the part of the motorists.

5. [Appendix 11A](#) lists study elements that Engineering Districts should consider when determining if the speed limit on a freeway should be reduced.

Installation of Signs

Speed limit signs should be posted beyond each interchange for both 55-mph and 65-mph speed limits. When a 55-mph section of highway is over 5 miles in length, Districts should install the “NEXT ____” (R2-9) sign beneath the first two 55-mph Speed Limit (R2-1) signs (typically one on the left and one on the right side of the roadway) to help motorists realize the extent of the restriction, e.g., “NEXT 15 MILES.”

Issuance of Permits

The 65-mph speed limit is a statutory speed limit on freeways. However, Districts are asked to continue the current practice of issuing permits for the 65-mph speed limits instead of the reduced 55-mph speed limits. This policy maintains the philosophy that without a speed limit permit, the speed limit is 55 mph.

In order to change a speed limit on any freeway, please perform the following:

- Complete an engineering and traffic study.
- Provide an informational copy of the speed limit permit to the Traffic Engineering and Operations Division (Bureau of Highway Safety and Traffic Engineering) and identify the approximate effective posting date.
- Update the information in the ExcelTM database at P:\bhste_shared\traffic_restrictions\speed_limits\65mph Speed Limits.xls.
- Modify the speed limit in the RMS “432 screen” using the offset breaks that are already defined in the system (Districts cannot change the offset breaks, which means that speed limits will essentially be “segment level” data).

11.3.6 Legal Challenges

On October 24, 1980, the Superior Court of Pennsylvania, in the Commonwealth of Pennsylvania v. James Aiello, ruled that a speed limit established and posted prior to July 1, 1977 (the effective date of Title 75, the basis for the current Vehicle Code does not need to be supported by an engineering and traffic study. (Note, this also applies to other engineering studies where regulations now exist.)

Other court decisions make it quite clear that warning signs are not necessary to advise that a specific speed detection device is in use to enforce speed limits. For example, there is no need for signs to indicate that aircraft, radar, VASCAR-plus, etc.

are used. However, in accordance with [§3368](#) of the Vehicle Code (relating to speed timing devices), all mechanical, electrical and electronic devices need to be of an approved type (see [67 Pa. Code Chapter 105](#)).

In addition, there is also a court decision indicating that municipalities do not need to enact an ordinance to use VASCAR-plus.

The FCC has modified the license requirements for radar units. If the organization using radar already holds a two-way radio license, an additional license for the radar units is not required. Although the FCC no longer requires the radar units to be calibrated every 6 months, it is recommended that the District calibrate radar units at least once a year in order to keep the equipment in its best possible condition.

11.4 No-Passing Zones

11.4.1 Warrants

National warrants for no-passing zones are included in Section 3B.02 of the *MUTCD*, but additional warrants for the Commonwealth are included in [§212.202](#) of Publication 212.

Additional PennDOT Warrants for No-Passing Zones

1. In advance of a divided highway or an obstruction such as a bridge support pillar, a channelizing island or a safety zone that separates the two lanes of traffic.
2. In advance of, and on or within any bridge, tunnel or underpass that is designated as a narrow bridge or underpass.
3. In advance of a STOP (R1-1) sign, YIELD (R1-2) sign, or traffic signal.
4. In advance of an intersection where passing may be undesirable due to the high number of crossing or turning movements.
5. In advance of and within a school zone.
6. In areas where an analysis of vehicle crashes shows an unusually high number of passing-related crashes.
7. In areas where the roadside development includes many driveways and intersections where passing would create frequent potential conflicts.
8. At locations where the roadway width is very restrictive, shoulders are nonexistent or in poor condition, the roadway cross-section has an excessive crown, or obstacles are close to the roadway.
9. In areas where traffic volumes are very heavy and there would be very limited opportunities for motorists to pass other vehicles.
10. At locations where a passing zone would otherwise be less than 600 feet in length.
11. At locations where engineering judgment indicates that allowing passing is undesirable because a better passing area exists farther ahead.

When establishing a no-passing zone under Warrants 1 through 5, start the no-passing zone at a minimum distance in advance of the specific physical feature in accordance with [Exhibit 11.4-A](#).

Exhibit 11.4-A Advance Distance for No-Passing Zones

Speed Limit or 85th-Percentile Speed (mph)	Advance Distance (feet)*
35 or less	300
40	350
45	400
50	450
55	500

* These advance distances apply to PennDOT's Warrants 1 through 5 on the previous page.

11.4.2 Establishment of No-Passing Zones through Intersections

Engineering Districts should review all passing zones through intersections and areas with major commercial driveways on a case-by-case basis to consider the establishment of a no-passing zone.

Two problems exist for the driver on a side road or driveway:

1. A passing vehicle from the left may be obscured by another vehicle from the left that is turning right.
2. If turning to the right, a driver may not anticipate a vehicle approaching in that lane.

In either case, if the driver on the side approach does not see the passing vehicle, there is a high potential for a crash.

When establishing a no-passing zone, in should begin at a minimum distance equal to the appropriate value in [Exhibit 11.4-A](#).

11.5 One-Lane and Narrow Bridges and Underpasses

11.5.1 General

The purpose of this section is to establish Department guidelines to provide safe, efficient and practical traffic control at narrow and one-lane bridges on two-lane, two-way highways. These guidelines apply not only to structures normally thought of as “bridges,” but also to any other restricted width areas such as tunnels, underpasses or culverts.

11.5.2 Suggested Traffic Control

An advance warning sign informing the driver of the type of restriction; i.e., a NARROW BRIDGE (W5-2) sign, NARROW UNDERPASS (W5-2A) sign, ONE LANE BRIDGE (W5-3) sign, or ONE LANE UNDERPASS (W5-3A) sign, should ordinarily be installed in advance of one-lane and narrow bridges and underpasses.

Install an Advisory Speed (W13-1) plaque when the safe speed on a narrow or one-lane structure is less than the legal speed limit. In these cases, determine the advisory speed by an engineering and traffic study.

The advance warning sign should ordinarily be located a minimum distance in advance of the structure in accordance with Condition B in Exhibit 2.5-A. Since drivers may need to stop at one-lane structures to allow on-coming traffic to clear the structure, use the advance distance to decelerate to “0 mph.” Increase the advance distances as necessary to fit field conditions.

Always install a Left Clearance (OM-3L) marker and Right Clearance (OM-3R) marker on the left and right bridge abutments, respectively. Position these markers to the greatest extent possible, so that the inside edge of the marker is in line with the inner edge of the obstruction.

On roadways with centerline pavement markings, verify that a no-passing zone exists in advance of the structure in accordance with [Section 11.4.1](#) and [Exhibit 11.4-A](#). On narrow structures, continue the no-passing zone across and at least 200 feet beyond the structure.

For one-lane bridges and underpasses, discontinue any centerline pavement markings between the two locations on the approaches where it is desirable for approaching vehicles to wait for oncoming traffic. These locations should typically be 100 to 150 feet before the abutments, but lesser distances may be necessary to improve visibility between the two approaches.

On roadways with centerline pavement markings but typically without edge lines, it is desirable to install edge lines on both sides of the roadway on the bridge and on both approaches beginning a distance near the advance warning sign.

11.5.3 Optional Traffic Control

- (a) Install additional delineation on the approaches to a narrow or one-lane bridge or underpass to create a “funneling effect.” Base the need for the additional delineation on an engineering study that takes into account roadway width, approach width, ADT, traffic speed, crash experience, etc. In addition, if the bridge or underpass is greater than 100 feet in length, install barrier-mounted delineators on both sides of the bridge at 50-foot intervals.
- (b) Install Yield (R1-2) and TO ONCOMING TRAFFIC (R1-2A) signs on both ends of one-lane bridges or underpasses. When these signs are used, also install Yield Lines at the appropriate location.
- (c) Install traffic signals on the approaches of one-lane bridges or underpasses when the bridge/underpass carries an exceptionally heavy volume of traffic or when it is not always possible for a driver on one approach to see vehicle on the opposite approach. Use traffic detectors, Signal Ahead (W3-3) signs, and stop lines in conjunction with traffic signals.

11.6 Angle Parking

11.6.1 General Criteria

Although businesses generally like angle parking because it provides approximately twice as many parking spaces as provided by parallel parking, there are safety concerns because of limited sight distance when patrons are backing out of the stalls. This is especially difficult when there is a mix of vehicles such as small cars and large SUVs. [§212.114\(b\)](#) of Publication 212 establishes the following minimum criteria to authorize angle parking on any street or highway in the Commonwealth:

1. The parking and maneuver area equals or exceeds the distance indicated in [Exhibit 11.6-A](#).

Exhibit 11.6-A Diagonal Parking Minimum Maneuver Area

Parking Angle (degrees)*	Minimum Parking Maneuver Area (feet)**
30	26
45	30
60	37
90	43

* The angle that vehicles on the nearest travel lane need to turn to the right to park in the center of a parking stall.

** The perpendicular distance between the right edge of the nearest travel lane and the front edge of the parking stalls.

2. Parked vehicles do not adversely affect the available intersection sight distance.
3. Additional travel lanes are not required for the existing traffic volumes to achieve a satisfactory level of operation.
4. Parking stalls will be adequately marked and spaced.
5. Pedestrian/bicycle activity is minimal within the parking maneuver area.

11.6.2 Reevaluation of Existing Angle Parking

Federal policy currently allows the retention of angle parking if, in the judgment of the Division Administrator, there will not be adverse effects on street capacity and safety.

[§212.114](#) of Publication 212 (relating to stopping, standing and parking restrictions), it defines criteria for the establishment of new angle parking and the retention of existing angle parking. Based on the criteria, use the following procedure to evaluate the possible retention of angle parking within the limits of future federal-aid project:

1. Compare the parking-related crash rate within the area of existing angle parking to the rate on a similar portion of the same street or other similar streets with parallel parking. If the rate is substantially greater within the area with angle parking, eliminate the angle parking.
2. Perform a capacity analysis on the intersection or roadway section that includes the angle parking. If the extra width required to facilitate angle parking is needed by through and/or turning vehicles in order to allow the intersection or roadway to operate at LOS C or better, eliminate the angle parking.
3. If the project under consideration includes traffic signal interconnection, determine the effect of angle parking on vehicle progression. Eliminate the angle parking if the number of expected parking turnovers coupled with the design of the angle parking and maneuver area will create significant platoon dispersion during the peak hours.

11.6.3 Back-In Angle Parking

Because of safety concerns involving conventional angle parking, the following cities are using back-in angle parking:

Indianapolis, IN	Salem, OR	Tucson, AZ
New York, NY	Salt Lake, UT	Washington, DC
Portland OR	Seattle WA	Wilmington, DE
Pottstown, PA	Tacoma, WA	

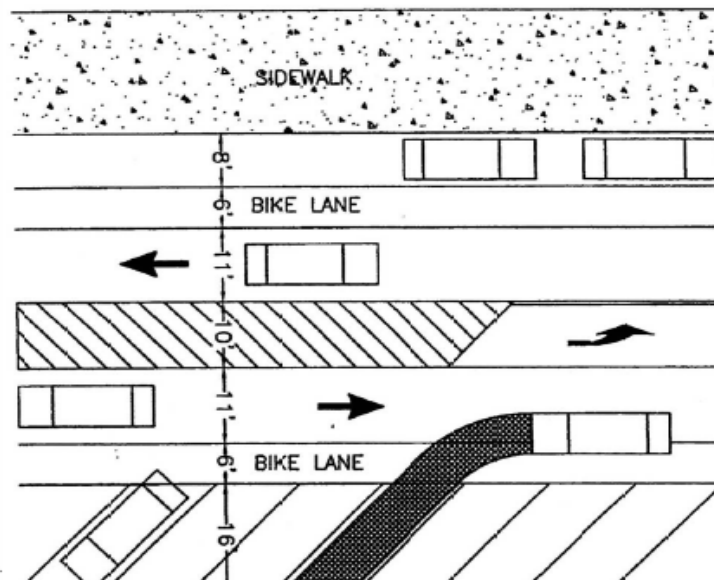
[Exhibit 11.6-B](#) shows an example of back-in angle parking as used in Pottstown. Although back-in angle parking is different, backing into stalls is common in parking lots, and it is much easier than parallel parking. Some of the reported advantages of back-in angle parking versus “pull-in” angle parking include:

- Less maneuver space required.

- No blind backing into traffic.
- Better visibility of bicycles.
- Curbside loading.
- Vehicle doors open towards curb.
- Easy and safe exit from space.

It is also true that back-in angle parking is similar to backing into parallel parking space, but with fewer movements.

Exhibit 11.6-B Back-In Angle Parking



Therefore, if a municipality requests to use back-in angle parking, the Engineering District should consider the option providing parked vehicles would not adversely affect the available intersection sight distance and additional travel lanes are not required for the existing traffic volumes to achieve a satisfactory level of operation.

If using back-in angle parking, install special BACK-IN ANGLE PARKING ONLY signs along the roadway. (This would be a special parking-series sign with green legend and border on a white background.)

11.7 Weight and Size Restrictions

11.7.1 References

Authorization to Use Highway Posted Due to Traffic Conditions ([67 Pa. Code Chapter 193](#)).

Engineering and Traffic Study Form TE-109 ([Restrictions as to Weight and Size Based on Condition of Highway or Bridge](#))

Hauling in Excess of Posted Weight Limit ([67 Pa. Code Chapter 189](#))

Official Traffic Control Devices (67 Pa. Code Chapter 212, Publication 212).

Maintenance Manual, Chapter 15, (Publication 23).

Special Hauling Permit Manual, (Publication 31).

Vehicle Code (75 Pa. C.S.)

- §4902, Restrictions on use of highways and bridges.
- §6109, Specific powers of Department and local authorities, specifically subsections (a)(7) and (a)(13).

11.7.2 Restrictions Based on Condition of Highway or Bridge

§§4902(a), 6109(a)(7), and 6109(a)(13) of the Vehicle Code authorizes the Department on State Highways, and local authorities on highways under their jurisdiction, to prohibit the operation of any types of vehicles or vehicles with designated loads or impose restrictions as to the weight or size of any vehicles whenever it is determined that a highway or bridge may be damaged or destroyed by vehicles. Weight restrictions may be temporary or seasonal, especially during the spring.

Studies for weight restrictions established due to the condition of a highway or bridge are conducted by District Maintenance personnel for highways and District Bridge personnel for bridges.

In order to establish a weight or size restriction based on the condition of a highway or bridge, an engineering and traffic study in accordance with [§212.117 of Publication 212](#) (relating to weight, size and load restrictions) must be conducted and

findings documented on an Engineering and Traffic Study Form, [TE-109](#) as also indicated in the Maintenance Manual, Publication 23, Chapter 15.

11.7.3 Restrictions Based on Traffic Conditions

§4902(b), and §6109(a)(7) and (a)(13), of the Vehicle Code authorizes the prohibition of vehicles on highway or bridge if they create a hazardous traffic condition or other safety factor. Further, [§212.117\(d\) of Publication 212](#) (relating to weight and size restrictions based on traffic conditions) indicates that traffic may be prohibited or restricted by weight or size of vehicle, or kinds or classes of vehicles, if any of the following indicate that the movement of certain vehicles constitutes a safety hazard:

1. Horizontal and vertical alignment.
2. Prevailing traffic speeds.
3. Compatibility of the various types of traffic.
4. History of vehicle crashes.
5. Vehicle characteristics.

Studies for size restrictions and for restrictions due to traffic or safety reasons are usually conducted by District Traffic Engineering personnel. The actual restriction may include any of the following:

- Weight.
- Height, width or length of vehicles or their loads.
- Types of cargo.
- Speed or gearing.
- Stopping requirements.
- Specified travel lanes.
- Hours of operation.

Because of the myriad reasons that could justify a restriction based on traffic or safety reasons, a suggested study summary sheet does not exist. Therefore, the District Traffic Unit should clearly describe the justification for the restriction. The District Traffic Unit should also prepare a TC 71-1 permit form (as included in Publication 23) for signature by the District Executive. Retain both the permit and the justification for future reference and possible use in legal proceedings.

11.7.4 Exceptions and Permits

Title 67, Chapter 189 regulates the use of highways with posted weight restrictions.

§4902(a) and (b) of the Vehicle Code authorizes the Department to exempt school buses, emergency vehicles, and vehicles making local deliveries or pickups from weight and size restrictions. Accordingly, the Department has defined “exempt vehicle” in [67 Pa. Code §193.2](#) (relating to definitions) as:

- (i) *Emergency vehicles.*
- (ii) *School buses operating on regular routes.*
- (iii) *Vehicles and combinations of governmental agencies and utilities or their contractors engaged in construction, maintenance or utility operations on or along a posted highway or in a location which can be reached only by a posted highway.*

The Department has also defined “local traffic” in [67 Pa. Code §193.2](#) (relating to definitions) as:

Vehicles and combinations going to or coming from a residence, commercial or industrial establishment, or farm located on or within the immediate vicinity of a posted highway or which can only be reached by a posted highway.

Further §4902(c) of the Vehicle Code (relating to permits and security) authorizes the Department to issue permits for movements of vehicles of weights and sizes in excess of the restrictions.

Chapter 15 of Publication 23 and Chapter 8 of Publication 31 address posted roads and bridges and discuss appropriate laws and regulations. As part of the discussion on regulations, the two chapters present information on exempt vehicles and issuance of permits for overweight or oversize vehicles. Normally, the following District personnel approve permits for overweight or oversize vehicles:

- District Maintenance Unit – Highways posted due to condition of roadway.
- District Bridge Unit – Bridges posted due to condition of bridge.
- District Traffic Engineering Unit – Highways posted due to traffic conditions.

District Traffic Engineering Unit Responsibilities

- (a) In conjunction with the issuance of permits for vehicles to use highways posted due to traffic conditions, the District Traffic Engineering Unit shall

provide the District Permit Unit with a current list of any traffic restrictions issued under §4902(b) of the Vehicle Code (relating to restrictions based on traffic conditions) each time a new restriction is approved or an existing restriction is revised or canceled. These restrictions will normally be prohibitions on hills but may involve prohibitions on other roadways due to non-compatibility of various types of traffic or cargo.

- (b) When the Traffic Engineering Unit receives an application for a permit to violate a restriction issued under §4902(b), they should:
- Determine if the vehicle is an “exempt vehicle.”
 - If it is an exempt vehicle, return the application with a note stating that in accordance with Chapter 193, a permit is not required.
 - If not an exempt vehicle, determine if the vehicle is “local traffic.” If the vehicle is “local traffic” and the application and check are in good order, the District Traffic Unit should review the application.
- (c) If it is determined that the application is to be reviewed, the Traffic Engineering Unit should review the justification for the permit as supplied by the applicant and use good engineering judgment to determine if special authorization should be granted. Factors that may be considered in the decision should include:
1. The driver’s thorough knowledge of the conditions.
 2. The presence of a vehicle retarder (engine brake).
 3. Field demonstration of the vehicle’s braking capabilities.
 4. The benefits derived from specific countermeasures included in [67 Pa. Code §193.3\(d\)](#) (relating to restrictions).
- (d) If the recommendation is to deny the application, return the application to the District Permit Unit with a written explanation so that the Permits Unit may notify the applicant.
- (e) If the recommendation is to approve the application, the District Traffic Engineer should sign and date the bottom of the application. Specify any expiration date and attach all permit conditions to the approved application before returning the application to the District Permit Unit for processing.
- (f) If approving the permit based on the driver’s knowledge of conditions, the permit should include the driver’s name and specify that it is only valid when the driver of the vehicle is the person whose name is on the permit.

- (g) If approving the permit because of the vehicle or the braking or operating capabilities of an individual vehicle, the permit should identify the specific vehicle and indicate that the permit is only valid for that specific vehicle.
- (h) If approving the permit because of both (f) and (g), the permit should include the driver's name and the vehicle's identification and should indicate that it is valid only for the identified vehicle and driver.

11.7.5 Oversize and Overweight Vehicle Traffic Control Plans (TCPs)

The District Traffic Engineering Unit is not normally involved in the issuance of a permit for oversize and overweight vehicles and loads; however, [67 Pa. Code §179.11\(b\)](#) (relating to movement of buildings) does require the District approval of the traffic control plan (TCP) for house moves.

11.7.6 Bridge Crawl Speeds

In 1998, the Department changed the policy on how District Bridge Units evaluate overweight permit applications. One of the changes was a requirement that the District Traffic Engineer grant a "safety approval" before approving any bridge crawl speed.

To assist the Districts, BHSTE developed two traffic control plans (TCPs) as standard set-ups to maximize traffic safety and to help maintain the structural integrity of bridges. When either TCP in [Appendix 11B](#) is included with the permit, the District Traffic Engineer's review and approval is no longer required. If a hauler wants to use an alternate TCP, the Bridge Unit must submit the alternate TCP to the District Traffic Engineer for approval.

This policy also places the following additional responsibilities on the hauler when bridge crawl speeds are involved:

- In addition to a private escort vehicle, provide a shadow vehicle with a sign to improve traffic safety.
- On freeways, provide a vehicle with an advance-warning message on a changeable message sign (CMS).
- Provide CB radios for communication between all vehicles (including any applicable Pennsylvania State Police officer), and at least one cellular telephone for emergencies.

Haulers will be responsible for making all arrangements of the specified traffic safety equipment. The Department will continue to arrange for PSP, but a minimum 48-hour advance notice is required.

If the Department or the PSP determines that a motor carrier has violated a bridge crawl speed, the Department should initiate administrative sanctions against the hauler.

11.8 Engine Brake Retarder Prohibition Policy

11.8.1 Prohibition Request Procedure

In accordance with [§4103](#) of the Vehicle Code (relating to the promulgation of vehicle equipment standards), the Department has the sole authority to regulate vehicle equipment. The use and/or prohibition of the use of engine brake retarders falls under this authority. Therefore, any municipality that plans to prohibit the use of engine brake retarders must first obtain the permission of the Department in accordance with the following procedure:

1. The municipality requests authorization to enact an engine-brake-retarder prohibition, in writing, to the appropriate District Executive. The request must indicate if the prohibition request is for specific roadway(s), portion(s) of a municipality or the entire municipality.
2. The District Traffic Unit reviews the request in accordance with the criteria in [Section 11.8.2](#), and prepares a proposed written response for the District Executive's signature, advising the municipality of the Department's decision.
3. The municipality must subsequently enact an ordinance — PennDOT will not enact any engine-brake-retarder prohibitions. The municipality is also responsible for procuring, erecting and maintaining the required signing, using official signs in accordance with the requirements of Publication 236M.
4. Enforcement of the engine-brake-retarder prohibition shall be the responsibility of the police agency that provides enforcement services for the municipality.

11.8.2 Criteria for Prohibition Approval

In order prohibit engine brake retarders; District Traffic Units shall ensure that all of the following are satisfied:

1. The roadway is not a limited access facility or a ramp exiting from a highway with a posted speed or 85th percentile speed of 55 mph or greater.
2. The average downhill grade does not exceed 4 percent within any 500-foot section of highway (i.e., the change elevation of the road is never greater than 20 feet in any 500-foot section of a downhill).

-
3. A reduced speed limit or reduced gear zone does not exist for trucks due to a hazardous grade determination.
 4. The posted speed limit or 85th percentile speed is less than 55 mph.
 5. There is no history of runaway truck crashes in the past 3 years.
 6. There is no discernible pattern of rear-end crashes in the past 3 years where a truck was the striking vehicle.

11.9 Unsignalized Midblock Crosswalks

11.9.1 Background

The installation of marked crosswalks has mixed reviews. While it is desirable to give guidance to pedestrians as to the safest locations to cross highways, studies have shown that pedestrian crash rates are sometimes higher in marked crosswalks than at other locations, perhaps because the markings give pedestrians a false sense of security.^{1 2}

§3542 of the Vehicle Code (relating to right-of-way of pedestrians in crosswalks), requires motorists to yield the right-of-way to pedestrians within any marked crosswalk, but this does not always happen.

In accordance with Section 3B.17 of the *MUTCD* indicates that an engineering and traffic study should be performed before crosswalks are installed at location away from highway traffic signals or STOP signs.

In a recent survey, a majority of state traffic engineers indicated that midblock crosswalks are highly discouraged in their state and are rarely installed. Currently, only a few states have any warrants or guidance for midblock crosswalks. Therefore, the purpose of this policy is to establish the direction for future unsignalized midblock crosswalks on State highways, but it is not necessary to reevaluate existing midblock crosswalks. Although the application of this policy on local roadways may be desirable, the Department currently has no authority to force municipalities to comply with this policy.

Department approval is required prior to the installation of any midblock crosswalk on a State highway; however, the installation and maintenance of the pavement markings and signs for crosswalks is the responsibility of the local authorities in accordance with [§212.5\(b\)\(1\)\(v\)\(T\) of Publication 212](#) (relating to the installation of pavement markings for midblock crosswalks).

11.9.2 Minimum Requirements for New Midblock Installations

1. Speed Limit. The posted speed limit is 35 mph or less.

¹ Herms, B. F., “Pedestrian Crosswalk Study: Accidents in Painted and Unpainted Crosswalks” (HRR 406). Highway Research Board, Washington, D.C., 1972.

² “City of Long Beach Crosswalk and Pedestrian Safety Study Final Report.” Prepared by Willdan and Associates, Industry California, February 1986.

2. Other Marked Crosswalks. The nearest marked crosswalk on the same roadway is over 300 feet from the proposed crossing.
3. Number of Pedestrian Crossings. To qualify for midblock crosswalks, the minimum number of pedestrians crossing the street within 150 feet of the proposed crossing during an average day should be 80 or more during any 1 hour, or 40 or more during each of any 4 hours. However, if there is a high concentration of children, elderly or disabled pedestrians crossing the roadway in the vicinity of the proposed crossing, then these pedestrian volume warrants may be reduced 50 percent.
4. Traffic Volume. The maximum traffic volume on the roadway is 10,000 ADT, except on two-lane roadways the maximum traffic volume may be 15,000 ADT. If a raised median or pedestrian refuge island exists where pedestrians are protected from vehicular traffic, the maximum traffic volume applies to each segment of the pedestrian crossing, but no more than three travel lane may be crossed without a raised median or pedestrian refuge island. In order to consider a refuge island, the minimum width of the refuge island is 4 feet from face-of-curb to face-of-curb, but the preferred minimum width is 6 feet. Islands should have a cut through ramp to accommodate wheelchair users.
5. Parking Restrictions. To improve visibility, parking is not permitted within 75 feet of the crosswalk, unless a 6- to 8-foot curb extension (sometimes referred to as bulb outs, bump outs, neck downs, sidewalk expansions, etc.) is in place to improve pedestrian visibility. If angle parking is present, any curb extension should place the curb at the inside edge of the parking lane. Curb extensions not only improve visibility between motorists and the pedestrians, but they also reduce the length of the crosswalk and the pedestrian exposure. However, curb extensions may impede drainage, street cleaning and winter maintenance operations, and create a formidable object.
6. Sight Distance. The available sight distance between an approaching driver and a person anywhere within the proposed crosswalk must satisfy the following minimum values, where both the eye and the object (i.e., the pedestrian) are assumed to be 3.5 feet above the roadway:

Speed Limit (mph)	Minimum Sight Distance for a Corresponding Grade (feet)		
	-6%	level	+6%
25	215	200	184
30	271	250	229
35	333	305	278

11.9.3 Design Considerations

Raised crosswalks may be used to improve the visibility of pedestrians. If used, the geometric design of raised crosswalks should be in accordance with Pennsylvania's [*Traffic Calming Handbook*](#) (Publication 383). Pavement markings in advance of raised crosswalks should conform to the "typical pavement markings for speed tables or speed humps with crosswalks" in the Pavement Marking Standard (TC-8600).

Use the Pedestrian (W11-2) sign with the Diagonal Downward Pointing Arrow (W16-7P) plaque immediately prior to the crossing. The sign and plaque may be fluorescent yellow green in color. If a curb extension is used, install the W11-2 sign in the "bumped out area."

Establish crosswalks at approximately 90 degrees to the roadway. For added visibility, the area of the crosswalk may be marked with wide white diagonal lines at a 45-degree angle to the edge of the crosswalk or with wide white longitudinal lines parallel to traffic. All crosswalk markings shall conform to the Pavement Marking Standard (TC-8600).

Consider using Yield Lines in advance of an unsignalized midblock crosswalk. If used, place Yield Lines 20 to 50 feet in advance of the crosswalk and install a YIELD HERE TO PEDESTRIANS (R1-5) sign immediately adjacent to the Yield Line. Yield Lines should conform to the Pavement Marking Standard (TC-8600).

Additional advance warning signs, internally illuminated overhead signs, In-Street Pedestrian Crossing (R1-6) signs, flashing beacons, etc., may supplement unsignalized midblock crosswalks. Yellow, pedestrian activated, in roadway warning lights may also be used providing traffic is not controlled by a STOP or YIELD sign (see Section 4L.02 of the *MUTCD*).

11.10 Mirrors

11.10.1 General

The purpose of this policy is to establish uniform guidelines governing the approval and installation of mirrors located within the Department's right-of-way to improve the effective sight distance at intersections.

Flat mirrors are unacceptable for use at highway intersections because they provide a very small field of vision and require every observer's eyes to be at the proper location to effectively use the mirror. On the other hand, convex mirrors overcome the alignment issue and have been used along highways with some success despite the following inherent problems:

1. The image is distorted and reversed.
2. Vehicles appear to be much farther away than they actually are. For example, the image of an approaching car when it is 100 feet away from the mirror will be only about 2 to 2.5 inches wide on a standard convex mirror.
3. Mirrors require routine cleaning and are subject to vandalism.
4. Mirrors are fairly expensive (approximately \$250).
5. Unfamiliar drivers require time to become oriented when attempting to use a mirror.

11.10.2 Minimum Warrants

The Department may authorize local officials, companies, associations or private individuals to install a mirror intended for traffic on a State highway, local roadway or a private driveway, providing the following conditions are met:

1. There are no viable alternatives for improving the sight distance or providing an alternate method of traffic control. Give consideration to redesigning or day-lighting the intersection; eliminating parking; reversing the stop control or providing multiway stop control, signalization, one-way streets or other appropriate action; etc.
2. The total volume of the traffic on the mainline and on the side road (or driveway) approach to the "intersection" with the mirror is less than 500 vehicles per day.

3. If a public highway, the approach is controlled by a STOP (R1-1) sign.
4. The municipality, company, association or private citizen has agreed, in writing, to purchase, erect and maintain the mirror and the VEHICLES ARE CLOSER THAN THEY APPEAR (W14-11) sign in satisfactory condition; save harmless the Department in the event of an crash; and remove the mirror and sign if directed by the Department.

11.10.3 Installation Guidelines

Convex mirrors are normally available at local glass dealers. As a minimum, they shall:

1. be designed for exterior use;
2. be made of “Plexiglas” or shatterproof glass;
3. have a minimum diameter of 36 inches; and
4. have a minimum radius of curvature of 80 inches.

If possible, position convex mirrors directly ahead of the intended user on the far side of the roadway, but it is best to experiment at other locations prior to permanently installing it. The angle formed between the observer, the mirror and the approaching vehicles should be 90 degrees or less. Typically, mount convex mirrors 10 to 15 feet above the roadway surface and brace it to reduce the chances of it becoming misaligned.

To determine the effectiveness of the mirror, have the stopped driver use the mirror and measure the elapsed time between the moment that a test car’s image becomes visible in the mirror and the moment the car reaches the intersection. Compare the elapsed times with the mirror at different locations and without the mirror. Always drive the test car at a constant speed for all comparisons.

Install the VEHICLES ARE CLOSER THAN THEY APPEAR (W14-11) sign beneath the convex mirror.

11.11 Rumble Strips, Speed Bumps, and Speed Humps

11.11.1 References

Pennsylvania Traffic Calming Handbook, January 2001, PennDOT, [Publication 383](#).

Work Zone Traffic Control Guidelines, January 2006, PennDOT, [Publication 213](#).

11.11.2 Transverse Rumble Strips

Highway agencies have successfully used transverse rumble strips for about 40 years to alert drivers of potential dangers. Although these devices are very useful, two common problems need emphasis:

1. Making depressions too deep and creating unnecessary driver panic.
2. Failing to consider the potential noise impacts in residential areas.

Section 6F.84 of the *MUTCD* and Application “PATA 42” in the *Work Zone Traffic Control Guidelines* ([Publication 213](#)) shows temporary bituminous rumble strips to alert drivers to unusual conditions on the approach to work zones. Districts may use transverse rumble strips for permanent installations, in advance of the following problem areas:

1. High crash intersections where STOP sign observance studies indicate that a high percentage of drivers do not stop.
2. Curves or other high crash areas where conventional traffic control has not been successful.

Although PATA 42 shows rectangular-shaped grooves that are recessed 1/2-inch, Districts are encouraged to reduce the depth of the rectangular-shaped grooves for permanent applications to 3/8-inch deep to reduce the panic factor, especially if the grooves are milled. Avoid installing transverse rumble strips on horizontal curves or over-vertical curves, or on shoulders where bicycles or pedestrian traffic may be common.

When used, in accordance with the standard, use five (Pattern A) or six (Pattern B) sets of rumble strip clusters in advance of the intersection or other condition, as indicated in PATA 42, with the closest set located approximately at the same location as the Intersection, Turn Curve or other advance warning sign.

Install a RUMBLE STRIP AHEAD (W8-15) sign in advance of transverse rumble strips.

11.11.3 Longitudinal Rumble Strips

The following types of longitudinal rumble strips alert drivers if their vehicle strays from its normal travel path:

1. Shoulder rumble strips (SRS) and Bicycle tolerable shoulder rumble strips” (BTSRS).
2. Centerline rumble strips (CLRS).
3. Edge line rumble strips (ELRS).



Shoulder Rumble Strips

In the mid-1980's, shoulder rumble strips (SRS) came into existence in the United States. Research at the Pennsylvania Transportation Institute indicates that shoulder rumble strips reduce the number of single-vehicle, run-off-the-road crashes by at least 20 percent based upon state and national crash data. Details for shoulder rumble strips are included in the RC-25M Standard, where Sheet 4/7 is for limited access highways, Sheet 5/7 is for conventional roads, and Sheet 6/7 is for gore areas.

However, because shoulder rumble strips can pose problems for bicyclists, “bicycle-tolerable shoulder rumble strips” (BTSRS) are now being used. Therefore, bicycle tolerable rumble strips are considered cost effective on rural highways with ADT of 1,500 or greater, and paved shoulders that are 6 feet or more in width.

Where the existing graded shoulder width is 6 feet or greater and is partially paved and the cost is not substantial, Engineering Districts should consider building a full width paved shoulder such that the bicycle tolerable shoulder rumble strip (BTSRS) can be incorporated into the design.

Engineering Districts may incorporate bicycle tolerable shoulder rumble strips (BTSRS) on 3R projects (100 percent state and federal-aid) using the criteria in [Appendix 11C](#).

Centerline Rumble Strips (CLRS)

Based upon an analysis of head-on and opposing crash data, centerline rumble strips produce a conservative crash reduction factor of 20 percent. Moreover, based on average installation costs in Pennsylvania, centerline rumble strips are cost-effective on rural two-lane highways where the ADT is 1,500 or greater.

Therefore, Engineering Districts should consider centerline rumble strips on all rural two-lane and four-lane undivided 3R (federal-aid and 100 percent state) projects where the ADT is greater than 1,500 and the pavement width is 20 feet or greater using the criteria in [Appendix 11D](#), except not on concrete pavements with an overlay of 2.5 inches or less. If the average surface life is 10 years, this initiative should save an estimated 275 lives.

Edge line rumble Strips (ELRS)

Edge line rumble Strips (ELRS) may be used in lieu of shoulder rumble strips when the travel lane and shoulders are both of sufficient width. ELRS may be more beneficial than shoulder rumble strips because they provide an earlier alert to a driver that is about to run off the road. All ELRS shall be in accordance with [Appendix 11E](#).

While the crash reduction potential is significant, the Department has insufficient experience with the combined impact of both centerline and edge line rumble strips on the same project to warrant full deployment. Of primary concern are the tight travel lane restrictions and the more frequent departures to one of the rumble strips. However, to gain that knowledge and experience, Engineering Districts may experiment on 3R projects, but should omit edge line rumble strips on the inside of moderate-to-sharp curves, which encompass dwellings.

Timely Placement of Pavement Markings

After installing CLRS or ELRS, apply pavement markings over CLRS and ELRS as soon as possible. Although one may think that these rumble strips would reduce nighttime retroreflectivity of the markings, highway agencies generally observe enhanced wet-night benefits of the milled surfaces, perhaps because water has an opportunity to drain off the pavement marking and pond in the recesses.

11.11.4 Speed Bumps

A speed bump is an abrupt pavement feature, 3 inches or more in height, and only 1 to 3 feet in length (measured in the direction of travel). State law and/or regulations do not prohibit speed bumps; however, the Department does not permit them on State highways and discourages their use on any local road or facilities open to the public.

Speed bumps can cause extreme discomfort to all motorists, including firefighters riding on the back of a fire truck, motorcyclists, and bicyclists. Studies have shown that in some cases, drivers of cars and trucks actually speed up to reduce the impact of the bump or to reduce the delay caused by slowing for the bump. Therefore, speed bumps generally have negative effects on safety.

Although not permitted on State highways, the Department has no authority or duty to require the removal of existing speed bumps on a local road, and the Department cannot withhold liquid fuels tax funds in an attempt to force a municipality to remove speed bumps from a local road. However, the Department may refuse to approve plans or proposals that would use liquid fuels monies on projects that include the installation of speed bumps.

11.11.5 Speed Humps

Speed humps are similar to speed bumps, but they are much longer and more gentle, and they are effective in reducing traffic speeds. They are normally 12 feet long, with a maximum 3-inch height at the mid-point, and are placed across the entire width of the roadway.

Chapter 5 in the Department's Pennsylvania Traffic Calming Handbook ([Publication 383](#)) discusses speed humps and other acceptable traffic calming measures. Although the Department does not permit the installation of speed humps on State highways, they may be beneficial in reducing speeds on local residential streets providing all of the following are true:

1. The posted speed limit is 25 mph or less.
2. The hump is located on a tangent section of roadway.
3. The hump is at least 150 feet from an unsignalized intersection, or 250 feet from a signalized intersection.
4. The hump is no greater than 3 or 4 inches high and at least 12 feet long, as measured in the direction of travel.
5. The hump extends across the entire roadway.
6. The hump does not impede drainage

11.12 Highway Occupancy

11.12.1 References

A Policy on Geometric Design of Highways and Streets, Fifth Edition, 2004, AASHTO. Also known as the “Green Book.”

Access to and Occupancy of Highway by Driveways and Local Roads ([67 Pa. Code, Chapter 441](#), Publication RR-441).

Design Manual, Part 2, PennDOT (Publication 13M).

Guidelines for the Design of Local Roads and Streets, PennDOT (Publication 70M).

Highway Occupancy Permit Handbook, PennDOT, (Publication 282).

Highway Occupancy Permit Manual. (Publication 170).

Roadway Construction Standards, PennDOT (Publication 72M).

Traffic Signal Design Handbook, PennDOT (Publication 149).

Traffic Signal Standards, PennDOT (Publication 148).

11.12.2 District Traffic Engineering Unit Responsibilities

The major responsibilities of the District Traffic Engineering Unit in highway occupancy are limited to the following:

- Review and approve TCPs.
- Review all applications to install or replace crash-damaged utility poles or those to be located in areas on the utility pole, crash-cluster list or in limited-access right-of-way.
- Review and recommend approval or denial of access requests, as indicated below.
- Review and recommend approval of traffic impact studies.

The Permit Unit is responsible for conducting all field measurement of sight distance for driveway applications.

District Traffic Engineers should review the following access applications:

1. Applications for medium-volume and high-volume driveways. Low-volume and medium-use driveways need not be reviewed by the Traffic Engineering Unit except in special instances where the District Permit Office requests assistance in resolving a possible traffic flow or safety problem. If assistance is requested, complete cooperation should be extended to the District Permit Unit.
2. Applications involving requests for auxiliary lane(s).
3. Applications where measured sight distances as determined by the County Permit Supervisor is marginal (i.e., plus or minus 25 feet of formula sight distance).
4. Applications where traffic control devices will be added or modified (i.e., signalization, signage or pavement markings).
5. Applications where work zone traffic control reviews are required.
6. Applications where the applicant has a waiver under Department regulation [67 Pa. Code §441.5\(e\)](#) (relating to waiver of design requirements).

11.12.3 Additional Review Guidelines

Unless indicated otherwise in [67 Pa. Code Chapter 441](#) or in Publication 170, design details and features on highway occupancy permit projects shall comply with the applicable provisions of:

- *A Policy on Geometric Design of Highways and Street.*
- *Design Manual, Part 2.*
- *Roadway Construction Standards.*
- *Guidelines for the Design of Local Roads and Streets.*
- *Traffic Signal Design Handbook.*
- *Traffic Signal Standards.*

Give particular attention to geometric modifications that modify the characteristics of through travel lanes and add new auxiliary turning lanes. It is important that the transitions (particularly for the through lanes), tapers, and storage requirements comply with the appropriate provisions in Design Manual, Part 2. In addition, design the proposed pavement structure for these lanes to be capable of withstanding the anticipated loads.

Pavement widening is often necessary to accommodate left-turn lanes. When this occurs, through travel lanes must transition from old to new pavement. The longitudinal joint created between old and new pavement crosses the through travel lane at an angle and can cause driver confusion, particularly at night. In order to eliminate this potential hazard, a full-width, thin overlay (normally 1.5" superpave) covering both new and old pavements is to be required at these sites. New pavement markings are also to be included.

Give emphasis to pedestrian and bicycle modes of transportation. Oftentimes, major developments are located in or on the fringe of major suburban areas and close to housing developments. Where these situations occur, request developers to consider incorporating provisions in the site plan to enhance these modes of transportation. Typical enhancements include sidewalks, pedestrian phases at signals, and bicycle storage accommodations.

Preliminary Reviews

Districts shall honor all requests made by developers for preliminary reviews. This obviously does not mean that all permit applications have to undergo a preliminary review process; but it does mean that if a developer asks the Department to conduct a preliminary review, the Department will do so.

Concurrent Reviews

The Central Permit Office, the Bureau of Highway Safety and Traffic Engineering, and the Bureau of Design will review all highway occupancy permit application packages that come into Central Office concurrently (rather than consecutively). FHWA will still perform any necessary review after the Central Office review so that they will be aware of the Department's comments or concurrence when conducting their review. Similarly, involvement by the Office of Chief Counsel will usually have to come after design features and other matters have been approved by the other reviewing offices before related agreements can be finalized.

11.13 Municipal Waste Facilities

11.13.1 General

The Department of Environmental Protection is responsible for the review and/or approval of all applications for the establishment of municipal and residual waste disposal facilities.

11.13.2 References

Highway Capacity Manual, Transportation Research Board, “Special Report 209.”

Highway Occupancy Permits, Commonwealth of Pennsylvania, [67 Pa. Code Chapter 441](#). Also available internally as Publication 441.

“*Municipal Waste Facilities Review Program*,” Executive Order No. 1996-5, dated August 29, 1996. (see http://www.oa.state.pa.us/oac/lib/oac/exec_orders/1996-5.pdf)

11.13.3 District Involvement with DEP Reviews

The following is a brief summary of the role that the Engineering District Offices will play in the Pennsylvania Department of Environmental Protection’s (DEP’s) Municipal Waste Facilities Review Program as a result of the Governor’s Executive Order Number 1996-5, “Municipal Waste Facilities Review Program” (dated August 29, 1996) and DEP’s related policies for the local municipality involvement process, the environmental assessment process (Phase I review), the municipal waste facility review – traffic analysis, and the process for evaluating daily volume.

PennDOT’s basic role in this program will be to serve as a reviewing agency for transportation-related studies (i.e., Transportation Impact Studies or TISs) involving State highways (SRs) that are submitted to DEP by an engineer for a municipal waste facility. PennDOT will forward comments and recommendations to DEP to assist them in making decisions regarding municipal waste facility permits. In addition, through its highway occupancy permit process ([67 Pa. Code Chapter 441](#)), PennDOT will need to review and approve plans and specifications for needed work within State highway right-of-way to accommodate waste facility traffic. These plans and specifications are to be developed by an engineer for the municipal waste facility, and the improvements will be implemented by the waste facility owner/operator in accordance with a PennDOT-issued HOP. PennDOT will also assist DEP, as necessary, in its review of existing waste facility permits and associated operations.

Listed below in chronological order are the roles that the Engineering District Offices will undertake regarding applications for new municipal waste facilities, applications for expansions of existing facilities that would result in additional waste volumes or capacity, and periodic reviews of existing waste facility permits and operations. Unless indicated otherwise, “PennDOT” means PennDOT’s appropriate Engineering District Offices, and “DEP” means DEP’s appropriate Regional Offices.

New Facilities or Major Modifications of Existing Facilities

1. Since PennDOT will be a reviewing agency, DEP will invite PennDOT to attend the pre-application meeting with the host municipality prescribed in DEP’s local municipality involvement process, if one is held. PennDOT attendance at this meeting will be optional. DEP forwards meeting minutes, or a summary of the meeting, to PennDOT.
2. DEP forwards the applicant’s responses to Form D, Section I – Traffic, and other pertinent sections, to PennDOT (e.g., a copy of the past TIS if major modifications are proposed for an existing facility, such as an increase in waste capacity or an increase in maximum daily waste volume). PennDOT reviews this information, makes a preliminary determination concerning whether a TIS (or a new TIS if modifications are proposed) will be required, and advises DEP of this preliminary determination and other preliminary comments regarding the scope of the study and the study area.
3. After receipt of a permit application, DEP schedules a meeting with interested local municipal officials and the permit applicant, and invites PennDOT to attend. This second meeting is described in DEP’s policy for the local municipality involvement process. At the meeting, the applicant’s proposal is explained, DEP explains its review process, and local officials can express public concerns. If its schedule permits, PennDOT will attend this local municipality involvement meeting in order to listen to the public concerns and to answer general transportation-related questions (if necessary). If PennDOT cannot attend this meeting, and if requested by DEP, PennDOT can provide a letter to be shared at the meeting regarding its views and current position on the application. DEP forwards meeting minutes, or a summary of the meeting, to PennDOT so that it is cognizant of any transportation-related public concerns.
4. Taking input from the local municipality involvement meeting into consideration, PennDOT makes a recommendation concerning whether a TIS (or a new TIS if modifications are being made) should be required, and advises DEP accordingly.
5. If requested by the applicant, PennDOT attends a preliminary review meeting on the TIS as indicated in PennDOT’s Transportation Impact Study

Guidelines for Municipal Waste Facilities. Issues pertaining to the required scope of the TIS are addressed.

6. In accordance with DEP's local municipality involvement process, it is necessary to negotiate a permit application review timeline. PennDOT attendance at meetings on this matter will be optional, but PennDOT will be required to give DEP an estimate of its anticipated review time requirements.
7. When requested, PennDOT will provide the following information to the engineer performing the TIS for the municipal waste facility:
 - Information on future PennDOT improvement projects within the study area.
 - Any readily available traffic volume data for the applicable locations.
 - Factors to use to project background traffic growth.
 - Reported traffic crash data for the study area as indicated in PennDOT's Transportation Impact Study Guidelines for Municipal Waste Facilities.
 - Any existing, readily available information that could be used in the approach route structural analyses.
8. PennDOT will review submissions and attend meetings related to the TIS, truck access route approval requests that may be needed in accordance with §4908 of the Vehicle Code (relating to the operation of certain combinations on interstate and certain other highways), and highway occupancy permit submissions for needed work within State highway right-of-way.

Periodic Review of Existing Facilities

1. DEP is required to review an existing municipal waste facility permit and associated operations at least every 5 years. When requested by DEP, PennDOT will provide comments (from a transportation system impact standpoint) to assist DEP in establishing a priority listing of sites for these reviews.
2. When DEP selects an existing facility for review, DEP will send PennDOT any previous TIS that was prepared for the site and any related information regarding any previous transportation-related concerns that have been received regarding the operation of the facility, and other pertinent information. PennDOT will determine if the facility has a highway occupancy permit, if there have been legitimate transportation-related public concerns, and if transportation-related problems appear to exist (or appear to have

occurred) due to the operation of the facility. To assist in this determination, PennDOT may conduct pavement testing, may ask DEP to have the owner/operator of the municipal waste facility submit updated Form D information, and may suggest that DEP employ its local municipality involvement process or other form of public input.

3. Based upon a review of this information, PennDOT will advise DEP whether a new TIS should be required, and the issues that need to be addressed within the scope of the TIS.
4. From this point, PennDOT involvement is similar to that described above, beginning in Step 3 of “New Facilities or Major Modifications of Existing Facilities.”

11.13.4 Transportation Impact Study (TIS) Guidelines

11.13.4.1 Purpose

When preparing a transportation impact study (TIS) for a municipal waste facility, follow these guidelines, which outline the procedures and criteria.

The purpose of the TIS is to define the impact of the waste facility on the transportation system – both at the waste facility site and along the approach routes to the site. The TIS must address the issues contained in these guidelines, must address transportation-related public concerns, and ultimately must identify improvements and other actions that are necessary (at the waste facility site and along its approach routes) in order to safely and efficiently accommodate traffic to and from the municipal waste facility without adversely affecting the transportation system.

These guidelines were prepared pursuant to the Governor’s Executive Order Number 1996-5, “Municipal Waste Facilities Review Program,” dated August 29, 1996. They also correlate with the Department of Environmental Protection’s (DEP’s) policies for the local municipality involvement process, the environmental assessment process (Phase I review), the municipal waste facility review – traffic analysis, and the process for evaluating daily volume.

11.13.4.2 Applicability

These guidelines apply for TISs involving State Routes (SRs), i.e., State highways under the jurisdiction of PennDOT.

PennDOT does not have jurisdiction over local highways. However, in many cases, municipal officials may require the municipal waste facility

owner/operator to conduct analyses of affected local highways in a manner consistent with these TIS guidelines.

11.13.4.3 Reviewers

A TIS must be coordinated with, and reviewed by, the appropriate federal, State, and local officials.

Upon receipt of a TIS, DEP will distribute it for review by the appropriate parties and/or their designated entities (consultants). DEP will consider these reviewer comments when determining whether to approve or deny a permit application for a new municipal waste facility or an expansion of an existing municipal waste facility that would result in additional waste volumes or capacity. DEP will use this TIS information to determine if specific permit conditions need to be imposed and/or whether the permit issuance must be contingent upon the satisfactory completion of specific transportation system improvements.

When a TIS is prepared for an existing facility that is not requesting an expansion (i.e., not requesting an increase in waste capacity or an increase in maximum daily waste volume), DEP will consider reviewer comments in a similar fashion to determine if:

1. The exiting permit should be renewed, revoked or suspended.
2. A modified permit should be issued with specific conditions and/or the satisfactory completion of specific transportation system improvements.

A TIS must identify needed transportation system improvements to accommodate municipal waste facility traffic. Proposed improvements to State highways require PennDOT approval, and proposed improvements to local highways require the approval of local authorities. Moreover, FHWA must review and approve any work within the right-of-way of any interstate highway. When the recommended improvements will require a commitment from a municipality for continued maintenance and/or operation (such as with traffic signals), the TIS must include written evidence from the municipality indicating whether or not the municipality has agreed to accept these responsibilities.

11.13.4.4 Other Transportation-Related Permits/Approvals

In accordance with [67 Pa. Code Chapter 441](#), PennDOT must provide a highway occupancy permit before constructing or altering a driveway, local road, or drainage facility/structure within State highway right-of-way.

Depending upon the length and width of combinations that will be traveling to and from a municipal waste facility, truck access route approval, in accordance with §4908 of the Vehicle Code (relating to the operation of certain combinations on interstate and certain other highways), may also need to be obtained from PennDOT and/or the local municipality.

Where applicable, the TIS submitted to DEP in accordance with these guidelines should contain the information necessary to satisfy PennDOT's HOP and truck access route requirements.

11.13.4.5 Determining the Need for a TIS

DEP will advise the owner/operator of the municipal waste facility whenever a TIS will be required after considering input from PennDOT and other appropriate reviewers.

Make the determination in the following manner.

New Facilities

For applications involving new municipal waste facilities, the applicant's responses to DEP's Form D, Section I – Traffic, and other pertinent sections, will be forwarded by DEP to PennDOT and other appropriate reviewers. In addition, relevant input from the public arising from DEP's local municipality involvement process and other sources will be forwarded to the same reviewers.

Based upon the comments of the reviewers of this information, DEP will advise the applicant whether a TIS will be required.

It is anticipated that a TIS will be required in conjunction with all applications for new municipal waste facilities.

Major Modifications of Existing Facilities – Increase in Capacity or Maximum Daily Waste Volume

For applications involving the expansion of an existing municipal waste facility that would result in additional waste volumes or capacity, DEP will forward PennDOT and other appropriate reviewers any previous TIS that may have been prepared for the site. The applicant's responses to Section I – Traffic, and other pertinent sections, and relevant input from the public arising from DEP's local municipality involvement process and other sources, will also be forwarded to the reviewers.

Based upon the comments of the reviewers of this information, DEP will advise the applicant whether a new TIS will be required. This determination will be based upon:

1. Whether there are transportation-related public concerns regarding the operation of the existing facility or the expanded volume or capacity facility.
2. Whether the operation of the existing facility has resulted in adverse transportation system impacts.
3. Whether the reviewers believe the proposed expansion in waste volumes or capacity may generate, or contribute to, adverse transportation-related impacts, and other factors.

It is anticipated that a TIS will be required in conjunction with most applications, for the expansion of an existing municipal waste facility that would result in additional waste volumes or capacity.

Periodic Review of Existing Facilities

DEP is required to review an existing municipal waste facility permit and associated operations at least every 5 years.

When DEP selects an existing facility for review, DEP will send PennDOT and other appropriate reviewers any previous TIS that was prepared for the site and related information, information regarding any previous transportation-related concerns that may have been expressed regarding the operation of the facility, and other pertinent information.

In cooperation with the other reviewers, DEP will determine if the facility has a highway occupancy permit, if there have been any transportation-related concerns, and if transportation-related problems exist or have occurred due to the operation of the facility. DEP's local municipality involvement process and other local input, as appropriate, will be used to help make these determinations.

DEP may request the owner/operator of the municipal waste facility to submit updated Form D information.

Based upon a review of this information, DEP will advise the municipal waste facility whether a new TIS will be required.

It is anticipated that a new TIS will only be required if there are legitimate transportation-related public concerns or (transportation-related problems are identified by the review agencies

11.13.4.6 Preparation of the TIS

When it has been determined that a TIS is required, it shall be the responsibility of the municipal waste facility owner/operator to ensure that the study is conducted in accordance with these guidelines. The study shall be prepared under the supervision of a registered professional engineer who possesses a license issued by the Pennsylvania State Registration Board for Professional Engineers. The final study report must contain the seal of the supervising engineer.

Unless specifically indicated otherwise, the study engineer is responsible for the collection of all information and data required to support the study effort. At the request of the municipal waste facility owner/operator, DEP and PennDOT will make available any existing, readily available information and data which may assist in the study effort. When additional traffic counts are required for the study, the study engineer shall collect such data.

Conduct the TIS using currently accepted engineering practices and procedures. The use of computer programs to conduct the required analyses is acceptable if the programs reflect the most current provisions of the analysis procedures. Recommended geometric, traffic operational, and transportation system improvements must meet or exceed all applicable federal, State, and local minimum design and operational criteria.

11.13.4.7 Preliminary Review/Scoping Meeting

Once it has been determined that a TIS is required, PennDOT will honor a request by the municipal waste facility owner/operator to have a scoping meeting to serve as a preliminary review of draft TIS materials and related criteria. Some of the items that may be appropriate to address in the scoping meeting and preliminary review stage include, but are not limited to, study area, scope of the study, intersections to be included in the level of service analysis, background traffic growth rate to be used, and the manner in which public concerns will be investigated/addressed.

This scoping meeting and preliminary review are done as a service to the waste facility owner, and are not considered to be binding upon PennDOT. The preliminary review will be limited to determining whether proper procedures will be used, and whether adequate documentation will be provided in accordance with the requirements of these guidelines. The intent of this preliminary procedural review is to provide the study preparer with guidance on the adequacy of the study in meeting the requirements of these guidelines. The adequacy or appropriateness of specific transportation system improvements will not necessarily be

addressed in this preliminary phase, since they depend upon the results of the study.

11.13.4.8 TIS Components and Report Requirements

Each TIS shall consist of the components described in the following subsections, which are necessary to measure the impact of the municipal waste facility on the transportation system.

These procedures and requirements are intended to provide a basic framework for the preparation of a municipal waste facility TIS. Additions or modifications to this framework may be made if such changes are approved by DEP, PennDOT, and other appropriate reviewers having jurisdiction. The preliminary review phase is the appropriate time to clarify issues pertaining to the required scope of a TIS for a specific facility.

The TIS report must document the results of the TIS and the recommended improvements that are necessary (at the waste facility site and along its approach routes) in order to safely and efficiently accommodate traffic to and from the municipal waste facility without adversely affecting the transportation system. The report must be presented in a format and context which can be understood by both technical and non-technical parties. Schematic diagrams, figures, charts, and/or tables should be used in the report for the presentation of data and the results of the various analyses. All data sources and study methodologies (including computer programs) must be properly referenced and documented. Any modifications to the referenced procedures must be properly documented to enable a review of the appropriateness of the modification.

The information and findings for each of the following subject areas must be contained in a separate section of the TIS report. An executive summary should also be included. Background calculations, computer printout, and supporting data should be included in a separate technical appendix.

Study Area – Site Location and Approach Routes

Unless indicated otherwise for a particular study component or during the preliminary review phase, the study area for the TIS shall consist of all portions of public highway, between the municipal waste facility site and the nearest limited access highway that will be used as an “approach route” by traffic traveling to or from the waste facility site.

For approach routes where the nearest limited access highway is a substantial distance from the waste facility, consideration may be given to using the nearest “major” highway as the study area boundary for that approach route, provided that all review agencies concur in advance to this modification. In some areas of the Commonwealth, “major” highway may be defined as a “major through traffic route” as shown on the Pennsylvania Official Transportation Map.

In addition to the above, the study area for a municipal waste facility TIS must include any section of public highway where, in the opinion of the reviewing agencies, legitimate transportation-related public concerns have been expressed as a result of DEP’s local municipality involvement process or other local input.

The proper study area to be used for a specific municipal waste facility TIS needs to be clarified in the preliminary review stage.

Site Description and Location

The TIS report shall contain a description of the municipal waste facility to include the type of facility and its operation. The name and address of the municipal waste facility owner/operator, as well as the study engineer, shall also be included with the submission.

Other required report information is as follows:

1. A location map showing the municipal waste facility site.
2. A site plan showing the location, geometry, and design features of all existing or proposed driveways to the facility.
3. Other related descriptive information, as required, to be in conformance with PennDOT’s HOP requirements (as outlined in [67 Pa. Code Chapter 441](#) and [Publication 282](#)).

Approach Routes

The TIS report shall contain maps, schematic drawings, and/or figures showing all approach routes to the site. Each approach route shall be labeled with State Route (SR) number, street name, and/or township road number.

A description of the approach routes shall also be included in this section of the report. Indicate type of roadway, number of lanes, speed limit, existing traffic signal locations, and other related information.

Waste Facility Traffic Volumes

Through the use of figures, schematic drawings, charts, and/or tables, indicate the daily and hourly volumes of traffic destined to and from the municipal waste facility on each approach route. Waste facility traffic volumes for each segment of every approach route shall be presented by direction (to and from the site).

In addition to trips by waste vehicles, the report shall separately identify the number of trips made to and from the site by other vehicles for other purposes (such as passenger car trips to and from work by the facility's employees).

The report shall provide a breakdown of waste facility traffic volumes by type of vehicle using AASHTO design vehicle designations (e.g., P for passenger car, SU for single unit truck, WB-50 for large semitrailer combination, WB-62 for interstate semitrailer with a 48-foot trailer, WB-67 for interstate semitrailer with a 53-foot trailer, etc.).

If vehicles transporting hazardous materials will be traveling to or from the site, the report shall identify their approach routes and traffic volumes by placard category.

Identify both existing (if applicable) and projected traffic volumes traveling to and from the municipal waste facility. The report shall include the maximum projected traffic volumes that are anticipated to travel along the approach routes to and from the site within the next 10 years for existing sites, or 10 years from the opening of a new site.

Programmed Transportation Improvements by Others

The TIS report shall identify all transportation system improvements within the study area that are programmed for completion within the next 10 years by PennDOT, local municipalities, developers, their contractors, or others.

In the TIS report, discuss the timing and nature of these improvements by others relative to their impact on any improvements or other actions that may be required by the municipal waste facility owner/operator.

Land Use and Environmental Impacts

The TIS report shall contain a description of existing land use at the municipal waste facility site, existing land uses along the approach routes, and anticipated land uses at the site and along the approach routes over the next 10 years. For new facilities, the report shall indicate whether the site

complies with existing zoning and land development ordinances, or whether a special exception, variance, or amendment will be required.

Identify land uses along the approach routes as residential, commercial, industrial, agricultural, or other appropriate descriptive category.

The report shall identify locations of schools, hospitals, nursing homes, and other sensitive buildings along approach routes. Also, identify locations where residences or other sensitive buildings are close to the roadway (e.g., 50-foot setback or less).

In addition, indicate the location of other sensitive areas (such as parks, playgrounds, recreation areas, forests, picnic areas, natural landmarks, wild areas, scenic rivers, wetlands, public water supplies, historic or cultural sites, etc.) along approach routes.

The report shall describe any existing or potential adverse impact on these sensitive areas that has resulted, or may result, from waste facility traffic, taking into account exhaust fumes, odors, noise, and other environmental factors. Address any public concerns regarding these issues.

Document any required measures that need to eliminate, mitigate, or minimize these existing or potential adverse impacts in the “Improvement Recommendations” section of the report.

Level of Service (LOS) Analysis

The TIS report shall include an analysis of existing and projected traffic flow conditions along the approach routes to the municipal waste facility. This analysis shall follow the procedures contained in the most recent edition of the Transportation Research Board’s “Highway Capacity Manual,” Special Report 209.

Complete analyses for the following six scenarios:

1. Existing approach route conditions without waste facility traffic.
2. Existing approach route conditions with waste facility traffic.
3. Future approach route conditions without waste facility traffic.
4. Future approach route conditions with waste facility traffic.
5. Future approach route conditions with waste facility traffic and needed improvements. (Complete these analyses for locations

where roadway and/or traffic operational improvements will be needed to satisfy LOS criteria.)

6. Existing approach route conditions with waste facility traffic and needed improvements. (Complete these analyses for locations where roadway and/or traffic operational improvements will be needed to satisfy LOS criteria.)

Unless specified otherwise by the reviewing agencies, base all “future” traffic analyses on a design year 10 years in the future, and shall reflect any growth in background and waste facility traffic during this period.

For each of the six above scenarios, levels of service must be determined for the AM peak hour of the approach route, the PM peak hour of the approach route, and the peak hour of the waste facility traffic (if it differs from the normal AM or PM peak hour of the approach route). All LOS evaluations must consider not only the overall intersection LOS but also the LOS for each approach and movement.

Complete LOS analyses for the following locations within the study area:

1. All driveways and site access points to the municipal waste facility.
2. Intersections and sections of highway along the approach routes where the reviewing agencies suspect that waste facility traffic may cause (or has caused) a degradation in the LOS or other undesirable traffic operations.
3. Intersections or sections of highway along the approach routes that are presently operating at LOS E or LOS F, or are expected to operate at LOS E or LOS F in the future design year, with or without the waste facility traffic.
4. Intersections or sections of highway along the approach routes with public concerns (because of DEP’s local municipality involvement process or other public input) about traffic flow, congestion, or other undesirable traffic operations.

Address specific intersections and locations to be included in the LOS analysis in the preliminary review phase.

In addition to the LOS analyses, the reviewing agencies may require the study engineer to complete other related engineering studies. For example, any new traffic signals that are proposed must be justified by a traffic signal warrant analysis in compliance with Publication 212 (i.e.,

[67 Pa. Code Chapter 212](#)). Queue length studies shall be completed, when required by the reviewing agencies, to evaluate needed lengths of turn lanes and whether there is the potential for a backup of traffic from controlled intersections that could impact other intersections, including access points to the waste facility. Complete gap studies when required by the reviewing agencies, to evaluate the need for signal control, turn prohibitions, or additional site access points.

The results of these LOS analyses and other engineering studies must be included in the TIS report. Whenever possible, include figures, schematic drawings, charts, and/or tables to clearly present the result. Also, include background calculations, computer printouts, and other supporting data in a technical appendix.

The following LOS criteria must be satisfied for locations within the study area:

1. The existing LOSs with the waste facility traffic must be no worse than the existing LOSs without the waste facility traffic.
2. The projected future LOSs with the waste facility traffic must be no worse than the projected future LOSs without the waste facility traffic.
3. For locations where either the existing or future design year LOS without the waste facility traffic is LOS F, needed improvements shall be identified which will provide estimated delay with the waste facility traffic that will be no worse than the delay for that time period without the waste facility traffic.
4. New intersections established to serve as driveways and site access points to a municipal waste facility shall be designed to operate at LOS C or better in the future design year where rural conditions exist and LOS D or better in the future design year where urban conditions exist.

Roadway and/or traffic operational improvements that are needed along the approach routes to satisfy the above LOS criteria shall be identified and included in the “Improvements Recommendations” section of the report. The TIS report shall contain the results of the LOS analyses indicated in Items 5 and 6 of the second paragraph of the “Level of Service (LOS) Analysis” in this subsection to document that these improvements will indeed satisfy the LOS criteria.

Crash Analysis

The TIS report shall contain a crash analysis of each approach route.

The study engineer shall obtain the following types of crash data from PennDOT's appropriate Engineering District Office(s):

1. The most recent 5 years of reported crash data for each approach route.
2. The "Homogeneous Report" (which provides a summary of the expected crash experience for particular types of roadway).
3. A printout listing approach route locations where there have been reported crashes involving refuse vehicles within the last 5 years.
4. A map (or maps) showing approach route locations on State highways where there have been reported crashes involving refuse vehicles within the last 5 years.

The study engineer shall analyze the crash data to determine if there are any existing high crash locations along the approach routes (in terms of the number of crashes, crash clusters, crash rates, severity, etc.). The crash history of large trucks and other vehicle types that will be traveling to and from the waste facility (or have been traveling to and from the waste facility) must be closely examined to determine if the waste facility traffic is expected to create or contribute (or has created or contributed) to a traffic safety problem.

Any public concerns about crash history, crash potential, and other safety issues along approach routes to the municipal waste facility (which have been expressed as a result of DEP's local municipality involvement process or other local input) shall be studied and compared to actual crash data.

If waste facility traffic is expected to create or contribute (or has created or contributed) to a traffic safety problem on a particular section of highway, that section of highway will not be permitted to serve as an approach route unless the safety problem is corrected to the satisfaction of the reviewing agencies. The "Improvement Recommendations" section of the report shall include the measures that need to be undertaken.

Approach Route Operational Assessment

Each approach route to a municipal waste facility must be reviewed by the study engineer to determine whether it is satisfactory to handle the volume

and type of waste facility traffic without having safety or operational problems. Therefore, an operational assessment of each approach route must be made, and the results must be contained in the TIS report.

The operational assessment shall examine the suitability of each approach route for waste facility traffic taking into consideration the size, type, volume, and operating characteristics of such traffic.

There are numerous factors that need to be considered when determining the suitability of a particular section of highway to serve as a municipal waste facility approach route. Some of the questions that need to be addressed for each approach route include:

1. Is there a more suitable alternate route that could be used as an approach route?
2. Are there any weight limits or vertical clearance restrictions that would affect waste facility traffic?
3. Are there any one lane or narrow bridges?
4. Are lane widths sufficient?
5. Are intersection turning radii sufficient to allow turns to be made within the physical boundaries of the roadway pavement and without encroaching on opposing travel lanes?
6. Are there locations where horizontal alignment, lane width, or other factor would result in encroachment into sidewalk areas, opposing/adjacent travel lanes, or onto shoulder areas? Are there shoulder drop offs, or locations where waste facility traffic may contribute to accelerated shoulder deterioration?
7. Are there locations where shoulders or a roadside clear zone are not present and a combination of factors such as curvature, lane width, etc., would result in off-tracking or run-off-the-road concerns?
8. Are there any long, steep downgrades? Are there any hazardous grade speed limits, truck pull off areas, or truck escape ramps?
9. For two-lane roadway sections, are the frequency and length of passing sections sufficient to provide for safe and efficient travel with the waste facility traffic?

10. Are there substantial lengths of grade, without climbing lanes or passing, which would impede truck speed?
11. Are there locations, which may present underclearance problems?
12. Are there locations with inadequate sight distance?
13. Are there any locations where turning, acceleration, or deceleration lane lengths are inadequate?
14. Would the municipal waste facility traffic have any adverse impact on school bus traffic?
15. Are there any other safety-related considerations relative to waste facility traffic?

The TIS report shall address the above questions for each approach route, and also identify specific locations where such features or conditions exist. In addition, the report must address any public concerns that have been expressed regarding these matters (from DEP's local municipality involvement or other public input processes).

DEP, PennDOT, or other reviewing agencies may require a waste facility owner/operator to provide test runs along a particular approach route using the types of vehicle that will be traveling to and from the waste facility to more conclusively demonstrate whether such movements can be made safely. The test runs may be videotaped for further study or use at a public meeting, and also to provide future documentation of the operation of the specific type of vehicle over the route.

If waste facility traffic is expected to create or contribute (or has created or contributed) to a traffic safety or operational problem on a particular section of highway, that section of highway will not be permitted to serve as an approach route unless the safety or operational problem is corrected to the satisfaction of the reviewing agencies. In this determination, existing waste facility traffic operations will be considered (if applicable), as well as existing operations by other vehicles of like size and operating characteristics. The "Improvement Recommendations" section of the report shall include the measures that need to be undertaken to address these issues.

Approach Route Structural Analysis

The TIS report shall include the results of a review, inspection, and analysis of the bridges and roadway pavement along the approach routes to determine if municipal waste facility traffic is expected to create or

contribute (or has created or contributed) to damage or destruction of the approach route highways and bridges. The report shall identify locations where rehabilitation or repair, or other actions (such as bonds or permits for excess maintenance), are necessary in order to accommodate municipal waste facility traffic. Address all public concerns regarding these issues.

The study engineer shall conduct engineering studies in accordance with [§212.117 of Publication 212](#) (relating to weight, size and load restrictions).

During the preliminary review stage, PennDOT and the other reviewing agencies will define the scope and methodology for the required approach route structural analysis, including any existing information that can be made available. If the reviewing agencies require the study engineer to conduct additional testing (such as pavement deflection testing using a falling weight deflectometer), the agencies will define the type of testing and analysis required, the locations to be tested, and improvement responsibilities.

An independent firm may be retained by the municipal waste facility owner/operator to conduct additional pavement testing and analysis. The specific firm and manner of testing must be approved in advance by PennDOT for test locations on State highways.

The TIS report shall identify highway and bridge locations along the approach routes that are structurally inadequate to accommodate projected waste facility traffic or are near the end of their useful service life. Such locations will not be permitted to serve as an approach route unless the structural problem is corrected to the satisfaction of the reviewing agencies. Methods of correction may include repair or rehabilitation by the municipal waste facility owner/operator. In some instances, bonds or permits for excess maintenance may be required in accordance with §4902 of the Vehicle Code (relating to restrictions on use of highways and bridges) and 67 Pa. Code Chapters [189](#), [191](#), and [193](#). The “Improvement Recommendations” section of the report shall include the measures that need to be undertaken to address these issues.

Public Concerns

All transportation-related public concerns that are expressed as a result of DEP’s local municipality involvement process or other public input sources shall be discussed in the TIS report.

The TIS report shall contain a separate section listing these concerns. The public concerns shall also be addressed in this separate section of the report, or via cross references to other appropriate sections of the report.

Measures or actions that need to be undertaken shall be included in the “Improvement Recommendations” section of the report.

Improvement Recommendations

The TIS report shall contain a separate section that lists the transportation improvements and other actions that will need to be implemented within the study area by the owner/operator of the municipal waste facility or others as a result of the land use and environmental impact review, the LOS analysis, the crash analysis, the approach route operational assessment, the approach route structural analysis, and public concerns.

Unless indicated otherwise by the reviewing agencies, the owner/operator of the municipal waste facility will be responsible for funding transportation system improvements that are needed to accommodate waste facility traffic.

Upon final approval of the TIS report by the reviewing agencies, the owner/operator of the municipal waste facility will be responsible for funding the development of plans and specifications for all required transportation improvements within the study area as well as their physical construction, unless indicated otherwise by the reviewing agencies.

Improvement plans and specifications will need to be reviewed and approved by the appropriate reviewing agencies as indicated in the “Reviewers” section of these guidelines. In accordance with [67 Pa. Code Chapter 441](#), the owner/operator of the municipal waste facility will need to obtain a highway occupancy permit from PennDOT prior to starting work within State highway right-of-way.

11.14 Snowmobile and All-Terrain Vehicle Roads

11.14.1 References

1. Vehicle Code (75 Pa. C.S.) – §§7721, 7722, and 7723.
2. Snowmobile and All-Terrain Vehicle Registration and Operation (17 Pa. Code, Chapter 51), (see the Department of Environmental Resources regulations at <http://www.pacode.com/secure/data/017/chapter51/chap51toc.html>).

11.14.2 Establishing Snowmobile and/or ATV Roads on State Highways

§7722 of the Vehicle Code (75 Pa.C.S. §7722, relating to the designation of snowmobile and ATV roads) allows the Department of Transportation on State highways and local authorities on any highway, road or street within its jurisdiction to:

- Designate any highway, road or street within its jurisdiction as a snowmobile road, an ATV road, or both.
- Determine whether such road shall be closed to vehicular traffic or whether snowmobiles and ATV's may share this designated road with vehicular traffic.

All-Terrain Vehicles (ATVs)

For safety reasons, Engineering Districts shall not designate any State highway as an ATV road, or authorize ATVs to operate along any State highway.

Snowmobiles

1. All requests for the designation of State highways as snowmobile roads must come from the local authorities and the request shall include a completed copy of the resolution shown in [Appendix 11F](#). The Engineering District shall maintain a copy of the completed resolution.
2. The following State highways shall not be designated as snowmobile roads:
 - Numbered traffic routes.
 - Any highways where the ADT is more than 200 vehicles.

3. Snowmobile roads should normally be limited to State highways that are not winter maintained. However, an Engineering District may establish a District policy to permit certain types of winter maintained highways to be designated as snowmobile roads.
4. Snowmobile roads shall not be established in such a manner that would deny access to adjacent properties such as homes, cabins, hunting camps and other properties used by the owners and/or occupants during the winter months.
5. Install signs or barricades in accordance with the Department of Environmental Resources Regulation, 17 Pa. Code, Chapter 51, Snowmobile and All-Terrain Vehicle Registration and Operation – see [17 Pa. Code §51.82](#) (relating to barricades for snowmobile or ATV roads) [17 Pa. Code §51.83](#) (relating to posting signs for snowmobile or ATV roads).

Snowmobile road designations shall remain in effect until PennDOT rescinds the designation, which would occur if requested to do so by the municipality or if the Engineering District determines that changes in conditions of the roadway, traffic or other factors indicate that the roadway no longer meets the criteria for designation as a snowmobile road.

11.15 Closing State Highways for Special Events

11.15.1 References

Vehicle Code (75 Pa.C.S.)

- §3367 – Racing on highways, specifically subsection (c), entitled “permits for special activities.”
- §6109 – Specific powers of Department and local authorities, specifically paragraphs (a)(3) and (a)(14).

Official Traffic Control Devices (67 Pa. Code Chapter 212, specifically [67 Pa. Code §212.701](#)).

11.15.2 Policy

[§212.701 of Publication 212](#) (relating to processions, assemblages and special activities) establishes minimum criteria for processions, assemblages and special activities for both local roadways and State highways. This section provides for the issuance of permits by the Department, and §212.701(b) identifies what information is required from the sponsors.

§212.701(b) stipulates that all requests must be received a minimum of 3 weeks prior to the date of an event, i.e., by the District Executive for conventional highways and expressways, and by the Secretary of the Department for freeways. However, the Department cannot allow these events to transpire without a fully executed permit, which means that the sponsors must provide the prescribed materials and the Department must concur that it meets the minimum criteria.

Therefore, since sponsors generally desire to advertise their event several months in advance, Engineering Districts are encouraged to contact sponsors of annual-type events and suggest that they submit their applications a minimum of 2 months prior to any proposed event.

11.16 Turnback of Traffic Restrictions

11.16.1 General

PennDOT’s “Road Turnback” Program is addressed in Publication 310, “*Transfer of State Highway (Road Turnback) Program, Policies and Procedures Manual*” (see <ftp://ftp.dot.state.pa.us/public/PubsForms/Publications/PUB%20310.pdf>). As part of this “Road Turnback” Program, it is necessary to send all approvals and study files for traffic restrictions to a municipality that accepts ownership of the highway.

11.16.2 Suggested Transmittal Letter

[Exhibit 11.16-A](#) provides a suggested letter that your District may wish to use when forwarding any type of traffic restriction approvals and study files to a municipality. While the letters are only suggested formats, the information contained in paragraphs two and three should be included as appropriate with any package of approvals that are sent to a municipality.

If there are current traffic restrictions that were established and posted prior to July 1, 1977, they were given “grandfather” status and do not require an engineering and traffic study. Therefore, if this “grandfather” status applies to any traffic restriction relative to the current turnback, it is important to note this in the letter.

In addition, if there is any special signing that is necessary for continuity, it is important to define the requirements in the letter. For example, if the turnback section is part of a numbered traffic route or BicyclePA route, it is essential that either the local authority commit to maintaining the signing or allow PennDOT to retain the right to enter the turnback section and maintain the signs.

11.17 Turn Lane Guidelines

11.17.1 General

The following turn lane guidelines have been developed to determine the warrants for turn lanes and to identify desirable length. The guidelines apply to highway occupancy permit projects, traffic signal permit projects, and Department construction projects. The guidelines apply for any one-hour period during the typical analysis periods. Typical analysis periods for projects are those for the future design year, as specified.

11.17.2 Traffic Engineering Software

In addition to the procedures outlined in this document, at the discretion of the Engineering District, an operational analysis may be conducted to determine whether turn lanes may be warranted and to identify associated storage lengths. The operational analysis shall be conducted utilizing traffic engineering software packages that are approved by the Department as prescribed in Chapter 12 of PennDOT Publication 46, *Traffic Engineering Manual*. The results of both procedures can then be compared, and the more conservative results may be used. For example, although the application of these turn lane guidelines may indicate a turn lane is not warranted, if the operational analysis identifies the need for a turn lane to achieve acceptable levels of service, then the incorporation of the turn lane into the design may be considered.

Additionally, in those cases involving closely spaced intersections and complex transportation systems when intersection function may be affected by adjacent locations, the operational analysis may yield more appropriate results based on site conditions and may be given more consideration than the results obtained from these guidelines.

Use the 95th percentile turn lane queue when estimating required storage length from traffic engineering software packages, unless directed otherwise.

11.17.3 Truck Adjustment Factors

To adjust for truck traffic, the following formula shall be used to develop a truck adjustment factor to be applied to the hourly volume to obtain a passenger-car equivalent volume for the advancing, opposing, and left turning volumes for both warrants and design:

$$T = 1 + P_T (E_T - 1)$$

where

E_T = passenger-car equivalent for trucks

P_T = proportion of trucks in the traffic stream (expressed as a decimal fraction)

T = truck adjustment factor

The value used for E_T shall be taken from the following table based on the terrain of the surrounding area:

Type of Terrain		
Level	Rolling	Mountainous
1.5	2.5	4.5

The hourly volume shall be multiplied by the truck adjustment factor (T), and the resulting passenger-car equivalent volume shall then be used in all subsequent calculations identified in these guidelines. The truck adjustment factors were taken from Reference 3.

11.17.4 Turn Lane Warrants

Use Figures 1 through 8 in Appendix 11G to determine whether a left turn lane is warranted on two-lane and four-lane roadways at unsignalized and signalized intersections. Use Figures 9 through 12 in Appendix 11G to determine whether a right turn lane is warranted on two-lane and four-lane roadways at unsignalized and signalized intersections. For unsignalized intersections, the warrants only apply to the free flow approach. For left turns, if the plotted point falls above the appropriate left turn percentage line (L), a left turn lane is considered warranted. Engineering District discretion should be used as to whether the actual left turn percentage should be rounded up or down to match one of the lines in the graph. A turn lane may be considered if the criteria is met for any one-hour period.

Figures 1 through 8 in Appendix 11G were adapted from References 1, 2, 4, and 9. Figures 9 through 12 in Appendix 11G were taken from Reference 5.

11.17.5 Turn Lane Storage Length

Use Tables 1 through 3 to compute storage length for left and right turn lanes at unsignalized and signalized intersections. Tables 1 through 3 were adapted from Reference 5. Minimum recommended storage length is 75 feet, and all results should be rounded the next highest 25-foot increment. Turn lane storage length does not include taper length.

Consider using dual left turn lanes at signalized intersections when the hourly left turn volume exceeds 300 vehicles per hour.

TABLE 1

TYPE OF TRAFFIC CONTROL	SPEED (MPH)					
	25 – 35		40 – 45		50 - 60	
	TURN DEMAND VOLUME					
	HIGH	LOW*	HIGH	LOW*	HIGH	LOW*
SIGNALIZED	A	A	B or C**	B or C**	B or C**	B or C**
UNSIGNALIZED	A	A	C	B	B or C**	B

* LOW is considered 10% or less of approach traffic volume

** Whichever is greater

TABLE 2

CONDITION A	
SPEED (MPH)	STORAGE LENGTH
Any speed	Length from Table 3
CONDITION B	
SPEED (MPH)	STORAGE LENGTH
40	75'
45	125'
50	175'
55	235'
60	295'
CONDITION C	
SPEED (MPH)	STORAGE LENGTH
40	61' + Additional length from Table 3
45	75' + Additional length from Table 3
50	93' + Additional length from Table 3
55	114' + Additional length from Table 3
60	131' + Additional length from Table 3

TABLE 3

Average no. of vehicles/cycle*	Length (feet)	Average no. of vehicles/cycle*	Length (feet)
1	75	17	600
2	100	18	625
3	150	19	650
4	175	20	675
5	200	21	725
6	250	22	750
7	275	23	775
8	325	24	800
9	350	25	825
10	375	30	975
11	400	35	1125
12	450	40	1250
13	475	45	1400
14	500	50	1550
15	525	55	1700
16	550	60	1850

* Average no. of vehicles/cycle = design hour volume of turning lane/cycles per hour

If cycles per hour are unknown, assume:

Unsignalized or 2 phase – 60 cycles per hour

3 phase – 40 cycles per hour

4 phase or more – 30 cycles per hour



At signalized intersections, consider dual left turn lanes and operational analysis.

11.17.6 Other Warranting Factors and Considerations

While the preceding sections present analytical procedures to determine whether turn lanes may be warranted, it must be recognized that there are other factors that may need to be considered. These factors may justify turn lanes at some locations where the numerical warrants are not satisfied.

Some of the factors, or combinations of factors, that may need to be considered and may justify turn lanes to preserve safe and efficient traffic flow include crash history; sight distance; deceleration requirements; the type and volume of turning traffic considering nearby land use; grades; locations on high-speed, multilane highways; 85th percentile and safe running speeds; and engineering judgment.

11.17.7 Example Problem

11.17.7.1 Problem Statement

Determine whether an exclusive left turn lane is warranted and its desired storage length on a signalized approach of a rural arterial highway (speed = 50 mph, 20% trucks, rolling terrain). Intersection approach volumes consist of 100 left turning vehicles per hour and 680 through vehicles per hour. Opposing traffic volume is 500 vehicles per hour. The traffic signal has a 90-second cycle length.

11.17.7.2 Determine Whether a Left Turn Lane Is Warranted

First, convert the hourly volume to a passenger-car equivalent volume:

$$T = 1 + .2 (2.5 - 1) = 1.3$$

Passenger-car equivalent volume (left) = $1.3 (100) = 130$ vehicles

Passenger-car equivalent volume (through) = $1.3 (680) = 884$ vehicles

Passenger-car equivalent volume (opposing) = $1.3 (500) = 650$ vehicles

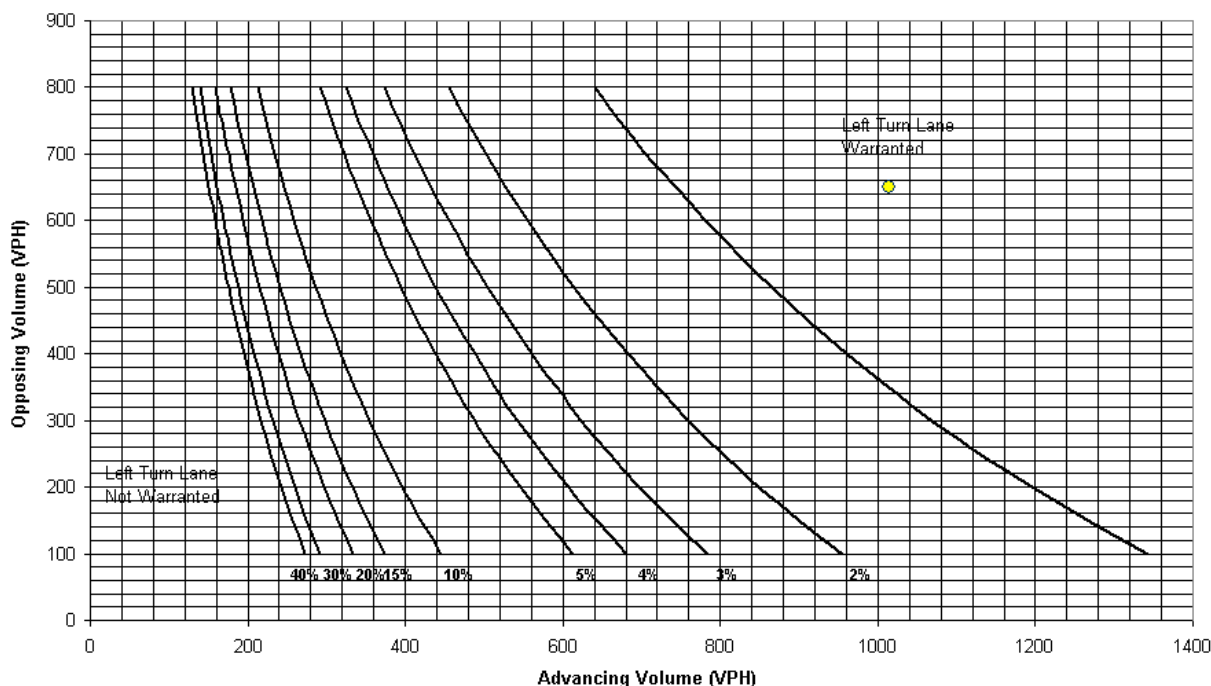
Passenger-car equivalent volume (advancing) = $130 + 884 = 1,014$ vehicles.

Next, identify the percentage of left turns contained in the advancing volume:

$$L = 130/1014 = 12.8\%.$$

Then, use the appropriate graph from Figures 1 through 8 in Appendix 11G to determine if a left turn lane is warranted. Therefore, based on site conditions, **Figure 4** will be utilized to plot the traffic information as follows:

Figure 4. Warrant for left turn lanes on two-lane highways (50 mph speed, unsignalized and signalized intersections)
(L = % Left Turns in Advancing Volume)



Since the plotted point falls above all of the left turn percentage lines, a left turn lane is warranted.

11.17.7.3 Determine Storage Length

Refer to the **Table 1** in Section 11.17.5. Using the given speed of 50 mph, enter the column with the speed “50-60”. Next, determine if the left turn demand volume is “high” or “low”. “Low” is considered 10% or less of the approach traffic flow. The demand is $130 / (884 + 130) = 12.8\%$. Therefore, the left turn demand is considered “high”. Based on a “signalized” intersection, the table indicates that Condition B or C (whichever is greater) should be used to calculate the storage length of the left turn lane.

Condition B, for the 50 mph speed from Table 2 in Section 11.17.5, gives a storage length of 175 feet.

Condition C is calculated by adding the 93 feet (for the 50 mph speed from Table 2 in Section 11.17.5) to the additional length determined from **Table 3** in Section 11.17.5. To determine the additional length, first calculate the number of cycles/hour ($3600 \text{ seconds/hour} \div 90 \text{ seconds/cycle} = 40 \text{ cycles/hour}$). Next, divide the hourly left turn approach volume by the number of cycles per hour ($130 \text{ left turning vehicles} \div 40 \text{ cycles/hour} = 3.25 = \text{round to } 3$). Using **Table**

3 in Section 11.17.5, the additional length is 150 feet. Adding the 150 feet to the 93 feet noted above equals 243 feet. A comparison of the values from Condition B and Condition C yields a storage length of 175 feet and 243 feet, respectively. Therefore, use the greater value of 243 feet. Rounding to the next highest 25-foot increment results in a recommended left turn storage length of 250 feet.

11.17.8 References

1. *Aspects of Traffic Control Devices, Volume Warrants for Left-Turn Storage Lanes at Unsignalized Grade Intersections*, M.D. Harmelink, Highway Research Record Number 211, Highway Research Board, National Research Council, National Academy of Sciences, Publication 1554, 1967.
2. *Guidelines for Left-Turn Lanes*, J.C. Oppenlander and J.C. Bianchi, 1990 ITE Compendium of Technical Papers, Pages 191 through 196.
3. *Highway Capacity Manual*, Transportation Research Board, National Research Council, 2000.
4. *Left-Turn Treatments at Intersections, NCHRP Synthesis 225*, Transportation Research Board, National Research Council, 1996.
5. *Location and Design Manual, Volume 1, Section 400, Intersection Design*, Ohio Department of Transportation, October 2004.
6. *Manual on Uniform Traffic Control Devices*, Federal Highway Administration, 2003 Edition.
7. *A Policy on Geometric Design of Highways and Streets, Fourth Edition*, American Association of State Highway and Transportation Officials, 2001.
8. *Design Manual Part 2 Highway Design, Publication 13M*, Pennsylvania Department of Transportation, July 2002.
9. *Engineering Study Guide for Evaluating Intersection Improvements, NCHRP Report No. 457*, Transportation Research Board, 2001.

Exhibit 11.16-A Sample Letter re the Turnback of Traffic Restrictions

Turnback of SR _____
by Act # _____

Dear _____:

Under the Department's "Road Turnback" Program, the above State Route was placed under your jurisdiction on _____, 20__ by Act _____, dated _____, 20__. The responsibility for traffic restrictions on this road now rests with your municipality.

Enclosed for your information and files are the previously established traffic restriction approval(s) for this road by the Department of Transportation. We are also including the engineering and traffic studies that we made to justify the establishment of each of the traffic restriction. Please retain these studies for future reference or possible presentation at judicial proceedings that may result from enforcement of the restrictions.

The only official action that must be taken by your municipality to officially adopt the enclosed traffic regulations is covered under 75 Pa. C.S. §6109 of the Vehicle Code (relating to specific powers of department and local authorities). Specifically, subsection (b) of §6109 specifies that actions taken by local authorities shall be:

- (1) by ordinance of the local body: or
- (2) by a commission or public official authorized to act on specified matters.

If you have any questions or comments, please contact _____.

Sincerely,

District Executive
District _____

Enclosure(s)

Chapter 11

Appendix

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Study Elements that Suggest a Speed Limit Below 65 mph

Design Considerations*

- Design speed less than 70 mph.
- Long sections where stopping sight distance is substandard, particularly on horizontal curves where sight distance is restricted by concrete median barrier.
- Numerous curves where the degree of curve or the superelevation is a problem for 70 mph.
- Grades where trucks typically travel at speeds of 50 mph or slower without a truck climbing lanes.
- Presence of bridges that are 200 feet or less in length and less than 34 feet wide, or are over 200 feet long and less than 31 feet wide.
- Short acceleration lanes that create substantial speed differentials, especially at locations where heavy traffic volumes or limited sight distance exist.
- Lane drop tapers that are less than 50:1.

Maintenance

- A road surface that encourages extensive speed differentials or may cause drivers to lose control at 65 mph.
- Skid numbers below 30.
- Frequent shoulder drop-offs of 2 inches or more, where drivers running onto the shoulder may be likely to lose control.
- Substandard size or condition of signs.

Miscellaneous

- Interchanges so close that excessive merging and diverging movements or excessive number of lane changes occur.
- An abnormally high crash or fatality rate.
- Engineering judgment suggests that physical characteristics would make the section unsafe for a 65-mph speed limit.

* If as-built construction plans are available, they may provide some geometric features.



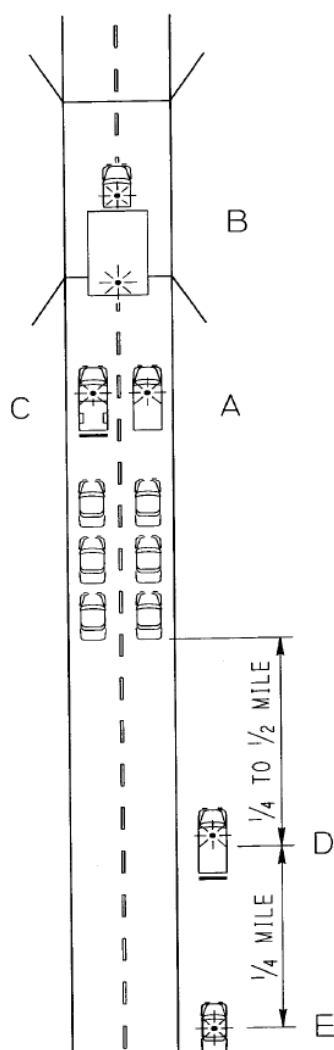
Traffic Control Plan for Superloads and Non-Superloads on Freeways with Bridge Crawl Speeds

Application ID#: _____

Bridge Structure ID#: _____

Motor Carrier Name: _____

Applicant's Signature: _____ Date: _____



- A. PRIVATE ESCORT VEHICLE.** The hauler provides this vehicle, equipped with a flashing or revolving yellow light. The vehicle should normally be on the right side of the roadway in advance of Vehicle B. However, prior to reaching the bridge, this vehicle may allow Vehicle B to pass in order to run parallel with Vehicle C and form a rolling blockade.
- B. SUPERLOAD OR NON-SUPERLOAD.** This vehicle shall be equipped with two flashing or revolving yellow lights, one on the front and one on the back. The vehicle should maintain a minimum distance behind Vehicle A of 200 feet, or one bridge span length, whichever is greater.
- C. SHADOW VEHICLE.** The hauler provides this vehicle, equipped with a flashing or revolving yellow light and a truck-mounted variable message sign (VMS). The VMS shall flash the preprogrammed message "DO NOT PASS" when appropriate. The Department strongly recommends using a truck-mounted attenuator (TMA), if used, it shall be a type approved by the Department as meeting NCHRP Level 3 requirements. When on the bridge, the shadow vehicle should maintain a minimum distance behind Vehicle B of 200 feet, or one bridge span length, whichever is greater.
- D. QUEUE-MONITORING VEHICLE.** The hauler provides this vehicle and positions it on the shoulder whenever possible. It shall be equipped with a flashing or revolving yellow light and a variable message sign (VMS) alternately displaying preprogrammed messages "PREPARE TO STOP" and "SLOW DOWN."
- E. STATE POLICE.** This vehicle is only required for superloads. When used, the vehicle shall be a marked vehicle with revolving or flashing lights. Whenever possible, the vehicle should be positioned on the shoulder.

NOTES:

1. VMSs shall be of a type approved by the Department and listed in Bulletin 15. The VMS on Vehicles C and D shall be capable of displaying a minimum of two lines of message with minimum 14-inch legend.
2. The hauler shall provide CB radios in all vehicles for communication, including one for Vehicle E. One of the operators shall also have a cellular telephone for emergency purposes.
3. When operating between locations with crawl speeds, all vehicles should normally be in the right lane. The VMSs should display messages such as "OVERSIZE LOAD" and "PASS WITH CARE."



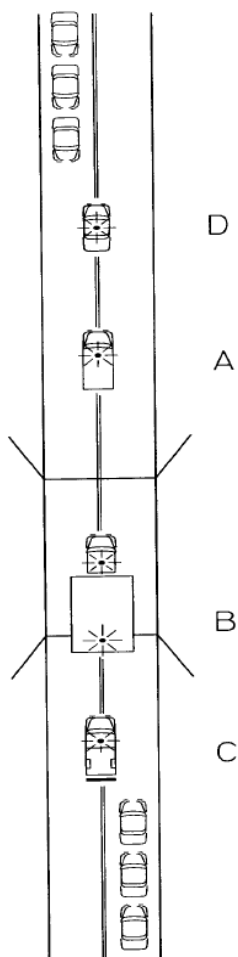
Traffic Control Plan for Superloads and Non-Superloads on Two-Lane, Two-Way Roadways with Bridge Crawl Speeds

Application ID#: _____

Bridge Structure ID#: _____

Motor Carrier Name: _____

Applicant's Signature: _____ Date: _____



- A. PRIVATE ESCORT VEHICLE.** The hauler provides this vehicle, equipped with a flashing or revolving yellow light. When Vehicle D is not used, this vehicle shall be used to stop the approaching traffic at a distance of at least 350 feet before the bridge, and then release the approaching traffic only after Vehicle B has cleared the bridge and is in the normal travel lane. In order to stop traffic, it may be necessary to have someone use a red flag.
- B. SUPERLOAD OR NON-SUPERLOAD.** This vehicle shall be equipped with two flashing or revolving yellow lights, one on the front and one on the back. This vehicle should be positioned in either the center or the right side of the roadway in accordance with the permit. When Vehicle D is used, this vehicle should maintain a minimum distance behind Vehicle A of 200 feet, or one bridge span length, whichever is greater.
- C. SHADOW VEHICLE.** The hauler provides this vehicle, equipped with a flashing or revolving yellow light and a sign with the message "DO NOT PASS." The sign may be either a variable message sign (VMS) or a static sign. If a VMS is used, it should flash the message, when applicable. If a static sign is used, the sign shall be removed when it is not applicable. This vehicle should be positioned either in the center or on the right side of the roadway similar to Vehicle B. When on the bridge, the vehicle should maintain a minimum distance behind Vehicle B of 200 feet, or one bridge span length, whichever is greater.
- D. POLICE VEHICLE.** This vehicle is only required for superloads. When used, the vehicle shall be a marked vehicle with activated revolving or flashing lights. The trooper in this vehicle should stop approaching traffic at a distance of at least 350 feet before the bridge. The trooper should release the approaching traffic only after Vehicle B has cleared the bridge and is in the normal travel lane.

NOTES:

1. If a VMS is used, it shall be of a type approved by the Department and listed in Publication 35, and it shall be capable of displaying two lines of message with minimum 14-inch legend. If a static sign is used, it shall use minimum 12-inch, black legend on a fluorescent yellow retroreflective sheeting material.
2. The hauler shall provide CB radios in all vehicles for communication, including one for Vehicle D. One of the operators shall also have a cellular telephone for emergency purposes.

Bicycle Tolerable Shoulder Rumble Strips (BTSRS) (For Non-Interstate and Non-Expressways Use)

Responsibilities

District Safety Engineer is the process owner.

District Pedestrian/Bicycle Coordinator shall jointly review the areas to be targeted.

Guidelines for Use

Consider Bicycle Tolerable Shoulder Rumble Strips (BTSRS) on the following locations and under following conditions:

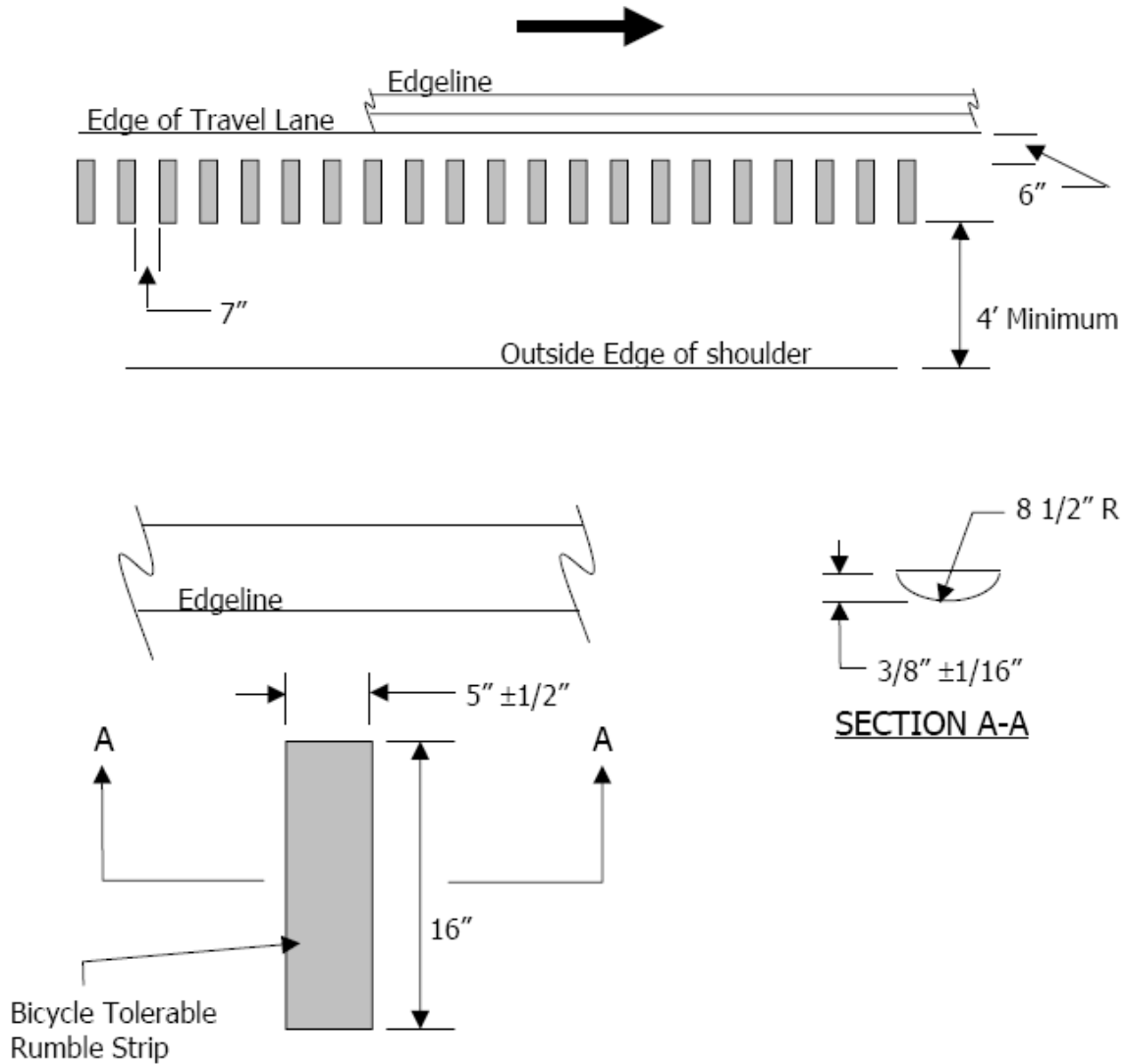
1. The purpose of BTSRS is to reduce run-off-the-road crashes on highways (except interstates and expressways).
2. Consider BTSRS for the following types of highways (except interstate and expressways):
 - Where the roadway lane width is 11 feet or greater, 6 feet or more of paved shoulder and the posted speed is 55 mph or greater, the BTSRS shall be installed on the shoulder as shown in Typical Drawing Detail #1.
 - Where the roadway lane width is 11 feet or greater, 6 feet or more paved shoulder and the posted speed is less than 55 mph, the BTSRS shall be installed on the shoulder as shown in Typical Drawing Detail #2.
 - If the shoulder width is less than 6 feet in any case, consider the Edge Line Rumble Strips (ELRS), outlined in [Appendix 11E](#).
3. The paved shoulder must be equal in smoothness to that of the adjacent travel lane.
4. Installing BTSRS on bituminous pavement requires an ID-2 or ID-3 surface with BCBC base or better.
5. If it is desired to retrofit BTSRS on existing pavement, the pavement and shoulder should be in sufficiently good condition, as determined by the District, to effectively accept the milling process without raveling or deteriorating the pavement. Otherwise both the pavement and shoulders need to be upgraded prior to milling BTSRS.
6. Do not install BTSRS on bridge decks.
7. BTSRS are to be broken for intersections. Also consider breaking for driveways according to engineering judgment. When breaking BTSRS pattern, discontinue BTSRS 25 feet from the Point of Curvature of any such highway or driveway (refer to Typical Detail #3).

8. Coordinate the milling of BTSRS with all necessary project phases. Do not mill the BTSRS until all appropriate construction phases are completed.
9. Take into consideration potential noise impacts when contemplating the installation of BTSRS in residential or urban areas.
10. As part of multi-modal transportation system planning, consult the District Pedestrian/Bicycle Coordinator where BTSRS are being planned for installation, and determine if the District Coordinator has any concerns. These concerns may include Bicycle PA routes, other local bike routes, Adventure Cycling Association, National Bike routes in Pennsylvania, proposed bike routes from MPO/LDD regional plans, potential ADA violations and others.

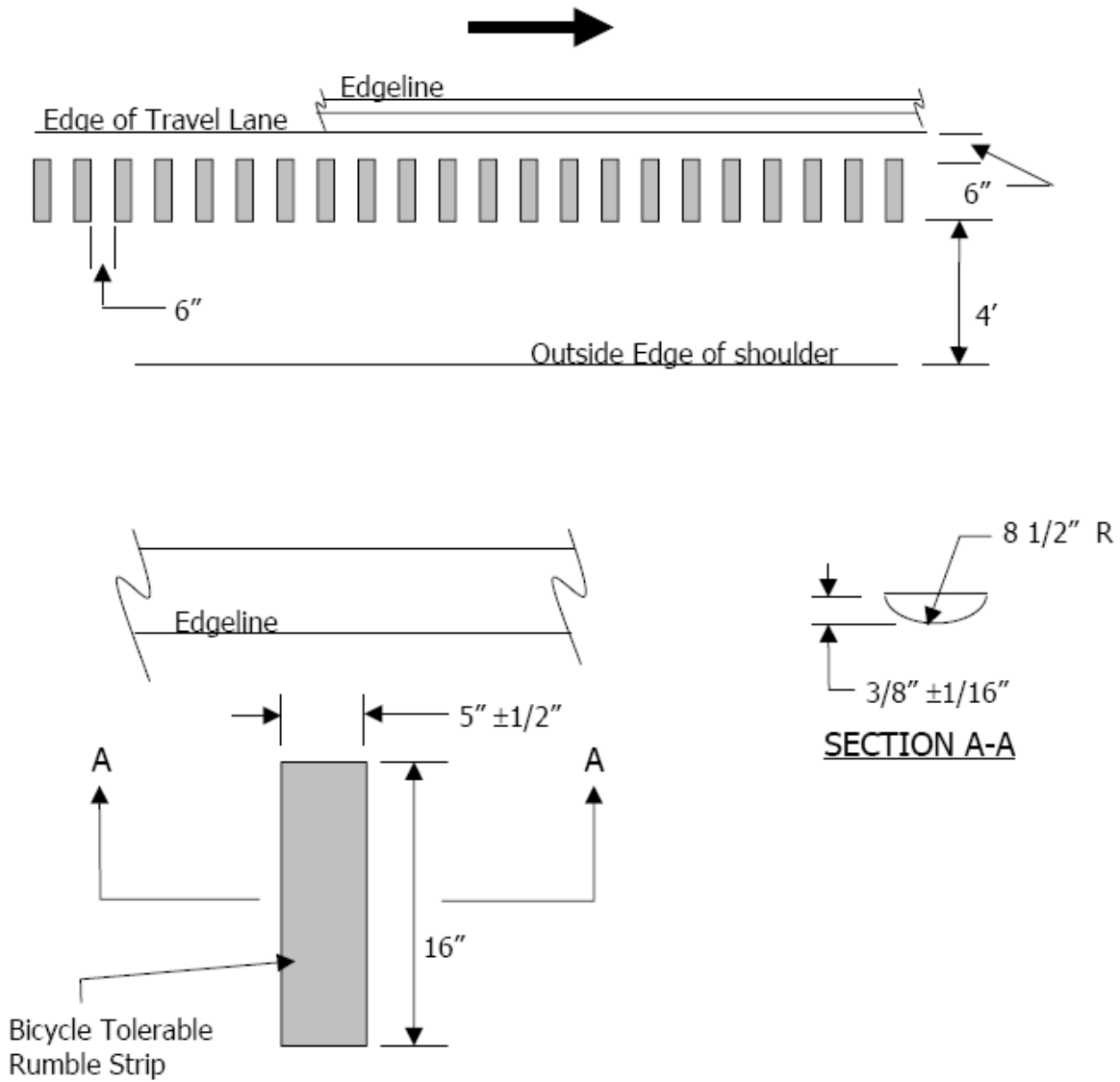
Design Deviations

Deviation from the above specifications and guidelines may be considered by the District; however, they must be approved by the Bureau of Highway Safety and Traffic Engineering prior to being implemented.

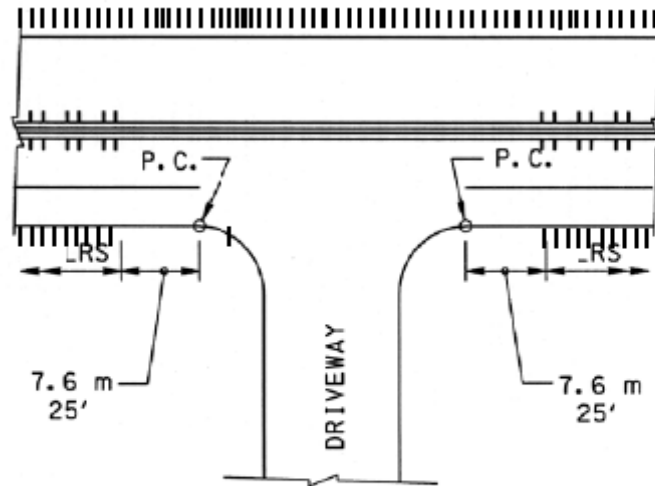
**BTSRS Detail (55 mph or Higher Posted Speed)
Typical Drawing Detail #1**



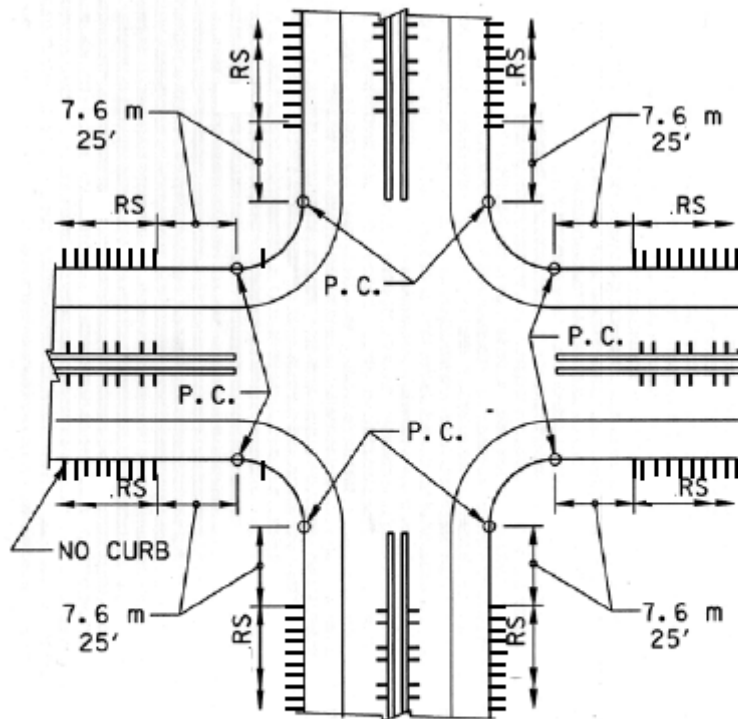
**BTSRS Detail (Less than 55 mph Posted Speed)
Typical Drawing Detail #2**



Bicycle Tolerable Rumble Strips Typical Drawing Detail #3



TYPICAL DRIVEWAY DETAIL



TYPICAL INTERSECTION DETAIL

Milled Centerline Rumble Strips (CLRS) (For Non-Interstate and Non-Expressways Use)

Responsibilities:

District Safety Engineer is the process Owner.

Guidelines for Use:

1. The purpose of milled Center Line Rumble Strips (CLRS) is to reduce the occurrence of head-on and/or sideswipe crashes on undivided two-lane or four-lane highways.
2. Consider CLRS on the following locations and under following conditions:

Roadway Description	Typical Drawing Detail
Roadway with 12 feet or greater lane width and minimum of 3 feet of paved shoulder.	Detail #1
Roadway with 11 feet lane width and minimum of 3 feet of paved shoulder.	Detail #1 or Detail #2
Roadway with 11 feet lane width and less than 3 feet of shoulder or no shoulder.	Detail #2
Roadway with 10 feet lane width with or without shoulder	Detail #2
Roadway with less than 10 feet lane width	Consult BHSTE

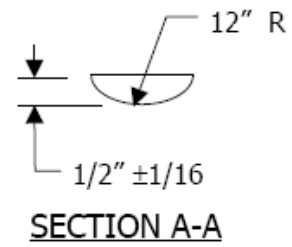
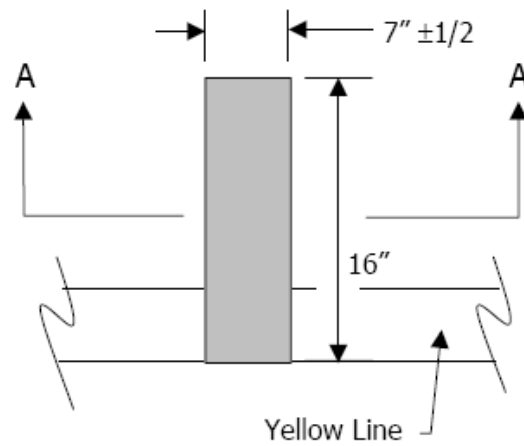
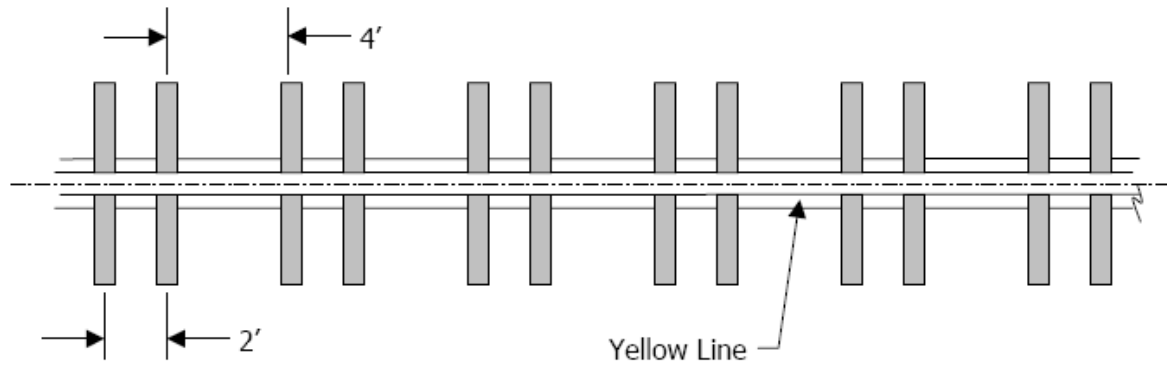
3. Milled centerline rumble strips (CLRS) are for use on bituminous pavement.
4. Installing CLRS on bituminous pavement requires an ID-2 or ID-3 surface with BCBC base or better.
5. If it is desired to retrofit CLRS on existing pavement, the pavement should be in sufficiently good condition, as determined by the District, to effectively accept the milling process without raveling or deteriorating. Otherwise the pavement needs upgraded prior to milling any desired CLRS
6. CLRS should not be installed on existing concrete pavements with overlay less than 2.5 inches in depth.
7. Do not install CLRS on bridge decks.
8. CLRS may be installed in passing zones where deemed appropriate by District safety personnel. Consider reducing depth of cut to 3/8" in areas where passing is permitted.

- If CLRS are being discontinued for a passing zone, use engineering judgment as to where to terminate CLRS in advance of a passing zone.
9. CLRS are to be broken for intersections. Also consider breaking for driveways according to engineering judgment. When breaking CLRS pattern, discontinue CLRS 25 feet from the Point of Curvature of any such highway or driveway (refer to Typical Detail #3).
 10. Coordinate the milling of CLRS with all necessary project phases. Do not mill the CLRS until all appropriate construction phases are completed.
 11. Coordinate the milling of CLRS with traffic line painting operations a) to avoid milling newly applied traffic lines and b) to install new yellow centerlines within two weeks of CLRS completion.
 12. Consult the Bureau of Highway Safety and Traffic Engineering before installing CLRS on highways with travel lane widths that are less than 10 feet.
 13. Take into consideration potential noise impacts when contemplating the installation of CLRS in residential or urban areas.

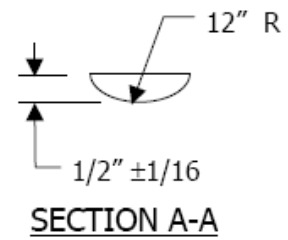
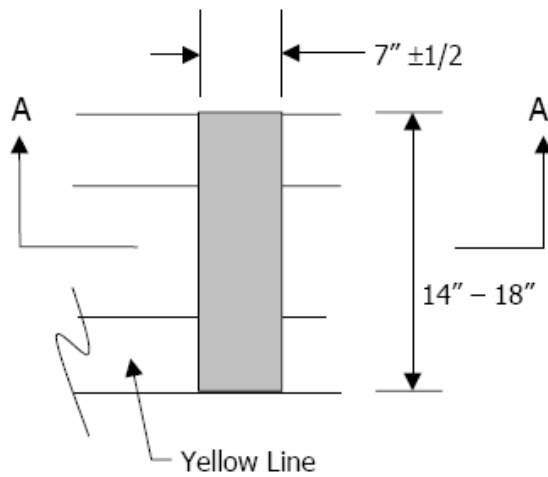
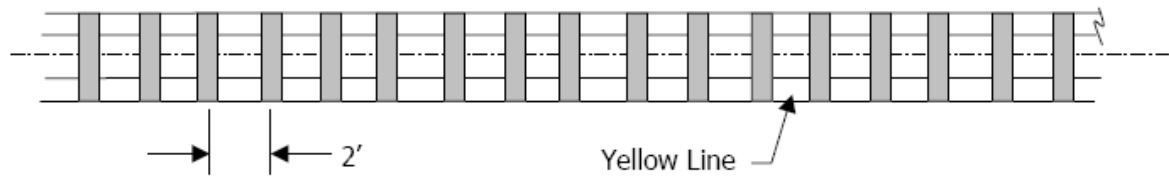
Design Deviation

Deviation from the above specifications and guidelines may be considered by the district; however, they must be approved by the Bureau of Highway Safety and Traffic Engineering prior to being implemented.

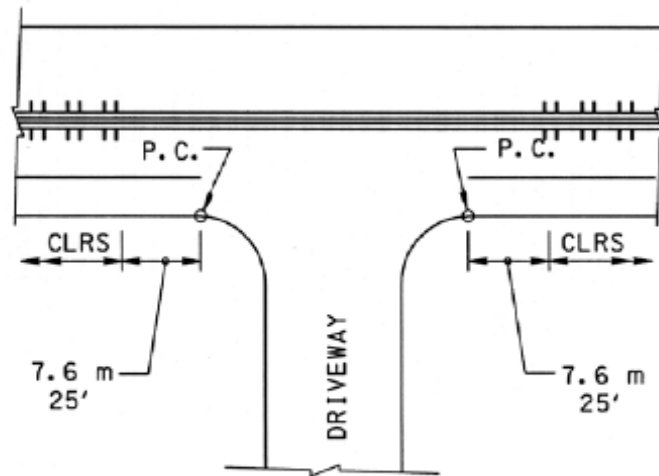
Milled Centerline Rumble Strips Typical Drawing Detail #1



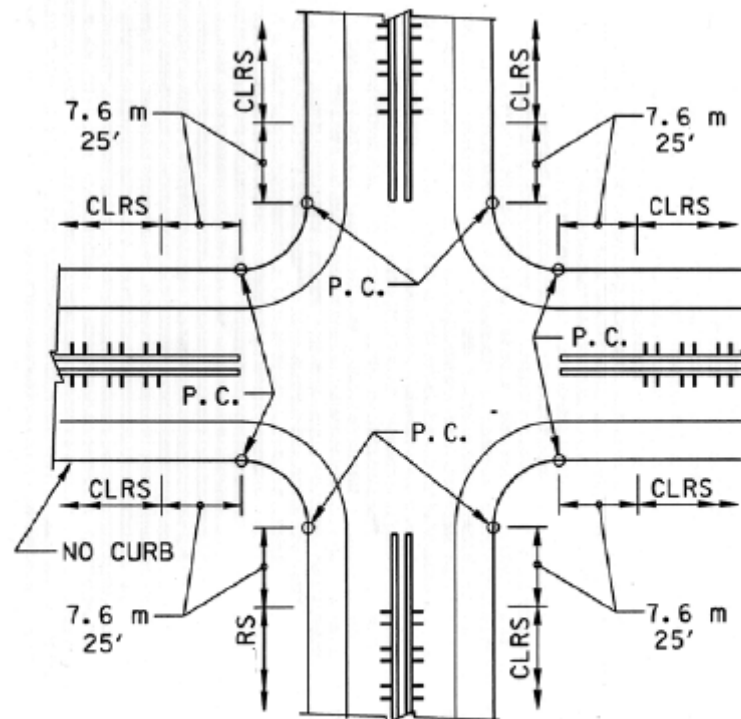
Milled Centerline Rumble Strips Typical Drawing Detail #2



Center Line Rumble Strips Typical Drawing Detail #3



TYPICAL DRIVEWAY DETAIL



TYPICAL INTERSECTION DETAIL

Edge Line Rumble Strips (ELRS) (For Non-Interstate and Non-Expressways Use)

Responsibilities:

District Safety Engineer is the process owner.

District Pedestrian/Bicycle Coordinator and the District Safety Engineer shall jointly review the areas to be targeted.

Guidelines for Use:

Consider milled Edge Line Rumble Strips (ELRS) on the following locations and under the following conditions:

The purpose of ELRS is to reduce run off the road crashes on highways (except interstates and expressways).

Consider ELRS for the following types of two-to-four lane highways (except interstate and expressways):

- On highways with 11 feet or greater lane width and 4 to 6 feet of paved shoulder, ELRS shall be installed on the edge of the roadway as shown in Typical Drawing Detail #1.
- If the shoulder width is greater than or equal to 6 feet, consider the Bicycle Friendly Shoulder Rumble Strips shown in [Appendix 11B](#).

If there is concern with the pavement joint between the roadway and the shoulder, district may consider the following options:

- Where shoulder width is between 5-6 feet, offset ELRS 2 to 4 inches from the joint into the shoulder surface.
- Where shoulder width is less than 5 feet, district may offset ELRS 2 to 4 inches from the joint into the travel lane surface.

The ELRS shall be discontinued 50 feet before and after adjacent guiderail, where the face of the guiderail is located less than 4 feet from the edge line of the roadway.

The paved shoulder and the adjacent travel lane should be of equal smoothness.

Installing ELRS on bituminous pavement requires an ID-2 or ID-3 surface with BCBC base or better.

If it is desired to retrofit ELRS on existing pavement, the pavement and shoulder should be in sufficiently good condition, as determined by the District, to effectively accept the milling process without raveling or deteriorating the pavement. Otherwise both the pavement and shoulders need to be upgraded prior to milling any desired ELRS.

Do not install ELRS on bridge decks.

ELRS are to be broken for intersections. Also, consider breaking for driveways according to engineering judgment. When breaking ELRS pattern, discontinue ELRS 25 feet from the Point of Curvature of any such highway or driveway (refer to Typical Detail #2).

Coordinate the milling of ELRS with all necessary project phases. Do not mill the ELRS until all appropriate construction phases are completed.

Coordinate the milling of ELRS with traffic line painting operations a) to avoid milling newly applied traffic lines and b) to install new white edge lines within 2 weeks of ELRS completion.

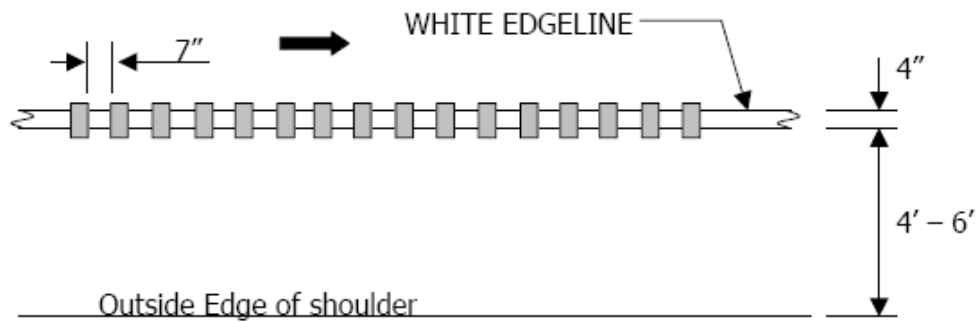
Take into consideration potential noise impacts when contemplating the installation of ELRS in residential or urban areas. Do not install ELRS on the inside of moderate to sharp curves that are in the immediate vicinity of any residence.

As part of multi-modal transportation system planning, consult the District Pedestrian/Bicycle Coordinator where ELRS are being planned for installation, and determine if the District Coordinator has any concerns. These concerns may include Bicycle PA Routes, other local bike routes, Adventure Cycling association, National Bike Route segments in Pennsylvania, proposed bike routes from MPO/LDD regional plans, potential ADA violations and others.

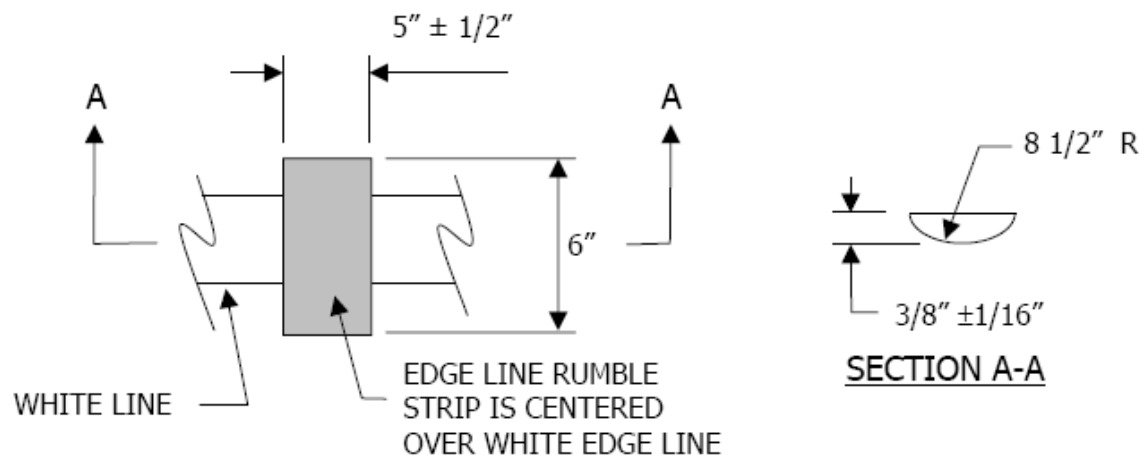
Design Deviations

Deviation from the above specifications and guidelines may be considered by the District; however, they must be approved by the Bureau of Highway Safety and Traffic Engineering prior to being implemented.

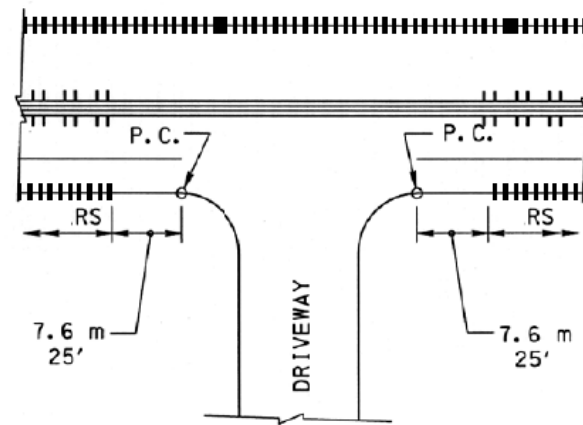
Edge Line Rumble Strips Typical Drawing Detail #1



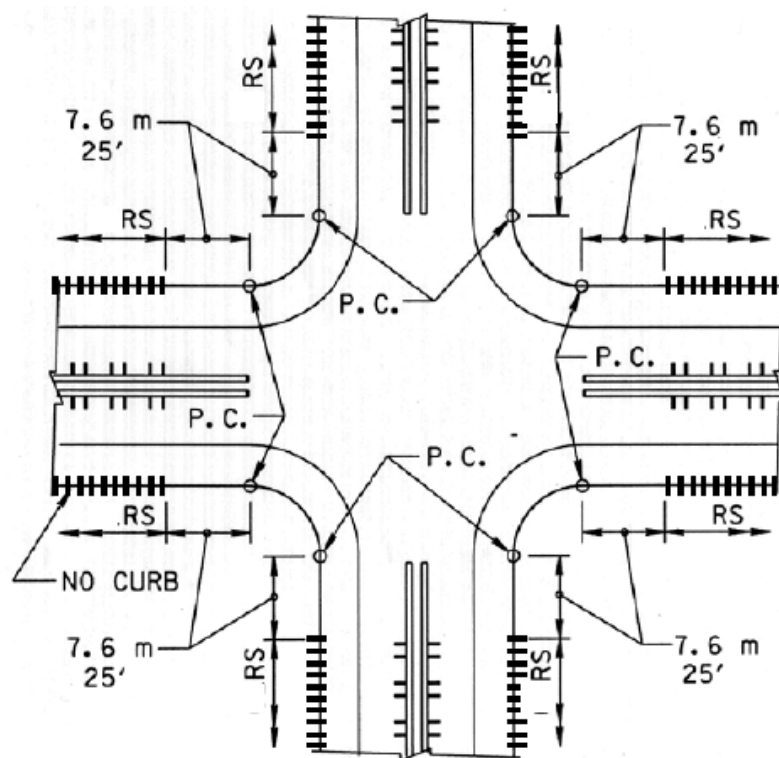
TYPICAL PLAN VIEW FOR MILLED EDGE
LINE RUMBLE STRIP PATTERN



Edge Line Rumble Strips Typical Drawing Detail #2



TYPICAL DRIVEWAY DETAIL



TYPICAL INTERSECTION DETAIL



Resolution of

BY:		
		(Title)
		(Title)
		(Title)

I certify the foregoing to be a true and correct copy of the Resolution of _____, _____ County,
adopted at a duly convened meeting on _____.

(Title)

(SEAL)

The above highway(s) is (are) hereby designated as a Snowmobile Road(s) in accordance with the terms and conditions expressed in the foregoing Resolution.

Date _____

By: _____
(District Executive)

Engineering District _____

For: _____
(Secretary of Transportation)

**Figure 1. Warrant for left turn lanes on two-lane roadways (speeds to 35 mph, unsignalized and
signalized intersections)**
(L = % Left Turns in Advancing Volume)

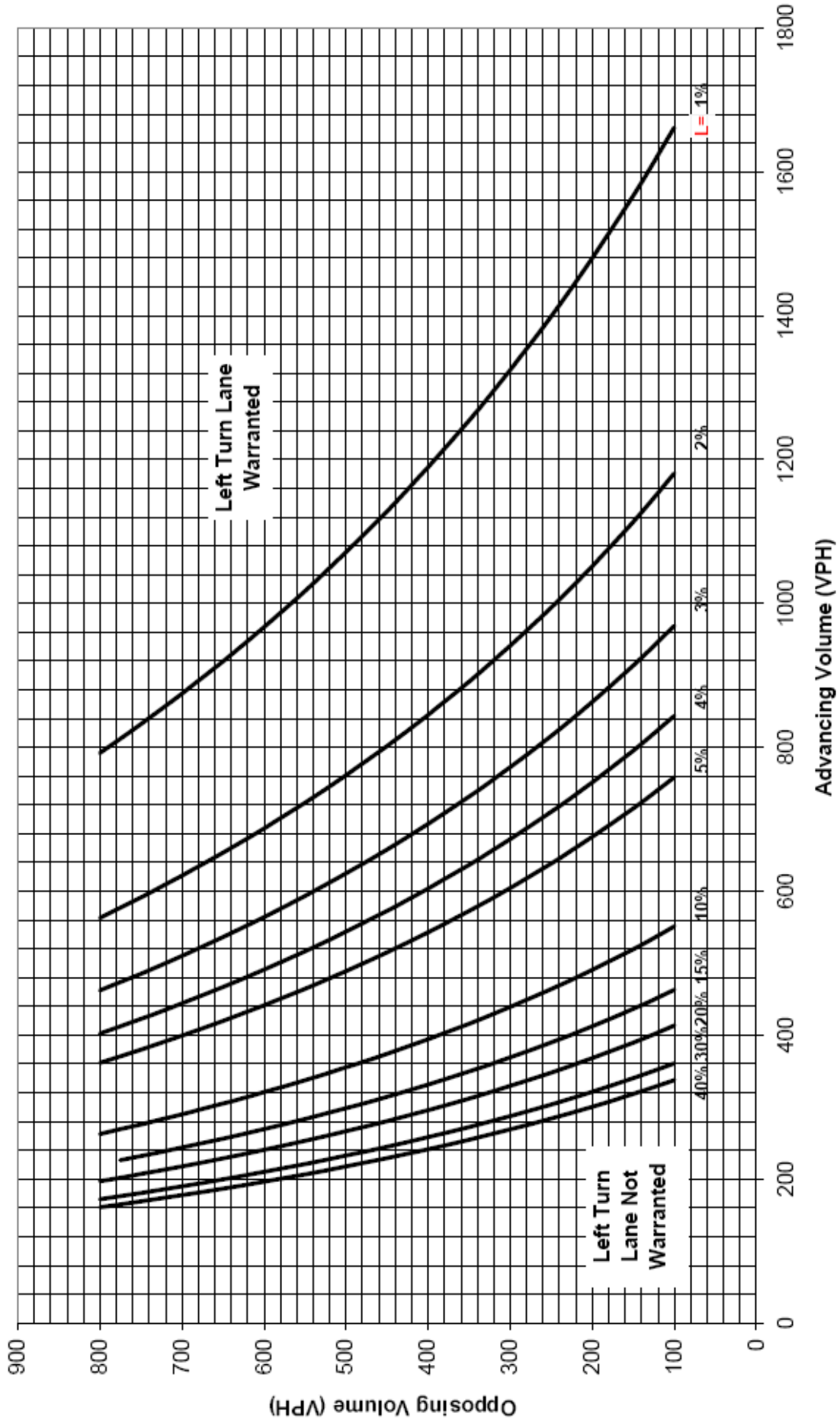


Figure 2. Warrant for left turn lanes on two-lane highways (40 mph speed, unsignalized and
signalized intersections)
(L = % Left Turns in Advancing Volume)

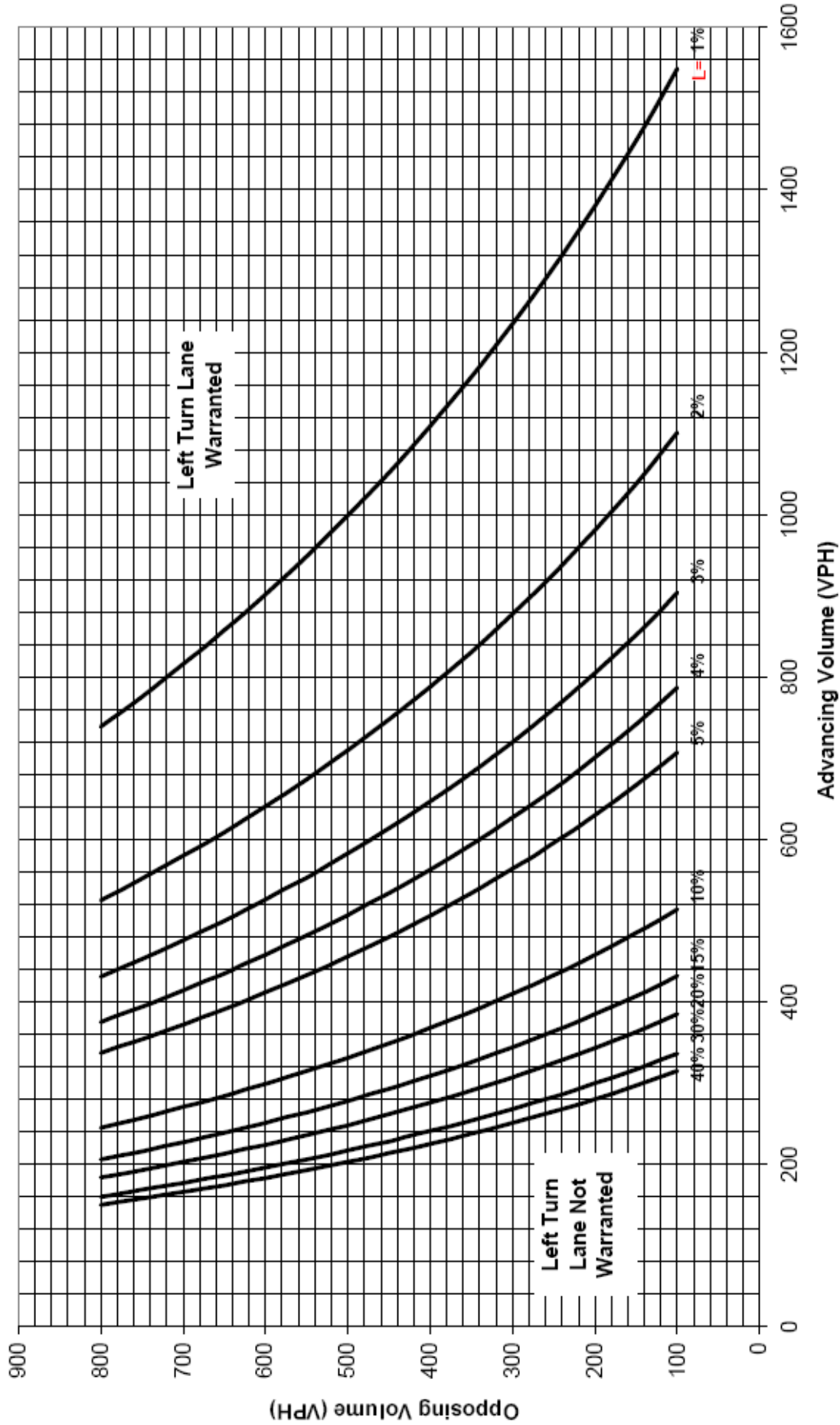


Figure 3. Warrant for left turn lanes on two-lane highways (45 mph speed, unsignalized and
signalized intersections)
(L = % Left Turns in Advancing Volume)

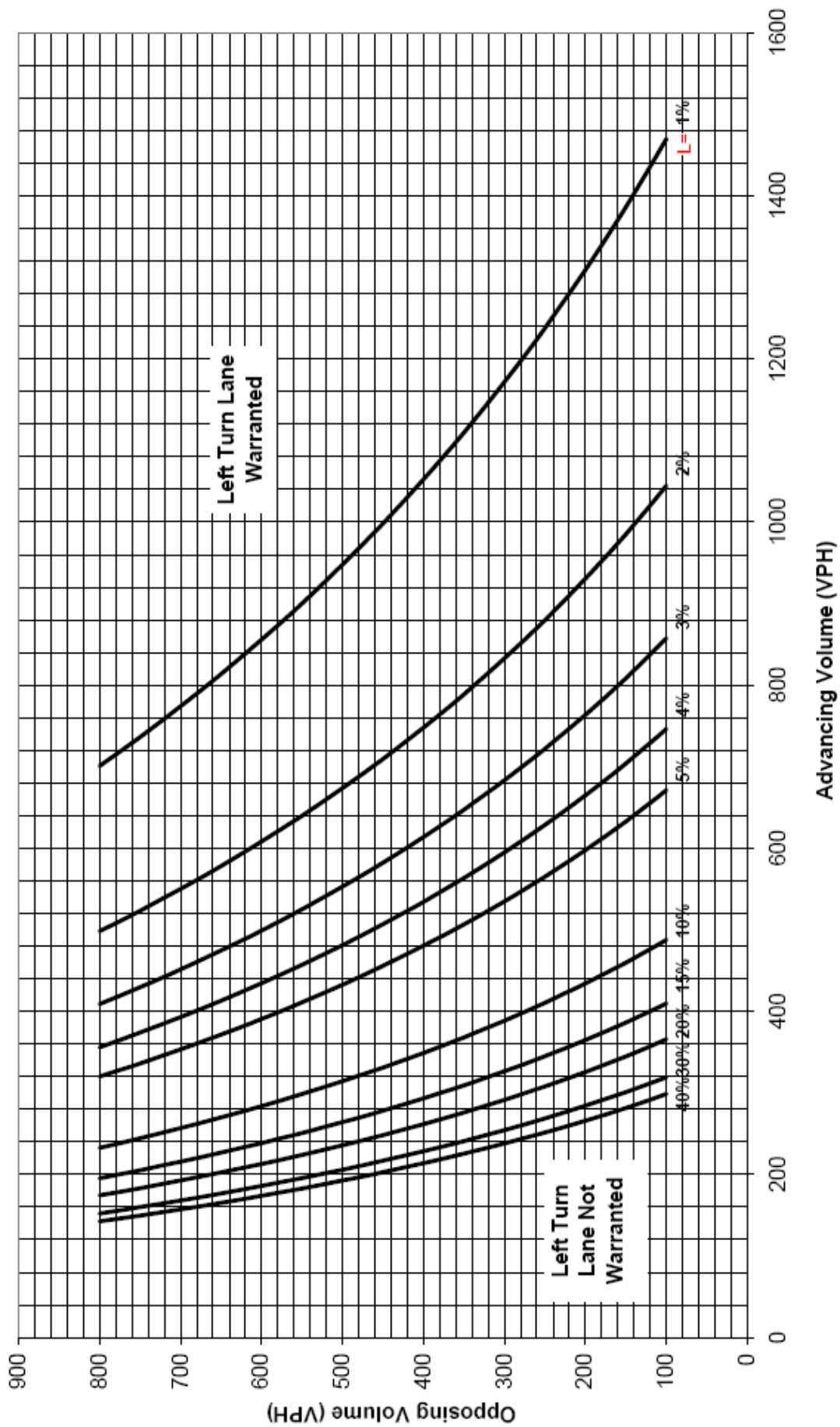
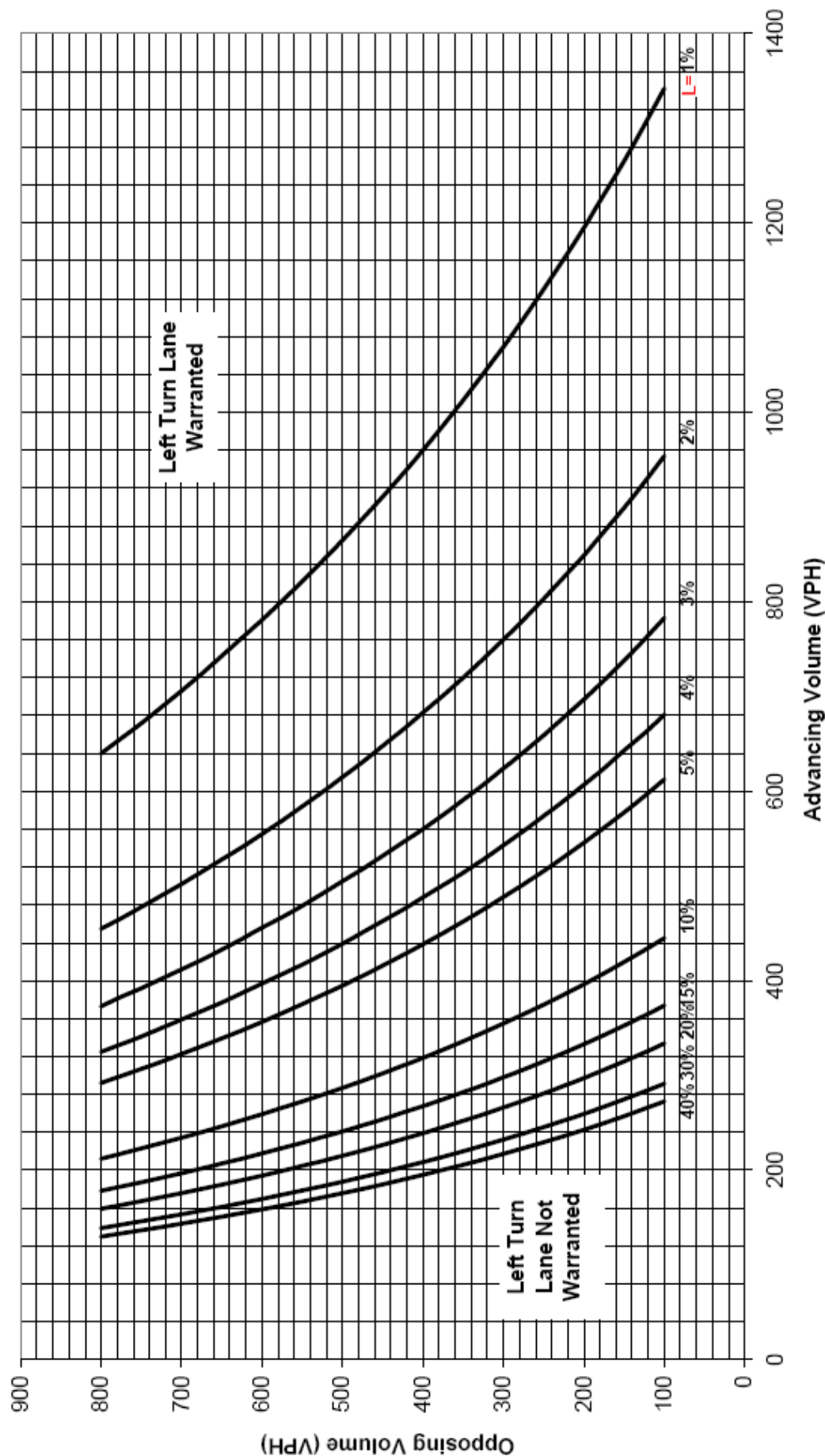


Figure 4. Warrant for left turn lanes on two-lane highways (50 mph speed, unsignalized and signalized intersections)
 (L = % Left Turns in Advancing Volume)



**Figure 5. Warrant for left turn lanes on two-lane highways (55 mph speed, unsignalized and
signalized intersections)**
(L = % Left Turns in Advancing Volume)

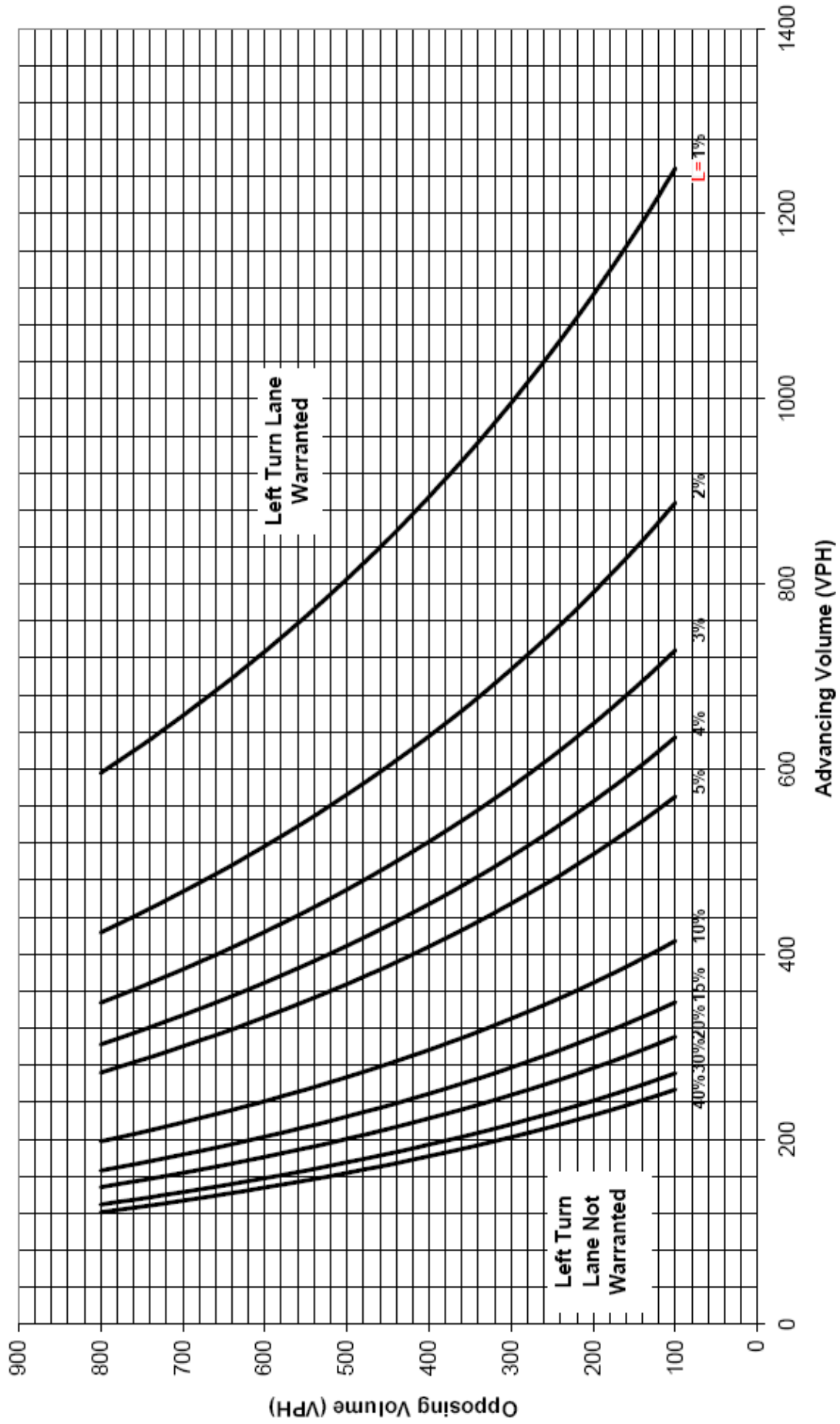


Figure 6. Warrant for left turn lanes on two-lane highways (60 mph speed, unsignalized and
signalized intersections)
(L = % Left Turns in Advancing Volume)

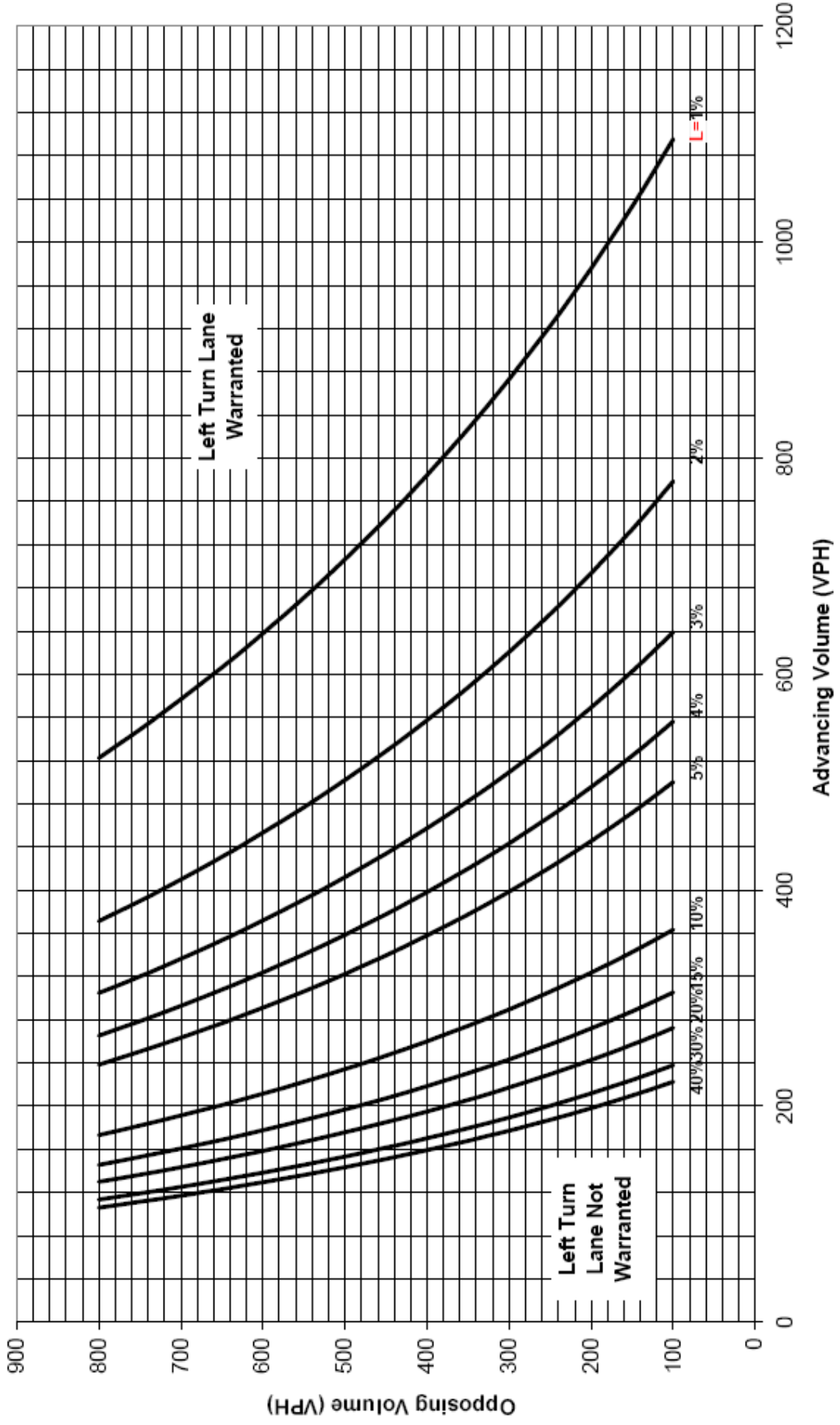


Figure 7. Warrant for left turn lanes on four-lane, undivided highways (unsignalized and signalized intersections)
(L = % Left Turns in Advancing Volume)

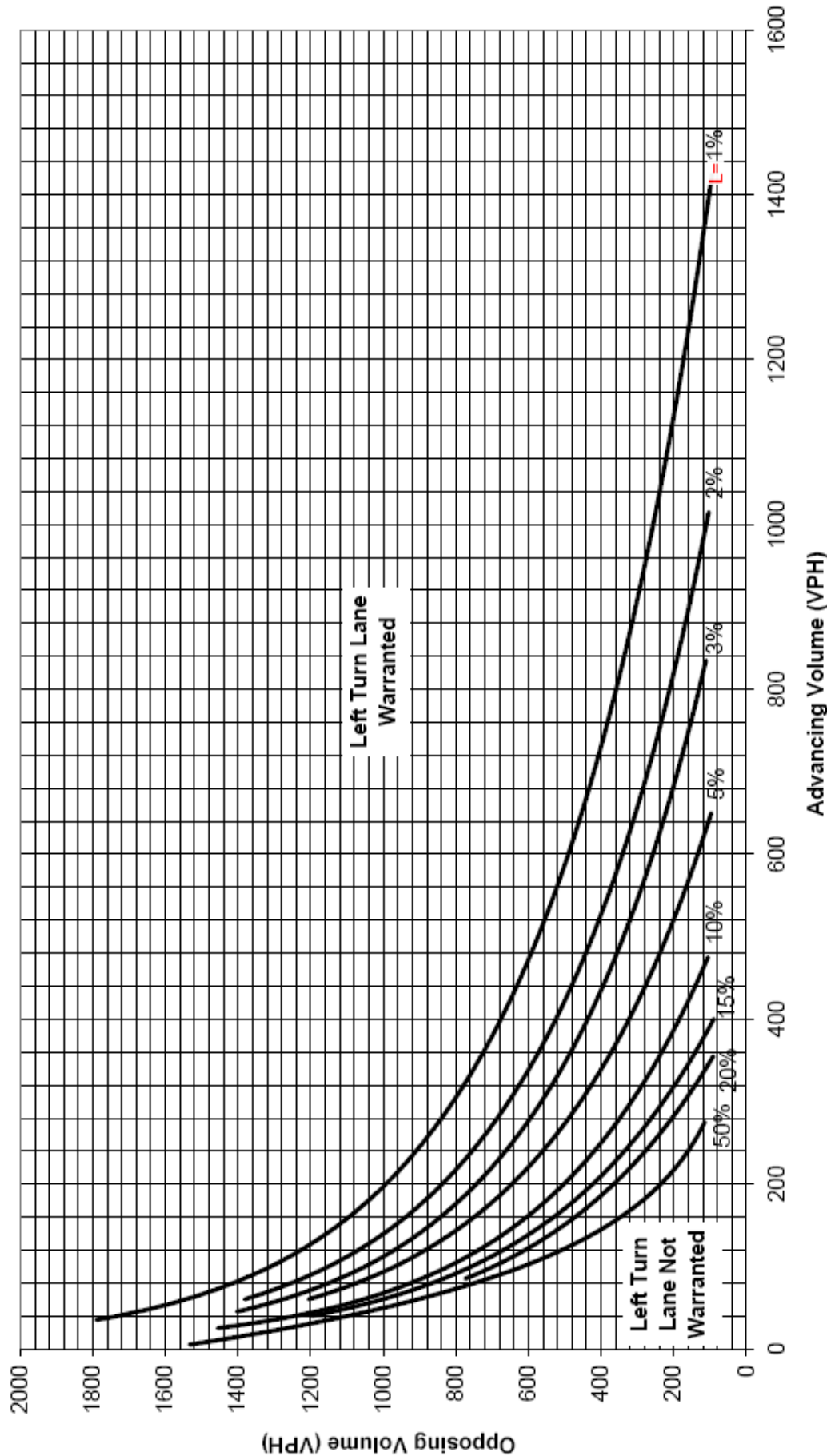


Figure 8. Warrant for left turn lanes on four-lane, divided highways (unsignalized and signalized intersections)
(L = % Left Turns in Advancing Volume)

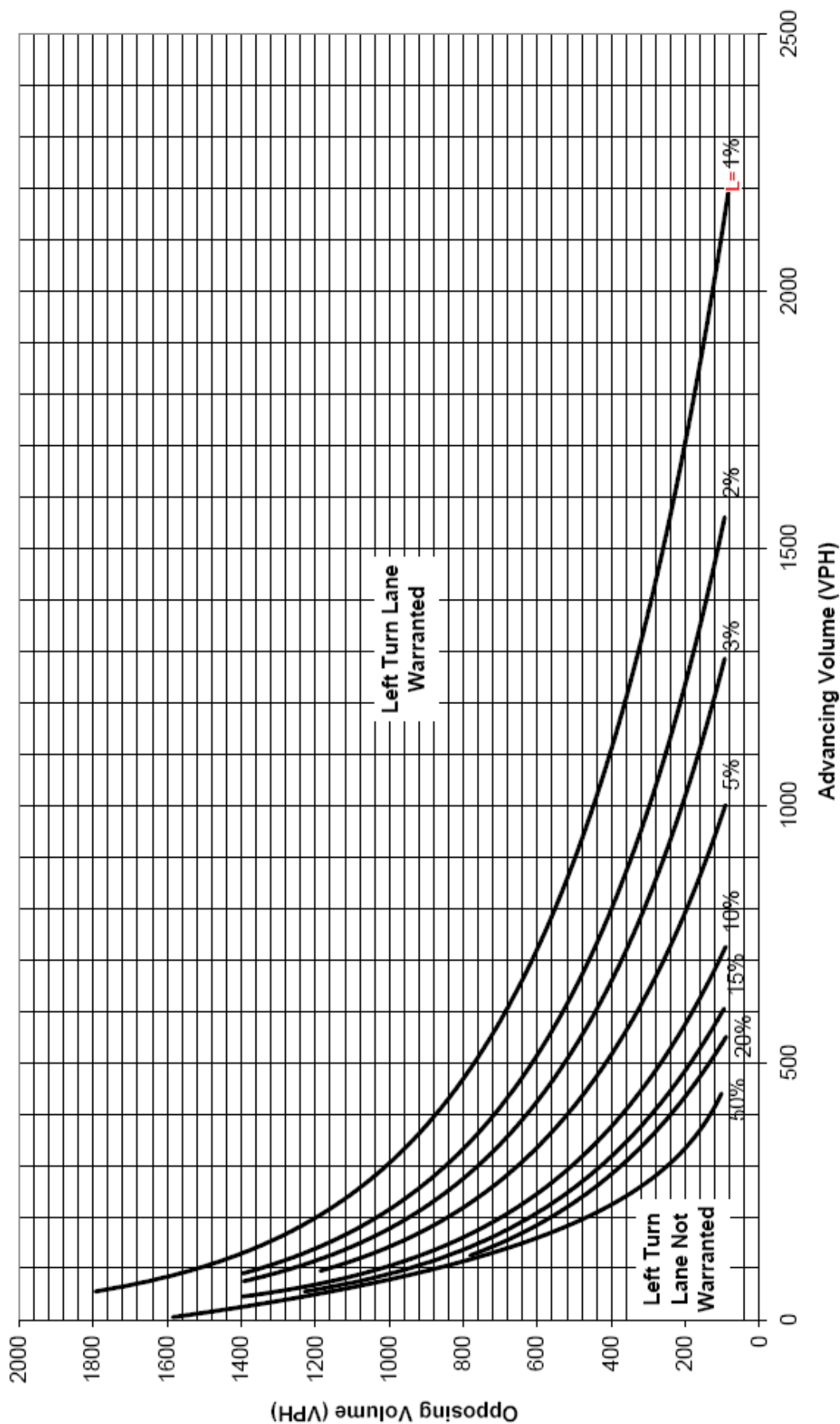


Figure 9. Warrant for right turn lanes on two-lane roadways (40 mph or lower speeds, unsignalized and signalized intersections)

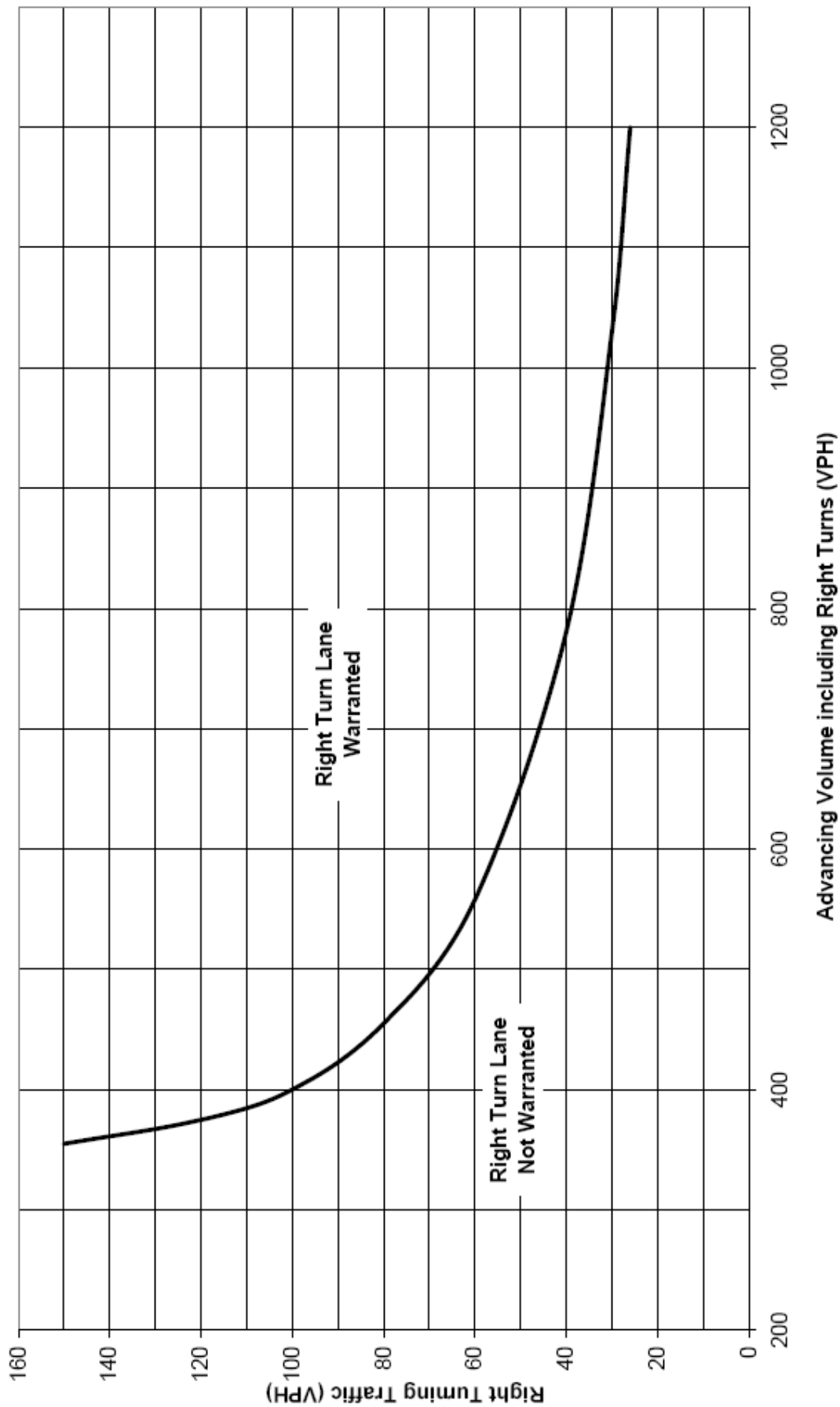


Figure 10. Warrant for right turn lanes on two-lane roadways (45 mph or greater speeds, unsignalized and signalized intersections)

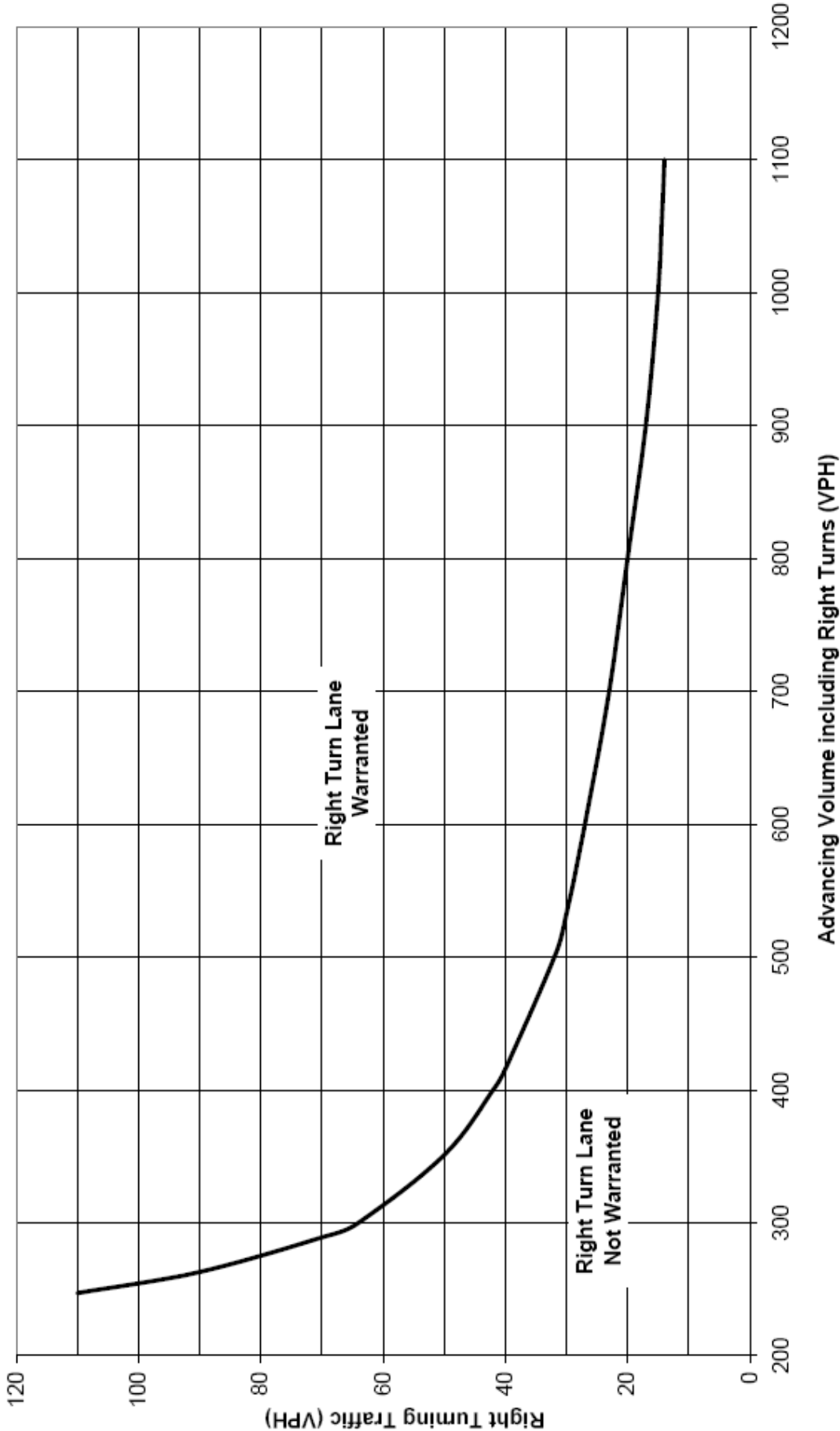


Figure 11. Warrant for right turn lanes on four-lane roadways (40 mph or lower speeds, unsignalized and signalized intersections)

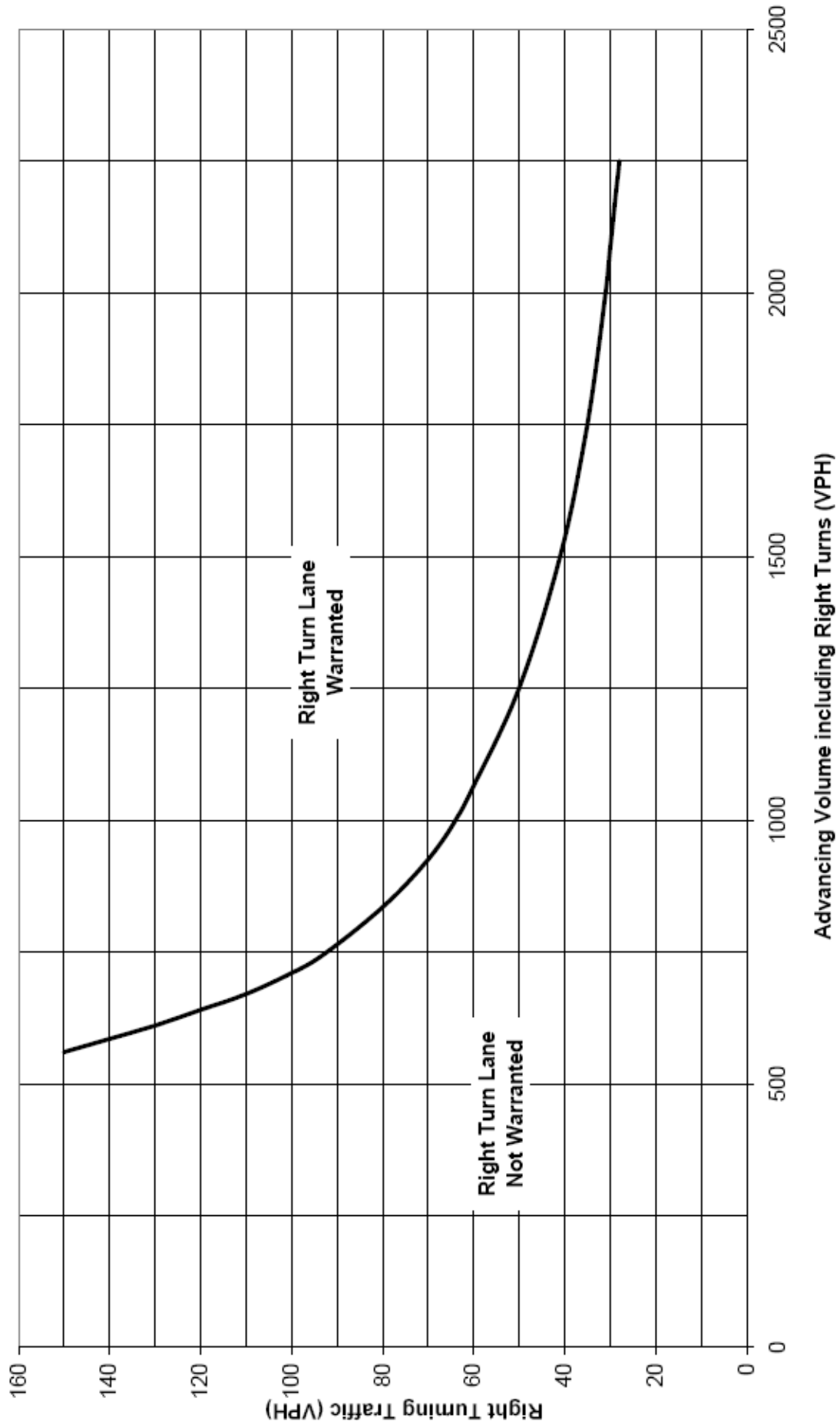
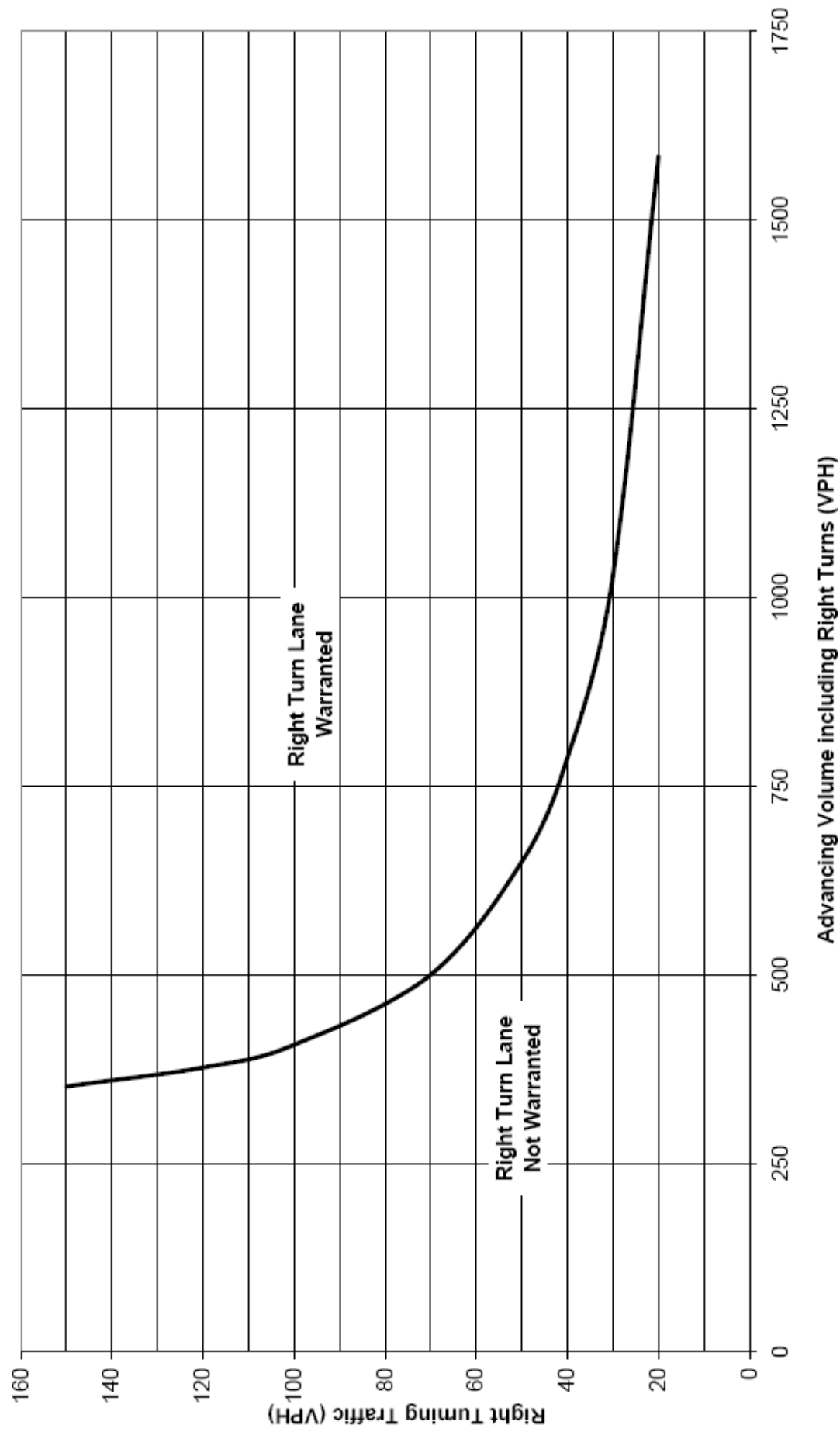


Figure 12. Warrant for right turn lanes on four-lane roadways (45 mph or greater speeds, unsignalized and signalized intersections)



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pennsylvania
DEPARTMENT OF TRANSPORTATION

**Publication 46
Chapter 12**

Traffic Engineering Software

Pennsylvania Department of Transportation
Chapter 12 of Publication 46
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12.1 General

12.1.1 Purpose

Various software packages are available for use by traffic engineers. Traffic analysis software packages are used to optimize traffic flows and capacity, or to simulate traffic flow. In general, the definition of simulation is the *“dynamic representation of some part of the real world achieved by building a computer model and moving it through time.”*

The purpose of this chapter is to summarize the Department’s position on the use of traffic engineering software.

12.1.2 Traffic Resources, Education, and Computing Support (TRECS) Group

The Department established the Traffic Resources, Education, and Computing Support (TRECS) Group to address computer hardware, software, and training issues as they pertain to the District Traffic Units and the Bureau of Highway Safety and Traffic Engineering (BHSTE). In addition, the TRECS Group procures needed traffic engineering reference materials. The TRECS Group is composed of representatives of every District Traffic Unit and BHSTE. The Group meets on a regular basis, and it has dedicated funding within BHSTE’s budget to fulfill its responsibilities.

One of the TRECS Group’s objectives is the review and evaluation of traffic engineering software packages to determine which software will be used and supported by the Department. Supported software means that the Department will continually purchase software upgrades for use by the Engineering Districts and Central Office, and that training has been, and will continue to be, provided for appropriate Department personnel.

12.2 Specific Software

12.2.1 Supported Software

The TRECS Group supports the following traffic engineering software:

- Highway Capacity Software (HCS)
- QuickZone
- Synchro
- SimTraffic
- TRANSYT-7F
- SignCAD

12.2.2 Required Use by Department and Consultants

Applicable work done by the Department, or by engineering consultants making submissions for Department review and approval for Department projects or Department Highway Occupancy Permit projects, should use the current version of one of the supported software packages, unless directed otherwise in writing by the Department.

Level-of-service (LOS) calculations and results should be completed using the standards established in the Highway Capacity Manual (HCM). This means LOS calculations should be completed and reported using HCS or the HCM calculation method in Synchro. In other words, LOS calculations should not be submitted using the Synchro calculation method unless directed otherwise in writing by the Department.